Exploring Social Metacognition

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This is where you will normally thank your advisor, colleagues, family and friends, as well as funding and institutional support. In our case, we will give our praises to the people who developed the ideas and tools that allow us to push open science a little step forward by writing plain-text, transparent, and reproducible theses in R Markdown.

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Ulrik Lyngs
Linacre College, Oxford
2 December 2018

Abstract

This *R Markdown* template is for writing an Oxford University thesis. The template is built using Yihui Xie's `bookdown` package, with heavy inspiration from Chester Ismay's `thesisdown` and the `OxThesis` LaTeX template (most recently adapted by John McManigle).

This template's sample content include illustrations of how to write a thesis in R Markdown, and largely follows the structure from this R Markdown workshop.

Congratulations for taking a step further into the lands of open, reproducible science by writing your thesis using a tool that allows you to transparently include tables and dynamically generated plots directly from the underlying data. Hip hooray!

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List of Abbreviations

1-D, 2-D . . . One- or two-dimensional, referring **in this thesis** to spatial dimensions in an image.

Otter . . . . . One of the finest of water mammals.

Hedgehog . . . Quite a nice prickly friend.

1
Introduction

!TODO[Style notes]:

- Graphs:
  - Continuous axes which show theoretical limits should have small expand, axes which do not show those limits should have arrowheads from broken_axis_*

!TODO[update evo models to use this terminology]

$a_{i,t}$ - advice agent i receives at time t

$b_{i,t}$ - bias of agent i at time t

$c_{i,t}^{I/A/F}$ - agent i's confidence in their initial/advisory/final estimate at time t

$e_{i,t}^{I/A/F}$ - agent i's initial/advisory/final estimate at time t

i, j - agent identifiers

$s_{i,t}$ - sensitivity of agent s at time t

u_i - fitness/utility of agent i

v - true value in the world

λ - learning rate

$\omega_{i,t}$ - weighting for agent i's estimate at time t

This thesis attempts to determine whether individual psychological processes in the seeking and taking of advice are sufficient to produce entrenched biases at the network level. The work includes empirical investigation into the individual psychological processes underlying advice seeking and advice taking using behavioural experiments; computational modelling and behavioural experimentation exploring the effects of contextual factors on advice-taking; agent-based computational modelling of the effects interactions between agents with psychological advisor evaluation processes that produce biases in assimilation of information and source selection; and a comparison between the structures of networks produced in these models and those naturally occurring social networks. The organisation is as follows: this introduction establishes the core concepts invoked in this thesis, and describes their treatment in the literature; the following sections each include a short chapter on the specific question addressed and the techniques used, a detailed description of the work conducted, and a short discussion of the conclusions drawn from the work; and a final section offers broader conclusions arising as a consequence of the presented work, alongside some suggestions for related research.

Overview

The first line of behavioural experiments report investigates whether people decide which advisors to consult using internal signals where external feedback is unavailable, extending a previous finding concerning the influence of advice (Pescetelli and Yeung 2021) into the domain of advisor choice. While little support is found for the metacognitive mechanisms underpinning the influence of advice, the frequency of agreement is shown to be a good indicator of advisor choice. Next, the effects of advisor agreement are explored in terms of influence using a different decision-making domain; a date estimation rather than perceptual decision task. Again, people denied objective feedback on the quality of advice are shown to attend to agreement. The work above demonstrates how the influence of advice depends upon the reputation of an advisor built up over time.

These results highlight the heterogeneity of individuals' tendencies to change their trust in advisors and select advisors on the basis of that trust. Agent-based modelling is used to explore the effects of this heterogeneity on how the whole population's opinions and advice-seeking behaviour evolve over time. This modelling work demonstrates that heterogeneity may stabilise these networks, slowing the emergence of polarised echo chamber-like structures.

In a second line of experiments, reputations of advisors are established at the outset and the responses to different pieces of advice are explored. Evolutionary modelling demonstrates that the presence of a few bad actors in a population can mean that distrusting all advice from all sources a little is adaptive. The evolutionary models are extended to show that this kind of general distrust, known as egocentric discounting, emerges as adaptive even where none of the agents deliberately mislead others, and even where all agents are equally skilled and fully cooperative.

Example behavioural experiments are reported which support the implications from the evolutionary models. Participants considered both the properties of advice and the properties of advisors, placing less trust in advice which appeared suspicious or which came from a suspicious source.

Advice

Advice is broadly defined as information which comes from a social source. Advice is therefore different from other sources of information in that it is the result of (at least) mental processing of other information. In some cases, it may additionally include discussions among different group members (e.g. advice from the International Advisory Panel on Climate Change). Throughout this thesis, the focus is primarily on advice which comes from a single, stable source, as when we see a post by an acquaintance on social media, or when a stranger provides us with advice.

Advice-taking

Advice occurs in the context of a decision, and forms a part of the information which is integrated during the decision-making process to produce a decision. To the extent that the decision reached differs from the decision that would have occurred had the advice not been presented, the advice has had an effect on the decision; to the extent that this difference changes the decision in a way consistent with the advice, the advice has been 'taken' (as opposed to 'rejected').

It is tacitly implied by many operationalizations of advice-taking that the informational content of the advice determines the extent to which it is taken or rejected !TODO[CITE]. Insofar as the identity of the advisor matters, it matters because it functions as a cue to the informational content of the advice. This is likely a major oversimplification, however, because in many real-world contexts advice-giving and advice-taking form part of a developing social relationship: being consulted for advice and having one's advice followed are inherently rewarding (Hertz and Bahrami 2018; Hertz, Palminteri, et al. 2017); and taking advice can serve as a (sometimes costly) social signal of valuing a relationship with a person or group (Byrne et al. 2016). Furthermore, some authors have argued that people may perceive taking advice as sacrificing their independence or autonomy !TODO[CITATIONNEEDED]. While this thesis follows previous literature in omitting to consider the wider social concerns influencing the taking of advice, it is nevertheless important to remember that the processes investigated herein take place in a variety of social contexts where complex social agents attempt to optimise over numerous goals over numerous timescales.

Three-factor model of trust

The degree to which advice is taken is proportional to the trust placed in the advisor by the decision-maker. Interpersonal trust, or the degree to which one is prepared to place one's fortune in the hands of another (e.g. by relying on their advice), is apportioned by Mayer et al. (1995) onto three properties of the advisor (as judged by the decision-maker): ability, benevolence, and integrity. To these three properties of the advisor we may add the decision-maker's general propensity to trust, as well as situational cues and task cues (e.g. the phenomenon that advice is more readily taken for hard tasks than easy ones, (Gino and Moore 2007; Yonah and Kessler 2021)).

Ability

Ability captures the expertise of an advisor: their raw ability to perform the task for which they are giving advice. In some cases this is relatively straightforward, as in the expertise of a General Practitioner in matters of health and disease, and in others more complex, as in the expertise of a hairdresser when deciding on a haircut (when matters of personal taste comingle with aesthetic considerations of facial structure, practical considerations of hair constitution, and social considerations of fashion). The greater the ability of an advisor, the greater the influence of their advice, as demonstrated by experiments showing that participants' decisions are more affected by the advice of advisors who are labelled as more expert in a relevant domain (Sah et al. 2013; Schultze, Mojzisch, et al. 2017; Sniezek, Schrah, et al. 2004; Sniezek and Van Swol 2001; Soll and Mannes 2011), or are shown to be more expert empirically (**pescetelliUseMetacognitiveSignals2017**; Sah et al. 2013; Yaniv and Kleinberger 2000).

Benevolence

Benevolence refers to the extent to which the advisor seeks to further the interests of the decision-maker. Where ability represents the absolute limit on the quality of advice, benevolence represents the extent to which the advice approaches this limit. The advice of even a renouned expert may be doubted if there is reason to believe their goal is to mislead, a vital lesson for medieval monarchs with their councils of politicking advisors. Experimental work has shown that psychology students relied more on the advice of their friends than on the advice of labelled experts !TODO[CITATIONNEEDED], and that participants are more inclined to reject advice when uncertainty is attributed to malice rather than ignorance (Schul and Peni 2015)}.

Integrity

Advisors with integrity exhibit adherence to principles which the decision-maker endorses. As with benevolence, integrity acts to determine the extent to which advice approaches the limit imposed by ability. While not mutually exclusive, integrity is typically important where relationships are less personal (e.g. we may place great trust in a General Practitioner because of their expertise in medical matters and their *integrity* in adhering to a set of professional ethical and conduct requirements). !TODO[Some description of the research]

Normative models of advice-taking

Advice-taking can be evaluated formally with reference to a normative model. The simplest and most common of these views the decision-making task as an estimation problem (or combination of estimation problems), and provides an approximately Bayesian variance-weighted integration of independent estimates. To borrow from Galton (1907)}, consider the task of judging the weight of a bullock. We can model any single guess (e) as the true weight (v) plus some error (ϵ):

$$e = v + \epsilon \tag{1.1}$$

The key insight is to observe that the error is drawn from a normal distribution ($\mathcal{N}(\mu = 0, \sigma^2)$)[1]. As the number of samples from this distribution increases, the mean of those samples tends towards the mean of the distribution. Thus, the more estimates are taken, the closer on average the sum of errors will be to 0.

[1]The normal distribution is well-supported by empirical evidence, but note that any symmetrical distribution around 0 will lead to the same conclusion.

$$\frac{\sum_i^N(e_i)}{N} = \frac{\sum_i^N(v + \mathcal{N}(\mu = 0,\, \sigma_i^2))}{N} \tag{1.2}$$
$$\frac{\sum_i^N(e_i)}{N} = \frac{\sum_i^N(v)}{N} + \frac{\sum_i^N(\mathcal{N}(\mu = 0,\, \sigma_i^2))}{N} \tag{1.3}$$
$$\frac{\sum_i^N(e_i)}{N} = \frac{Nv}{N} + \hat{0} \tag{1.4}$$
$$\frac{\sum_i^N(e_i)}{N} \approx v \tag{1.5}$$

Observe that this formulation is true no matter the value of N. On average, it is always better to have more estimates than fewer. This suggests that, even in the situation where there are only two estimates (the decision-maker's and the advisor's), the best policy will be to incorporate both estimates into the final decision.

The variance of the normal distribution from which errors are derived (σ_i^2) is, in the example above, drawn from a normal distribution itself ($\sigma_i^2 \sim \mathcal{N}(\mu = 0,\, \mathrm{sd}^2)$meaning that it is also cancelled out on average over repeated samples). Where few estimates are taken, weighting those estimates by the variance of the error distributions will increase the accuracy of the estimates in proportion to the difference between the variances:

$$e^F = \frac{\frac{1}{N}\sum_i^N \omega_i e_i^I}{\sum_i^N \omega_i}$$

Where ω_i is $1/\sigma_i^2$.

Many experimental implementations of this model avoid weighting issues by calibrating decision-makers and advisors to be equally accurate on average ($\sigma^2_{\text{decision-maker}} = \sigma^2_{\text{advisor}}$). The result of this constraint is that the optimal policy is simply to average all estimates together:

$$e^F = \frac{1}{N}\sum_i^N e_i^I \approx v$$

Egocentric-discounting

From the perspective of the normative model above, decision-makers should weigh their own estimate equally with each other estimate they receive in the process of coming to their decision. One of the most robust findings in the literature on advice-taking is that people routinely underweight advisory estimates relative to their own estimates, a phenomenon known as *egocentric discounting*(Dana and Cain 2015; Gino and Moore 2007; Hütter and Ache 2016; Liberman et al. 2012; Minson and Mueller 2012; Rader et al. 2017; Ronayne and Sgroi 2018; See et al. 2011; Soll and Mannes 2011; Trouche et al. 2018; Yaniv and Kleinberger 2000; Yaniv and Choshen-Hillel 2012; Yaniv and Milyavsky 2007). Egocentric discounting occurs in both feedback and no-feedback contexts (Yaniv and Kleinberger 2000). Explanations for egocentric discounting are usually framed in terms of personal-level psychology: decision-makers have better access to reasons for their decision (Yaniv and Kleinberger 2000); overrate their own competence (Sniezek, Schrah, et al. 2004); may have a desire to appear consistent (Yaniv and Milyavsky 2007); may see opinions as possessions (Soll and Mannes 2011); may be loss-averse to providing a worse final estimate due to advice-taking (Soll and Mannes 2011); or have difficulty avoiding anchoring (Schultze, Mojzisch, et al. 2017) or repetition bias effects (Trouche et al. 2018). None of these explanations has survived rigorous empirical testing, however, and recently suggestions have widened to include consideration of aggregate-level rather than personal-level causes, with Trouche et al. (2018) arguing that the potential for misaligned incentives between decision-maker and advisor motivate discounting of advice. In the course of this thesis `crossref needed`, I demonstrate that egocentric discounting may be a stable metastrategy which protects against exploitation, carelessness, incompetence, and miscommunication. From this perspective, the normative model pertains to a particular instantiation of a problem with questionable ecological validity given the typical ethology of advice-taking in humans. While such considerations affect the conclusions one draws from egocentric discounting relative to the normative model, they do not detract substantially from the practice of using the normative model as an optimum 'set point' from which to evaluate advice-taking behaviour.

Homophily and echo-chambers

Homophily is the ubiquitous phenomenon that individuals more closely connected to one another within a social network tend to be more similar to one another than would be expected by chance across numerous dimensions, from demographics to attitudes (McPherson et al. 2001). Whether homophily in virtual social networks is responsible for increases in polarisation is debated.

Proponents point to increases in polarisation (e.g. in politics: !TODO['Pew Research Center, 2014']), to empirical studies demonstrating homophily in virtual social networks (Cardoso et al. 2017; Colleoni et al. 2014), and to studies examining selective exposure online (Kobayashi and Ikeda 2009), and to echo chambers: egregious examples of highly homophilous networks with pathological polarisation (Sunstein 2002; Sunstein 2018). The empirical components of the argument are contested, with evidence that virtual social networks are less homogenous than offline social networks (and hence depolarising, Barberá 2015), and that selective exposure, and especially selective avoidance, is a somewhat dubious finding which does not show up clearly online (Garrett 2009a; Garrett 2009b; Nelson and Webster 2017; Sears and Freedman 1967). Modelling work demonstrates, however, that where there is a bias in assimilation of information, homophily exacerbates polarisation (Dandekar et al. 2013). Where polarisation in turn increases homophily, for example through selective exposure or avoidance, a self-reinforcing spiral emerges wherein social connections become increasingly homogenous and attitudes increasingly extreme (Song and Boomgaarden 2017).

Source selection and information weighting

Context-dependency of epistemic processes

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Psychological mechanisms of advisor evaluation

Where feedback is unavailable, people may use their own sense of certainty as a yardstick for evaluating advice (**pescetelliUseMetacognitiveSignals2017**; Pescetelli and Yeung 2021): advisors who agree when one is confident are perceived as more helpful; while those who disagree when one is confident are perceived as less helpful. Confidence serves as a proxy for objective feedback, and functions well in this role insofar as the judge has high metacognitive resolution (i.e. higher confidence is indicative of a greater probability of being correct).

Use of advice

Normative models of advice taking state that averaging estimates minimises errors. As discussed at length in Section III, the assumptions underlying the normative model do not always hold in the real world. The performance of the normative model can be characterised according to differences between the advisor and the decision-maker on ability and bias (Soll and Larrick 2009) !TODO[how does PAR relate to what's going on here?]. Recall that the normative model states that advice should contribute to the final decision in proportion to the ability of the advisor compared to the decision-maker.

$$e_i^F = \frac{\omega_i e_i^I + \omega_j e_j^I}{2} \tag{2.1}$$

Where agent i is the decision-maker and agent j is the advisor. This weighting can be simplified to be expressed only in terms of the decision-maker's weighting of the advisor because the two are constrained to sum to 1 by virtue of being relative to one another:

$$\omega_i + \omega_j = 1 \tag{2.2}$$

$$\omega_j = 1 - \omega_i \tag{2.3}$$

$$\therefore e_i^F = \frac{(1 - \omega_j)e_i^I + \omega_j a_i}{2} \tag{2.4}$$

Where a_i is the advice received (i.e. agent j's initial estimate - $a_i \equiv e_j^I$). In the normative model, the weighting is equivalent to the ratio of variance of the errors made by each agent:

$$\omega_j = \frac{\sigma_j^2}{\sigma_j^2 + \sigma_i^2} \tag{2.5}$$

The normative model thus represents weighting by relative ability. Precise knowledge of the ability of others relative to oneself is rarely available in the real world, however, and, as discussed in Section III, other assumptions concerning the trustworthiness or interpretability of advice may be violated.

The normative model can be adapted to provide a more psychologically-realistic account of advice usage by substituting the three factor model of trust into the equations in place of the ability variable. We start with the statement within the three factor model that trust (ω) is proportional to ability (a), benevolence (b), and integrity (g).

$$\text{trust} \propto \text{ability} + \text{benvolence} + \text{integrity} \tag{2.6}$$

We can thus replace the measure of accuracy in the normative model with the measure of trust in order to calculate the relative weighting:

$$\omega_j = \frac{\text{trust}_j}{\text{trust}_j + \text{trust}_i} \tag{2.7}$$

At this point we may question whether the variable trust_i is a meaningful property or simply an artefact of mathematical symbol manipulation. Mathematically it provides a fixed point against which trustworthiness of advisors can be measured, allowing for scaling weightings meaningfully across different advisors in different decisions. In real world terms, while it is generally unlikely that benevolence_i and integrity_i will be anything less than maximal, perceptions of one's own ability (ability_i) are likely to allow for others to exceed it. I make no strong claims on the relationship between trust and its component variables other than proportionality, and within this conception it is meaningful to consider weighting as a property of trust in another's judgement relative to one's own, adjusted in some manner for the perception of that other's benevolence and integrity. If the concept of self-trust still appears untenable, note that trust_i can be replaced with a constant without compromising the equations.

We thus return to the normative model of advice-taking, but with an alternative derivation of the weighting between advice and initial estimate:

$$e_i^F = \frac{(1-\omega_j)e_i^I + \omega_j a_i}{2} \tag{2.8}$$

2.0.1 Critique of the aggregation model

This conception of advice-taking as a weighted aggregation process between an initial estimate and advice underpins both the modelling and the experiments presented in this thesis. It is thus worth taking a little space to highlight areas in which this model is known to depart from reality.

Generality beyond judge-advisor systems

Firstly, the model is an idealised situation approximated by the experimental method: a decision-maker makes an explicit initial estimate, then receives advice, then makes an explicit final decision. Yaniv and Choshen-Hillel (2012) showed that preventing decision-makers from making initial decisions resulted in very different advice weighting, suggesting that this may be a model of a specific scenario rather than of advice integration per se. The model presented here could in principle explain an integration process where an initial estimate can only be made after the advice is known, but empirically performs poorly. At best, it could be argued that pre-exposure to the advice either anchors the initial estimate (thus moving e_i^I systematically closer to a_i), or that having to trust advice because one cannot make one's own decision inflates the weighting of the advisor.

Multiple advisory estimates

Secondly, the model does not perform well when multiple advisors are consulted. The normative model, and the psychological derivative, predicts that a decision-maker's estimate ought to be weighted in conjunction with the other estimates. In other words, as the number of advisory estimates increases, the weight of the initial estimate should decrease. Hütter and Ache (2016) presented evidence that this does not happen: the weight of the initial estimate stays relatively constant while the weights of the advisor estimates are reduced. This implies that if a decision-maker were to average evenly their initial estimate with an advisor estimate ($\omega_i = .5$; $\omega_j = .5$), adding an extra advisor estimate would result in the weights of the advice being halved while the weight of the initial decision remained constant ($\omega_i = .5$; $\omega_{j \neq i} = .25$), rather than the more transparently optimal policy of weighting all estimates evenly ($\omega_i = \omega j \neq i = 1/3$).

Individual trial data

The model is supported by patterns in averages. Analysis of individual trials shows that the aggregate patterns of advice-taking appear to be roughly distributed between an averaging strategy and a picking strategy (Soll and Mannes 2011; Soll and Larrick 2009). The model, derived from these patterns, approximates the contribution of an individual trial to the overall average rather than the actual advice-taking strategy on any given trial.

2.0.2 Justification for use of the aggregation model

The criticisms above are important, but they do not invalidate the model for use in the present project. Here we seek to establish how differences in advice taking manifest according to properties of advisors. These differences are well characterised by the model, especially in the judge-advisor system used for the experiments. All models are inexact descriptions of reality, and inclusion of a more complex model capable of handling the cases outlined above would require greatly increased complexity for relatively little gain in explanatory power. For studying the questions at hand, the psychological model is an appropriate and useful approximation of human behaviour.

Updating advisor weights

The weights assigned to the advisors (relative to the decision-maker themself) are subject to change as the result of experience. This experience can be exogenous or endogenous to the decision-making task. In the exogenous case, advisors may be labelled in a particular way (Önkal, Gönül, et al. 2017; Tost et al. 2012; Schultze, Mojzisch, et al. 2017) or have some summary of their performance displayed (Gino, Brooks, et al. 2012; Yaniv and Kleinberger 2000). Endogenous experience refers to the information that advice on a given trial carries about the trustworthiness of an advisor. Exogenous experience is relatively straightforward, but endogenous experience requires some explanation.

Endogenous experience of advice means that the weighting of an advisor is in part dependent upon the past advice offered by that advisor. As each piece of advice is evaluated, the overall weighting of the advisor is updated accordingly. For clarity, two simplifiying assumptions are made in the explanation below. Firstly, while it is probable that properties of the advice are used to inform the dimensions of ability, integrity, and benevolence simultaneously, the examples below will deal with ability in isolation. Another project could explore in detail how experience of advice on any given trial updates an advisor's position in 3-dimensional trust space in a Bayesian manner according to the relative certainties about each dimension. This would capture the task of assigning blame for erroneous advice (e.g. was it unintentionally poor - a failure of ability - or deliberately misleading - a failure of benevolence?). Such an undertaking is beyond the scope of this project.

Secondly, it is assumed that advice is judged on its own merit as an estimate rather than on its usefulness as advice. The former means that advice is assessed in terms of the optimality of the decision recommended by the advice itself. The latter assesses advice based on the optimality of the decision based on advice relative to the optimality of the decision which *would have been made had the advice not been received*. There is some evidence that people alter their advice-giving behaviour in anticipation of discounting on the part of the decision-maker !TODO[CITE; for a case in human-machine teaming see @azariaStrategicAdviceProvision2016], somewhat akin to starting negotiations with a higher demand than one is hoping to settle for. There is no evidence as yet as to whether decision-makers anticipate and adjust for this adjustment on the part of the advisor. For the questions considered here, conclusions obtained under these simplifying assumptions are likely to hold even when the additional complexity is restored. The effects in the real world of interactivity between trust dimensions and game theoretic adjustments in the giving and interpretation of advice are likely to be small in comparison to general effects of advisor updating.

Evaluation of advice

A single piece of advice can be evaluated using its own properties and the properties of the advisor giving the advice. Furthermore, that evaluation can serve to update the properties of the advisor. A piece of advice's own properties will include its plausibility (e.g. participants in estimation tasks discount advice which is distant from their own initial estimates more heavily (Yaniv 2004)), while the properties of the advisor will include the advisor's trustworthiness (see above). The updating of trust following experience of advice is likely to be largely in the domain of ability, although other domains may be affected where the advice is particularly egregious.

Updating advisor evaluations

While a single piece of advice must be taken on its own terms, people can construct relatively accurate estimates of advisors' advice when provided with feedback on the decisions they use the advice to make (**pescetelliUseMetacognitiveSignals2017**; Sah et al. 2013; Yaniv and Kleinberger 2000). This likely happens as an analogue of reinforcement learning, where feedback allows an error signal to be used to update the estimate of the advisor's ability ($\hat{s}^a$) rather than one's own beliefs about the world, according to some learning rate (λ).

$$\text{this is wrong; need to check RL models for an analogue} \tag{2.9}$$

$$\hat{s}_{t+1} = \hat{s}_t + |e^a_t - v| \cdot \lambda \tag{2.10}$$

2.0.3 Criticism of the advisor evalutation model

Ecological validity

While many experiments have established the existence reinforcement learning in humans and other animals, it is unclear whether reinforcement learning operates in the social domain in which advising takes place. It is not obvious that there are many situations in the course of everyday relationships which can be characterised by the rapid advice-feedback cycle required to learn about advisor ability in the manner modelled above. FeldmanHall and Dunsmoor (2019) argued in a review that a wide variety of social phenomena could be explained via reinforcement learning processes, and Behrens et al. (2008) demonstrated that a Bayesian reinforcement learning model provided a good fit to behavioural data from a social influence task. Additionally, Heyes et al. (2020) have argued that social learning is wholly explicable in terms of general reinforcement learning processes paired with attentional biases to social stimuli. Reinforcement learning in the social domain operates on the basis of rapid feedback, just as in the non-social domain. Below, the advisor evaluation model is extended to cases where objective feedback is not available by substituting the decision-maker's confidence for objective feedback. While not foolproof, the method allows better-than-average approximation of the quality of advisors provided several plausible assumptions are met !TODO[Discuss assumptions somewhere: independence of errors, better-than-chance accuracy, trying to help, rough metacognitive calibration/resolution].

Advisor evaluation without feedback

Where feedback is not available, participants in experiments continue to demonstrate an ability to respond rationally to differences in advisor quality (**pescetelliUseMetacognitiveSignal**
This is evidently not done through access to the correct real-world values, because feedback providing those values is unavailable, and, were participants aware of those values themselves, it stands to reason they would have provided those values (and thus not require advice!). Pescetelli and Yeung (**pescetelliUseMetacognitiveSignals2017**) suggest the mechanism for this ability to discriminate between advisors in the absence of feedback is performing updates based on confidence-weighted agreement.

Agreement

Consider first the non-weighted agreement case, where the advisor's estimate at time t (e^a_t) and the decision-maker's estimates (e^d) are binary ($\in 0, 1$). The estimate of the advisor's ability ($\hat{s}^a$) is updated positively if the advisor and decision-maker agree, and negatively otherwise, according to the learning rate λ.

$$\hat{s}^a_{t+1} = \hat{s}^a_t + (-2\,|e^d_t - e^a_t| + 1) \cdot \lambda \tag{2.11}$$

Confidence-weighted agreement

The updating of advice contingent on agreement may be weighted by confidence in the initial decision (c^d), such that agreement and disagreement are considered more informative about the quality of the advice when the decision with which they agree or disagree is more certain.

$$\hat{s}^a_{t+1} = \hat{s}^a_t + (-2\,|e^d_t - e^a_t| + 1) \cdot c^d_t \cdot \lambda \tag{2.12}$$

Continuous estimate case

2.1 Measuring advice-taking

Judge-advisor system

Table of experiments

This is now outdated. Update or remove?

Advisors	Choice	Task	Feedback	Result
AiC vs AiU	Yes	Perceptual, binary (MAT-LAB)	No	Suggestive
AiC vs AiU	Yes	Perceptual, binary	No	Inconclusive
In/accurate	Yes	Perceptual, binary	No	Accuracy selected more often
Low/High agreement	Yes	Perceptual, binary	No	Agreement selected more often

Advisors	Choice	Task	Feedback	Result
Accurate vs Agreement	No	Estimation, continuous	Yes and No	Accurate preferred given feedback, agreement preferred without feedback

3
Behavioural experiment method

The behavioural experiments reported in this thesis share a common structure. This structure is detailed here to reduce repetition elsewhere in the thesis. Individual experiments have truncated methods sections in which the specific deviations from the general method are noted.

3.1 General method

The experiments take place using a judge-advisor system. Participants give an **initial estimate** for a decision-making task, receive **advice**, and then provide a **final decision**. The advice is always computer-generated, although the specifics of the generating procedure vary between experiments.

3.1.1 Participants

Recruitment

Human participants were recruited from the online experiment participation platform Prolific (`https://prolific.co`). Participants were prevented from taking the study if they had participated in one of the other studies in the thesis, or if they had an overall approval rating on Prolific of less than 95/100.

Payment

Participants were paid approximately GBP10-15/hour pro rata. Experiments took the average participant between 10 and 30 minutes to complete.

Some participants encountered technical problems, prompting them to contact me via the Prolific platform. These participants were thanked, and additional information about the problem sought if necessary. These participants were paid on an ad hoc basis depending upon the time taken before the errors emerged and the detail of the error reports.

Later studies introduced attention checks which terminated the study as a consequence for failure. It is not clear whether this technique constitutes best practice on Prolific because automatic termination means participants may return the study rather than having their participation explicitly rejected (and thus affecting their Prolific participation rating). Participants who returned the studies were not paid. Participants who attempted to complete the study with an invalid code after having their participation terminated for failing attention checks were also not paid, and their completion attempt was rejected on the Prolific platform. There is an ongoing ethical debate concerning non-payment of participants who fail attention checks in online studies. The studies in this thesis used a mixture of paying for and not paying for participation attempts with failed attention checks. In online studies, where low-effort participation is a serious and enduring concern, platforms such as Prolific make clear to participants and researchers that payment is only expected for responses which are given with satisfactory effort. Participants are thus fully aware of and consenting to the process of screening results for adequate effort prior to payment. It is important to note the difference between low effort responses, for which payment may ethically be withheld, and atypical responses, which may represent genuine engagement with the task. It is unethical, in my view, to withhold payment from participants for atypical responses within an experiment (including low or high response times, accuracy, etc).[1] Participants should only be denied

[1] Data may be excluded from *analysis* for these reasons, but the participants should still be paid.

payment for failing to provide adequate responses to explicit attention checks.

Demographics

Demographic information on participants, such as age and gender, was not collected. While there is a robust case for collecting these data and conducting sex-disaggregated analyses (Criado Perez 2019), initial concerns over General Data Protection Regulation resulted in a cautious approach to the collection of data concerning protected characteristics of participants. !TODO[move to discussion]Gender differences, whether due to socialisation, biological factors, or their interactions, may well alter advice-taking and expressed confidence in decisions. I suspect, although I can offer no evidence, that gender differences in the results presented in this thesis will at most show overlapping distributions. I do not think it highly plausible that different strategies are wholly the preserve of any particular gender, or that egocentric discounting is markedly stronger in any particular gender.

Participants were at least 18 years of age, confirmed by the requirements for possessing an account on the Prolific platform and by explicit confirmation when giving informed consent.

3.1.2 Ethics

Ethical approval for the studies in the thesis was granted by the University of Oxford Medical Sciences Interdivisional Research Ethics Committee (References: R55382/RE001; R55382/RE002).

3.1.3 Procedure

Participants visited the Uniform Resource Locator (URL) for the study by following a link from Prolific using their own device (Figure 3.1). Early studies only supported computers, but later studies included support for tablets and smartphones. After viewing an information sheet describing the study and giving their consent to participate, participants began the study proper. The study introduced the software to the participant interactively, demonstrating the decision-making task and how

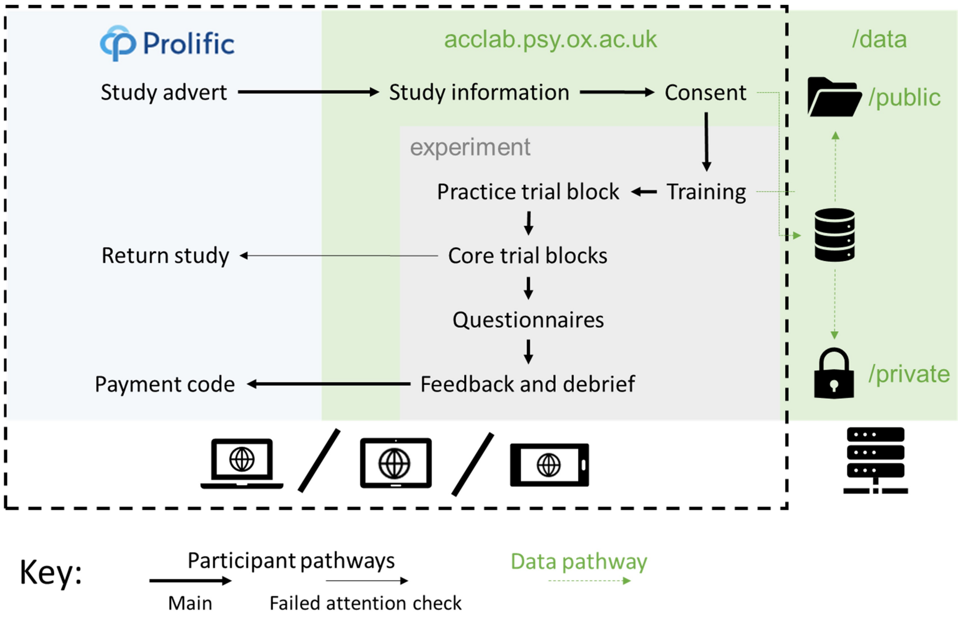


Figure 3.1: Participant pathway through the studies.
Participants used their own devices to complete the study, which was presented on a website written in HTML, CSS, and JavaScript. The data were saved on the server using PHP.

responses could be made. Next, participants were given a block of practice trials to familiarise them with the decision-making task. Participants were then introduced to advice, and given a block of practice trials in which they received advice. The core experimental blocks followed the practice with advice. Finally, debrief questions were presented and feedback provided concerning the participant's performance, including a stable link to the feedback and a payment code. The participant entered the payment code into the Prolific platform and their participation was at an end.

On each trial, participants were faced with a decision-making task for which they offered an initial estimate. They then received advice (on some trials they were able to choose which of two advisors would provide this advice). They then made a final decision. On feedback trials, they received feedback on their final decision. The schematic for this trial structure is shown for the Dots Task in Figure 3.2.

!TODO[Check feedback duration and style in image caption]

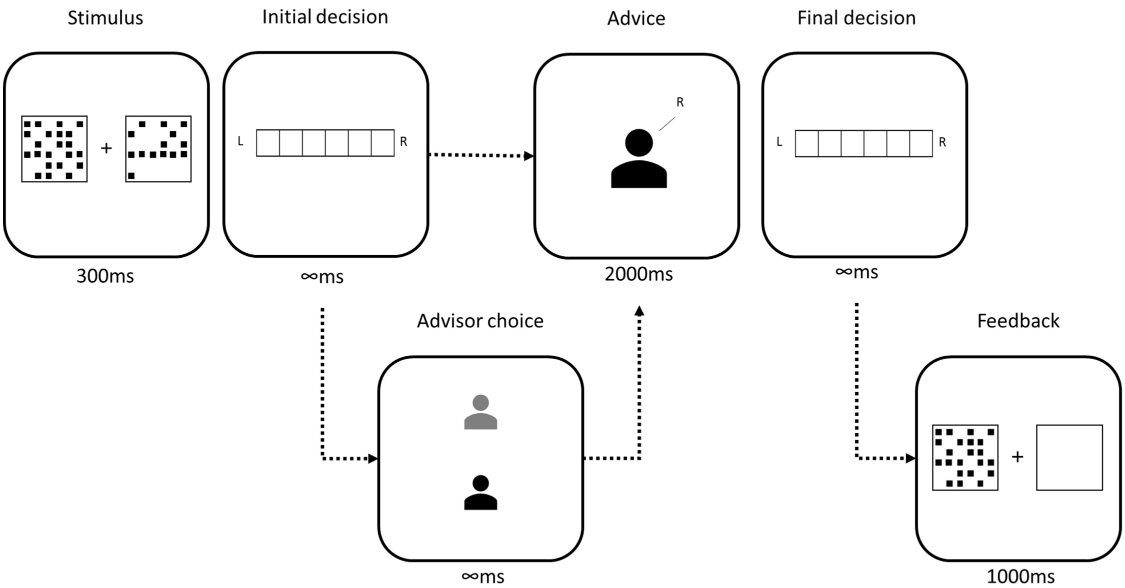


Figure 3.2: Trial structure of the Dots Task.
In the initial estimate phase, participants saw two boxes of dots presented simultaneously for 300ms. Participants then reported whether there were more dots on in the left or the right box, and how confident they were in this decision. Participants then received advice, sometimes being offered the choice of which advisor would provide the advice. The advice was displayed for 2000ms before participants could submit a final decision, again reporting which box they believe contained more dots and their confidence in their decision. On feedback trials, feedback was presented by redisplaying the correct box while showing the other box as empty.

Perceptual decision (Dots Task)

Stimuli in the Dots Task consisted of two boxes arranged to the left and right of a fixation cross (Figure 3.3). These boxes were briefly (300ms) and simultaneously filled with an array of non-overlapping dots, and the participant was instructed to identify the box with the most dots. The participant submitted their response by selecting a point on a horizontal bar: points to the left of the midpoint indicated the participant thought the lefthand box had more dots, and points to the right that they thought the righthand box had more dots. The further away from the midpoint the participant selected on the bar, the more confidence they indicated in their response.

The number of dots was exactly determined by the difficulty of the trial: the box with the !TODO[check this description] least dots had 200 - the difficulty, while the box with the most had 200 + the difficulty. The dots did not move during the

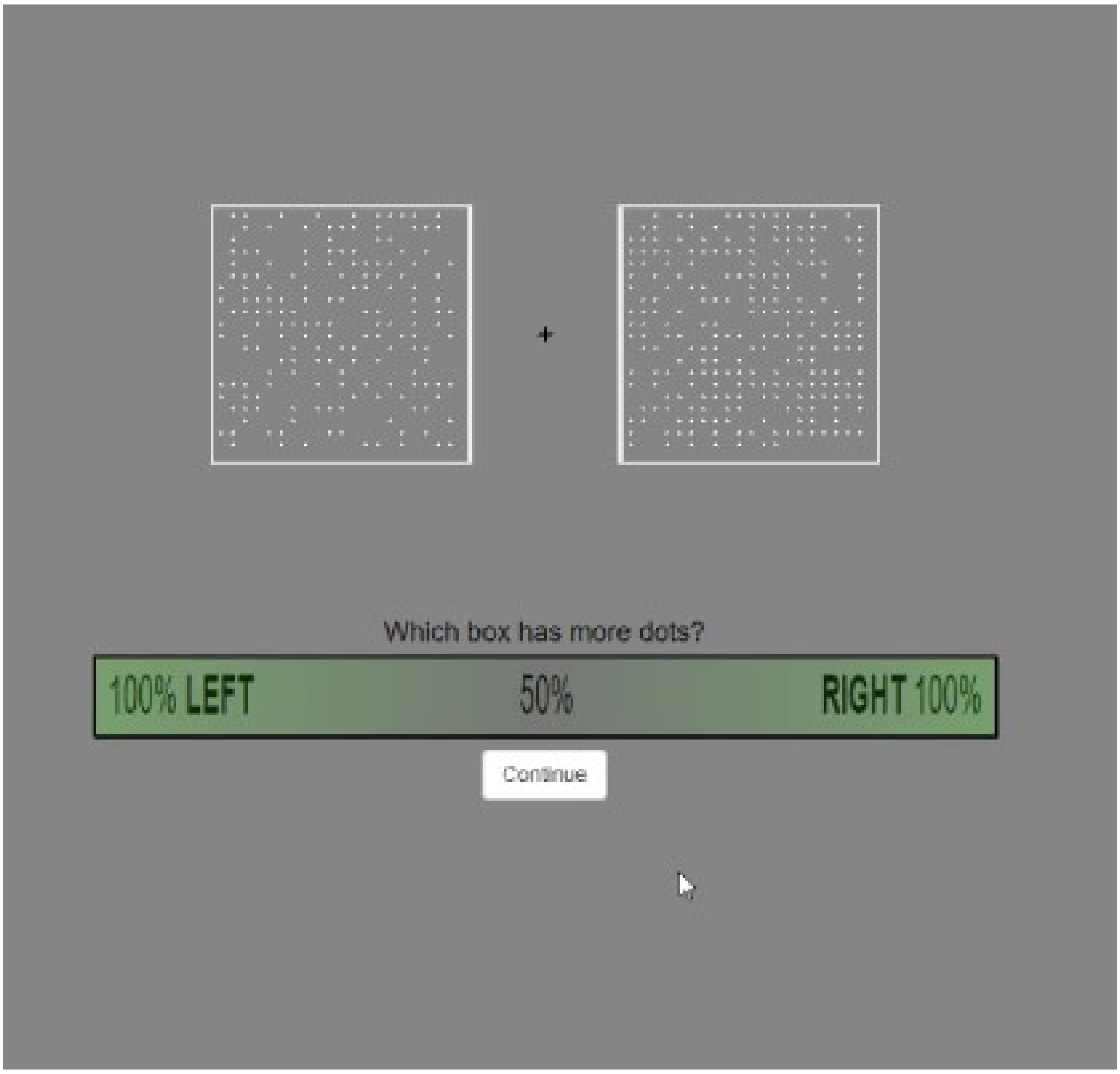


Figure 3.3: Dots Task stimulus.

presentation of the stimulus. There was thus an objectively correct answer to the question which, given enough time, could be precisely determined from the stimulus.

The Dots Task stimuli can be customised to make the discrimination easier or more difficult. This means that the stimuli can be adjusted to maintain a specific accuracy for each individual participant, allowing confidence to be examined in the absence of confounds with the probability of being correct. Stimuli were continually adjusted throughout the experiment to maintain an initial estimate accuracy of around 71% using a 2-down-1-up staircase procedure. There were a substantial number of trials in the practice block so that participants could eliminate practice effects and thus experience a more stable objective difficulty during the core trial blocks. The specific number of practice trials differed between experiments as we sought to balance minimising the participant time requirement with minimising practice effects.

After the practice there were blocks where participants received advice on their initial estimates and made a final decision using the same response process as their initial estimate. Advice was presented with a representation of an advisor with a text bubble stretching out to the side the advisor endorsed, containing the text "I think it was on the RIGHT" or LEFT as appropriate. The advisors were represented in various manners in various versions of the task, but predominantly in a format with a central box containing a generic blank portrait icon and text at the bottom with "Advisor ##" where ## was a number between 10 and 60.

On some trials participants could choose between potential advisors. In these cases, the potential advisors were positioned vertically in the centre of the screen and the participant clicked on the advisor from whom they wished to receive advice. There was never a time limit for selecting an advisor.

Throughout the experiment a progress bar provided a graphical indication of the number of trials remaining in the experiment. After each block participants were told what percentage of the final decisions they had provided were correct and allowed to take a short, self-paced break.

At the end of the experiment participants were presented with a questionnaire asking them to rate their advisors' likeability, ability, and benevolence (Mayer et al. 1995), and offering them the opportunity to provide free-text comments on the advisors and the experiment in general. These questionnaire responses are not analysed because they were generally uninformative. Some free text responses may be redacted in the open data to prevent participants being identifiable.

Specific limitations of the Dots task The Dots task uses a perceptual decision with a high number of trials. This structure makes it plausible that participants respond to the advice primarily through simple reinforcement learning rather than through specific social processes thought to be at work in advice-taking and advisor evaluation.

Whether or not the Dots task taps into social processes, both the task itself and the experimental structure are very different from most advice-taking and advisor evaluation in the real world. Perceptual decisions are rarely the subject of advice, and thus the central task is an unusual one for joint decision-making. The amount of exposure to advisors also greatly exceeds that which would be obtained over a far longer period of time in most real world situations. This much greater level of exposure risks investing effects with artificial importance: while it is a strength of experimental designs to magnify the effects they aim to study we must not let such magnification blind us to the real relevance of these effects within the complex and dynamic context of real life.

Estimation (Dates Task)

The Dates task presented participants with historical events that occurred in a specific year in the 20th century. Participants were then asked to either: a) drag a marker onto a timeline to indicate a range of years within which they thought the event occurred (continuous version); or b) state whether the event occurred before or after a specific date (binary version). Once they had made this initial estimate, they were presented with advice from an advisor and then asked to make a final decision.

During practice trials (either with or without advice), participants received feedback on their answers. Some participants were placed in a Feedback condition where they received feedback on all trials except those where they were asked to choose an advisor to provide advice.

After completing all the trials, participants were presented with a feedback form for each advisor, requiring them to rate the advisor on their level of knowledge, helpfulness, and likability, chosen to reflect the three aspects of trust identified by Mayer et al. (1995) (ability, integrity, and benevolence). Participants could also provide free-text responses containing further comments on the advisors. Participants then completed a general debrief form that told them: “There was a difference between the advisors. What do you think it was?" They also had the opportunity to add free text questions or comments about the experiment. The latter responses are not included in the shared data.

Finally, participants were given a screen that allowed them to inspect their performance and send links to the study to others (only participants directly invited on the Prolific platform were included for analysis and in the shared data). This final screen also contained the payment code participants should enter on Prolific.

Rationale There were several reasons behind the development of the Dates task. Firstly, we believed the theory that advisors are evaluated on the basis of agreement when objective information concerning accuracy is not available (Pescetelli and Yeung 2021) to be a general property of advice-taking and not constrained to the domain of perceptual decision-making. We therefore wanted to replicate the effects see by Pescetelli and Yeung (2021) in a different task domain.

Secondly, we wanted to replicate the effects in a task domain that was more similar to the kinds of decisions that people might seek advice about. While people do occasionally consult one another concerning perceptual decisions ("is that a bird or a plane?"), such consultation is rare, especially in comparison to the ubiquity of perceptual decision-making. We thus selected a more deliberative task.

Thirdly, much of the judge-advisor system and advice-taking literature has used tasks based on estimation (Sniezek, Schrah, et al. 2004; Gino and Moore 2007; See et al. 2011; Soll and Mannes 2011; Gino, Brooks, et al. 2012; Liberman et al. 2012; Minson and Mueller 2012; Tost et al. 2012; Yaniv and Choshen-Hillel 2012; Bonner and Cadman 2014; Schultze, Rakotoarisoa, et al. 2015; Hütter and Ache 2016; Schultze, Mojzisch, et al. 2017; Trouche et al. 2018; Wang and Du 2018), and often specifically estimation of dates (Yaniv and Kleinberger 2000; Yaniv and Milyavsky 2007; Gino 2008). We felt that being able to replicate the effects seen by Pescetelli and Yeung (2021) in a task domain commonly used for evaluating advice-taking behaviour would be especially useful.

Lastly, we wanted to design a task in a way that would be engaging for participants. This decision did not point us towards date estimation specifically in the way the previous decisions did, but we did feel that it would be possible to make an engaging task along those lines. Subsequent to designing the task we discovered two tabletop games with similar designs, both of which are simple to learn and engaging to play (TimeLine, 2012; CONFIDENT?, 2018). In keeping with this approach, we decided to make the task as compact as possible, for which estimation questions are useful, as demonstrated by the typical number of questions in the estimation tasks in the literature.

Continuous In the continuous version of the Dates task, participants were shown a timeline below the event description (Figure 3.4). Below the timeline were the markers that they used to indicate their responses. In earlier versions of this task, multiple markers were available, each with a point value that decreased with the width of the marker. Markers widths were chosen to span odd numbers of years, meaning that they could mark an even number of years in an inclusive manner (e.g. in the example shown, a 3-year marker marks the years 1916-1918 inclusively). Once participants had placed their marker they clicked the tick button in the bottom-right to confirm their choice.

Decisions in the continuous version of the Dates task did not have an explicit confidence rating. The participant's choice of a narrower or wider marker was taken as an implicit indication of their confidence. In order to score points, the participant's marker had to contain the correct year on the timeline, and thus the more confident a participant was about the answer the more likely they would be able to score points using a smaller marker, and consequently the higher their average return from using a smaller marker.

Advice in the continuous version of the task was presented using a one-second animation wherein the advisor's avatar slid along the timeline to the middle of the advisory estimate. Attached to the avatar was the advisor's marker, and this indicated a range of years into that the advisor signalled included the year of the event. Once the advisor's advice had reached its target location, the advisor's advice remained visible throughout the remainder of the trial, and the participant could enter a final decision.

To enter a final decision the participant could either simply click their existing marker, confirming their initial estimate as their final decision, move the marker along the timeline, or, where multiple markers were available, drag a different marker onto the timeline. Once again, to confirm their response they clicked the tick button in the bottom-right.

Advice in the continuous version of the task was still conceptualised along dimensions of accuracy and agreement. While it is possible to operationalise these as binary properties, with accurate advice including all advice where the advisor's marker covered the correct year and agreeing advice including all advice where the advisor's maker intersected with the participant's initial estimate marker, we decided that this approach was prohibitively difficult to implement, especially where initial estimate markers were very wide. Instead, accuracy and agreement were taken as continuous properties, measured from the centre of the advisor's marker to the correct year (for accuracy) or the centre of the participant's initial estimate marker (agreement).

Advisors in these experiments typically placed their advice markers by identifying a target year, either the correct year or the centre of the participant's initial estimate marker, and then placing the centre of their advice marker on a year sampled from a normal distribution around that target year. On occasion, advisors would offer 'off-brand' advice, designed to neither be close to the participant's initial estimate nor the correct year.

When a choice of advisors was offered, the two advisors were shown next to one another vertically on the lefthand side of the screen, and the participant clicked on the advisor from whom they wished to receive advice. There was never a time limit for making this choice.

On trials that contained feedback, the correct year was identified by the placement of a gold star placed over the timeline with a line pointing to the correct point on

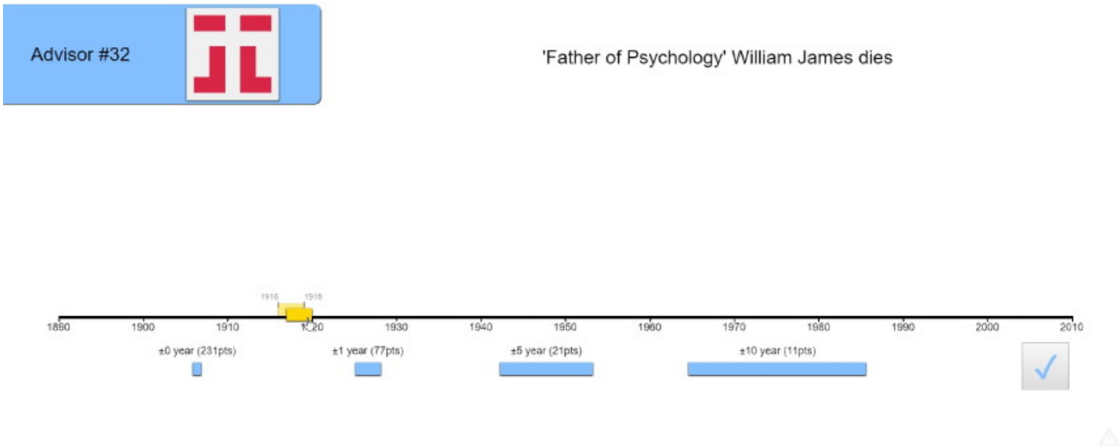


Figure 3.4: Dates Task with continuous responses.

the timeline. The correct year was also displayed in text. This allowed participants to see both their own marker placement and the advisor's advice marker placement (if it was a trial with advice), and to compare both easily to the correct answer. The feedback screen lasted 2 seconds before the next trial began.

Catch trials in the continuous Dates task consisted of an instruction to use the smallest marker to include a specific year, written out in words (e.g. 1942 would be "nineteen fourty-two"). Participants who failed in this attention check task immediately failed the experiment, and were not allowed to continue.

Binary In the binary version of the Dates task, participants were shown an event and given an 'anchor' year. Their task was to correctly identify whether the event occurred before or after the date shown (Figure 3.5).

Below the event and anchor display were the answer bars used to make responses. Participants selected a point on the bar of their choice (the lefthand bar indicating they believed the event was before the anchor year, and the righthand bar indicating they believed the event occurred after the anchor year). The higher the point they selected on the bar, the more confident their response. A blue bar appeared while they were choosing the height of their response, and the bar remained through the trial as a reference.

On trials with advice, an advisor's avatar would appear in the middle between the two bars with an arrow pointing to the bar that the advisor endorsed. On some versions of the paradigm, the advisor would provide an estimate with a measure of confidence. In these cases, the advisor's avatar appeared close to the bar they endorsed, and then slid up or down the bar to indicate how confident they were in the advice.

Once the advisor's avatar had indicated the advised response, participants again selected a height within a bar to enter their final decision. On trials with feedback, a gold star would appear next to the correct bar, along with the actual year of the event. The feedback lasted 2s.

As in the Dots task§3.1.3, advice was conceptualised along the binary dimensions of accuracy (whether the indicated bar was the correct one) and agreement (whether the indicated bar was the one selected in the participant's initial estimate). Where advisors gave estimates with a confidence, they first calculated an internal estimate of when they believed the event took place by drawing from a normal distribution around the correct year. They then determined their answer and their confidence by comparing their belief about the year the event took place with the anchor year, and scaling their confidence as a function of the distance between those years. This method is a somewhat rationalist version of how participants themselves may generate confidence judgements in this task.

When a choice of advisors was offered, advisor avatars were placed vertically in the centre of the screen, between the two bars, and participants clicked on the avatar of the advisor from whom they wished to hear advice. There was never a time limit for this choice.

Attention check trials in the binary Dates task prompted the participant to enter a specific response with a specific confidence. For example, the prompt that usually contained an event might instruct the participant to "enter 'Before' with

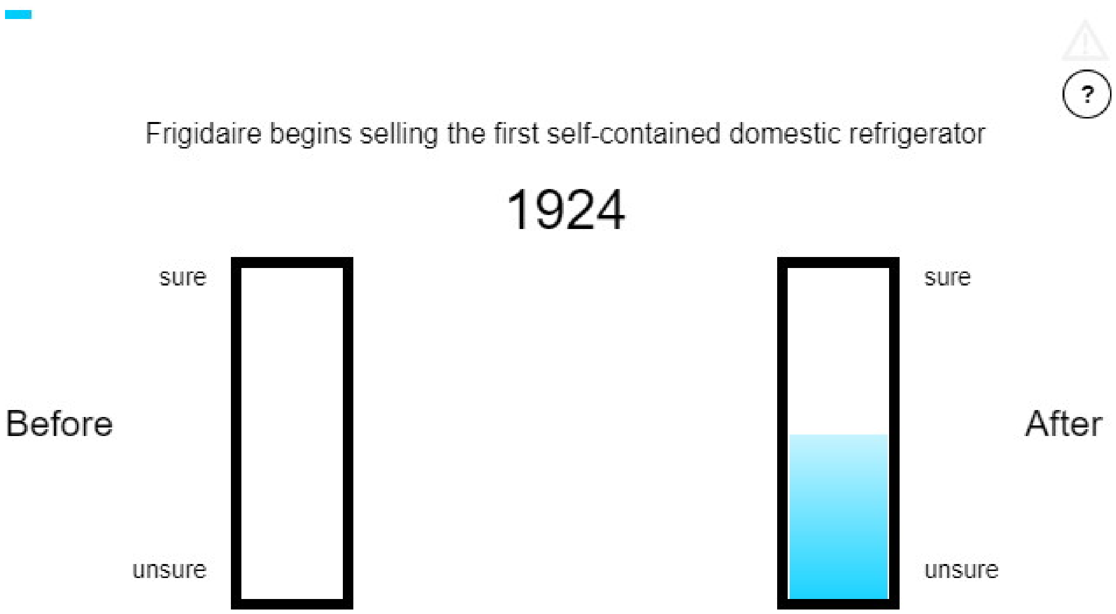


Figure 3.5: Dates Task with binary responses.

high confidence". As with the continuous version, entering an incorrect response would result in immediate termination of the experiment.

Selection of events for the Dates task The events selected for inclusion were determined through an iterative process of trial and error. Initially, a selection of events between 1850 and 1950 were compiled primarily from Wikipedia's timelines of the 1800s (*Timeline of the 19th Century* 2021) and 1900s (*Timeline of the 20th Century* 2021). Events were chosen that were felt by me to be somewhat guessable, and to have occurred on a specific year. A small website was created to allow people to enter a range of years within which they believed an event occurred, and these events were piloted by recruiting participants from Prolific. During piloting, each respondent saw each event in turn, and entered the year they believed the event to have occurred, along with years they were 90% sure the event occurred *after* and 90% sure the event occurred *before*. The data from the piloting indicated that people generally performed extremely poorly on the task, with a few exceptions for particularly famous dates. Performance was especially dire for events from the nineteenth century.

A new date range of 1900-2000 was chosen, and further events were added by revisiting the Wikipedia timeline for the 1900s (*Timeline of the 20th Century* 2021), as well as the Oxford Reference timeline of the 1900s (*20th Century* 2012). The new list of questions performed better during piloting. Questions where respondents' answers were particularly accurate or inaccurate were removed, leaving a final list of around 80 events.

QUESTION: I'm pretty sure I can put my hands on the experimental code, screenshots of the estimation task, and the brief write-ups or analysis of the responses. Likely I have the raw data for the responses, too. Is it worth including that stuff, either here in brief or in an Appendix?

Specific limitations of the Dates task There are several specific limitations of the Dates task. Firstly, and most importantly, we were unable to control participants' accuracy in the way we were able to control their accuracy in the Dots task. The advisor advice profiles' advice was determined through varying the probability of agreement contingent on the accuracy of the participant's initial estimate. Being unable to fix the overall accuracy of participants' initial estimates meant that the overall accuracy and agreement rates of the advisor advice profiles could differ quite substantially across participants and also between advisors for the same participant.

The inability to control overall accuracy rates is in part a consequence of the second limitation: the difficulty of any given question was idiosyncratic. The questions concerned dates of real historical events, and some of these may have had particular relevance to individual participants, or were perhaps encountered by those participants recently or memorably, allowing precise, accurate, and high-confidence responses to what would be for the average participant very difficult questions. The average performance of all participants on any given question is not necessarily a good guide to the individual experience of any given participant.

Lastly, participants found the task difficult overall. People tend to be more willing to take advice concerning difficult tasks (Gino and Moore 2007), perhaps as a way of diluting responsibility for the decision (Harvey and Fischer 1997). This meant that there was potential for some ceiling effects whereby advisors were not differentiated because all advisors were seen as useful, even if this was because of their potential for sharing blame rather than the informational content of their advice.

General limitations

There are several limitations common to both task designs. The most obvious limitation is that the advisors are not organically-interacting humans. There are other ecological validity limitations in the presentation of advice, the structure of the experiment, the absence of other cues, and the types of tasks used.

The use of artificial advisors means that advice can be carefully specified, and the experiments can be run easily, cheaply, and quickly. Transparently artificial advisors may, however, limit the generalisability of the experimental results in two ways. Firstly, if different integration processes exist for social and non-social information, it is plausible that, for at least some participants, the advice information is perceived as non-social information. While social and non-social information processing would not invalidate any findings (because such a factor would be unlikely to be systematically related to manipulations of interest), they may harm the ability of experimental results to inform us about the processes by which social information is integrated. Secondly, artificial advisors may not trigger a number of human-centred processes such as equality bias (Mahmoodi et al. 2015), meaning that effects revealed in these experiments may be much more difficult to observe in real human advice exchanges.

The advice presented to participants in the experiments is specific and impersonal. During real-life advice-taking, advice is often provided within a discussion, with estimates accompanied by reasons and points responded interactively. Although studies have indicated that advice-taking behaviour remains similar where discussion is allowed (Liberman et al. 2012; Minson, Liberman, et al. 2011), these experiments placed discussion within the context of advice exchanges over a decision made by both dyad members individually, and not with distinct roles for the advisor and judge as used in these experiments.

Further ecological validity limitations arise from the structure of the experiment. The task presents a series of trials sequentially, with a rapid procession through each. This structure is intended to condense a real-world relationship with an advisor, built up over repeated interactions over time, into as narrow a time window as possible. It is likely that this temporal compression does not fundamentally alter the processes of advisor evaluation and trust formation, but we have little positive evidence to support this supposition.

Another major difference between real-life advice and the advice offered in these experiments is in the richness of the relationship between advisor and judge. In a real-life relationship there would be numerous other factors at play which may alter or overwhelm any advice-taking and advisor evaluation processes revealed by these experiments. How well a person gets along with another could matter beyond simply increasing the perceived benevolence of the other, for example.

Lastly, both the Dots task and the Dates task had an objectively correct answer on every question. A substantial portion of everyday real-world advice-seeking behaviour concerns questions on which there is no readily-determinable objectively correct answer, such as where might be a good holiday destination, or whether one should pursue postgraduate education. These more subjective questions have received some attention by Van Swol (2011), but are seldom studied in the advice-taking cognitive psychology literature.

TODO: explain No gender breakdown (simplicity/privacy vs data gap)

3.1.4 Analysis

Dependent variables

Pick rate Pick rate provides a measure of source selection behaviour. In most experiments there are some trials that offer participants a choice of which advisor they would like to hear from. There are always two choices, and a choice must always be made. The two choices are consistent within the experiment. Pick rate is the proportion of choice trials in which a specified advisor was chosen.

A participant's pick rate is an aggregate over a number of trials, and expresses the observed probability of picking the specified advisor. The mentally represented preference for that advisor is not measured directly (if such a thing even exists), and cannot be determined from the observed pick rate without knowing the mapping function for each individual participant. Mapping functions (such as the logistic sigmoid function) produce stochastic choice behaviour from a preference marked on a continuous scale. The relationship between preference and pick rate is non-linear and idiosyncratic, but it is likely monotonic for all participants: the stronger the preference the higher the pick rate. Despite this monotonic relationship, it is important not to infer that one participant's preference for an advisor is stronger than another's on the basis of the former picking that advisor more frequently: it may be instead that, for the former participant, smaller differences in preference lead to more consistent picking behaviour.

Weight on Advice Weight on Advice, and its compliment Advice Taking, are commonly used to quantify the relative contributions of advice and initial estimates in making final decisions. It is obtained by dividing the amount an initial estimate was updated by the amount the advisor recommended adjusting the initial estimate. It thus expresses the amount the estimate changed as a proportion of the advised change.

Formally, Weight on Advice is given by $(e^F - e^I)/(a - e^I)$ where e^I and e^F are the initial estimate and final decision, respectively, and a is the advice.

Where the final answer moves away from advice (i.e. the final decision is further from the advice than the initial estimate), the value of Weight on Advice is negative, and where the adjustment towards advice exceeds the advice itself (i.e. the advice falls between the final decision and the initial estimate) the value of Weight on Advice is greater than 1. These values are typically truncated to 0 and 1, respectively.

In cases where the advice is exactly equal to the initial estimate, the denominator is equal to zero and the value for Weight on Advice is thus undefined. Trials in which the advice is exactly equal to the initial estimate are consequently discarded when calculating Weight on Advice.

Influence

Capped influence Influence, the dependant variable in some analyses, is calculated as the extent to which the judge's initial decision is revised in the direction of the advisor's advice. The initial (C_1) and final (C_2) decisions are made on a scale stretching from -55 to +55 with zero excluded, where values <0 indicate a 'left' decision and values >0 indicate a 'right' decision, and greater magnitudes indicate increased confidence. Influence (I) is given for agreement trials by the shift towards the advice:

$$I|\text{agree} = f(C_1) \begin{cases} C_2 - C_1 & C_1 > 0 \\ -C_2 + C_1 & C_1 < 0 \end{cases} \tag{3.1}$$

And by the inverse of this for disagreement trials:

$$I|\text{disagree} = -I|\text{agree} \tag{3.2}$$

The confidence scale excludes 0, and thus the final decision can always be more extreme when moving against the direction of the initial answer than when moving further in the direction of the initial answer. A capped measure of influence was used to minimise biases arising from the natural asymmetry of the scale. This measure was calculated by truncating absolute influence values which were greater than the maximum influence which could have obtained had the final decision been a maximal response in the direction of the initial answer (Figure 3.6).

The capped influence measure I_{capped} is obtained by:

$$I_{\text{capped}} = \min(I, |S_{\text{max}} - C_1|) \tag{3.3}$$

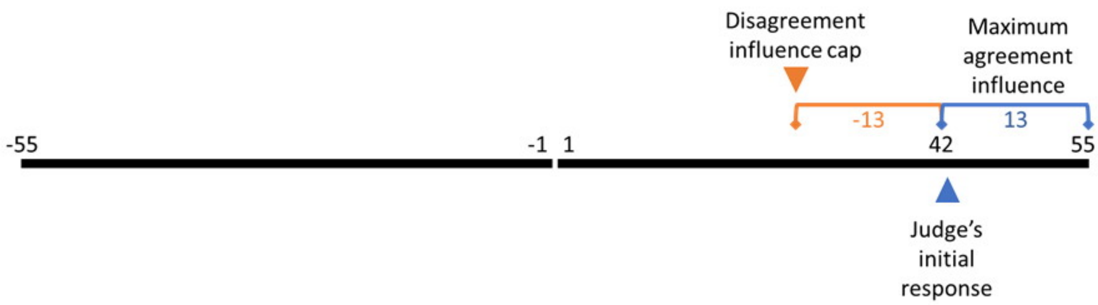


Figure 3.6: Capping influence to avoid scale bias.
In this example the judge's initial response is 42, meaning that their final decision could be up to 13 points more confident or up to 97 points less confident. Any final decision which is more than 13 points less confident is therefore capped at 13 points less confident.

Where C_1 and C_2 are the initial estimate and final decision, respectively, I stands for influence, and $S_{\max}$ indicates the maximum value of the scale. Thus, to follow the example in Figure 3.6, the initial estimate was 42, and if disagreeing advice led to a final decision of 2 the raw influence measure would be 40. This would then be compared to the absolute value of the scale width minus the initial estimate (55 - 42 = 13) and the minimum of the two would be the capped influence value, in this case 13.

There are benefits to and limitations of this approach. A benefit is that it places agreement and disagreement on the same potential scale. A limitation is that the truncation happens most extremely where initial estimates are made with very high confidence (because there is almost no increase that could have happened following agreement, so the cap is very low on the influence of disagreeing advice). Conceptually, these changes where a participant adjusts a view previously held with near certainty may in fact be the most interesting of all.

An alternative approach to capping, not used here, is to scale the observed adjustment by the potential space for adjustment. This approach often has the effect of overinflating influence of agreement, and can make very subtle differences in the use of the top end of the scale highly important. If, for example, a participant using a 50-point rating scale selected 48 for their initial estimate and received agreeing advice, a final decision of 49 would constitute an influence rating of 50% and a final decision of 50 an influence rating of 100%. It is not at all clear that participants differentiate the positions on the scales with this decree of precision, and so we considered this approach to capping influence inappropriate.

Inferential statistics

Statistical analyses are conducted using a both Frequentist and Bayesian statistics.

Frequentist statistics Frequentist statistics, often in psychology referred to simply as 'statistics', are a family of statistical tests which minimise the long-term error rates rates of the conclusions they invite. To perform one of these tests, the user first specifies a null hypothesis which the test will seek to reject. Next, the user selects a rate of false-positives that is acceptable, termed an alpha (α). In psychology α is almost universally .05, meaning that 5% of results for tests where there is no true effect will be positives. The principle statistic of interest from these tests is the p-value: the chance that a sample equivalent to the user's sample would be as extreme or more extreme than the observed sample *assuming that the null hypothesis is true.* The p-value quantifies how expected the sample observations are where the null hypothesis is the model: its complement quantifies how surprising the data are given the null hypothesis is true. If the p-value is lower than the α the null hypothesis may be rejected on the reasoning that such an unlikely sample as the current one is more consistent with some other hypothesis. Where the null hypothesis is rejected the sample is labelled as being *significantly different* from that expected from the null model, and this general feature of the difference is known as statistical significance.

It is common practice when using frequentist statistics to set up experiments such that two, and only two hypotheses can possibly explain the data. One of these is deemed the null hypothesis and tested as described above. The other hypothesis is termed the 'alternate' hypothesis, and, if the null hypothesis is rejected, is accepted in place of the null hypothesis because there were only two possible explanations and the null hypothesis has been ruled out.

Frequentist statistics are useful, better understood by psychologists than alternatives, and capable of delivering genuine insights when used correctly. A great many caveats surround their use and interpretation, however. Frequentist statistics cannot express positive evidence for the null hypothesis (because the null hypothesis is an assumption of the tests): p-values are uniformly distributed among samples drawn from the null hypothesis (Murdoch et al. 2008) and so no argument can be made from any particular value that it constitutes more or less evidence for the null. Likewise, p-values below the chosen α cannot strictly be interpreted as expressing more or less evidence for the alternate hypothesis (or against the null hypothesis). Frequentist statistics control long-term error rates rather than capturing relative likelihoods of theories, and thus the only legitimate inferences from a frequentist test are to accept or reject the null hypothesis. The uniform distribution of p-values under the null also means that repeated sampling from the null distribution without adjusting the α will increase the rate at which false-positive conclusions are drawn. This repeated sampling can happen in many ways such as running related tests on the same data, using slightly different data (e.g. by adjusting exclusions), and running tests at multiple time points in data collection (especially if a significant result terminates data collection).

In this thesis a range of frequentist statistical tests are used, most frequently t-tests and analyses of variance (ANOVA). The α is always set at .05 unless otherwise stated. Null hypotheses are always the expected distribution if the effect being tested is nil. Where the level of influence exerted by two different advisors is studied, for example, the null hypothesis would be that there were no systematic differences in influence exerted by those advisors.

Bayesian statistics Bayesian statistics consist in a statistical approach that attempts to capture the (posterior) likelihood of a hypothesis being true on the basis of its (prior) plausibility and the strength of the evidence. These tests are frequently adapted to quantify the relative likelihood of two competing hypotheses given their relative prior plausibilities and how consistent the observed evidence is with each theory. To perform these Bayesian statistical tests the user specifies the hypotheses to be compared, their relative likelihoods, and the evidence sampled.

The principle statistic in Bayesian tests is the Bayes Factor (BF). BF quantifies the posterior likelihood of one hypothesis over another. In the notation used here, BF always quantifies a more complex model over a simpler one: in the case of a Bayesian t-test it therefore quantifies the alternate hypothesis (in which there is an effect) over the null hypothesis (in which there is not an effect). The BF takes a value between 0 and infinity. Values below 1 indicate a greater likelihood for the simpler model, and that greater likelihood is given by 1/BF. Where BFs below 1 are reported in this manuscript the notation 1/BF is used to allow an intuitive reading of the strength of evidence in favour of the simpler model.

Bayesian tests produce a continuous measure of relative likelihood which simply describes the data and prior beliefs. In order to draw categorical inferences, thresholds are placed on this continuous outcome. Here these thresholds are $1/3 < \text{BF} < 3$, meaning that a BF of less than 1/3 constitutes evidence in favour of the simpler model while a BF greater than 3 constitutes evidence in favour of the more complex model. These values are those suggested by !TODO[CITE whoever has that neat little table. Jeffreys?] as representing !TODO[which word does the paper use?] evidence in a given direction. Where the BF lies between these thresholds it is labelled as 'uninformative'. An uninformative result supports neither the simpler nor the more complex model, and indicates that the data are insufficient to distinguish the hypotheses.

The Bayesian tests used here rely on the priors specified by the BayesFactor R package !TODO[CITE BayesFactor]. These priors govern the expected distributions of observed differences between samples where there is or is not a genuine effect creating systematic differences. The use of the same, weakly-informative priors for all tests means the approach used here is an 'objective Bayesian' approach. This objective Bayesian approach can be contrasted with a 'subjective Bayesian' approach in which the goal is to specify the exact amount of belief one should have in one hypothesis over another. Neither the objective nor subjective approach is clearly superior. The subjective approach is used here because it is simpler. There is some risk that results will be a poor fit to reality because the priors are inappropriate, but this risk is fairly low and somewhat mitigated by the additional inclusion of frequentist statistics.

Integrating statistical results In most cases, Bayesian and frequentist statistics produce the same conclusion !TODO[pretty sure there's a paper which states/demonstrates this]. Where this is not the case, results should be interpreted very cautiously: a significant frequentist test with an uninformative or null-favouring Bayesian test can indicate that the result may be a false-positive, while clear Bayesian support for the alternate hypothesis in the absence of a significant frequentist test can indicate that the priors in the Bayesian test are inappropriate.

Where null conclusions are to be drawn, i.e. the null hypothesis is to be retained, only Bayesian statistics can be considered informative. In these cases Bayesian statistics will be interpreted, with the caveat that the safeguard of using two independent approaches to draw statistical conclusions has lapsed.

Software Data analysis was performed using R (R Core Team 2018). For a full list of packages and software environment information, see !TODO[figure out where to include this stuff. Appendix? Also link to a containerized version of this.]

Unanalysed data

Early versions of several experiments had bugs in the experiment code made the results unreliable. The data collected during these runs is available alongside the data collected in the final versions of experiments. For details on what went wrong with the experiments in which bugs were found, see the description column of the data files in the `esmData::` R package. While not useful for the hypotheses of the experiments for which the data were collected, some excluded data can be used for other purposes such as analysing responses to advice.

!TODO[Appendix these?]

Experiment 0 This branch of experiments was the core of the date task. As such, most experimentation and piloting happened on this branch, meaning that there were many versions of the study where data were collected for which analysis is not included here.

Experiment 1A The earliest versions contained a bug where advisors instructed to agree with a participant instead provided advice identifying the correct answer. Other versions had a bug in the staircasing code used to titrate the difficulty of the task was converging on too high a value (74% initial decision accuracy as opposed to 71%). Once the staircasing bug was fixed, two more experiments were run, one with 60 practice trials in which participants did not quite reach the desired accuracy before the beginning of the main experiment, and one with 120 practice trials which constitutes the data analysed below.

Experiments 1B and 2B Early versions included a bug which prevented feedback from being shown during the familiarisation phase even to participants in the Feedback condition. These participants could theoretically be included in the No feedback condition regardless of their condition label in the data, but this is not done here.

Experiment 3A Pilot data was collected to ensure the study functioned properly, and so the data are not analysed with regard to the hypotheses. The v1 Mixed design data was conducted and run as a proper experiment in which participants learned about both advisors simultaneously (preregistered at https://osf.io/5z2fp), but there were no effects in the data.

Experiment 3B The first version included a bug in which the advisor choice options were not recorded, making it difficult or impossible to work out which trials included a choice of advisor. Data for these participants could be included in a study that was agnostic about advisor choice.

Experiment 4A The first version had a bug which meant that the advisors were identical in the familiarisation phase. Both the first and second versions had a bug in which advisors who were supposed to agree with the participant's initial estimate gave the correct answer rather than agreeing. These participants' data could be included in an analysis which used a participant's actual experience of advice to predict their advice-taking and source selection behaviour, provided appropriate care was taken to reconstruct the advice data from the raw values instead of relying on the reported summaries.

Experiment 5 Several versions of this study were run in the course of developing the manipulation. The initial version had a bug that prevented the groups from being visually distinct. Later versions introduced a clearer manipulation, equivalent rather than genuinely misleading advice for the Sometimes misleading advisor, and rating of advice deceptiveness, respectively.

Experiment 6 Two short pilot versions of this experiment were run, and participants' data were not analysed due to bugs in the experiment code.

3.2 Open science approach

3.2.1 Open science

Nullius in verba ("take nobody's word for it") is written in stone above the entrance to the Royal Society's library. This fundamental principle of science, that it proceeds on evidence rather than assertion, has frequently been forgotten in practice. Concerns about sloppy, self-deluding, or outright fraudulent science have existed since at least the time of Bacon. The modern open science movement in psychology dates from the early 2010s. Simmons et al. demonstrated how easily false positive results could emerge from unconstrained researcher degrees of freedom in analysis (Simmons et al. 2011), Nosek and colleagues published a roadmap for improving the structure and function of academic research and publishing (Nosek and Bar-Anan 2012; Nosek, Spies, et al. 2012), and the Open Science Collaboration began

(Collaboration 2015). In the years following, a deluge of papers, movements, and practical changes have emerged. The meaning of open science varies within each sub-discipline, and this section outlines how the experiments comprising this thesis have been conducted in a reproducible and transparent manner.

3.2.2 Badges

Following the Center for Open Science (`https://cos.io`), this thesis uses a series of badges to indicate adherence to particular aspects of open science. Three badges, *preregistration*, *open materials*, and *open data*, are adopted directly from the Centre and used according to the Centre's rules. Studies which qualify for a badge will have the badge displayed immediately below their title. Each badge will contain a link to online resources which provide the content for which the badge is awarded.

Preregistration

Preregistration of a study means that information about the study has been solidified prior to the analysis of the data. This means that hypotheses cannot be changed to represent unanticipated or overly-specific findings as a priori predicted (Kerr 1998). In practice in this thesis, preregistration means describing in detail the design and analysis plan for an experiment and depositing the description with a reputable organisation prior to data being collected. The links which accompany the preregistration badge will point to the preregistration document. These measures help to prevent presenting a highly selected and biased interpretation of the data as the result of a natural analytical process.

The preregistration badge also appears within results sections to designate those statistical investigations which were included in the preregistration. Some analyses are exploratory. These exploratory analyses are not included in the preregistration, because they are inspired by the data themselves. They are reported after the preregistered analyses, or are clearly designated as exploratory in the text.

Open materials

A foundational principle of science is that findings can be reproduced by other people. Open materials facilitate reproduction by making it easier to rerun an experiment. Open materials also increase the likelihood that errors can be identified. In the case of the behavioural experiments reported here, the open materials include computer code necessary to run the experiment. The links accompanying the open materials badge points to this code.

Open data

Theories are the output of science as a whole, but data are the output of any individual study. Sharing data directly allows other scientists to check and extend the data analysis conducted, to reuse the data in meta-analyses, and to repurpose the data for other investigations. This increases the robustness of the results, and increases the efficiency of science as a whole. The links accompanying the open data badge point to online storage where the data can be obtained for a study, along with appropriate metadata.

3.2.3 Thesis workflow

This thesis is written in RMarkdown, with the data fetched and analysed at the time the document is produced using the publicly available pipeline - the entire document can be reproduced locally using the source code in an appropriate environment. Some parts of the thesis use computational modelling or model fitting. These parts use cached datasets, also available online, in order to reduce the time required to create the thesis from its source files from hours to minutes using a high-end desktop personal computer. The caching behaviour can be prevented for those wishing to evaluate the validity of the modelling components by setting the value of the R option `ESM.recalculate` to higher values (documented in the `index.Rmd` file). A Docker environment copying the environment used to produce this document is available at !TODO[the containerisation thing]

4 Psychology of advice-taking

The model of advice-taking presented earlier§2 has some empirical support from previous work in our lab (Pescetelli and Yeung 2021). In those experiments, participants completed a perceptual decision-making task within a judge-advisor system. Participants provided a judgement (including a confidence rating) concerning which of two briefly presented boxes of dots contained more dots. They then received advice on the answer from an advisor and provided a final decision, again including a confidence judgement.

Pescetelli and Yeung (2021) showed that participants who received feedback on their decisions, which could in turn be used to evaluate the advisors, showed larger shifts in their confidence in the direction of advice from more accurate advisors. Participants who did not receive feedback showed similarly higher influence from advisors who tended to agree more often (i.e. to suggest as advice the same side participants chose in their initial estimate). Both effects, accuracy and agreement, were present regardless of feedback, although the agreement effect was far more pronounced in the no feedback group than the feedback group, and the accuracy effect was more pronounced in the feedback than the no feedback group. Participants' subjective assessments of the trustworthiness of the advisors roughly[1] followed their

[1] In some cases the results were only numerically compatible, and not statistically significant.

behaviour: advisors who were more influential were rated as more trustworthy.

We aimed to replicate these results in a new paradigm. This new paradigm, the Dates task, was newly-implemented for this project. The Dates task was designed with several appealing features in mind: it was to be more similar to the tasks performed by participants in previous advice-taking experiments; it was to be shorter for participants to complete, and suitable for online delivery; and it was supposed to be more engaging for participants.

For this replication, we chose to directly contrast the tendency to prefer agreeing advisors over accurate ones where feedback was withheld with the tendency to prefer accurate advisors over agreeing ones where feedback was available.

4.1 Experiment 0: Extending results to the Dates task

Open scholarship practices

This experiment was preregistered at `https://osf.io/fgmdw`. The experiment data are available in the `esmData::` package for R from !TODO[Zenodo esmData], and also directly from !TODO[OSFify Dates task data]. A snapshot of the state of the code for running the experiment at the time the experiment was run can be obtained from `https://github.com/oxacclab/ExploringSocialMetacognition/blob/f90b6f9266a901211a4ddb7b5ee1de1c74e8df57/ACv2/index.html`.

This is a replication of a study of identical design that produced the same results. The data for the original study can be obtained from the `esmData::` R package or from !TODO[OSFify].

Method

37 participants each completed 42 trials over 3 blocks of the continuous version of the Dates task§3.1.3. On each trial, participants were presented with an historical event that occurred on a specific year between 1900 and 2000. They were asked to drag one of three markers onto a timeline to indicate the date range within which they thought the event occurred. The three markers each had different widths,

and each marker had point value associated with it, with wider markers worth fewer points. The markers were 1, 3, and 9 years wide, being worth 27, 9, and 3 points respectively. Participants then received advice indicating a region of the timeline in which the advisor suggested the event occurred. Participants could then mark a final response in the same manner as their original response, and could choose a different marker width if they wished.

Participants started with 1 block of 10 trials that contained no advice to allow them to familiarise themselves with the task. All trials in this section included feedback for all participants indicating whether or not the participant's response was correct.

Participants then did 2 trials with a practice advisor to get used to receiving advice. They also received feedback on these trials. They were informed that they would "get advice on the answers you give" and that the feedback they received would "tell you about how well the advisor does, as well as how well you do". Before starting the main experiment they were told that they would receive advice from multiple advisors and that "advisors might behave in different ways, and it's up to you to decide how useful you think each advisor is, and to use their advice accordingly".

Participants then performed 2 blocks of trials that constituted the main experiment. In each of these blocks participants had a single advisor for 14 trials, plus 1 attention check.

Participants were split into four conditions that produced differences in their experience of these blocks. These conditions were whether or not they received feedback, and which of the two advisors they encountered first. For each advisor, participants saw the advisor's advice on 14 trials. On each trial there was an 80% chance the advisor gave advice according to the advice profile (detailed below). Otherwise, the advisors issued the same kind of advice as one another, chosen to neither agree with the participant's answer nor indicate the correct answer. This "off-brand" advice was used to control for the effects of advice when the influence of advice was the dependent variable.

Advice profiles The High accuracy and High agreement advisor profiles defined marker placements based on the timeline based on the correct answer and the participant's initial estimate respectively. Both advisors used markers that spanned 7 years, and both placed the markers in a normal distribution around the target point with a standard deviation of 5 years. The target point for the High accuracy advisor was the correct answer, and the target point for the High agreement advisor was the participant's initial estimate. Neither advisor ever placed their marker exactly on the midpoint of the participant's marker (because doing so means the weight on advice statistic is undefined).

Each advice trial had a 20% chance of being designated "off-brand". On these trials advisors neither indicated the correct answer nor agreed with the participant. This was achieved by picking a target point of the participant's answer reflected around the correct answer. Thus, if the centre of the participant's initial estimate was 1955, and the correct answer was 1945, the target point would be 1935. In some cases there was not enough of the scale left once this reflection had taken place. If the centre of a participant's initial estimate was 1960, and the correct answer was 1990, the target point could not be 2020, because the scale only went up to 2010. Where this occurred, advisors issued answers that were twice as wrong as the participant's answer, in the same direction. In this case, that would mean the new target point was 1930.

Results

Exclusions In line with the preregistration, participants' data were excluded from analysis where they failed attention checks, fail to complete the entire experiment, or had more than 2 outlying trials. Outlying trials were calculated after excluding participants who failed to complete the experiment, and were defined as trials for which the total trial time was greater than 3 standard deviations away from the mean of all trials from all participants.

Table 4.1: Participant exclusions for Dates task advice influence experiment

Reason	Participants excluded
Attention check	6
Unfinished	2
Too many outlying trials	1
Missing offbrand trial data	1
Total excluded	**8**
Total remaining	**29**

A browser compatibility issue in this study meant that any participants completing the study using the Safari family of browsers had to be excluded because the advice was not presented appropriately.

Task performance

```
## Warning: Removed 1 rows containing non-finite values (stat_boxplot).
## Warning: Removed 1 row(s) containing missing values (geom_path).
## Interaction calculation: Initial - Final
```

Participants had lower error on final decisions than on their initial estimates ($F(1,28) = 102.57$, $p = <.001$; $M_{\text{Initial}} = 16.45$ [14.51, 18.39], $M_{\text{Final}} = 11.26$ [9.72, 12.81]), indicating improved performance after seeing advice. They also had lower error on their answers with the High accuracy advisor ($F(1,28) = 44.75$, $p = <.001$; $M_{\text{HighAgreement}} = 16.89$ [14.70, 19.09], $M_{\text{HighAccuracy}} = 10.82$ [9.24, 12.41]). As expected, there was an interaction: participants reduced their error much more following advice from the High accuracy advisor ($F(1,28) = 74.50$, $p = <.001$; $M_{\text{Reduction|HighAgreement}} = 1.06$ [0.45, 1.67], $M_{\text{Reduction|HighAccuracy}} = 9.32$ [7.38, 11.25]; Figure 4.1).

Generally, we expect participants to be more confident on trials on which they are correct compared to trials on which they are incorrect. Confidence can be measured by the width of the marker selected by the participant. Where participants are more confident in their response, they can maximise the points they receive by selecting a thinner marker. Where participants are unsure, they can maximise their chance of getting the answer correct by selecting a wider marker. Participants'

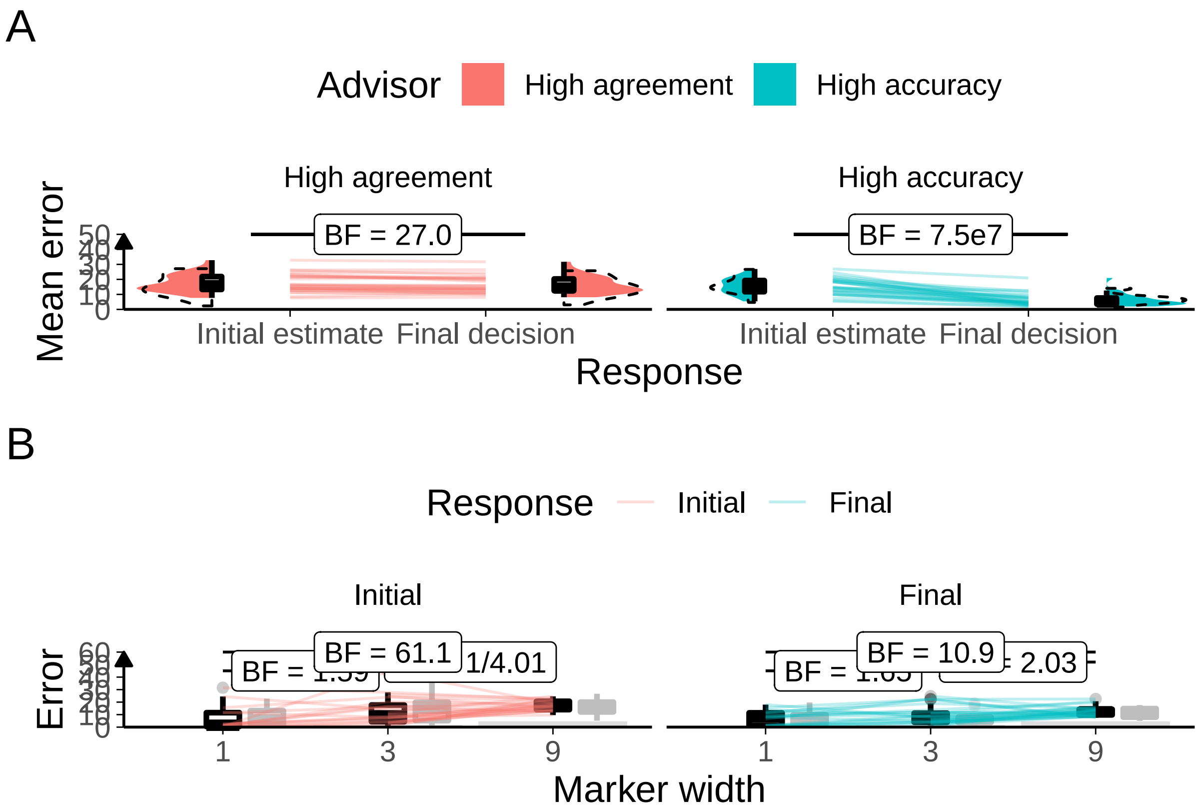


Figure 4.1: Task performance for Experiment 0.
A: Response error.
Faint lines show individual participant mean error (the absolute difference between the participant's response and the correct answer), for which the violin and box plots show the distributions. The dashed line indicates chance performance. Dotted violin outlines show the distribution of participant means on the original study which this is a replication. The dependent variable here is error, the distance between the correct answer and the participant's answer, and consequently lower values represent better performance. The theoretical limit for error is around 100.
B: Error by marker width.
Faint lines show individual participant mean error (distance from the centre of the participant's marker to the correct answer) for each width of marker used, and box plots show the distributions. Some participants did not use all markers, and thus not all lines connect to each point on the horizontal axis. The dashed box plots show the distributions of participant means in the original experiment of which this is a replication. The faint black points indicate outliers.
Grey bars show half of the marker width: mean error scores within this range mean the marker covers the correct answer.

error was lower for each marker width in final decisions than initial estimates (Figure **??**). For both initial estimates and final decisions, error was higher for wider markers than for narrower ones.

Advisor performance The advice is generated probabilistically so it is important to check that the advice experienced by the participants matched the experience we designed. On average, the High accuracy advisor had lower error than the High agreement advisor ($t(28)$ = -12.95, p < .001, d = 3.37, BF = 3.0e10; $M_{\text{HighAccuracy}}$ = 3.88 [3.49, 4.27], $M_{\text{HighAgreement}}$ = 16.99 [14.93, 19.05]), and their advice was further away from the participants' initial estimates than the High accuracy advisor's ($t(28)$ = 9.66, p < .001, d = 2.00, BF = 5.0e7; $M_{\text{HighAccuracy}}$ = 17.80 [15.58, 20.02], $M_{\text{HighAgreement}}$ = 7.77 [6.24, 9.29]). 29/29 (100.00%) participants experienced the High accuracy advisor as having lower average error than the High agreement advisor, and 27/29 (93.10%) participants experienced the High agreement advisor as offering advice closer to their initial estimates than the High accuracy advisor. Overall, this indicates that the manipulation was implemented as planned.

Hypothesis test

```
## Warning: Data is unbalanced (unequal N per group). Make sure you specified a
## well-considered value for the type argument to ezANOVA().
```

```
## Interaction calculation: Accurate - Agreeing
```

There were systematic differences in the influence of advice on the key trials where the advice itself was balanced between advisors (Figure 4.2. The Accurate advisor was more influential than the Agreeing advisor (F(1,27) = 5.10, p = .032; M_{Accurate} = 0.63 [0.51, 0.76], M_{Agreeing} = 0.47 [0.33, 0.62]), and this was more extreme in the Feedback than the No feedback condition (F(1,27) = 9.72, p = .004; $M_{\text{Accurate-Agreeing|Feedback}}$ = 0.38 [0.19, 0.58], $M_{\text{Accurate-Agreeing|NoFeedback}}$ = -0.05 [-0.28, 0.17]). There was no significant difference between the Feedback and No feedback groups overall (F(1,27) = 2.02, p = .167; M_{Feedback} = 0.48 [0.34, 0.61], $M_{\text{NoFeedback}}$ = 0.62 [0.45, 0.80]). As shown in the Bayesian statistics on the graph (Figure 4.2),

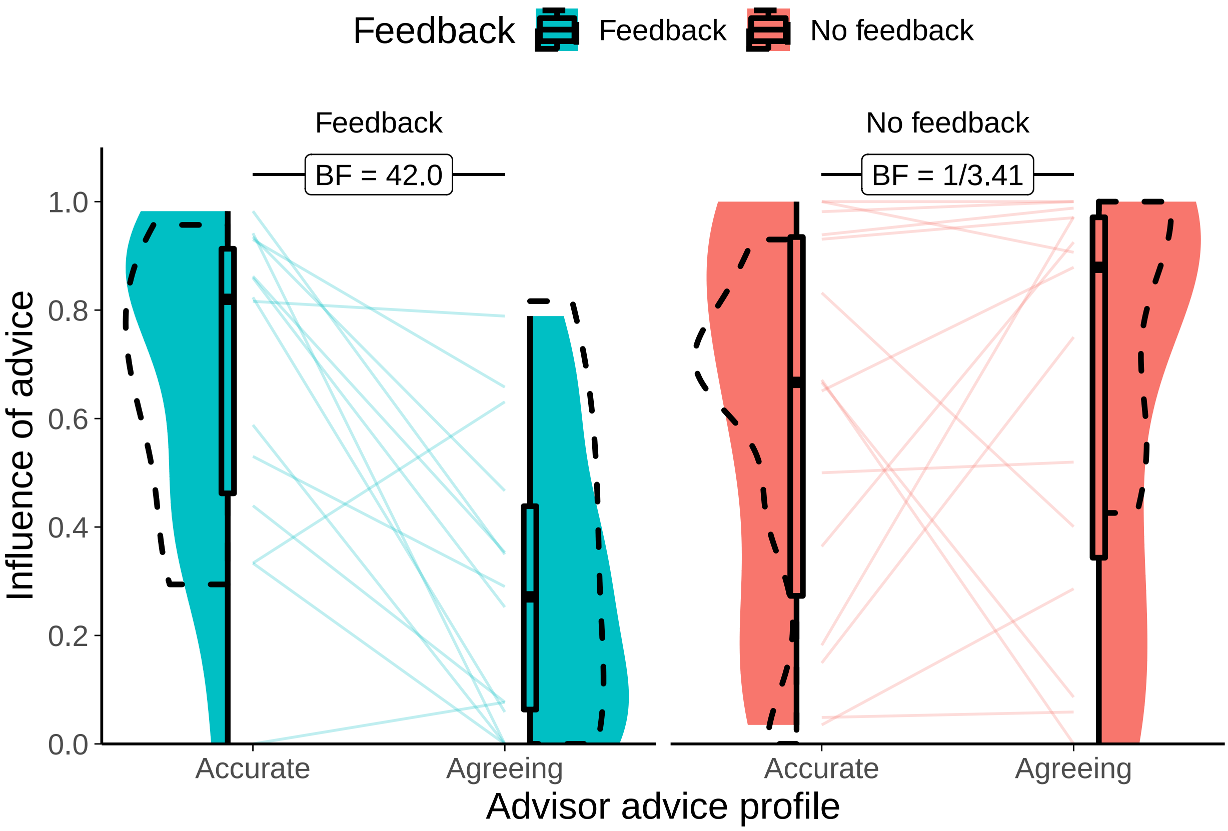


Figure 4.2: Dates task advisor influence for high accuracy/agreement advisors. Shows the influence of the advice of the advisors. The shaded area and boxplots indicate the distribution of the individual participants' mean influence of advice. Individual means for each participant are shown with lines in the centre of the graph. The dashed outline shows the distribution of participant means in the original study of which this is a replication.

there was good evidence that the advisors were differently influential in the Feedback condition ($t(13) = 4.27$, $p < .001$, $d = 1.34$, BF = 42.0; $M_{Accurate|Feedback} = 0.67$ [0.50, 0.84], $M_{Agreeing|Feedback} = 0.29$ [0.13, 0.44]), and adequate evidence that they were not different in the No feedback condition ($t(14) =$ -0.50, $p = .626$, $d = 0.14$, BF = 1/3.41; $M_{Accurate|!Feedback} = 0.60$ [0.40, 0.80], $M_{Agreeing|!Feedback} = 0.65$ [0.44, 0.86]).

4.1.1 Discussion

The experiments show that people can learn to disregard advice which provides support but no information, provided that feedback is available. Where feedback is unavailable, people do not appear to distinguish between useful and supportive advice. These results are not wholly consistent with an account of advice-taking in which people use agreement to evaluate advisors in the absence of feedback. Under such an account, we would expect participants in the No feedback condition to have shown a greater susceptibility to advice from the Agreeing advisor, but this did not happen.

The equivalence of the influence of the advisors in the No feedback condition may be a consequence of relatively high levels of advice influence overall, producing a ceiling effect (the numerical advantage for the Agreeing advisor is as predicted by the theory). The relatively high levels of advice influence are a feature of the Dates task; the questions are difficult for most participants and consequently participants are likely to take more advice (Gino and Moore 2007; Yonah and Kessler 2021). Another possible explanation for the equivalence of influence between the advisors is that participants may not have had enough exposure to the advisors to properly learn about the value of their advice. The limitation on exposure is another feature of the Dates task: while the Dots task provides participants with many tens of trials in which to update their assessment of advisors, the Dates task provides a level of exposure more similar to normal social interaction (although not necessarily very similar).

It is also arguable that the absence of differences in the No feedback condition is a consequence of the hypothesised mechanism itself. While the most plausible account of the equivalence between advisors in the No feedback condition must be that the participants were not able to distinguish between them, one might argue that the results are a consequence of averaging participants who came to different conclusions (as shown in Figure 4.2 where the individual participants' lines have steep slopes). If participants who did not receive feedback distinguished advice using two features, high-confidence agreement and variability of advice, participants whose Accurate advisors happened to agree where the participants were most confidence (i.e. the participants were well-calibrated) would be interpreted as better because they would agree when the participant had high confidence and provide visible correction where they did not. In a somewhat similar vein, the infrequent but necessary off-brand trials may have occurred at inopportune moments for some advisors, e.g. when the participant was particularly sure of an answer. It would be expected that these errors would seriously damage a participant's trust in the advisor who offered off-brand (neither agreeing nor correct) advice on those high-confidence trials. There is no reason to suspect that this happened systematically for one advisor over another, but by its unsystematic nature it could have overshadowed the effects we were looking for.

The design of this experiment deliberately violated an assumption which may be generally true in real life, that the advice is sufficiently independent as to convey at least some information regarding the correct answer. Had we told participants that the agreeing advisor would agree with them no matter what they the participants said, the participants may have disregarded the advice. Nevertheless, the results show that, even where they would have performed objectively better by preferring accurate over agreeing advice, participants were not able to detect the more accurate advice without objective feedback.

The results were only partially compatible with those of Pescetelli and Yeung (2021). In the Feedback condition, we saw, as expected, a much greater influence of advice from the Accurate advisor. In the No feedback condition, however, the influence of the advisors appeared to be equivalent. We did demonstrate that this paradigm is capable of measuring influence differences.

In the experiments that follow§(chapter-advisor-choice), both the Dots and Dates tasks are used to investigate differences in source selection behaviour.

4.2 Confidence-contingent advice

We had hoped to investigate confidence-contingent advice, as used by Pescetelli and Yeung (2021). Confidence-contingent advice depends upon the participant's confidence in their initial estimate. A Bias-sharing advisor would be more likely to agree where the participant was confident in the initial estimate, and less likely to agree where the participant was less confident. An Anti-bias advisor would do the opposite, and the advisors could be matched for overall agreement and accuracy behaviour.

We designed and ran a prototype experiment along these lines, but were unable to get the manipulation to work, or to determine a way to appropriately balance overall agreement and accuracy behaviour of the advisors given the wide variability of participants' initial estimate performances.

5

Psychology of source selection

Advice is relied upon to a different extent depending upon a variety of markers for its trustworthiness, including its plausibility and the reputation of its source. The question at the core of this chapter is whether people preferentially seek out advice from advisors they believe to be more trustworthy. At first glance, it may seem a foregone conclusion that people will seek out advice from more trustworthy advisors: who after all wants to be advised by fools or liars? Nevertheless, empirical evidence for this kind of source selection behaviour has been somewhat mixed. The evidence that people do tend to seek out information from sources likely to agree with them is moderate ('selective exposure', Hart et al. 2009). The evidence that people avoid information likely to disagree is poor ('selective avoidance', Jang 2014; Weeks et al. 2016), with evidence becoming less persuasive as tasks become more ecologically valid (Sears and Freedman 1967; Nelson and Webster 2017).

There are also intuitive arguments for a range of potential findings. It would make sense that people seek out information they are more likely to use, because all information acquisition comes with some kind of cost, even if only attentional and opportunity costs, and rational actors should maximise their benefit-cost trade-off. It may make sense for people to seek out information they are likely to agree with, regardless of usefulness, because they may be exercising critical vigilance over their own side in a debate, ensuring that bad arguments are not used to support their ideological position. It would also make sense, however, for people to seek out information from sources they disagree with: perhaps those we disagree with have access to evidence or reasons we had not considered; or perhaps learning about others' views will allow us to better counter them and convert their adherents (Freedman 1965). People may even prefer a balanced or random diet of information because they feel unable to judge relative quality, or because all the reasons above are pulling them in different directions.

The vast majority of the source selection literature uses surveys or browsing tasks with stimuli being realistic politically-charged media items. Measurements are either active interest ratings or passive activity monitoring (usually on links clicked or reading time, but eye tracking is a recent innovation: Marquart 2016; Schmuck et al. 2020). The experiments here are more traditional cognitive psychology experiments: the complex contextual factors suspected of driving selectivity are removed (Festinger 1957; Knobloch-Westerwick 2015) and only the informational motive remains. While it is not impossible that a preference for agreement which makes sense from a self-image preservation perspective bleeds into a context where accuracy is key, the experiments here at least provide a context where a correct answer exists.[1]

5.0.1 Similar work

The source selection literature is largely from the domains of social and personality, in which the constructs that produce the phenomena are attitudes and self-concepts. This work is grounded in the cognitive psychology domain, and consequently uses a model of advisor evaluation (Advisor Evaluation without Feedback§2.0.3) that posits measurable variables and mathematically-describable processes. This model is based on a similar model from Pescetelli and Yeung (2021). It takes as a starting point the observation that, given objective feedback, people can use that feedback to learn about the trustworthiness of advisors (Yaniv and Kleinberger 2000; Pescetelli

[1]It is arguable that people may seek out politically-concordant information because they believe it to be more accurate rather than because they do not want their self-image to be challenged.

and Yeung 2021; Behrens et al. 2008). The extent to which advice is taken (§3.1.4) is commonly used as a measure of a participant's trust in an advisor, on the argument that the participant seeks to maximise task performance and task performance is maximised by taking more advice from more trustworthy advisors. As expected, people make greater use of advice they believe will be more accurate compared to less accurate (Gino, Brooks, et al. 2012; Rakoczy et al. 2015; Sniezek, Schrah, et al. 2004; Soll and Larrick 2009; Tost et al. 2012; Schultze, Mojzisch, et al. 2017; Wang and Du 2018; Önkal, Gönül, et al. 2017).

When objective feedback is unavailable, it is still possible for people to demonstrate a greater dependence upon advice from more as opposed to less accurate advisors. This is a consequence of agreement: where the base probability of being correct is greater than chance, the independent estimates of people who are more accurate will agree more often (leading to 100% agreement on the correct answer for two independent decision-makers of perfect accuracy). In the absence of feedback, therefore, agreement can be used as a proxy for accuracy, as formalised in the model.

Pescetelli and Yeung (2021) demonstrated that advice is more influential from (equally accurate) advisors who tend to agree with a participant more frequently when objective feedback is not provided. This is despite the fact that advice is generally more influential when it disagrees with the participant's initial estimate. [2] Their data suggest that people may be using agreement as a proxy for accuracy, although they may also simply prefer agreement over disagreement when there is no accuracy cost to be paid. In this chapter, we partially replicate the findings of Pescetelli and Yeung (2021), and explore the consequences of pitting accurate advisors against agreeing advisors.

5.0.2 Overview of Experiments

We conducted a series of experiments to explore whether the advice-taking behaviour observed previously (Pescetelli and Yeung 2021) would translate into preferential

[2] This is partly due to the nature of the judge-advisor system: there is always room for disagreement to be more extreme than agreement, because agreement is lower-bounded by the participant's initial estimate.

source selection behaviour. In these experiments participants were familiarised with different pairs of advisors, and then given the opportunity to select which advisor they would like to get advice from (General Method§3.1).

Each experiment was repeated using two different tasks: a perceptual decision-making "Dots task", extended from Pescetelli and Yeung (2021); and a historical date estimation "Dates task" newly built for this project. To reduce expenditure, all participants in the Dots task experiments received no feedback on their answers while learning about advisors, because contrasting the presence and absence of feedback was done by Pescetelli and Yeung (2021). In the Dates task, due to its novelty, participants were split into conditions based on whether or not they received feedback while learning about the advisors.

The first of the advisor pairs was a high accuracy advisor and a low accuracy advisor (Experiment 1A§5.1.1 and Experiment 1B§5.1.2). We predicted that the high accuracy advisor would be selected more often, even where feedback was not available. This experiment would demonstrate the minimum phenomenon of interest – sensitivity to advisor accuracy in the absence of feedback translating into preferential advisor selection.

The second advisor pair was a high agreement and a low agreement advisor (Experiment 2A§5.2.1 and Experiment 2B§5.2.2). We predicted that the high agreement advisor would be selected more frequently, because we expect agreement to be the method by which the accurate advisors were detected in the previous task. This experiment would constitute a test of the purported mechanism.

The third advisor pair was a high agreement advisor and a high accuracy advisor (Experiment 3A§5.3.1 and Experiment 3B§5.3.2). We predicted that the high agreement advisor would be selected more frequently than the high accuracy advisor, but only where feedback was withheld. Where feedback was provided, we expected the high accuracy advisor to be picked more often. This experiment would test whether the absence of feedback invites using agreement as a substitute for accuracy.

The final advisor pair were confident contingent advisors like those used in Pescetelli and Yeung (2021) (Experiment 4A§5.4.2 and the Lab Study§5.4.1). These advisors agree at the same rate and are similarly accurate, but one agrees more when the participant expresses high initial confidence and less when the participant expresses low initial confidence, and the other vice-versa. We predicted that the former 'bias-sharing' advisor would be selected more often because participants would use their own sense of confidence to weight the value of agreement. This experiment would explore whether metacognitive processes are able to finesse the basic agreement-for-accuracy substitution.

5.1 Effects of Advice Accuracy on Advisor Selection

Pescetelli and Yeung (2021) demonstrated that more accurate advisors are more trusted and more influential (regardless of the presence of feedback) in a lab-based perceptual decision-making task. We attempted to extend this finding to the domain of advisor selection in two online tasks: a 'Dots task' requiring similar perceptual decision-making, and an estimation-based 'Dates task'. We predicted that participants would choose a more accurate advisor over a less accurate one, and would do so even in the absence of objective feedback (based on the hypothesis that they can infer accuracy from differing agreement rates).

The ability to distinguish between accurate advisors in these experiments is important because discrimination based on accuracy is crucial to the phenomenon we are attempting to explain: rational advice-seeking behaviour in the absence of feedback. Pescetelli and Yeung (2021) demonstrated that people could identify and exploit more accurate advice, and these experiments seek to determine whether people will use that ability to obtain advice from a more reliable source. Experiment 1A§5.1.1 addressed this issue using the same perceptual decision task as use by Pescetelli and Yeung (2021); Experiment 1B§5.1.2 extended the approach in a task asking participants to estimate historical dates.

5.1.1 Experiment 1A: Advice accuracy effects in the Dots Task

Open scholarship practices

This experiment was preregistered at https://osf.io/u5hgj. The experiment data are available in the esmData:: package for R from !TODO[Zenodo esmData], and also directly from https://osf.io/kn23p/. A snapshot of the state of the code for running the experiment at the time the experiment was run can be obtained from https://github.com/oxacclab/ExploringSocialMetacognition/blob/9932543c62b00bd96ef7c AdvisorChoice/index.html.

Method

59 participants each completed 368 trials over 7 blocks of a perceptual decision-making task. Each trial consisted of three phases: participants gave an initial estimate (with confidence) of which of two briefly presented boxes contained more dots; received advice on their decision from an advisor; and made a final decision (again, with confidence).

Participants started with 2 blocks of 60 trials that contained no advice to allow them to familiarise themselves with the task; to allow the staircasing process to titrate the difficulty to their ability in order to maintain approximately 71% initial estimate accuracy; and to allow estimating of participants' idiosyncratic confidence reporting style. The first three trials were introductory trials that explained the task. All trials in this section included feedback indicating whether or not the participant's response was correct.

Participants then did 5 trials with a practice advisor to get used to receiving advice. They were informed that they would "get **advice** from an advisor to help you make your decision", and that "advice is not always correct, but it is there to help you: if you use the advice you will perform better on the task."

Participants then performed 2 sets of 2 blocks each. These sets consisted of 1 familiarisation block of 60 trials in which participants were assigned one of two advisors. The familiarisation block was followed with a test block of 60 trials

Table 5.1: Advisor advice profiles for Dots task Accuracy experiment

	Probability of agreement			
Advisor	Participant correct	Participant incorrect	Overall	Overall accuracy
High accuracy	.800	.200	.626	.800
Low accuracy	.600	.400	.542	.600

Table 5.2: Participant exclusions for Dots task Accuracy experiment

Reason	Participants excluded
Accuracy too low	0
Accuracy too high	0
Missing confidence categories	3
Skewed confidence categories	6
Too many participants	0
Total excluded	**9**
Total remaining	**50**

in which participants could choose between the advisors they encountered in the familiarisation block. The participants saw different pairs of advisors in each set, with each pair consisting of one advisor with each of the advice profiles.

Advice profiles The two advisor profiles (Table 5.1) used in the experiment were High accuracy and Low accuracy. The advisors' advice was stochastically generated according to the participant's response. The High accuracy advisor predominantly agreed with correct participant responses and disagreed with incorrect ones. The Low accuracy advisor did likewise, but was less likely to agree with correct responses and more likely to agree with incorrect ones. Overall, given an expected participant accuracy of 71% obtained by the staircasing procedure, the High accuracy advisor was correct 80% of the time while the Low accuracy advisor was correct 60% of the time. The advisor profiles were not balanced for overall agreement rates.

Results

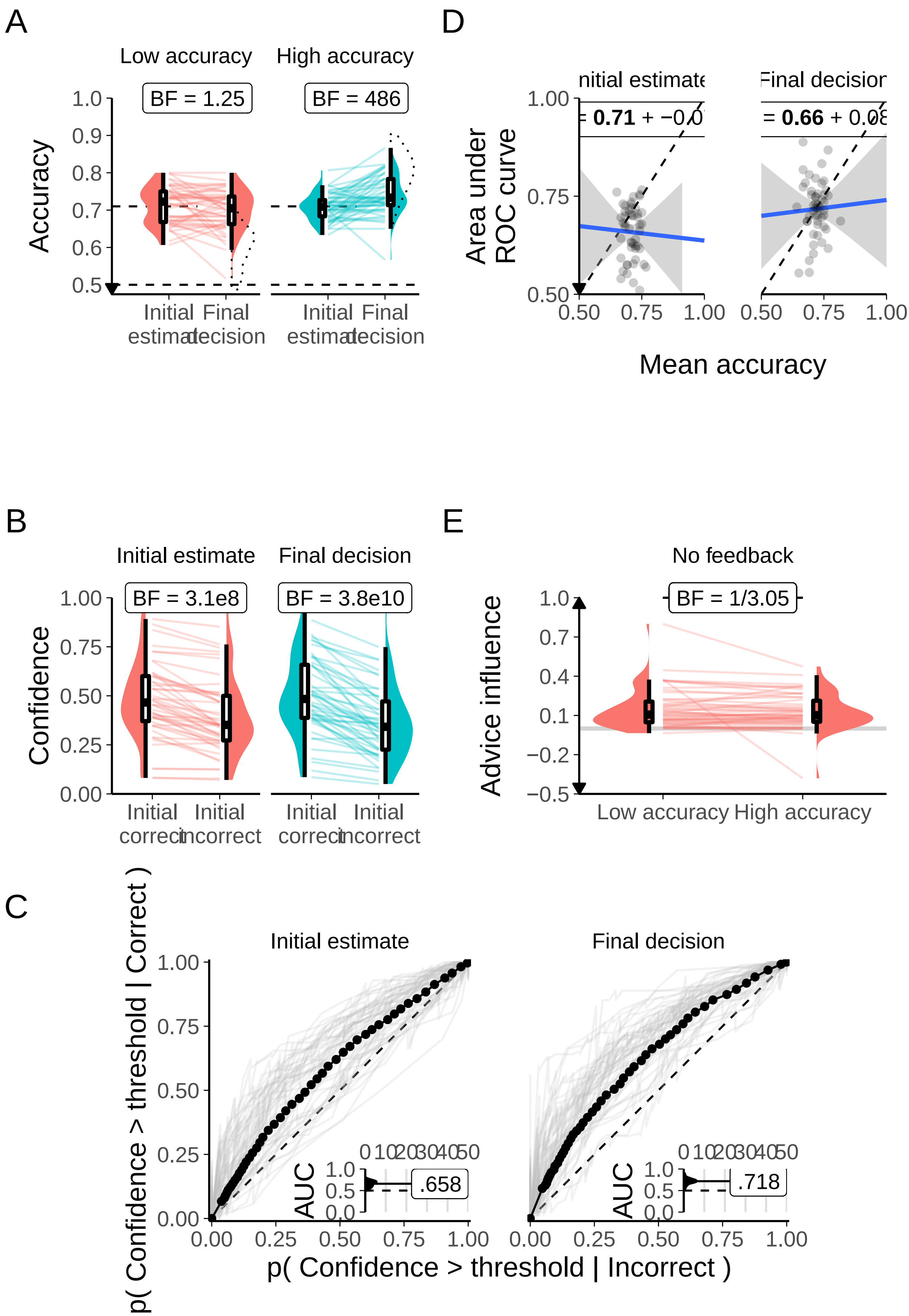
Exclusions In line with the preregistration, participants' data were excluded from analysis where they had an average accuracy below 0.6 or above 0.85, did not have choice trials in all confidence categories (bottom 30%, middle 40%, and top 30% of prior confidence responses), had fewer than 12 trials in each confidence category, or had completed the experiment after the preregistered amount of data has already been collected. Overall, 9 participants were excluded, with the details shown in Table 5.2.

Task performance Before exploring the interaction between the participants' responses and the advisors' advice, and the participants' advisor selection behaviour, it is useful to verify that participants interacted with the task in a sensible way, and that the task manipulations worked as expected. In this section, task performance is explored during the Familiarization phase of the experiment where participants received advice from a pre-specified advisor on each trial. There were an equal number of these trials for each participant for each advisor.

Accuracy

```
## Interaction calculation: Final - Initial
```

Accuracy of initial decisions was controlled by a staircasing procedure which aimed to pin accuracy to 71%. The accuracy of final decisions was free to vary according to the ability of the participant to take advantage of the advice on offer. As Figure 5.1A shows, participants' accuracy scores for initial decisions were close to the target values (partly because participants whose accuracy scores diverged considerably were excluded). Participants tended to improve the accuracy of their responses following advice from High accuracy advisors, while the evidence was unclear as to whether there was any difference in response accuracy with Low accuracy advice. This is supported statistically by an ANOVA of response accuracy by Advisor and Time: there was no discernible effect of Advisor ($F(1,49) = 3.86$, $p = .055$; $M_{LowAccuracy} = 0.71$ [0.69, 0.72], $M_{HighAccuracy} = 0.72$ [0.71, 0.73]), but there was an interaction between Advisor and Time ($F(1,49) = 17.32$, $p = <.001$; $M_{Improvement|LowAccuracy} = -0.01$ [-0.03, 0.00], $M_{Improvement|HighAccuracy} = 0.03$ [0.02, 0.05]); and there was an effect of Time $F(1,49) = 4.80$, $p = .033$; $M_{Final} = 0.72$ [0.71, 0.73], $M_{Initial} = 0.71$ [0.70, 0.72].


A
Low accuracy
High accuracy
BF = 1.25
BF = 486
Accuracy
Initial estimate
Final decision
D
Initial estimate
Final decision
Area under ROC curve
0.71 + −0.0
0.66 + 0.08
Mean accuracy
B
Initial estimate
Final decision
BF = 3.1e8
BF = 3.8e10
Confidence
Initial correct
Initial incorrect
E
No feedback
BF = 1/3.05
Advice influence
Low accuracy
High accuracy
C
Initial estimate
Final decision
p(Confidence > threshold | Correct)
p(Confidence > threshold | Incorrect)
AUC
.658
.718

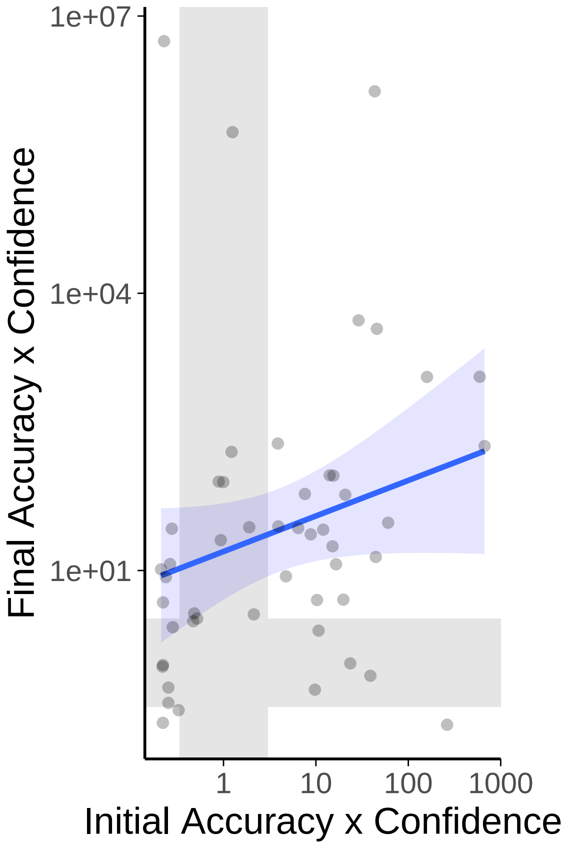


Figure 5.2: Accuracy x Confidence correlations for Dots task with in/accurate advisors. Each point marks the Bayes factor for a participant's correlation between their initial accuracy and confidence (horizontal axis) and final accuracy and confidence (vertical axis). Shaded bands show areas of no information ($1/3 < \text{BF} < 3$), with evidence for a correlation rightwards and upwards of the area and evidence against below and leftwards. Note that the Bayes factor is a measure of the likelihood the correlation is not 0, not a direct measure of the strenght of the correlation. The blue line indicates the overall pattern, with shaded area giving the 95% confidence intervals. Axes use $\log_{10}$ scale.

Confidence Generally, we expect participants to be more confident on trials on which they are correct compared to trials on which they are incorrect. Participants were systematically more confident on correct as compared to incorrect trials for both initial estimates and final decisions (Figure 5.1B; Figure 5.2). There is considerable variation between participants both on their baseline confidence and on its variability (not shown), despite all participants being roughly matched for accuracy. The narrow accuracy range, and its random nature mean there was no evidence of a correlation between participant accuracy and confidence for initial ($\text{BF}_{10} = 1/2.80$) or final ($\text{BF}_{10} = 1/2.08$), although neither of these correlations indicated good evidence that no correlation existed. Variation between individuals' confidence reports is expected (Ais et al. 2016; Navajas et al. 2017).

ANOVA of Time, Advisor, and Correctness of the decision indicated that there was a main effect of Correctness, with participants being more confident on correct vs incorrect decisions ($F(1,49)=111.44$, $p=<.001$). This effect showed interactions with Advisor ($F(1,49)=6.13$, $p=.017$) and Time ($F(1,49)=21.50$, $p=<.001$), summed up in the three-way Advisor-Correctness-Time interaction ($F(1,49)=9.44$, $p=.003$). Confidence increased for correct responses and decreased for incorrect responses, and this confidence change difference was greater for the High accuracy advisor (4.11) than the Low accuracy advisor (1.76). This indicates that confidence is behaving as in a sensible manner.

Metacognitive ability Where performance on the underlying task is held constant, as here at least for participants' initial pre-advice decisions, metacognitive sensitivity can be measured in a bias-free way by plotting Receiver Operating Characteristic (ROC) curves for metacognitive responses (Fleming and Lau 2014).[3] ROC curves are obtained by calculating at each of a number of different points on a confidence scale, the probability that the confidence is at least that high for correct versus incorrect answers. The area under the ROC curve gives a measure of the ability of confidence ratings to distinguish correct and incorrect responses. An area under the ROC curve of .5 indicates chance performance, and a value of 1 indicates perfect discrimination.

As shown by Figure 5.1C, almost all participants showed above-chance metacognitive sensitivity for initial estimates and final decisions. Participants generally showed higher metacognitive sensitivity for final decisions, in line with their improved performance on these trials. Participants' metacognitive sensitivity was not particularly high, reflecting the difficulty of the task, and in line with previous datasets with this task (Pescetelli, Hauperich, et al. 2021). There was no evidence of

[3]The constant underlying task performance is only true for initial estimates in the paradigm used here, and thus the ROC curves for final decisions should be interpreted with caution because they cannot be proven to be unaffected by metacognitive bias.

Table 5.3: Advisor agreement for Dots task Accuracy experiment

Advisor	Target|correct	Actual|correct	Target|incorrect	Actual|incorrect
High accuracy	.800	.800	.200	.203
Low accuracy	.600	.608	.400	.422

Table 5.4: Advisor accuracy for Dots task Accuracy experiment

Advisor	Target accuracy	Mean accuracy
High accuracy	.800	.799
Low accuracy	.600	.601

participants' metacognitive sensitivity being correlated with their task performance (Figure 5.1D). This is expected when task performance is tightly controlled, because under these conditions variation in task performance reflects variation in ability within a participant rather than between participants.

Advisor performance The advice is generated probabilistically from the rules described previously in Table 5.1. It is thus important to get a sense of the actual advice experienced by the participants.

The advisors' performance was stochastic, with the advisors agreeing or disagreeing with set probabilities depending upon whether the participant was correct or incorrect in their initial decision. The performance of the advisors in practice was as specified (Table 5.3). The participants' accuracy rates were controlled with an adaptive staircase, meaning that the advisors' agreement strategies produced overall advice accuracy at target rates. The advisors' actual accuracies matched the target accuracies (Table 5.4), with 49/50 participants experiencing the planned relationship wherein the High accuracy advisor's advice was more accurate than the Low accuracy advisor's advice.

Advisor influence During the Familiarization stage of the experiment, participants were assigned advisors by fiat. Although participants were primarily learning about the advisors at this time, we can nonetheless look at differences in the influence of the advisors without the confound of advisor choice.

```
## Interaction calculation: Agree - Disagree
```

As expected from participants who are paying attention to the task and attempting to maximize the calibration and accuracy of their final decisions, when participants receive agreeing advice they tend to increase their confidence in their initial response. Likewise, where participants receive disagreeing advice, they tend to reduce their confidence in their answer. This influence of advice is stronger for disagreeing than agreeing trials (even when corrected to account for the scale asymmetry; F(1,49) = 12.19, p = .001; M_{Agree} = 0.07 [0.04, 0.10], M_{Disagree} = 0.14 [0.11, 0.17]). This difference did not appear to be different between advisors (F(1,49) = 0.08, p = .782; $M_{\text{LowAccuracy}}$ = 0.11 [0.08, 0.13], $M_{\text{HighAccuracy}}$ = 0.10 [0.08, 0.13]), and as shown in Figure 5.1E, however, the level of influence was equivalent between advisors, with most participants being almost exactly equally influenced by both advisors (F(1,49) = 0.63, p = .430; $M_{\text{Agree-Disagree|LowAccuracy}}$ = -0.06 [-0.10, -0.02], $M_{\text{Agree-Disagree|HighAccuracy}}$ = -0.07 [-0.11, -0.03]).

NICK:Also, not to change now, but to flag for the viva: The figure brings out some of the questionable features of how you cap influence. It does not make sense to me that the maximum confidence change for disagreement with high initial confidence is small. I realise why you do it mathematically, but it is very odd conceptually when these are arguably the most influential changes – when a person shifts their view initially held with near certainty. MATT: Should we keep the off-diagonal figure thing to illustrate this?

Hypothesis test With basic task performance as expected, our key analysis focused on participants' choice of advisors. As predicted, and as shown in Figure 5.3, participants selected the High accuracy advisor at a rate greater than would be expected if their choosing were random ($t(49) = 3.09$, $p = .003$, $d = 0.44$, BF = 9.96; $M = 0.57$ [0.52, 0.61], $\mu = 0.5$). The modal choice remained at chance level (.5), but almost all participants manifesting a preference preferred the High accuracy advisor.

While this effect is interesting, it is substantially smaller than participants' preference for picking the top advisor regardless of identity ($t(49) = 5.47$, $p < .001$, $d = 0.77$, BF = 1.0e4; $M_{\text{P(PickFirst)}}$ = 0.65 [0.60, 0.71], $\mu = 0.5$), an effect that we

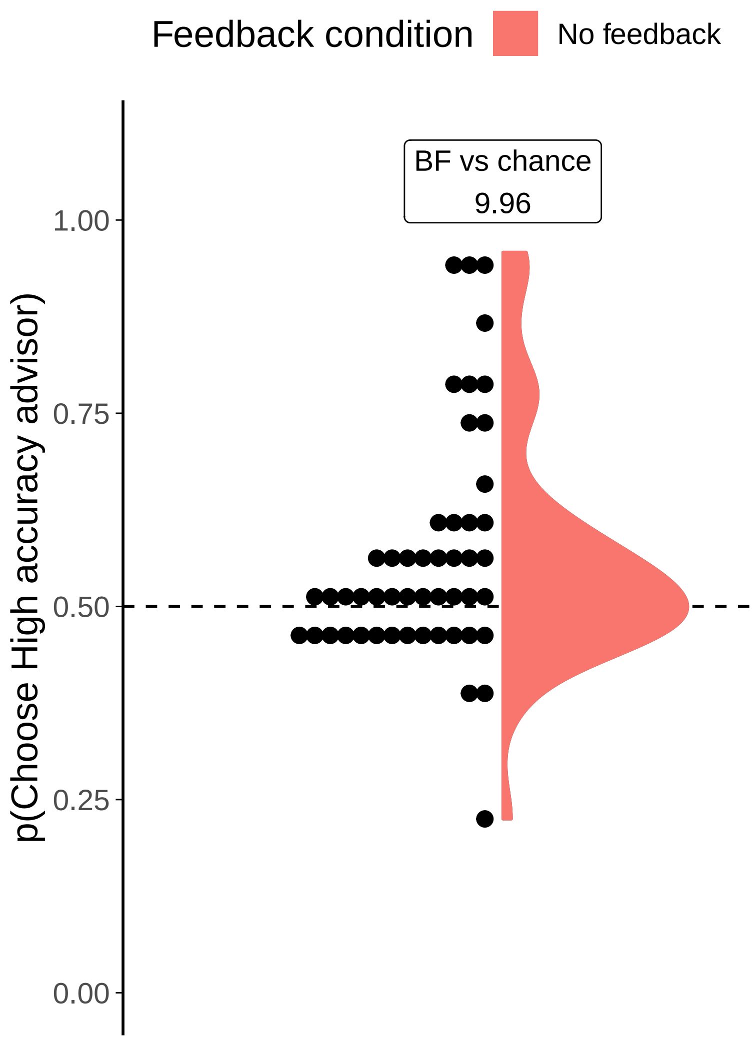


Figure 5.3: Dot task advisor choice for in/accurate advisors.
Participants' pick rate for the advisors in the Choice phase of the experiment. The violin area shows a density plot of the individual participants' pick rates, shown by dots. The chance pick rate is shown by a dashed line.

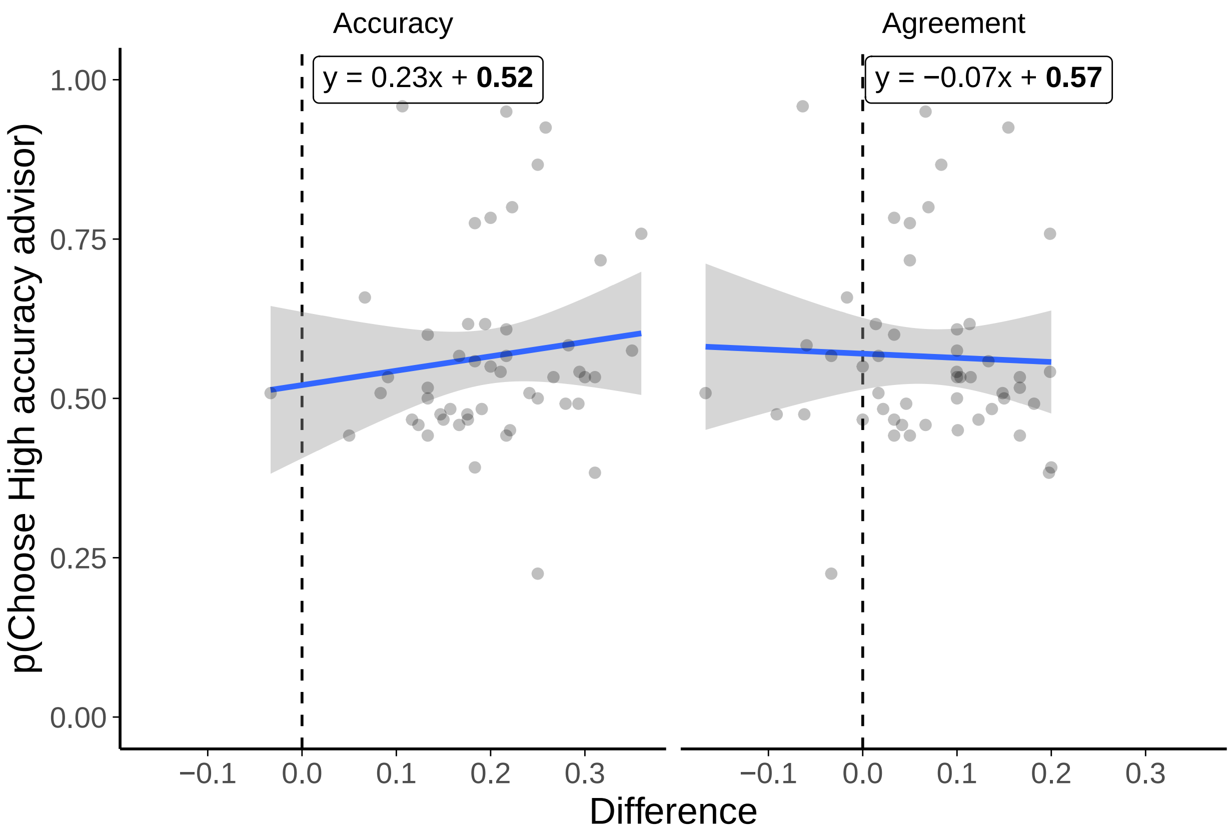


Figure 5.4: Advisor choice by experience in the Dots task with in/accurate advisors. Each dot is a participant's proportions. The difference in accuracy rates is calculated as the proportion of correct answers seen from the High accuracy advisor minus the proportion of correct answers seen from the Low accuracy advisor, and the difference in agreement rates similarly for agreement. The participants did not receive feedback on the correct answers

would hope would be random and even out across participants. Note that because the advisor position is well balanced (BF = 1/6.29; $\mathrm{M}_{\mathrm{P(HighAccuracyFirst)}} = 0.50$ [0.48, 0.51], $\mu = 0.5$) across advisors the presence of a preference for advisor by position would not cause a preference for an individual advisor.

Follow-up tests As noted above, the stochastic nature of the advisors' advice meant that there was some variation in the participants' experience of the advisors. Despite this difference, there was no evidence of a relationship between participants' advisor preference and their experience of either advisor accuracy ($BF_{10} = 1/2.28$) or advisor agreement ($BF_{10} = 1/3.05$), with the latter indicating an absence of a relationship. This is not entirely surprising because, as with the accuracy correlations discussed above, there was relatively little variation in experience of advisors so effects might be expected to be small and difficult to detect.

We might also expect advisor preference to vary as a function of initial confidence. Perhaps, for example, participants may have a strong preference for the High accuracy advisor but only exercise that preference where they are unsure about the answer themselves (i.e. where advice is most valuable to them). This appeared not to be the case: we split participants' trials into high and low confidence based on their idiosyncratic median confidence, and conducted a paired t-test of the pick rates for the High accuracy advisor in each confidence category. The Bayes factor for this t-test indicated good evidence of no difference in pick rates ($BF_{10} = 1/6.49$).

Discussion In the absence of feedback, it should be possible for a person to evaluate advice using the proxy of whether or not the advice accords with their initial opinion, at least given some reasonable assumptions about the independence of the initial opinion and the advice. In this experiment, we tested whether participants would be able to exploit this heuristic to detect that one advisor was more useful than another, and whether they would choose to hear advice from the more useful advisor. Participants showed a tendency, where they had a preference, to prefer the High accuracy advisor. In the next experiment, we aimed to replicate the results in a different task.

5.1.2 Experiment 1B: Advice accuracy effects in the Dates Task

This experiment attempted to replicate the results of the previous experiment using a different task. The replication used a binary version of the Dates task reported in Experiment 0§4.1. Unlike in the Dots task above, and because the study was newly designed for this work, participants in the Dates task were split into conditions so that half received feedback while learning about the advisors and half did not.

QUESTION: Perhaps explain some of the appeal of the Dates task, or is that done well enough in the General Method/Taking Advice sections?

Open scholarship practices

This experiment was preregistered at https://osf.io/5xpvq. The experiment data are available in the `esmData::` package for R from !TODO[Zenodo esmData], and also directly from !TODO[OSFify Dates task data]. A snapshot of the state of the code for running the experiment at the time the experiment was run can be obtained from https://github.com/oxacclab/ExploringSocialMetacognition/blob/master/ACBin/acc.html.

Method

62 participants each completed 52 trials over 4 blocks of the binary version of the Dates task§3.1.3. On each trial, participants were presented with an historical event that occurred on a specific year between 1900 and 2000. They were given a date and asked whether the event occurred before or after that date, indicating their confidence in their decision by selecting an appropriate point on the relevant answer bar. Participants then received advice indicating which of the two bars (before or after) was supposedly the correct answer. Participants could then mark a final response in the same manner as their original response.

Participants started with 1 block of 10 trials that contained no advice to allow them to familiarise themselves with the task. All trials in this section included feedback for all participants indicating whether or not the participant's response was correct.

Participants then did 2 trials with a practice advisor to get used to receiving advice. They also received feedback on these trials. They were informed that they would "receive advice from advisors" to "help you complete the task". They were told that the "advisors aren't always correct, but they are quite good at the task", and informed that they should "identify which advisors are best and to weigh their advice accordingly".

Participants then performed 3 blocks of trials that constituted the main experiment. The first two of these were familiarization blocks where participants had a single advisor in each block for 14 trials, plus 1 attention check.

Participants were split into four conditions that produced differences in their experience of these familiarization blocks. These conditions were whether or not they received feedback, and which of the two advisors they were familiarized with first.

Finally, participants performed a test block of 10 trials that offered them a choice on each trial of which of the two advisors they had encountered over the last two blocks would give them advice. No participants received feedback during the test phase.

Advice profiles The High accuracy and Low accuracy advisor profiles issued binary advice (endorsing either the 'before' or 'after' column) probabilistically based on whether or not the participant had selected the correct column in their initial estimate. The High accuracy advisor was agreed with the participant's initial estimate on 80% of the trials where the participant was correct, but only 20% of the trials on which the participant was incorrect, meaning that the High accuracy advisor was correct 80% of the time. Using an analogous setup, the Low accuracy advisor was correct 59% of the time. These figures were chosen to balance agreement at 50% accuracy, i.e. at chance level. To the extent that a participant was better than chance in answering the questions, the High accuracy advisor profile would agree more frequently. This mimics the hypothesised relationship wherein agreement between advisors and judges is driven by shared access to the truth.

QUESTION: The 50 percent agreement thing still doesn't explain why 59 and 41. I imagine it came from some modelling or something during the development of the task but I can't remember what.

Results

Exclusions Individual trials were screened to remove those that took longer than 60s to complete. 4 participants had a total of 5 trials removed in this way,

Table 5.5: Advisor advice profiles for Dates task Accuracy experiment

	Probability of agreement (%)			
Advisor	Participant correct	Participant incorrect	Overall[a]	Overall accuracy[a]
High accuracy	.800	.200	.500	.800
Low accuracy	.590	.410	.500	.590

[a] Where participants' initial estimate accuracy is 50%

representing 0.21% of all trials. Participants were then excluded for having fewer than 11 trials remaining, fewer than 10 trials on which they had a choice of advisor, or for giving the same initial and final response on more than 90% of trials. These criteria led to no participants being excluded from this experiment.

Task performance Before exploring the interaction between the participants' responses and the advisors' advice, and the participants' advisor selection behaviour, it is useful to verify that participants interacted with the task in a sensible way, and that the task manipulations worked as expected. In this section, task performance is explored during the Familiarization phase of the experiment where participants received advice from a pre-specified advisor on each trial. There were an equal number of these trials for each participant for each advisor, although as mentioned above a small number of trials were dropped from analysis where response times were overly long.

For the purposes of exploring participants' performance on the task, the conditions are pooled together. The participants were randomly assigned to conditions, and thus we know any differences in performance between conditions would be random.

Response times Participants made two decisions during each trial. Neither of these decisions had a maximum response time. Each participant's response times for both initial estimates and final decisions can be seen in Figure 5.5. The distribution of these response times helps characterise some differences between the Dots task and the Dates task. In the former, decisions for both initial estimates and final decisions are tightly clustered, with a clear structure and pattern to the responses for all participants. In the Dates task however, response times are not only longer, but they are also much more varied within participants. Some increase in variance is expected with an increase in mean, especially with fewer trials for each participant, but the extent of the differences clearly shows that the tasks provide participants with different experiences: the Dots task is tightly rhythmic and repetitive, while the Dates task is more heterogeneous.

Accuracy

```
## Interaction calculation: Final decision - Initial estimate
```

Unlike in the Dots version of the task, participant accuracy is not controlled because it depends on participants' existing knowledge (and guesses) across a relatively small and varied set of questions. Correspondingly, accuracy varied substantially across participants (Figure 5.6A). Figure 5.6A also shows that participants managed to improve their performance from their initial estimates to their final decisions with both advisors ($F(1,61) = 36.40$, $p = <.001$; $M_{FinalDecision} = 0.68$ [0.65, 0.70], $M_{InitialEstimate} = 0.60$ [0.57, 0.63]). This is likely because the advisors themselves were more accurate than the participants, so following their advice was generally a good strategy, and the difficulty of the task meant that participants were very willing to be influenced by advice.

As would be expected from participants following advice, the improvement in accuracy from initial estimates to final decisions was greater for the High accuracy advisor than the Low accuracy advisor ($F(1,61) = 32.46$, $p = <.001$; $M_{Improvement|HighAccuracy} = 0.15$ [0.11, 0.18], $M_{Improvement|LowAccuracy} = 0.01$ [-0.02, 0.04]).

Confidence

```
## Interaction calculation: Final decision - Initial estimate
```

Generally, we expect participants to be more confident on trials on which they are correct compared to trials on which they are incorrect (Figure 5.6B). Participants' initial estimates and final decisions were both systematically more confident when

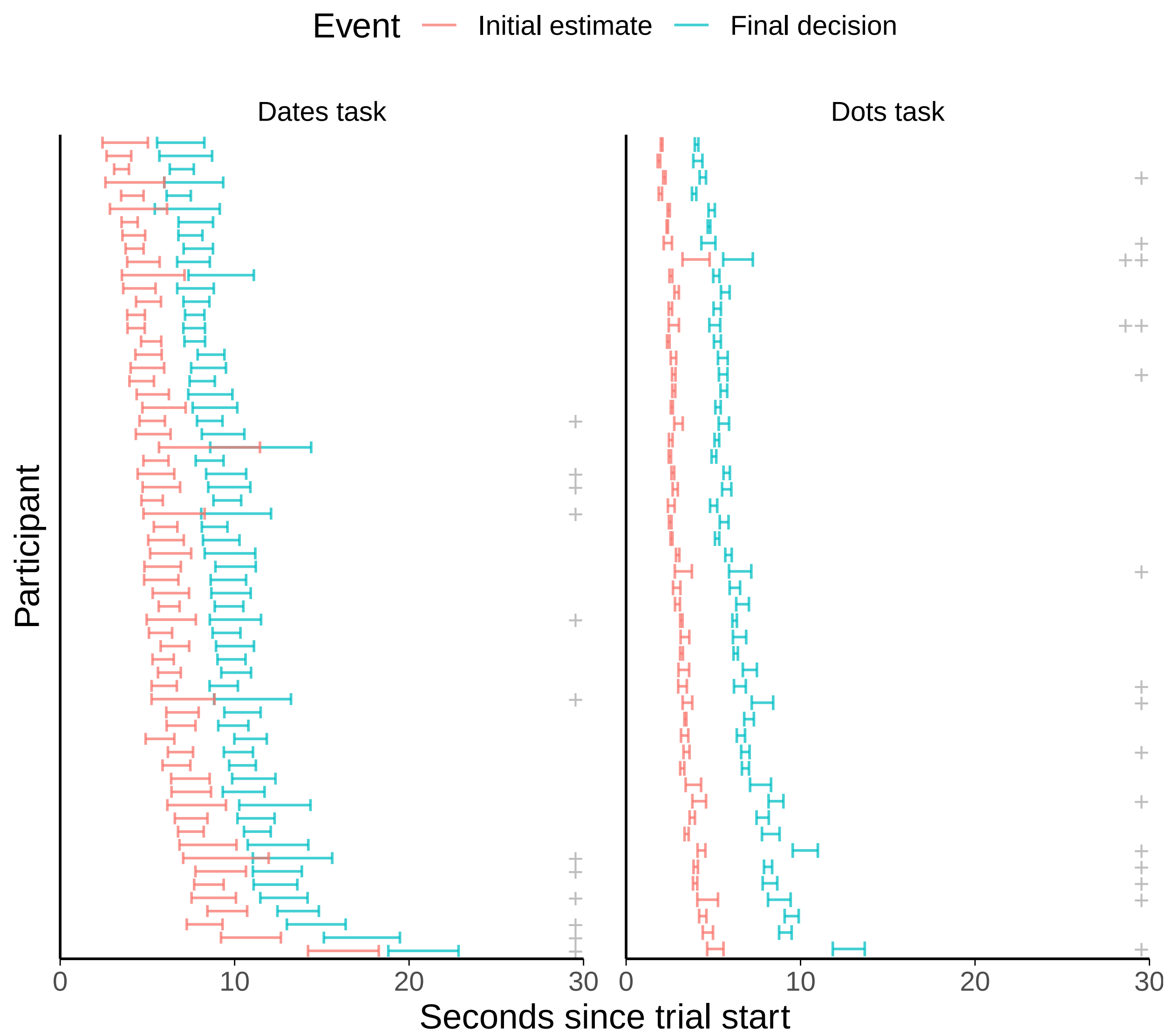
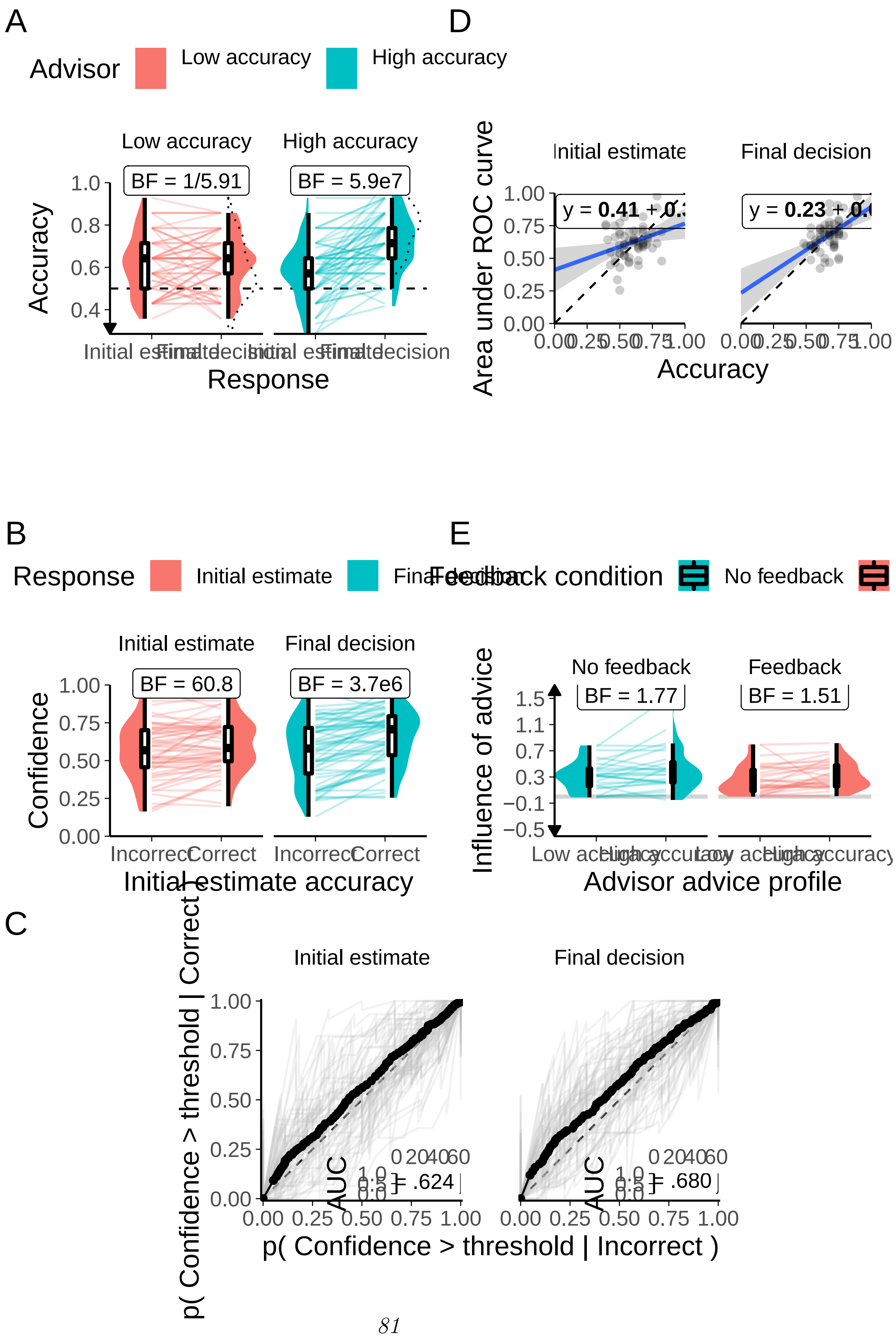


Figure 5.5: Response times for the Dates task with in/accurate advisors.
Each row indicates a single participant's trials. The error bars show the 95% confidence intervals of the mean response time for each decision. The plusses on the right show the number of trials where response times were more than 3 standard deviations away from the mean of all Dates task final response times (rounded to the next 10s): + = 1-5 trials, ++ = 6-10 trials.


A
Advisor
Low accuracy
High accuracy
BF = 1/5.91
BF = 5.9e7
Accuracy
Response
D
Initial estimate
Final decision
Area under ROC curve
Accuracy
B
Response
Initial estimate
Final decision
BF = 60.8
BF = 3.7e6
Confidence
Initial estimate accuracy
E
Feedback condition
No feedback
Feedback
BF = 1.77
BF = 1.51
Influence of advice
Advisor advice profile
C
p(Confidence > threshold | Correct)
p(Confidence > threshold | Incorrect)
AUC

the initial estimate was correct as compared to incorrect (). Participants were also more confident on final decisions than on initial estimates (), and the increase over time only appeared for the trials where the initial estimate was correct ().

Metacognitive ability Estimates of the participants' metacognitive abilities were highly variable, with many participants displaying below-chance metacognitive ability (Figure 5.6C). While this may appear concerning, recall that metacognitive sensitivity and bias vary substantially and cannot be reliably estimated using ROC curves where performance accuracy on the underlying task is highly variable, so it is not necessarily the case that these values give cause for alarm.

The correlation between performance on the underlying task and metacognitive ability (Figure 5.6D) shows that, as one might expect, participants with a greater ability to perform the Dates task have a greater insight into their performance on the Dates task. This in turn suggests that, despite the low number of trials on the task, we are able to obtain meaningful insights into participants' metacognitive abilities, albeit without being able to precisely estimate the metacognitive sensitivity or bias for an individual participant.

Response to advice QUESTION: What to do about this section. The observation that confidence change/influence might be a bit of a broken measure is important, but that means it needs explanation, meaning this section is quite long. We could leave the actual analysis of this out and just mention the thing in the discussion?

The extent and manner of decision and confidence changes is an important indicator of the extent to which participants treat advice as informative (Figure 5.7). As with the Dots task (Figure **??**, when participants receive agreeing advice they tend to increase their confidence in their initial response, suggesting they are paying attention to the task and attempting to maximize the calibration and accuracy of their final decisions. Likewise, where participants receive disagreeing advice, they tend to reduce their confidence in their answer. As before, when participants

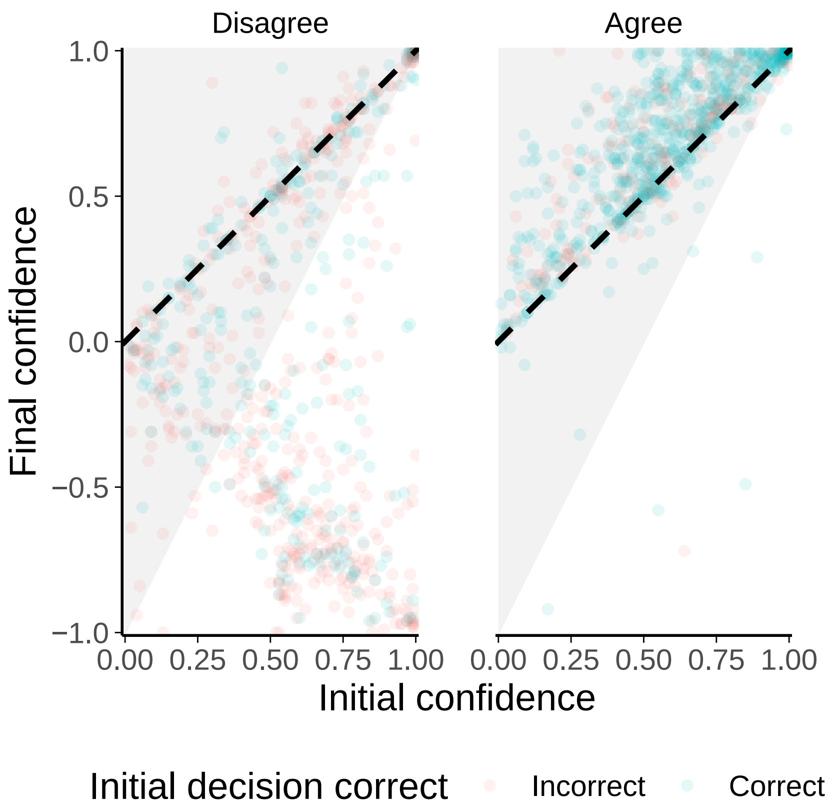


Figure 5.7: Advice influence on the Dates task with in/accurate advisors.
Each point shows the initial and final confidence on a single trial. Final confidence is coded relative to initial confidence so that increasing confidence in the opposite decision (i.e. confidence on trials where the participant changed their mind) is increasingly negative. Points above the dashed y = x line represent increased confidence, while those below it give decreased confidence. Points close to the y = x line indicate relatively little change, while points further away indicate relatively greater change. The shaded grey area shows the zone outside which influence is capped (by moving vertically towards the grey zone boundary) when using the capped influence measure. Agreement and disagreement trials are plotted seperately, with trials coloured according to whether the initial decision was correct.

change their decision (a relatively uncommon event) they quite often make their final decision with a confidence equivalent to their initial estimate, producing the distinctive off-diagonal pattern along the $y = -x$ line. This pattern was explained as largely an artefact of aggregating data from multiple individual participants in the Dots task, but the same explanation is less tenable here.

The individual participant data suggest that some participants may actually be porting their confidence from the initial estimate directly into the final decision,

Table 5.6: Advisor agreement for Dates task Accuracy experiment

Advisor	Target|correct	Actual|correct	Target|incorrect	Actual|incorrect
High accuracy	.800	.787	.200	.216
Low accuracy	.590	.606	.410	.438

Table 5.7: Advisor accuracy for Dates task Accuracy experiment

Advisor	Target accuracy	Mean accuracy
High accuracy	.800	.797
Low accuracy	.590	.585

despite the fact that they have changed their mind (Figure **??**). Whereas in the Dots task the off-diagonal pattern in the aggregate data looked to be an artefact of combining data from individual participants with relatively little variation within their confidence responses placed at various points along the off-diagonal (Figure **??**), data from this task suggested that several participants were using a range of values for their final decisions and displaying a clear positive correlation between initial and final confidence.

This increased tendency for participants to response in this way on the Dates task may be a consequence of the layout of the response bars on the screen. In the Dots task, the bars were oriented horizontally, such that the extreme values for one were furthest away from extreme values for the other (Figure 3.3). In the Dates task, the bars were oriented vertically, meaning that the shortest distance from one point on one bar was to the same point on the other bar (Figure 3.5). Participants changing their minds may well have felt that the important feature was the change of response bar, and not concerned themselves with reporting their confidence. These participants would naturally have taken the shortest route from their initial estimate to the other response bar, which would be the same confidence as for the initial estimate, producing the positive correlation between initial and final confidence seen in these data.

Advisor performance The advice is generated probabilistically from the rules described previously (Advice profiles§5.1.2). This advisors agreed with participants

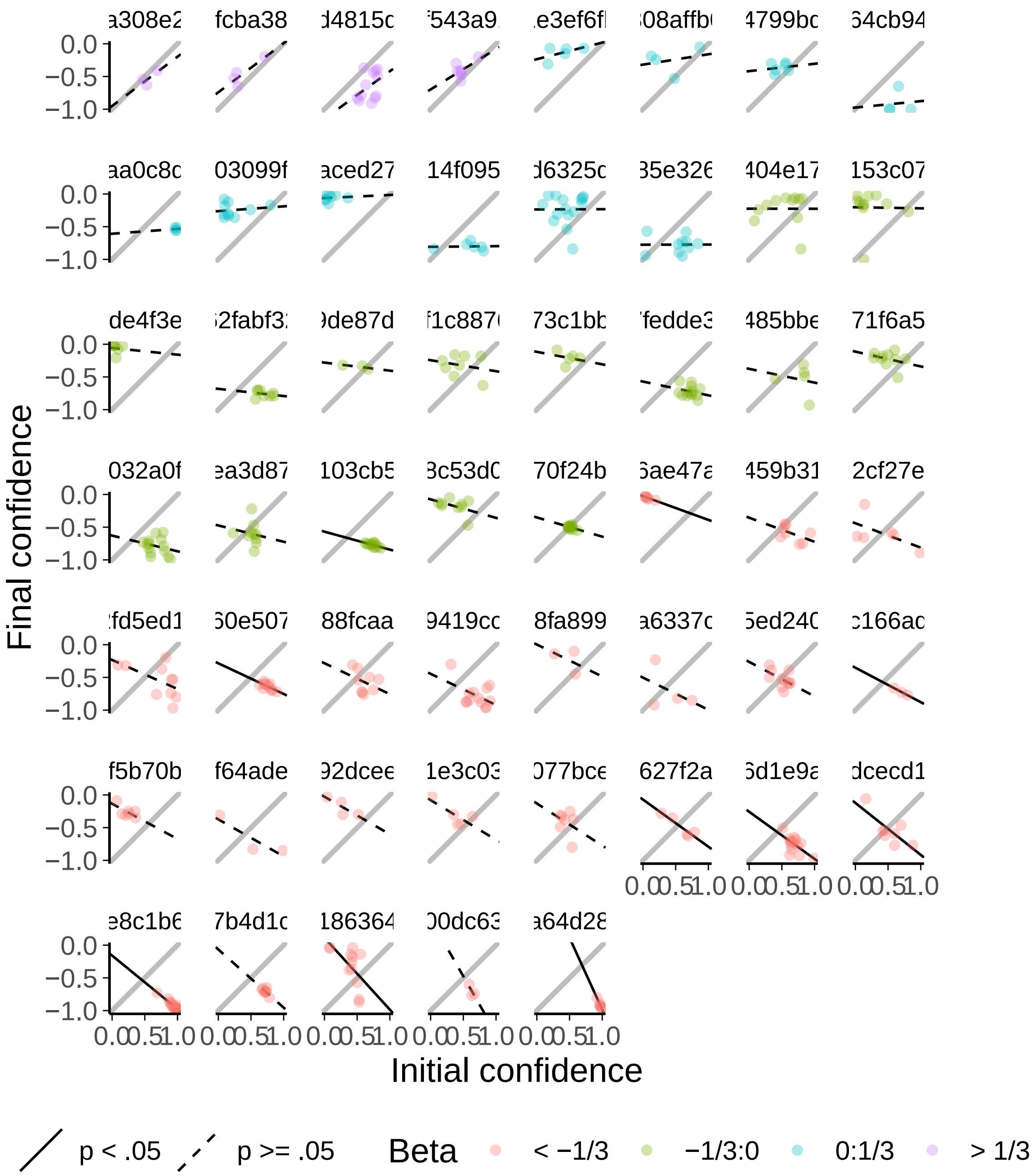


Figure 5.8: Change-of-mind confidence updating on the Dates task.
Each facet shows data from a single participant for trials where they changed their mind on the categorical decision between the initial estimate and the final decision. Participants who never changed their mind are not included. Each point shows the initial and final confidence on a single trial. All final confidence scores are negative because they are coded relative to initial confidence; increasing confidence in the final decision is increasingly negative.
Lines show the best fit for a linear prediction of final from initial confidence, with solid lines indicating that the slope is significantly different from zero at alpha = .05. Points are coloured according to the value of the slope parameter. The grey line is the y = x - 1 line that shows the expected fit line according to the intuitive model of confidence updating.

contingent on the accuracy of the participants' initial estimates at close to the target rates (Table 5.6). This meant that advisors were as accurate overall as they were intended to be in the familiarization phase (Table 5.7). Most (57/62, 91.94%) participants experienced the High accuracy advisor as providing more accurate advice than the Low accuracy advisor.

Advisor influence

```
## Interaction calculation: High accuracy - Low accuracy
```

The High accuracy advisor was substantially more influential than the other Low accuracy advisor (F(1,60) = 9.98, p = .002; $\mathrm{M}_{\mathrm{HighAccuracy}}$ = 0.36 [0.29, 0.43], $\mathrm{M}_{\mathrm{LowAccuracy}}$ = 0.28 [0.22, 0.34]). This tendency did not differ significantly between the groups (F(1,60) = 0.20, p = .652; $\mathrm{M}_{\mathrm{High\text{-}LowAccuracy|NoFeedback}}$ = 0.09 [0.01, 0.18], $\mathrm{M}_{\mathrm{High\text{-}LowAccuracy|Feedback}}$ = 0.07 [0.01, 0.14]), and nor did the participants in the feedback and no feedback conditions appear to differ in the extent to which they took advice (F(1,60) = 1.79, p = .186; NA).

These influence measurements are calculated on the Familiarization phase trials in which participants are not offered a choice of advisor. It is during this phase that participants are learning about the value of the advice (especially in the Feedback condition), and thus any influence on later trials may be diluted by low influence on trials which occur before an advisor has had time to develop a reputation as reliable. This means that influence cannot be used as a reliable outcome measure for this experimental design, but it is nevertheless useful to explore to get a sense of how participants responded to the advice. An inspection of the individual participants' data shows that very few participants had large influence differences between advisors (Figure 5.6E).

Hypothesis test The key analysis in this experiment explores the participants' preferences for picking the High accuracy advisor over the Low accuracy advisor. In the No feedback condition the mean of the distribution of participant picking preferences between the advisors was equivalent to chance ($t(27)$ = -0.93, p =

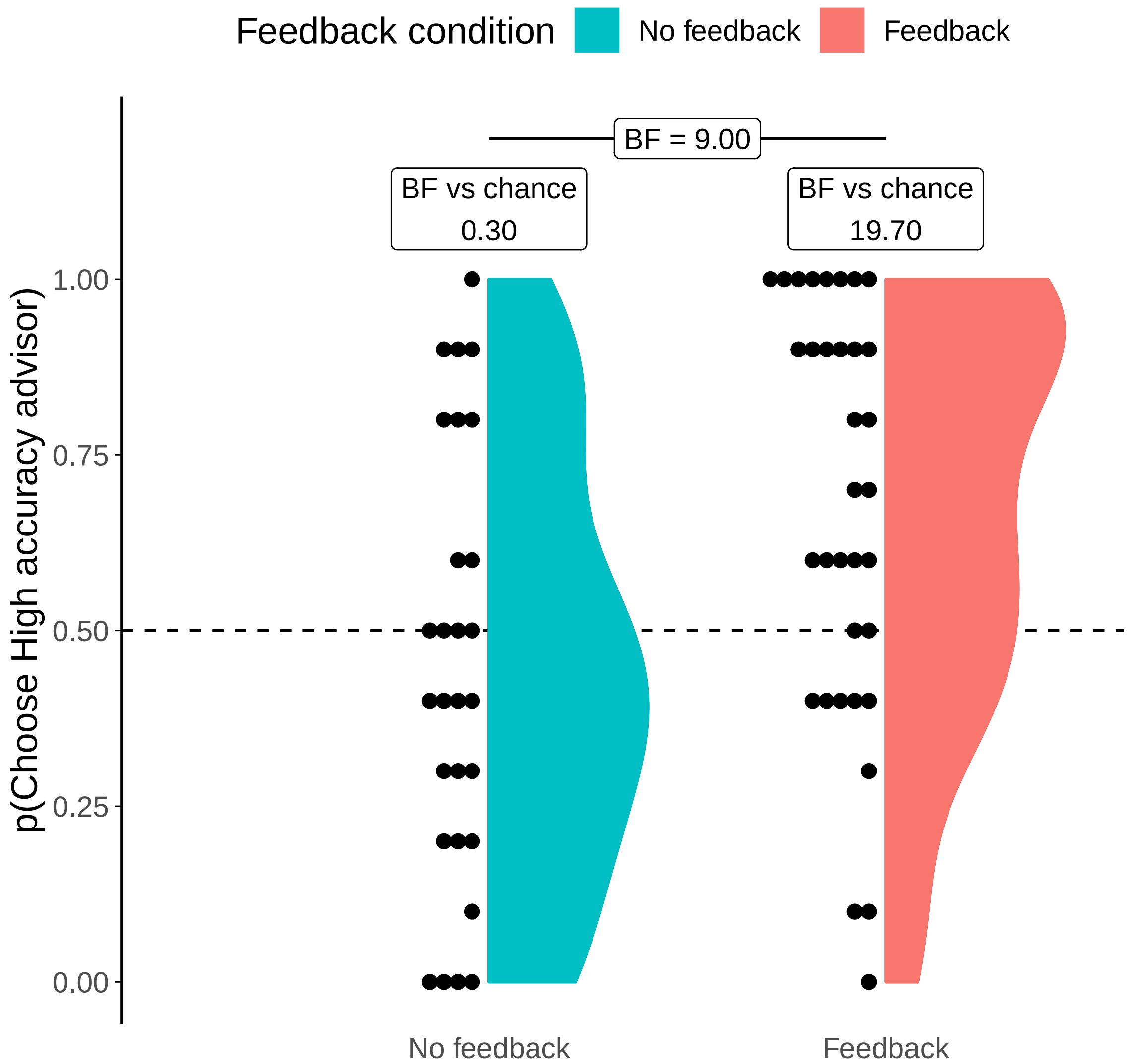


Figure 5.9: Dates task advisor choice for in/accurate advisors.
Participants' pick rate for the advisors in the Choice phase of the experiment. The violin area shows a density plot of the individual participants' pick rates, shown by dots. The chance pick rate is shown by a dashed line. Participants in the Feedback condition received feedback during the Familiarization phase, but not during the Choice phase.

.363, d = 0.18, BF = 1/3.37; M~No feedback~ = 0.45 [0.33, 0.57], μ = 0.5). This is a different result to that observed in the Dots task§5.1.1, which also had no feedback. Preferences were quite evenly distributed across the full range of directions and strengths, with a slight numerical advantage for the Low accuracy advisor (Figure 5.9).

In the Feedback condition the mean of the distribution of selection rates was clearly different from chance. The High accuracy advisor was preferred by more participants, and preferred more strongly ($t(33) = 3.41$, $p = .002$, $d = 0.58$, BF = 19.7; $M_{\text{Feedback}} = 0.67$ [0.57, 0.78], $\mu = 0.5$). The modal selection strategy was to select the High accuracy advisor at every opportunity. This indicates that participants could identify the more accurate advisor when feedback was provided and preferred to receive advice from that advisor.

Interestingly, this meant that there was a difference in participants' preference for picking the High accuracy advisor according to their experimental condition ($t(57.02) = 2.95$, $p = .005$, $d = 0.75$, BF = 8.99; $M_{\text{Feedback}} = 0.67$ [0.57, 0.78], M~No feedback~ = 0.45 [0.33, 0.57]).

It was discovered after the completion of experiments that the advisor position (whether the advisor appears on the top or bottom of the advisor choice panel) was not counterbalanced between advisors. It had been decided during development of the tests to keep the advisors in the same position for every trial so that participants did not get mixed up between them. Together, this meant that the High accuracy advisor always appeared at the top and the Low accuracy advisor always appeared at the bottom, for every trial for every participant. We are thus unable to confirm that pick rate differences, or the absence of those differences, are caused by participants' preferences for the advice the advisor would provide or the position the advisor was in on the screen. Furthermore, it is possible that any genuine preference for one advisor over the other was *induced* by the position, rather than independent of it (Zajkowski and Zhang 2021).

Follow-up tests

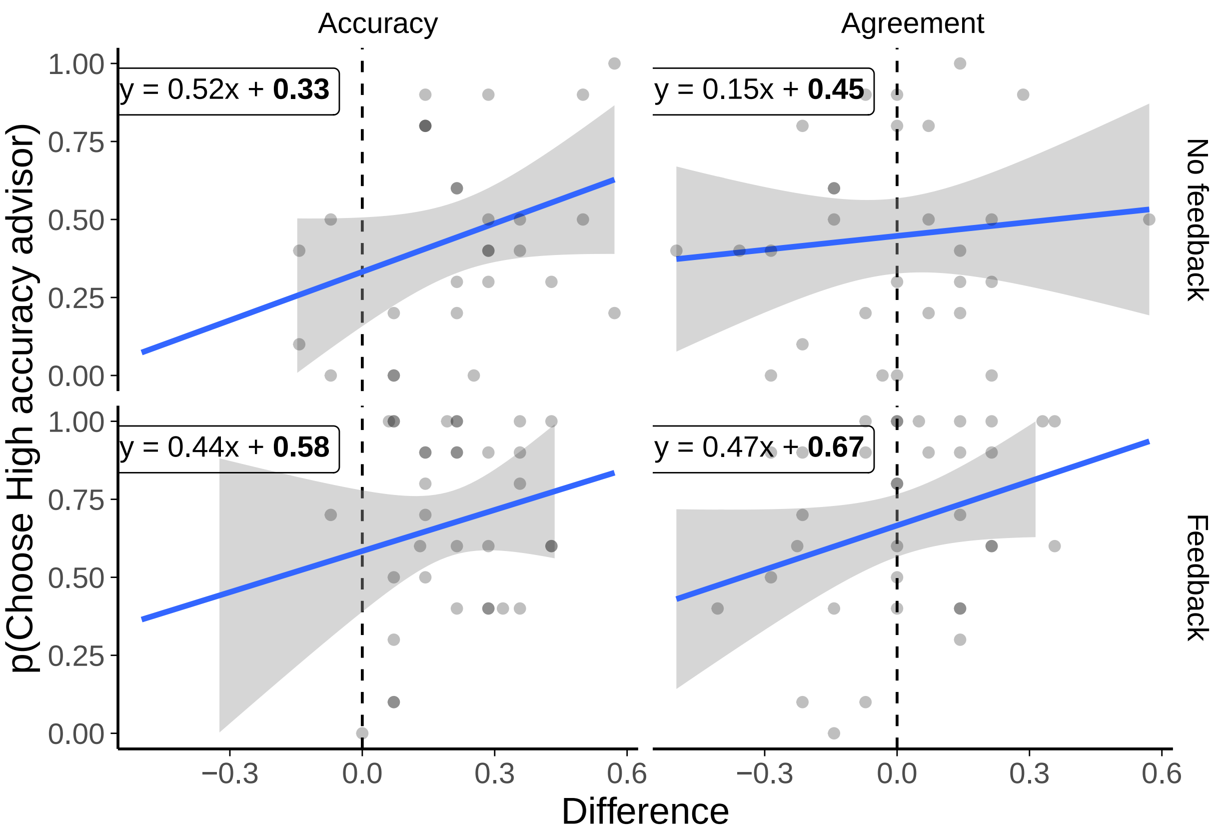


Figure 5.10: Preference predictors in the Dates task with in/accurate advisors. Scatter plots of participants' experience with advisors in terms of agreement or accuracy rates. Differences are expressed as the experienced rate for the High accuracy advisor minus the experienced rate for the Low accuracy advisor during the Familiarization phase. Numbers in bold in the regression equations are significant at $p < .05$.

Ability of participants It is plausible that participants who were better at the task had more insight into which of their advisors was more accurate. There was not enough evidence to determine whether participants in the No feedback condition selected the High accuracy advisor more frequently where they were more accurate themselves (r(26)=-.108, p=.586, BF=1/2.14) or more well calibrated (as measured by area under the Receiver Operator Characteristics curve for initial estimates; r(26)=.157, p=.424, BF=1/1.85).

Experience of Advisors The stochastic nature of the advisors' advice meant that there was some variation in the participants' experience of the advisors. Linear models were run predicting advisor selection behaviour based on experienced

Table 5.8: Experienced accuracy difference effects in the Dates task with in/accurate advisors

Effect	Estimate	SE	t	p	
(Intercept)	0.33	0.09	3.91	< .001	*
Accuracy	0.52	0.29	1.78	.080	
FeedbackFeedback	0.25	0.13	1.97	.053	
Accuracy:FeedbackFeedback	-0.08	0.49	-0.16	.875	

Model fit: F(4.5, 3) = 58; p = .007; R^2_{adj} = .147

Table 5.9: Experienced agreement difference effects in the Dates task with in/accurate advisors

Effect	Estimate	SE	t	p	
(Intercept)	0.45	0.06	7.96	< .001	*
Agreement	0.15	0.26	0.58	.565	
FeedbackFeedback	0.22	0.08	2.88	.006	*
Agreement:FeedbackFeedback	0.32	0.37	0.88	.385	

Model fit: F(4.1, 3) = 58; p = .011; R^2_{adj} = .134

differences in accuracy (Table 5.8) and agreement (Table 5.9). Bayesian linear models were run obtaining Bayes Factors of leaving each component out of a model containing experienced agreement or accuracy difference, feedback condition, and their interaction, as well as a random factor for the participant's identity.

In aggregate, the models indicated that the feedback group had a stronger preference for the High accuracy advisor, regardless of the actual agreement or accuracy experienced ($BF_{Accuracy}$=13.6; $BF_{Agreement}$=8.98). There was no evidence of a relationship between participants' advisor preference and their experience of either advisor accuracy (BF=1.62) or advisor agreement (BF=1.15), but neither of these had evidence strong enough to suggest the absence of such a relationship. The strongest evidence for the absence of an effect was for the interaction between feedback condition and experienced accuracy (BF=1/2.94) or agreement (BF=1/2.08), but this was also not beyond the stated threshold of 1/3.

The vagueness of the results is is not entirely surprising because, as with the accuracy correlations discussed above, there was relatively little variation in experience of advisors so effects might be expected to be small and difficult to detect.

5.1.3 Discussion

We investigated whether more accurate advisors would be preferentially selected by participants when participants were unable to use feedback to evaluate the quality of advisors. We performed two experiments using different tasks, and found mixed results. In the branch of the Dates task where feedback was provided, participants had a clear preference for the more accurate advisor. This preference was also seen in the Dots task, in which no participants received feedback. Contrary to these results, however, participants in the Dates task who did not receive feedback did not show a systematic preference for either advisor.

The difference between the task results where feedback was denied to participants is probably due to the Dates task being a generally more difficult task for participants than the Dots task. This extra difficulty likely meant that participants in the Dates task were unable to tell the advisors apart. If the Pescetelli and Yeung (2021) model of metacognitive evaluation of advice is accurate, participants may have been subjectively very unsure of whether their answer was correct or incorrect, and thus in the absence of feedback they cannot glean insight into the accuracy of advice by attending to whether or not the advice contradicts their initial estimate. Alternatively, if this process is not driving performance in the Dates task, the additional difficulty may simply have meant that participants' strong desire to see advice (Gino and Moore 2007) may have rendered the relative quality of the advice unimportant.

In both tasks the advisor preferences included many participants whose preference was either neutral (all tasks) or in favour of the Low accuracy advisor (Dates task). This variability in pick rates from the participants in the No feedback condition of the Dates task, suggests that preferences are diverse both in terms of direction and strength in the absence of any systematic effects. There is substantial variability in pick rates for participants in the Feedback condition of the Dates task, too, indicating that, compared to the No feedback condition, everyone might have nudged their preference a bit towards the High accuracy advisor.

Pick rates in the Dots task were also varied, but rather than being evenly spread (as in the No feedback Dates task participants) or massed in favour of the High accuracy advisor with a long, fat tail including exclusive selection of the Low accuracy advisor, Dots task participants were massed in the centre with a long tail out to strong preference for the High accuracy advisor. The Dots task data may have reduced variability because there were more trials that offered the participants a choice of advisor, and the novelty value of the ignored advisor may have increased relative to the chosen advisor as the test phase progressed. Alternatively, participants making repeated choices may have eventually felt that continuing to ignore one advisor was unfair, and that pragmatic reasons for including the opinion of a less expert voice outweighed the performance-maximisation reasons for not including that voice (Mahmoodi et al. 2015).

Another explanation for the difference may be the level of engagement with the tasks. If participants in the Dates task were more engaged with the more challenging and (subjectively but consistent with participant feedback) more enjoyable task, picking of advisors may have been more deliberative than in the Dots task, where the repetitive nature of the trials could have led to disengagement and random advisor selection behaviour for some participants.

The results of these studies were mixed in terms of supporting our hypothesis that more expert advisors would be discriminated and preferentially picked by participants even in the absence of feedback. The underlying mechanism we believe to be responsible for evaluating advisors in the absence of feedback is agreement, and thus a more powerful test of the mechanism is to move from demonstrating the phenomenon (detection of accuracy differences without feedback) to demonstrating the mechanism (discrimination based on agreement differences). This investigation of agreement as a mechanism for driving advisor evaluation in the absence of feedback is the subject of the next experiments.

5.2 Effects of Advisor Agreement on Advisor Selection

Experiments 1A§5.1.1 and 1B§5.1.2 revealed differences in how participants selected the advisors between the Dots task (which has no feedback) and the No feedback condition of the Dates task for High versus Low accuracy advisors. We may expect more pronounced effects in the absence of feedback when contrasting High versus Low agreement advisors, because we expect that agreement is the driving force behind the accuracy differences where feedback is not provided. Pescetelli and Yeung (2021) demonstrated that advisors who agree more frequently are more influential (regardless of the presence of feedback, but especially without it) in a lab-based perceptual decision-making task. Here we explored the impact of agreement on choice of advisor. Experiment 2A§5.2.1 looks at this effect in the Dots task, while Experiment 2B§5.2.2 does the same for the Dates task.

5.2.1 Experiment 2A: Advisor agreement effects in the Dots Task

Open scholarship practices

This experiment was preregistered at NA. The experiment data are available in the `esmData::` package for R from !TODO[Zenodo esmData], and also directly from https://osf.io/8cnpq/. A snapshot of the state of the code for running the experiment at the time the experiment was run can be obtained from https://github.com/oxacclab/ExploringSocialMetacognition/blob/9932543c62b00bd96ef7ddb3439e6c2d5bdb AdvisorChoice/index.html.

Method

68 participants each completed 368 trials over 7 blocks of a perceptual decision-making task. Each trial consisted of three phases: participants gave an initial estimate (with confidence) of which of two briefly presented boxes contained more dots; received advice on their decision from an advisor; and made a final decision (again, with confidence).

Participants started with 2 blocks of 60 trials that contained no advice to allow them to familiarise themselves with the task; to allow the staircasing process to titrate the difficulty to their ability in order to maintain approximately 71% initial estimate accuracy; and to allow estimating of participants' idiosyncratic confidence reporting style. The first three trials were introductory trials that explained the task. All trials in this section included feedback indicating whether or not the participant's response was correct.

Participants then did 5 trials with a practice advisor to get used to receiving advice. They were informed that they would "get **advice** from an advisor to help you make your decision", and that "advice is not always correct, but it is there to help you: if you use the advice you will perform better on the task."

Participants then performed 2 sets of 2 blocks each. These sets consisted of 1 familiarisation block of 60 trials in which participants were assigned one of two advisors. The familiarisation block was followed with a test block of 60 trials in which participants could choose between the advisors they encountered in the familiarisation block. The participants saw different pairs of advisors in each set, with each pair consisting of one advisor with each of the advice profiles.

Advice profiles The two advisor profiles (Table 5.10) used in the experiment were High agreement and Low agreement. These advisors were defined in terms of their likelihood of agreement with participants' correct and incorrect initial estimates, while being matched for objective accuracy. The High agreement advisor gave advice that endorsed the same answer side as the participant's initial estimate 77.3% of the time while the Low agreement advisor agreed with the participant 51.8% of the time. These overall agreement rates were split based on on the participants' initial estimate accuracy to achieve balanced overall accuracy rates between advisors.

Results

Exclusions In line with the pre-registration, participants' data were excluded from analysis where they had an average accuracy below 0.6 or above 0.85, did

Table 5.10: Advisor advice profiles for Dots task Agreement experiment

	Probability of agreement			
Advisor	Participant correct	Participant incorrect	Overall	Overall accuracy
High agreement	.840	.610	.773	.709
Low agreement	.660	.170	.518	.709

Table 5.11: Participant exclusions for Dots task Agreement experiment

Reason	Participants excluded
Accuracy too low	0
Accuracy too high	0
Missing confidence categories	7
Skewed confidence categories	12
Too many participants	0
Total excluded	**18**
Total remaining	**50**

not have choice trials in all confidence categories (bottom 30%, middle 40%, and top 30% of prior confidence responses), had fewer than 12 trials in each confidence category, or finish the experiment after 50 participants have already submitted data which passed the other exclusion tests. Overall, 18 participants were excluded, with the details shown in Table 5.11.

Task performance

```
## Interaction calculation: Final - Initial
```

```
## Interaction calculation: Final - Initial
```

Basic behavioural performance was similar to that observed with the same Dots task in Experiment 1A§??. Initial decision accuracy converged on the target 71%, and participants may have benefited from advice in terms of their final decisions being more accurate than their initial decisions ($F(1,49) = 5.48$, $p = .023$; $M_{Final} = 0.73$ [0.72, 0.74], $M_{Initial} = 0.72$ [0.71, 0.73]; Figure 5.11A). There was no evidence of a general difference in participants' overall accuracy between advisors ($F(1,49) = 0.66$, $p = .420$; $M_{LowAgreement} = 0.72$ [0.71, 0.73], $M_{HighAgreement} = 0.73$ [0.72, 0.74]), nor was there evidence of a difference in participants' improvement in accuracy

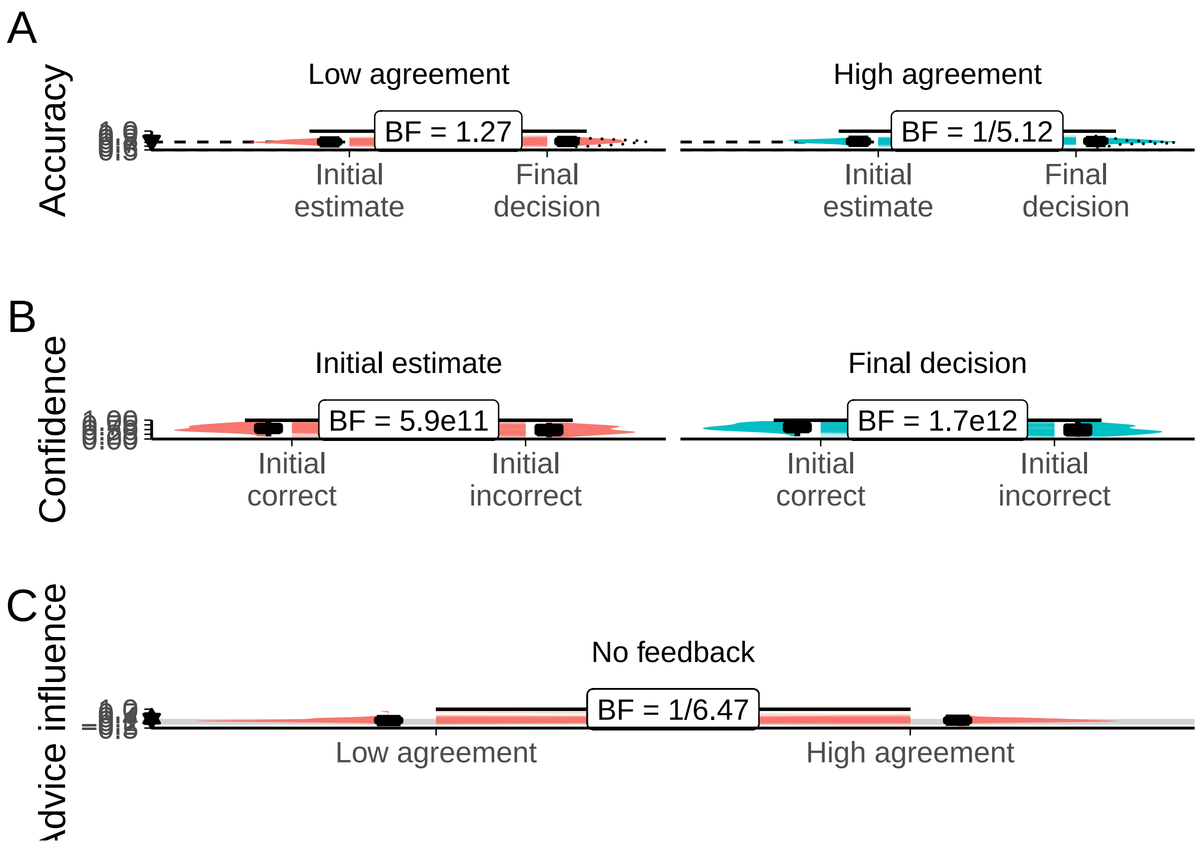


Figure 5.11: Task performance for the Dots task with dis/agreeing advisors.
A: Response accuracy. Faint lines show individual participant means, for which the violin and box plots show the distributions. The half-width horizontal dashed lines show the level of accuracy which the staircasing procedure targeted, while the full width dashed line indicates chance performance. Dotted violin outlines show the distribution of actual advisor accuracy.
B: Confidence. Faint lines show individual participant means, for which the violin and box plots show the distributions.
C: Influence. Participants' weight on the advice for advisors in the Familiarization stage of the experiment. The shaded area and boxplots indicate the distribution of the individual participants' mean influence of advice. Individual means for each participant are shown with lines in the centre of the graph. The theoretical range for influence values is [-2, 2].

Table 5.12: Advisor agreement for Dots task Agreement experiment

Advisor	Target|correct	Actual|correct	Target|incorrect	Actual|incorrect
High agreement	.840	.832	.610	.631
Low agreement	.660	.651	.170	.166

Table 5.13: Advisor accuracy for Dots task Agreement experiment

Advisor	Target accuracy	Mean accuracy
High agreement	.709	.705
Low agreement	.709	.704

between advisors ($F(1,49) = 1.33$, $p = .255$; $M_{\text{Improvement|LowAgreement}} = 0.02$ [0.00, 0.03], $M_{\text{Improvement|HighAgreement}} = 0.00$ [-0.01, 0.02]).

Figure 5.11B and ANOVA indicated that participants were more confident in their correct answers than their incorrect ones ($F(1,49) = 145.02$, $p = <.001$; $M_{\text{Correct}} = 29.51$ [26.96, 32.07], $M_{\text{Incorrect}} = 24.28$ [21.49, 27.06]), and less confident in their final decisions than their initial estimates ($F(1,49) = 5.31$, $p = .025$; $M_{\text{Final}} = 27.62$ [24.89, 30.35], $M_{\text{Initial}} = 26.17$ [23.47, 28.86]). These two factors interacted ($F(1,49) = 42.26$, $p = <.001$; $M_{\text{Increase|Correct}} = 2.73$ [1.43, 4.02], $M_{\text{Increase|Incorrect}} = 0.18$ [-1.18, 1.54]).

Perhaps surprisingly, there was no correlation between initial estimate accuracy and confidence (1/3.06), and no evidence for a correlation between final decision accuracy and confidence (1/1.06).

Neither advisor appeared to be more influential than the other (Figure 5.11C).

Advisor performance The advice is generated probabilistically from the rules described previously in Table 5.10. It is thus important to get a sense of the actual advice experienced by the participants.

The advisors agreed with the participants' initial estimates at close to target rates (Table 5.12), and were as accurate on average as expected (Table 5.13). Nevertheless, some participants experienced in practice 10-20% differences in advisor accuracy (although neither advisor was systematically more accurate across participants).

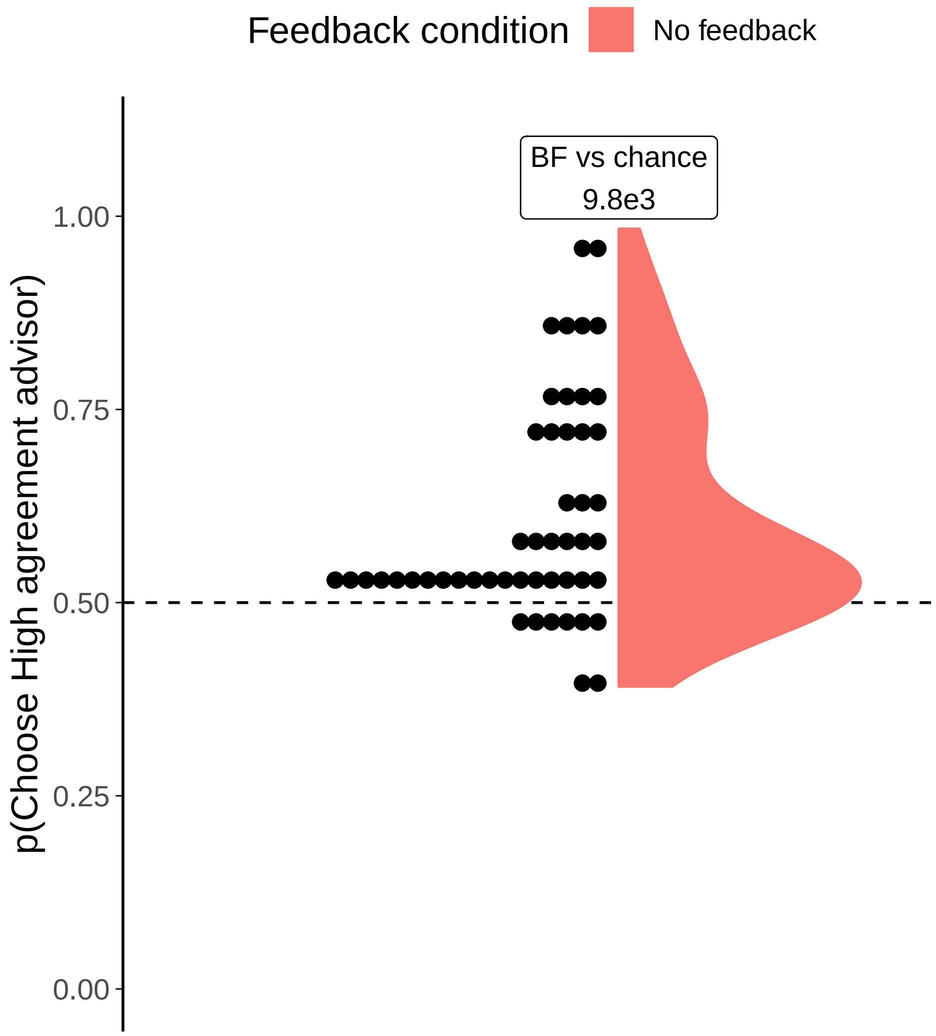


Figure 5.12: Dot task advisor choice for high/low agreement advisors.
Participants' pick rate for the advisors in the Choice phase of the experiment. The violin area shows a density plot of the individual participants' pick rates, shown by dots. The chance pick rate is shown by a dashed line.

All participants experienced the intended relationship wherein the High agreement advisor agreed with them more than the Low agreement advisor.

Hypothesis test Our key analysis concerned whether participants would have a systematic preference for choosing the High agreement advisor when they were given a choice of advisor. Consistent with the key prediction of this experiment, advisor choice varied significantly as a function of advisor agreement rate (Figure

5.12): The High agreement advisor was preferred at a rate greater than that expected by chance ($t(49) = 5.43$, $p < .001$, $d = 0.77$, BF = 9.8e3; $M = 0.61$ [0.57, 0.65], $\mu = 0.5$). The modal preference remained at chance, but almost all participants who manifested a preference preferred the High agreement advisor.

While this effect is interesting, it is substantially smaller than participants' preference for picking the top advisor regardless of identity ($t(49) = 7.26$, $p < .001$, $d = 1.03$, BF = 4.4e6; $\mathrm{M}_{\mathrm{P(PickFirst)}} = 0.66$ [0.62, 0.71], $\mu = 0.5$), an effect that we would hope would be random and even out across participants. Note that because the advisor position is well balanced across advisors (BF = 1/2.70; $\mathrm{M}_{\mathrm{P(HighAgreementFirst)}} = 0.51$ [0.50, 0.52], $\mu = 0.5$) the presence of a preference for advisor by position would not cause a preference for an individual advisor.

Follow-up tests As noted above, the stochastic nature of the advisors' advice meant that there was some variation in the participants' experience of the advisors. This allowed us to investigate whether either objective accuracy or agreement rate differences were predictive of pick rates. This did not appear to be the case: there was no evidence of a relationship between participants' advisor preference and their experience of either advisor accuracy ($BF_{10} = 1/2.56$) or advisor agreement ($BF_{10} = 1/1.28$).

We would not expect a relationship to accuracy here, because we hypothesise that the work done by accuracy differences in the previous experiment was actually driven by agreement, which we manipulated in this experiment. We might have expected a correlation between experienced agreement rates and preference strength, but these are between-participant differences and we suspect there is a great deal of noise in participant preferences that would mask any differences over the rather limited agreement rate difference range.

We might also expect advisor preference to vary as a function of initial confidence. Following the logic in the previous experiment (Experiment 1A§5.1.1), participants may have a strong preference for the High agreement advisor but only exercise that preference where they are unsure about the answer themselves (i.e. where advice is

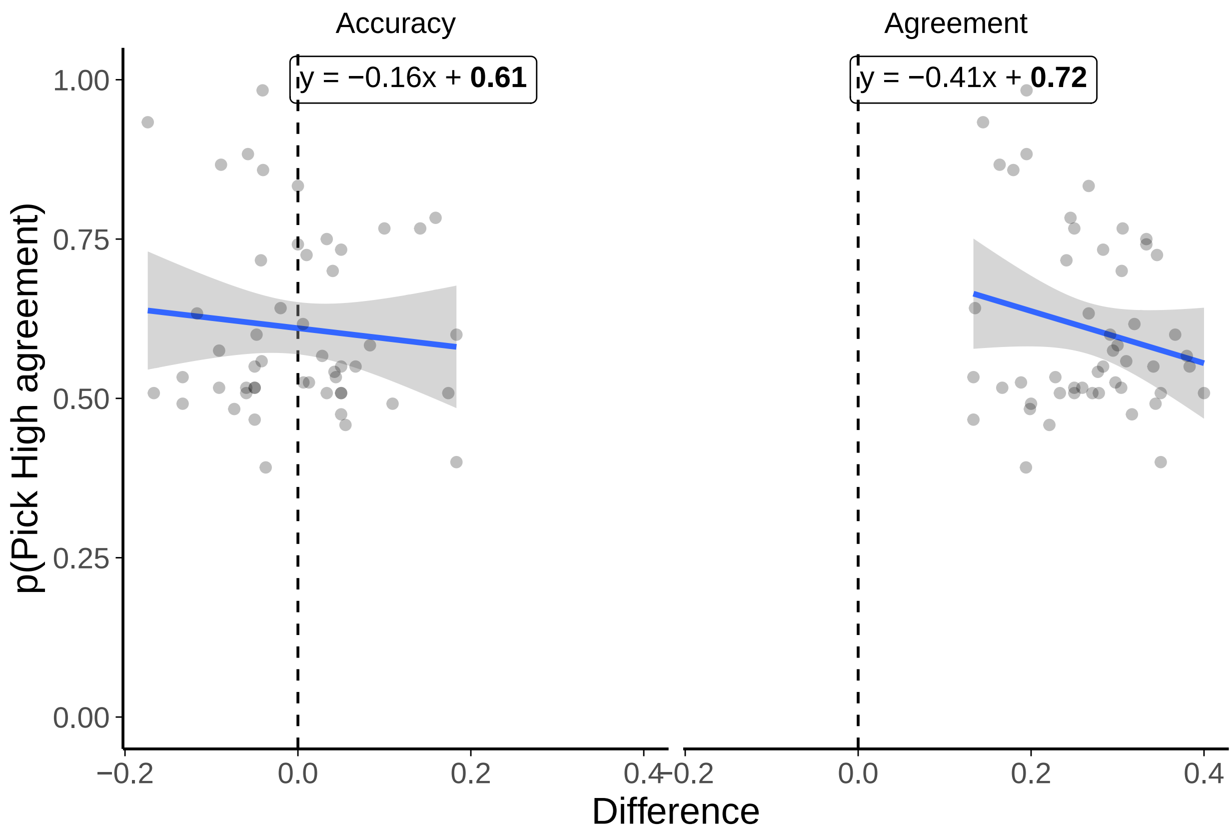


Figure 5.13: Advisor choice by experience in the Dots task with dis/agreeing advisors. Each dot is a participant's proportions. The difference in accuracy rates is calculated as the proportion of correct answers seen from the High accuracy advisor minus the proportion of correct answers seen from the Low accuracy advisor, and the difference in agreement rates similarly for agreement. The participants did not receive feedback on the correct answers

most valuable to them). This appeared not to be the case: we split participants' trials into high and low confidence based on their idiosyncratic median confidence, and conducted a paired t-test of the pick rates for the High accuracy advisor in each confidence category. The Bayes factor for this t-test indicated good evidence of no difference in pick rates ($BF_{10} = 1/6.34$).

Summary/Discussion Participants who had a preference for one of the two advisors almost universally preferred the High agreement advisor. These results are in line with the effects of advisor accuracy in the same task as found in Experiment 1A§5.1.1. They are also consistent with our hypothesis that agreement is used as a proxy for feedback when objective feedback is unavailable. We next explored whether this pattern would also be apparent in the Dates task.

5.2.2 Experiment 2B: Advisor agreement effects in the Dates Task

As with Experiment 1B§5.1.2, we attempted to replicate the result using the Dates task. Participants in this task were split into conditions depending upon whether or not they received feedback, allowing a direct exploration of the effect of feedback on advisor preference.

Open scholarship practices

This experiment was preregistered at `https://osf.io/8d7vg`. The experiment data are available in the `esmData::` package for R from !TODO[Zenodo esmData], and also directly from !TODO[OSFify Dates task data]. A snapshot of the state of the code for running the experiment at the time the experiment was run can be obtained from `https://github.com/oxacclab/ExploringSocialMetacognition/blob/master/ACBin/acc.html`.

Method

76 participants each completed 52 trials over 4 blocks of the binary version of the Dates task§3.1.3. On each trial, participants were presented with an historical event that occurred on a specific year between 1900 and 2000. They were given a date and asked whether the event occurred before or after that date, indicating their confidence in their decision by selecting an appropriate point on the relevant answer bar. Participants then received advice indicating which of the two bars (before or after) was supposedly the correct answer. Participants could then mark a final response in the same manner as their original response.

Participants started with 1 block of 10 trials that contained no advice to allow them to familiarise themselves with the task. All trials in this section included feedback for all participants indicating whether or not the participant's response was correct.

Participants then did 2 trials with a practice advisor to get used to receiving advice. They also received feedback on these trials. They were informed that they would "receive advice from advisors" to "help you complete the task". They were told that the "advisors aren't always correct, but they are quite good at the task", and informed that they should "identify which advisors are best and to weigh their advice accordingly".

Participants then performed 3 blocks of trials that constituted the main experiment. The first two of these were familiarization blocks where participants had a single advisor in each block for 14 trials, plus 1 attention check.

Participants were split into four conditions that produced differences in their experience of these familiarization blocks. These conditions were whether or not they received feedback, and which of the two advisors they were familiarized with first.

Finally, participants performed a test block of 10 trials that offered them a choice on each trial of which of the two advisors they had encountered over the last two blocks would give them advice. No participants received feedback during the test phase.

Advice profiles The High agreement and Low agreement advisor profiles issued binary advice (endorsing either the 'before' or 'after' column) probabilistically based on which column the participant had selected in their initial estimate and whether that was the correct answer. Unlike in the Dots task above (Experiment 2A§5.2.1), the accuracy of the advisors was not controlled because we were unable to control the participants' accuracy, and advisor accuracy depends upon participant accuracy when agreement rates are fixed.

Results

Exclusions Individual trials were screened to remove those that took longer than 60s to complete. 3 participants had a total of 3 trials removed in this way,

Table 5.14: Advisor advice profiles for Dates task Agreement experiment

	Probability of agreement (%)			
Advisor	Participant correct	Participant incorrect	Overall[a]	Overall accuracy[a]
High agreement	.900	.650	.775	.625
Low agreement	.750	.350	.550	.700

[a] Where participants' initial estimate accuracy is 50%

representing 0.11% of all trials. Participants were then excluded for having fewer than 11 trials remaining, fewer than 10 trials on which they had a choice of advisor, or for giving the same initial and final response on more than 90% of trials. These criteria led to no participants being excluded from this experiment.

Task performance Before exploring the interaction between the participants' responses and the advisors' advice, and the participants' advisor selection behaviour, it is useful to verify that participants interacted with the task in a sensible way, and that the task manipulations worked as expected. In this section, task performance is explored during the Familiarization phase of the experiment where participants received advice from a pre-specified advisor on each trial. There were an equal number of these trials for each participant for each advisor. As before (Experiment 1B§5.1.2), the conditions are pooled together while exploring participants' performance on the task.

```
## Interaction calculation: Final decision - Initial estimate

## Interaction calculation: Final decision - Initial estimate

## Interaction calculation: High agreement - Low agreement
```

There were some similarities to and some differences from the basic behavioural performances compared to the same Dates task in Experiment 1B§??.

Participants' accuracy (Figure 5.14A), which was uncontrolled in this task, was greater on final decisions than on initial estimates ($F(1,73) = 32.19$, $p = <.001$; $M_{FinalDecision} = 0.66$ [0.63, 0.68], $M_{InitialEstimate} = 0.60$ [0.57, 0.62]). There was no significant difference between advisors ($F(1,73) = 3.28$, $p = .074$; $M_{HighAgreement} = 0.61$ [0.57, 0.64], $M_{LowAgreement} = 0.65$ [0.62, 0.68]), but the increase in final decision

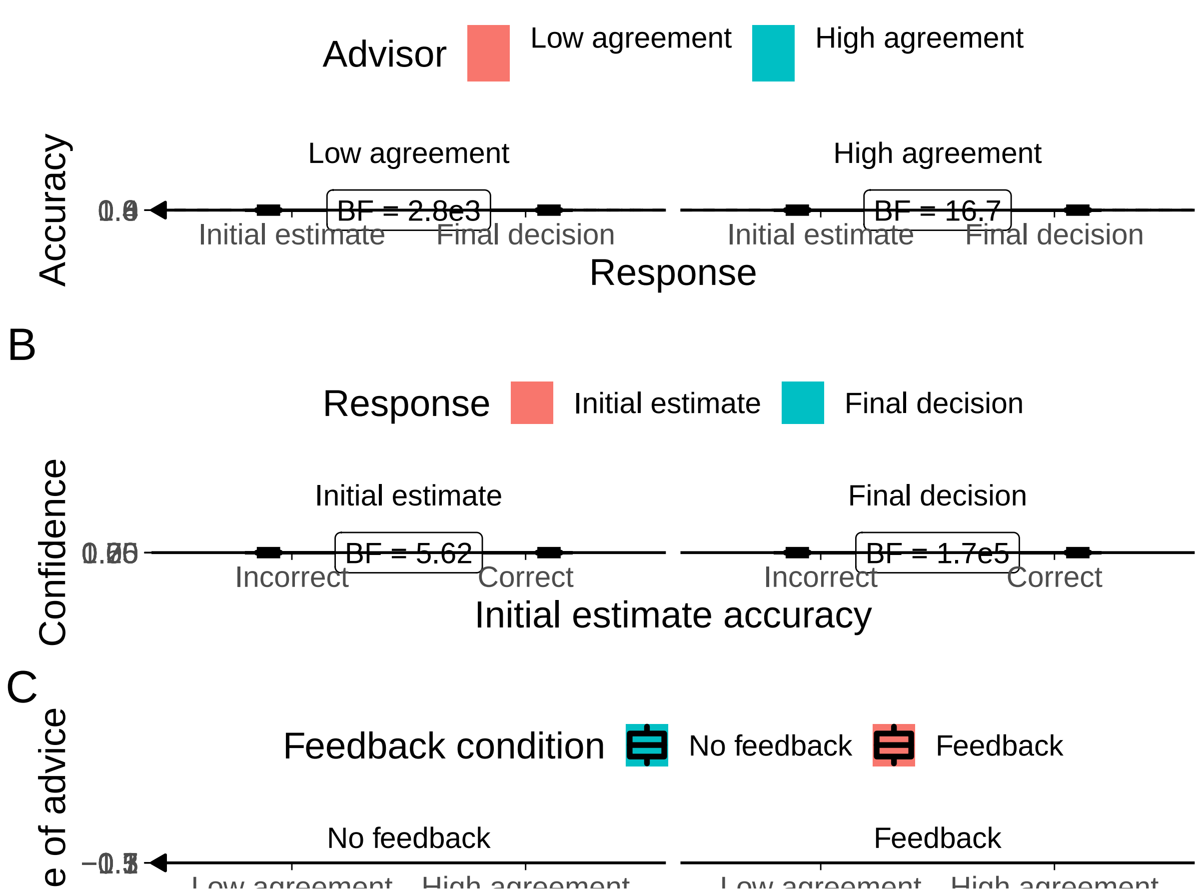


Figure 5.14: Task performance for the Dates task with dis/agreeing advisors.
A: Response accuracy. Faint lines show individual participant means, for which the violin and box plots show the distributions. The dashed line indicates chance performance. Dotted violin outlines show the distribution of actual advisor accuracy.
Because there were relatively few trials, the proportion of correct trials for a participant generally falls on one of a few specific values. This produces the lattice-like effect seen in the graph. Some participants had individual trials excluded for over-long response times, meaning that the denominator in the accuracy calculations is different, and thus producing accuracy values which are slightly offset from others'.
B: Confidence. Faint lines show individual participant means, for which the violin and box plots show the distributions.
C: Influence. Participants' weight on the advice for advisors in the Familiarization stage of the experiment. The shaded area and boxplots indicate the distribution of the individual participants' mean influence of advice. Individual means for each participant are shown with lines in the centre of the graph. The theoretical range for influence values is [-2, 2].

Table 5.15: Advisor agreement for Dates task Accuracy experiment

Advisor	Target|correct	Actual|correct	Target|incorrect	Actual|incorrect
High agreement	.900	.874	.650	.612
Low agreement	.750	.761	.350	.334

Table 5.16: Advisor accuracy for Dates task Accuracy experiment

Advisor	Target accuracy	Mean accuracy
High agreement	.625	.665
Low agreement	.700	.715

accuracy was greater for the Low agreement advisor than the High agreement advisor ($F(1,73) = 5.30$, $p = .024$; $M_{Improvement|HighAgreement} = 0.04$ [0.02, 0.06], $M_{Improvement|LowAgreement} = 0.09$ [0.05, 0.12]).

As expected, and as shown in Figure 5.14B participants were systematically more confident when their initial estimate was correct as compared to incorrect ($F(1,73) = 28.55$, $p = <.001$; $M_{Correct} = 0.68$ [0.64, 0.72], $M_{Incorrect} = 0.63$ [0.58, 0.68]). Participants were also more confident on final decisions than on initial estimates ($F(1,73) = 24.92$, $p = <.001$; $M_{FinalDecision} = 0.67$ [0.63, 0.72], $M_{InitialEstimate} = 0.63$ [0.58, 0.68]). This increase in confidence was greater when the initial estimate was correct as compared to incorrect ($F(1,73) = 25.02$, $p = <.001$; $M_{Increase|Correct} = 0.07$ [0.05, 0.09], $M_{Increase|Incorrect} = 0.02$ [-0.01, 0.04]).

The High agreement advisor was more influential (Figure 5.14C) than the Low agreement advisor overall ($F(1,72) = 14.17$, $p = <.001$; $M_{HighAgreement} = 0.20$ [0.16, 0.24], $M_{LowAgreement} = 0.30$ [0.25, 0.35]), and participants in the No feedback condition were more influenced than participants in the Feedback condition ($F(1,72) = 5.74$, $p = .019$; $M_{NoFeedback} = 0.30$ [0.24, 0.35], $M_{Feedback} = 0.21$ [0.16, 0.26]). There was no evidence of an interaction between these differences, however ($F(1,72) = 0.26$, $p = .614$; $M_{High\text{-}LowAccuracy|NoFeedback} = -0.08$ [-0.15, -0.02], $M_{High\text{-}LowAccuracy|Feedback} = -0.11$ [-0.19, -0.03]).

Advisor performance The advice is generated probabilistically from the rules described previously (Advice profiles§5.1.2). This advisors agreed with participants contingent on the accuracy of the participants' initial estimates at close to the target rates (Table 5.15). This meant that advisors were distinguished by their overall agreement rates as they were intended to be in the familiarization phase. The accuracy of participants' initial estimates was not much above 50%, meaning that the overall accuracy rates of the advisors were similar to those projected (Table 5.16). Most (66/74, 89.19%) participants experienced the High accuracy advisor as providing more accurate advice than the Low accuracy advisor.

Hypothesis test Consistent with the result from the Dots task, the key analysis demonstrated that in the No feedback condition participants' preferences for receiving advice from the High agreement advisor were greater than chance ($t(34) = 2.62$, $p = .013$, $d = 0.44$, BF = 3.39; M~No feedback~ = 0.63 [0.53, 0.73], $\mu = 0.5$). The modal preference was to select the High agreement advisor on every Choice trial, and although some participants still showed a preference for hearing advice from the Low agreement advisor, preferences for the High agreement advisor were generally stronger and more frequent (Figure 5.15).

In the Feedback condition, the mean of the participants' selection rates was equivalent to random picking ($t(38) = 0.46$, $p = .648$, $d = 0.07$, BF = 1/5.24; $\text{M}_{\text{Feedback}} = 0.52$ [0.42, 0.62], $\mu = 0.5$). This is consistent with a strategy which attempts to maximise the accuracy of final decisions, because neither advisor would help with this task systematically. The null result here does indicate, however, that there is no strong and clear preference for agreement over and above its accuracy benefits.

Interestingly, despite different patterns of preferences when compared to chance, there was not enough evidence to demonstrate whether the preference patterns for the two conditions were or were not different from one another. This may be a consequence of the variability of preferences: as with the Dates task using High accuracy and Low accuracy advisors (Experiment 1B§5.1.2), the participants in both Feedback and No feedback conditions spanned the entire gamut of preference strengths and directions. One participant in the No feedback group here, for example,

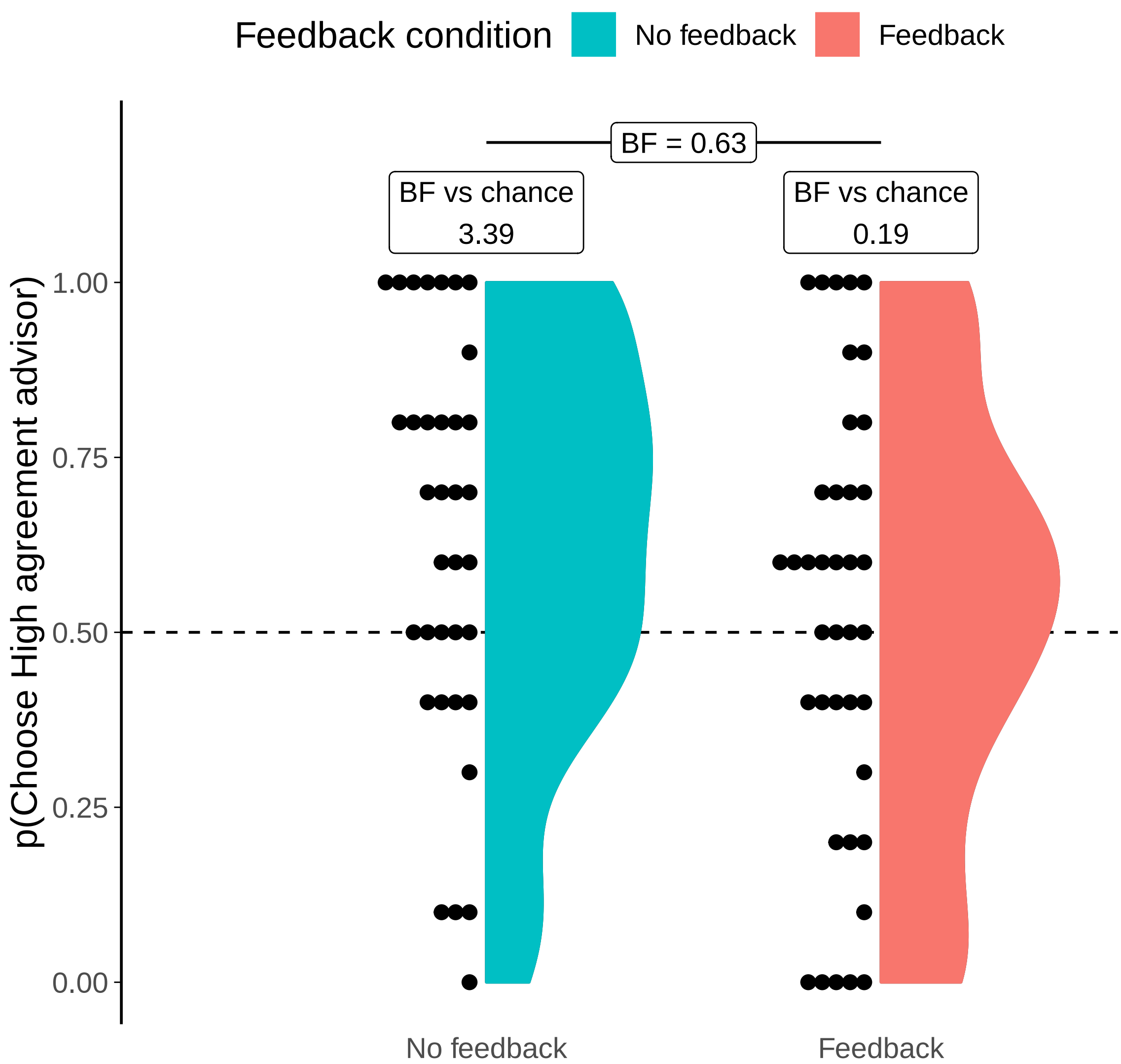


Figure 5.15: Dates task advisor choice for High/Low agreement advisors. Participants' pick rate for the advisors in the Choice phase of the experiment. The violin area shows a density plot of the individual participants' pick rates, shown by dots. The chance pick rate is shown by a dashed line. Participants in the Feedback condition received feedback during the Familiarization phase, but not during the Choice phase.

never chose the High agreement advisor. Even where there were no systematic effects (No feedback condition in Experiment 1B, Feedback condition here), participants still had a range of preferences with some picking one or the other advisor exclusively. A substantial minority (0.0, 100.0, 100.0, 100.0, 100.0, 0.0, 0.0, 0.0, 100.0, 0.0, 0.0, 100.0, 0.0, 0.0, 0.0, 100.0, 100.0, 100.0, 0.0, 100.0, 0.0, 0.0, 0.0, 100.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 100.0, 100.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 100.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 100.0, 100.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 100.0, 100.0, 0.0, 0.0, 100.0, 100.0, 0.0, 100.0, 0.0, 0.0, 100.0, 100.0, 100.0, 0.0, 100.0, 0.0, 0.0%) of participants had preference strengths beyond those expected by chance when picking randomly.[4] As noted previously (Experiment 1: Discussion§5.1.3), this is in contrast to the behaviour in the Dots task, where preferences tend to mass towards even pick rates with a long tail differentiating the population preferences from chance.

It was discovered after the completion of experiments that the advisor position (whether the advisor appears on the top or bottom of the advisor choice panel) was not counterbalanced between advisors. It had been decided during development of the tests to keep the advisors in the same position for every trial so that participants did not get mixed up between them. Together, this meant that the High accuracy advisor always appeared at the top and the Low accuracy advisor always appeared at the bottom, for every trial for every participant. We are thus unable to confirm that pick rate differences, or the absence of those differences, are caused by participants' preferences for the advice the advisor would provide or the position the advisor was in on the screen. Furthermore, it is possible that any genuine preference for one advisor over the other was *induced* by the position, rather than independent of it (Zajkowski and Zhang 2021).

Follow-up tests There was variation in the relative agreement and accuracy rates of the advisors as experienced by each participant during the Familiarization phase. This experienced difference could have led to detectable variation in picking

[4] 5% would be expected to have had strengths of these magnitudes if picking randomly.

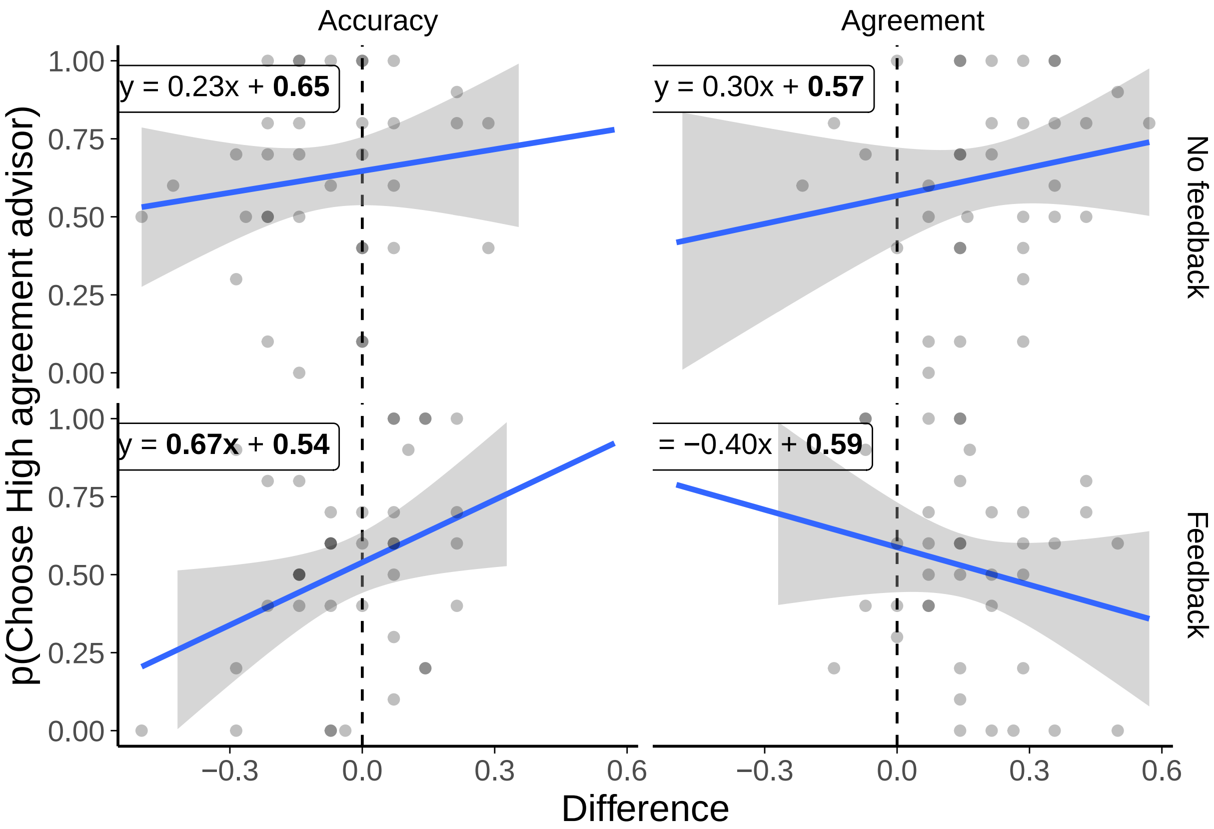


Figure 5.16: Preference predictors in the Dates task with dis/agreeing advisors. Scatter plots of participants' experience with advisors in terms of agreement or accuracy rates. Differences are expressed as the experienced rate for the High agreement advisor minus the experienced rate for the Low agreement advisor during the Familiarization phase. Numbers in bold in the regression equations are significant at p < .05.

Table 5.17: Experienced accuracy difference effects in the Dates task with dis/agreeing advisors

Effect	Estimate	SE	t	p	
(Intercept)	0.65	0.05	11.89	< .001	*
Accuracy	0.23	0.28	0.84	.406	
FeedbackFeedback	-0.11	0.07	-1.49	.141	
Accuracy:FeedbackFeedback	0.44	0.40	1.09	.281	

Model fit: \$F\$(2.8, 3) = 70; \$p\$ = .048; \$R^2_{adj}\$ = .068

Table 5.18: Experienced agreement difference effects in the Dates task with dis/agreeing advisors

Effect	Estimate	SE	t	p	
(Intercept)	0.57	0.08	7.22	$< .001$	$*$
Agreement	0.30	0.30	1.01	.315	
FeedbackFeedback	0.02	0.10	0.19	.850	
Agreement:FeedbackFeedback	-0.70	0.43	-1.65	.104	

Model fit: \$F\$(1.7, 3) = 70; \$p\$ = .181; \$R^2_{adj}\$ = .027

behaviour because relative agreement rate is the mechanism we suspect to be driving picking behaviour differences.

This did not appear to be the case either for accuracy (Table 5.17) or agreement (Table 5.18). As shown in Figure 5.16, experienced accuracy difference was a significant predictor of picking preference for participants in the Feedback condition. This effect did not show up in the linear regression with agreement rate difference and feedback condition as predictors (Table 5.17), and the evidence in the Bayesian equivalent was sub-threshold (BF=2.55). Despite the possibility that this result is a false positive (running many individual regressions without familywise error correction will tend to indicate some significant relationships by chance alone), it does make sense because this relationship was argued to lie behind the result in Experiment 1B§5.1.2. Experiment 1B showed a preference for the High accuracy advisor in the picking behaviour, but not a relationship between that picking preference strength and experienced accuracy difference. Experienced accuracy differences were tightly controlled in Experiment 1B, however, whereas here they have more room to vary and thus may be more likely to reveal relationships.

Comparing the overall model fits to Experiment 1B, it is notable that the adjusted R^2, or the amount of variance in the dependent variable explained by the predictors, is much lower here. Whatever is driving picking preference variation, it does not seem to be well captured by the variation in experienced agreement or accuracy rates, or by the difference between the experimental conditions.

5.2.3 Discussion

These two experiments investigated the impact of advisor agreement on choice of advice. In both Dots and Dates tasks, where the absence of feedback means advisors' performances cannot be evaluated objectively, participants preferred high agreement advisors to low agreement advisors. This is consistent with our underlying theory that agreement is used as a mechanism for evaluating advisors in the absence of objective feedback.

When feedback was provided in the Dates task, advisors were selected at rates equivalent to chance overall. Despite this overall chance level of preference in the sample, individual participants had a wide range of preference strengths and directions, and this was true for both the experimental conditions. This contrasts with what we see in the Dots task data, for both Experiment 1A§5.1.1 and 2A§5.2.1, where the majority of participants' preferences are moderate, with a minority of participants' more marked preferences producing the systematic effects.

In the Dates task, as we saw in Experiment 1B§5.1.2, systematic effects show up as a general nudging of preferences in one direction rather than as every participant developing similar preferences. This wide variability with some systematic nudging is explicable in terms of the heterogeneity of the Dates task. Participants will have a different level of knowledge on different questions, and we might expect advisors to occasionally offer implausible advice to questions participants consider easy. If this were to happen, participants might weight these occurrences highly, in line with the Pescetelli and Yeung (2021) theory of metacognitive evaluation of advice. More frequently disagreeing advisors will be more likely to disagree on these subjectively easy questions and suffer the reputational consequences. This greater likelihood could mean that the wide spread of behaviour from participants is due to the greater frequency of these important events for the Low accuracy advisor, rather than a consistent effect of agreement alone.

A challenge for this explanation is that in Experiment 1B§5.1.2 we observed participants in the No feedback condition demonstrating a range of preferences that were not systematically different from chance. We would expect, provided participants' performance on subjectively easy questions is above chance, that the High accuracy advisor would have a higher probability of agreement on those questions, just as the High agreement advisor had a higher probability of agreement in Experiment 2B§5.2.2. It is unlikely, but not impossible, that participants' accuracy even on subjectively easy questions was sufficiently bad as to render these effects undetectable.

Another reason why advisor preferences are so varied is that there are reasons why participants might prefer to see advice from the disagreeing advisor. In both Dots and Dates tasks, it was harder to predict what the Low agreement advisor would say, potentially making the advice more interesting. Furthermore, if participants thought they were unlikely to be correct, the Low agreement advisor's advice was more diagnostic of the correct answer because the Low agreement advisor was far less likely to endorse incorrect responses. It may be, therefore, that people evaluate advisors on the basis of agreement, but that what they do on the basis of that evaluation is a matter of personal preference.

In the previous two studies we looked at the effects of advisor accuracy and agreement on the preference for picking those advisors when offered a choice. In order to isolate the effects, agreement was balanced in the accuracy experiments, and accuracy was balanced (as well as we were able) in the agreement experiments. Next, we directly contrast these domains by providing participants with a pairing consisting of a High accuracy advisor and a High agreement advisor, introducing a clear performance cost of preferring to hear advice from the High agreement advisor.

5.3 Effects of accuracy vs. agreement (Date estimation)

The High versus Low agreement advisor experiment showed that participants tended to prefer to receive agreeing advice when they did not receive feedback. The advisors did not differ in their accuracy (by design), which meant that participants could not increase their performance by selecting one advisor over another. Here we introduce a discrepancy between advisors' objective performance (accuracy) and their subjective performance from the judge's perspective (agreement). By playing off accuracy against agreement we can explore whether participants continue to prefer agreement when there is a cost associated with agreement through a reduction in overall accuracy. We expect that participants will gravitate towards the agreeing advisor who, from their perspective, should appear more accurate, despite the poorer objective performance of that advisor.

5.3.1 Experiment 3A: Accuracy vs. agreement effects in the Dots Task

Open scholarship practices

This experiment was preregistered at https://osf.io/f3k4x. The experiment data are available in the `esmData::` package for R from !TODO[Zenodo esmData], and also directly from https://osf.io/y47ec/. A snapshot of the state of the code for running the experiment at the time the experiment was run can be obtained from https://github.com/oxacclab/ExploringSocialMetacognition/blob/c18c26b5da3622988e226 AdvisorChoice/ava.html.

Unanalysed data

An initial version of this study was conducted and run as a proper experiment in which participants learned about both advisors simultaneously (preregistered at https://osf.io/5z2fp), but there were no effects in the data. We suspected the absence of effects was because participants had difficulty distinguishing the advisors when they were presented together. The version of the experiment reported here presented one advisor per block in the familiarisation phase. Data for the unreported null study can be found in the `esmData::` R package and at https://osf.io/26yut/.

Method

89 participants each completed 277 trials over 6 blocks of a perceptual decision-making task. Each trial consisted of three phases: participants gave an initial estimate (with confidence) of which of two briefly presented boxes contained more dots; received advice on their decision from an advisor; and made a final decision (again, with confidence).

Participants started with 2 blocks of 60 trials that contained no advice to allow them to familiarise themselves with the task; to allow the staircasing process to titrate the difficulty to their ability in order to maintain approximately 71% initial estimate accuracy; and to allow estimating of participants' idiosyncratic confidence reporting style. The first three trials were introductory trials that explained the task. All trials in this section included feedback indicating whether or not the participant's response was correct.

Participants then did 4 trials with a practice advisor to get used to receiving advice. They were informed that they would "get **advice** from advisors to help you make your decision", and that "advice is not always correct, but it is supposed to help you perform better on the task."

Participants then performed 3 blocks of trails that made up the core experiment. There were 2 familiarisation blocks of 60 trials, with participants seeing one of the two advisors for an entire block in random order. The familiarisation block was followed with a test block of 30 trials in which participants could choose between the advisors they encountered in the familiarisation blocks.

Advice profiles The two advisor profiles used in the experiment were High accuracy and High agreement. The advisors' advice was stochastically generated according to the participant's response. The advisor profiles were not balanced for overall agreement or accuracy rates.

The High accuracy advisor predominantly agreed with correct participant responses and disagreed with incorrect ones. The High agreement advisor agreed with the participant at the same rate regardless of the accuracy of the participant's

Table 5.19: Advisor advice profiles for Dots task Accuracy/agreement experiment

Advisor	Probability of agreement			Overall accuracy
	Participant correct	Participant incorrect	Overall	
High accuracy	.800	.200	.626	.800
High agreement	.800	.800	.800	.626

Table 5.20: Participant exclusions for Dots task Accuracy vs Agreement experiment

Reason	Participants excluded
Accuracy too low	1
Accuracy too high	0
Missing confidence categories	6
Skewed confidence categories	18
Too many participants	14
Total excluded	**39**
Total remaining	**50**

initial estimate. In practical terms, the difference between the advisors is that the High agreement advisor continued to agree with participants where their initial estimates were incorrect, while the High accuracy advisor did not.

Results

Exclusions In line with the pre-registration, participants' data were excluded from analysis where they had an average accuracy below 0.6 or above 0.85, did not have choice trials in all confidence categories (bottom 30%, middle 40%, and top 30% of prior confidence responses), had fewer than 12 trials in each confidence category, or finish the experiment after 50 participants have already submitted data which passed the other exclusion tests. Overall, 39 participants were excluded, with the details shown in Table 5.20. This number is somewhat higher than in the previous experiments, but this is largely due to collecting data in larger batches than previously: a large number of participants were excluded because their data was in excess of the preregistered sample size.

Task performance

```
## Interaction calculation: Final - Initial
```

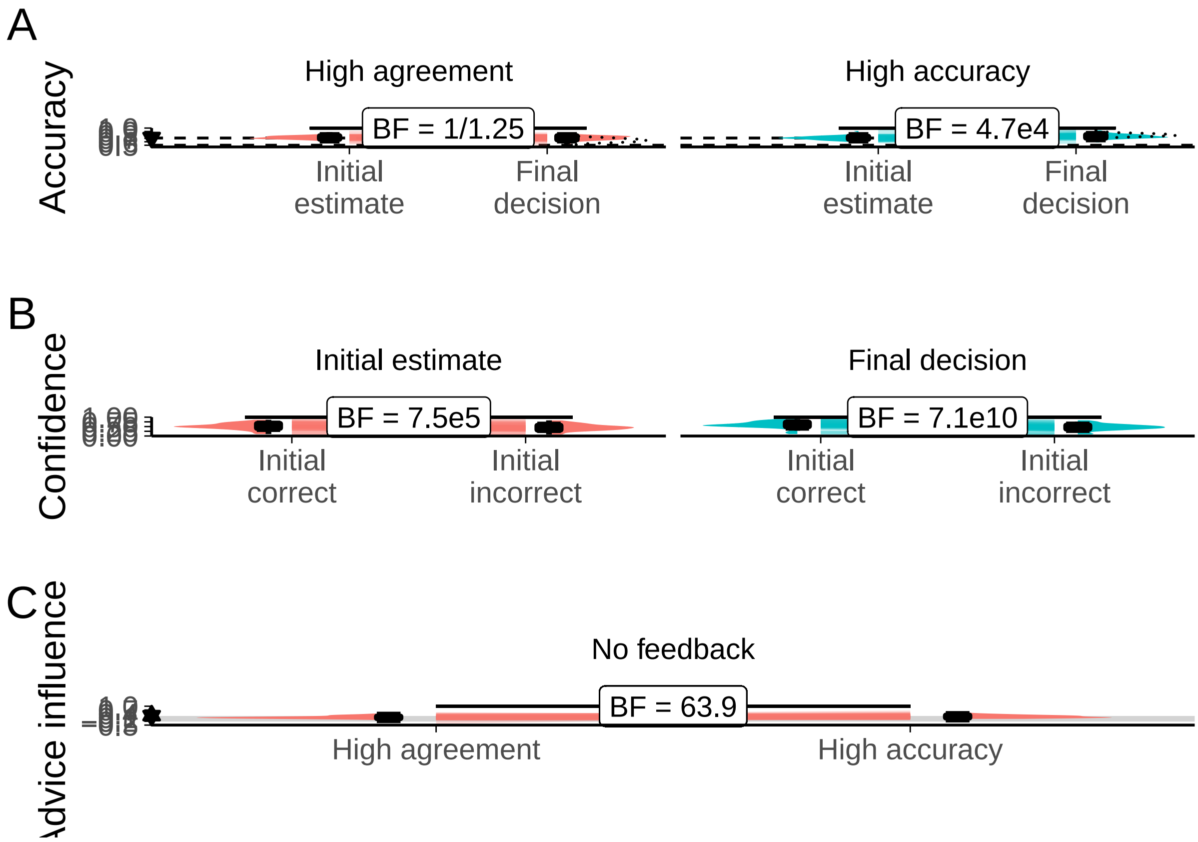


Figure 5.17: Task performance for the Dots task with agreeing vs. accurate advisors.
A: Response accuracy. Faint lines show individual participant means, for which the violin and box plots show the distributions. The half-width horizontal dashed lines show the level of accuracy which the staircasing procedure targeted, while the full width dashed line indicates chance performance. Dotted violin outlines show the distribution of actual advisor accuracy.
B: Confidence. Faint lines show individual participant means, for which the violin and box plots show the distributions.
C: Influence. Participants' weight on the advice for advisors in the Familiarization stage of the experiment. The shaded area and boxplots indicate the distribution of the individual participants' mean influence of advice. Individual means for each participant are shown with lines in the centre of the graph. The theoretical range for influence values is [-2, 2].

```
## Interaction calculation: Final - Initial
```

Basic behavioural performance was similar to that observed with the same Dots task in Experiments 1A§**??** and 2A§**??**. Initial decision accuracy converged on the target 71%, and, as shown in Figure 5.17A, participants benefited from advice in terms of their final decisions being more accurate than their initial decisions (ANOVA main effect of Time: F(1,49) = 16.63, p = <.001; M_{Final} = 0.74 [0.72, 0.75], M_{Initial} = 0.72 [0.71, 0.73]), especially following advice from the High accuracy advisor (interaction of Time and Advisor: F(1,49) = 33.88, p = <.001; $M_{\text{Improvement|HighAgreement}}$ = -0.01 [-0.02, 0.00], $M_{\text{Improvement|HighAccuracy}}$ = 0.05 [0.03, 0.06]). There was no main effect of Advisor (F(1,49) = 3.42, p = .070; $M_{\text{HighAgreement}}$ = 0.72 [0.70, 0.73], $M_{\text{HighAccuracy}}$ = 0.74 [0.72, 0.75]).

Figure 5.17B and ANOVA indicated that participants were more confident in their correct answers than their incorrect ones (F(1,49) = 77.24, p = <.001; M_{Correct} = 27.28 [24.90, 29.66], $M_{\text{Incorrect}}$ = 23.65 [21.34, 25.96]), and more confident in their final decisions than their initial estimates (F(1,49) = 37.39, p = <.001; M_{Final} = 26.74 [24.42, 29.06], M_{Initial} = 24.19 [21.82, 26.56]), and that these two factors interacted (F(1,49) = 41.82, p = <.001; $M_{\text{Increase|Correct}}$ = 3.74 [2.71, 4.77], $M_{\text{Increase|Incorrect}}$ = 1.37 [0.58, 2.15]).

The advisor also mattered; confidence decreased for incorrect answers and increased for correct answers with the High accuracy advisor, and increased regardless of correctness following advice from the High agreement advisor. This indicates that in general the confidence changed in the direction of the advice (because only the High accuracy advisor disagreed frequently, and then only on trials where the participant's initial estimate was incorrect), meaning the High accuracy advisor was more influential (Figure 5.11C).

There was no evidence of a correlation between initial estimate accuracy and confidence (1/1.69), and no evidence for a correlation between final decision accuracy and confidence (1/2.30).

Table 5.21: Advisor agreement for Dots task Accuracy vs Agreement experiment

Advisor	Target|correct	Actual|correct	Target|incorrect	Actual|incorrect
High accuracy	.800	.804	.200	.208
High agreement	.800	.812	.800	.804

Table 5.22: Advisor accuracy for Dots task Accuracy vs Agreement experiment

Advisor	Target accuracy	Mean accuracy
High accuracy	.800	.800
High agreement	.626	.640

Advisor performance The advice is generated probabilistically from the rules described previously in Table 5.19. It is thus important to get a sense of the actual advice experienced by the participants.

The advisors agreed with the participants' initial estimates at close to target rates (Table 5.21), and were as accurate on average as expected (Table 5.22). 49/50 participants experienced the intended relationship wherein the High agreement advisor agreed with them more than the High accuracy advisor and the High accuracy advisor gave more accurate advice than the High agreement advisor.

Hypothesis test Despite the influence differences observed above, and counter to our predictions, Figure 5.18 shows that there was no consistent picking preference in favour of either the High accuracy or the High agreement advisor. While several participants did develop very strong preferences, picking one or the other advisor nearly all the time, these preferences were not systematically oriented towards either advisor ($t(49)$ = 0.95, p = .345, d = 0.13, BF = 1/4.23; M = 0.54 [0.45, 0.63], μ = 0.5).

For context, note that we did see a significant effect of picking the advisor in the first position on the screen ($t(49)$ = 2.65, p = .011, d = 0.37, BF = 3.49; $\mathrm{M}_{\mathrm{P(PickFirst)}}$ = 0.54 [0.51, 0.56], μ = 0.5), an effect that we would hope would be random and even out across participants. In this experiment, as a function of chance, the High accuracy advisor appeared in the favoured top position less frequently than we would expect (BF = 3.49; $\mathrm{M}_{\mathrm{P(HighAccuracyFirst)}}$ = 0.46 [0.44, 0.49], μ = 0.5). If there

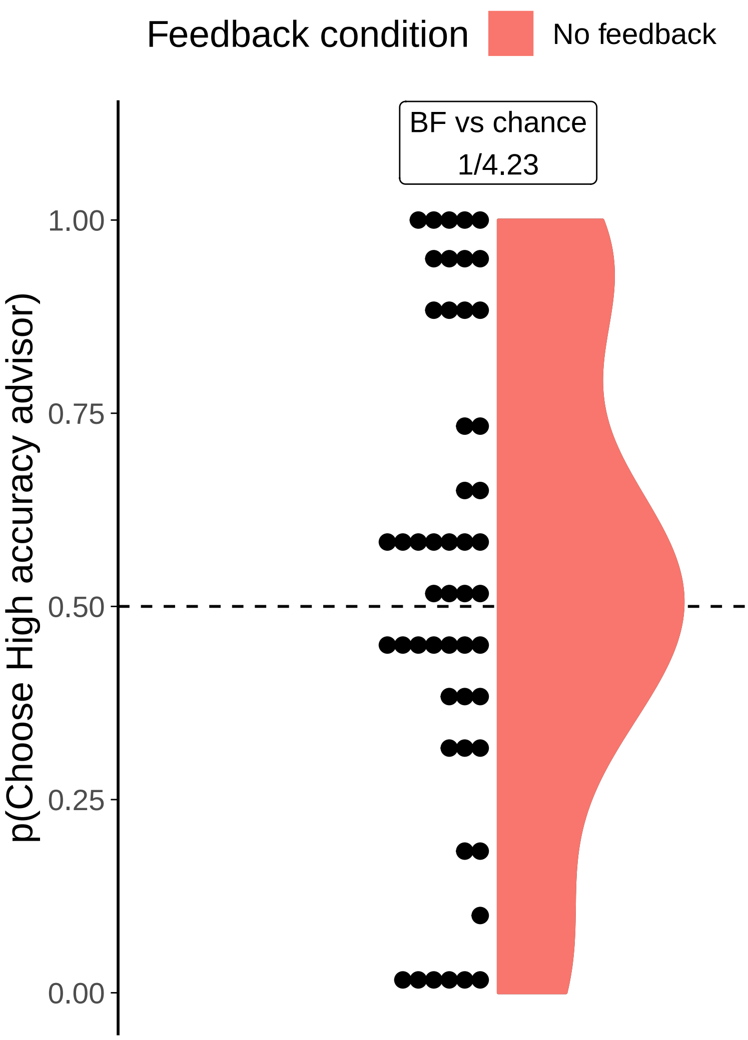


Figure 5.18: Dot task advisor choice for high accuracy/agreement advisors. Participants' pick rate for the advisors in the Choice phase of the experiment. The violin area shows a density plot of the individual participants' pick rates, shown by dots. The chance pick rate is shown by a dashed line.

were a general preference for the High agreement advisor, as per our prediction, this would be *increased* by that advisor appearing more often in the top position. Thus, even if the position of the advisors affected the results, it would not be confounding results in the direction of our prediction.

Follow-up tests

Ability of participants Although participants did not have a clear preference for the agreeing over the accurate advisor, it may be the case that participants who were well-calibrated were able to detect the usefulness of the accurate advisor, and therefore tended to prefer to hear that advisor's advice. The evidence was not sufficient to draw firm conclusions, but there was little indication of a correlation between preference for the High accuracy advisor and participant accuracy ($r(48)=-.071$, $p=.623$, $BF=1/2.81$) or confidence calibration ($r(48)=.106$, $p=.463$, $BF=1/2.46$).

Experience of advisors The lack of systematic preference from the participants was surprising. Each participant's data was tested against a null hypothesis that their picking was random, and 52.0% of participants demonstrated a statistically significant preference. As seen in Figure 5.18, however, these preferences were quite evenly split between the High accuracy and the High agreement advisors, both in terms of frequency and strength (although with a slight advantage for the High accuracy advisor).

The wide variation in preferences was not significantly correlated with either experienced agreement ($r=.035$, $p=.810$) or experienced accuracy ($r=.162$, $p=.262$).

Summary Contrary to expectations, participants did not appear to use advice agreement as a proxy for advice accuracy when objective feedback was not available. Whereas in the previous experiment (2A§**??**) this did happen, in the current experiment there was a cost to seeking advice from the High agreement advisor: the advice was less accurate.

It is not clear whether some participants were aware of the accuracy difference by some mechanism other than agreement. It is possible that, as suggested by Pescetelli and Yeung (2021), participants were able to use a combination of their subjective confidence and advice agreement to determine advisors' accuracy. This is explored more in Experiment 4A§5.4.2.

The next experiment explores the contrasted High agreement and High accuracy advisors in the Dates task. QUESTION: Did Niccolo do an agreement vs accuracy experiment? Was the feedback/no feedback between or within participants. He might have found something by using within participants feedback so they could have figured out what was coming...

5.3.2 Experiment 3B: Accuracy vs. agreement effects in the Dates Task

This experiment explored preferences for accuracy versus agreement using the continuous version of the Dates task introduced in Experiment 0§??. The continuous task was used because we aimed to investigate both advisor influence and advisor choice as dependent variables. Once again, participants were separated into Feedback and No feedback conditions.

Open scholarship practices

This experiment was preregistered at `https://osf.io/nwmx5`. The experiment data are available in the `esmData::` package for R from !TODO[Zenodo esmData], and also directly from !TODO[OSFify Dates task data]. A snapshot of the state of the code for running the experiment at the time the experiment was run can be obtained from `https://github.com/oxacclab/ExploringSocialMetacognition/blob/ed13951c488e1996df7ff53d48629843bacfd074/ACv2/ac.html`.

This is a replication of a study of identical design. The data for the original study can be obtained from the `esmData::` R package or from !TODO[OSFify].

Method

49 participants each completed 52 trials over 4 blocks of the continuous version of the Dates task§3.1.3. On each trial, participants were presented with an historical event that occurred on a specific year between 1900 and 2000. They were asked to drag one of three markers onto a timeline to indicate the date range within which they thought the event occurred. The three markers each had different widths, and each marker had point value associated with it, with wider markers worth fewer points. The markers were 7, 13, and 21 years wide, being worth 25, 10, and 5 points respectively. Participants then received advice indicating a region of the timeline in which the advisor suggested the event occured. Participants could then mark a final response in the same manner as their original response, and could choose a different marker width if they wished.

Participants started with 1 block of 10 trials that contained no advice to allow them to familiarise themselves with the task. All trials in this section included feedback for all participants indicating whether or not the participant's response was correct.

Participants then did 2 trials with a practice advisor to get used to receiving advice. They also received feedback on these trials. They were informed that they would "get advice on the answers you give" and that the feedback they received would "tell you about how well the advisor does, as well as how well you do". Before starting the main experiment they were told that they would receive advice from multiple advisors and that "advisors might behave in different ways, and it's up to you to decide how useful you think each advisor is, and to use their advice accordingly".

Participants then performed 3 blocks of trials that constituted the main experiment. The first two of these were familiarization blocks where participants had a single advisor in each block for 14 trials, plus 1 attention check.

Participants were split into four conditions that produced differences in their experience of these familiarization blocks. These conditions were whether or not they received feedback, and which of the two advisors they were familiarized with first. For each advisor, participants saw the advisor's advice on 14 trials. On most of these trials, including the first two, the advisor gave advice according to the advice profile (detailed below). On between 2 and 3 trials, the advisors issued the same kind of advice as one another, chosen to neither agree with the participant's answer nor indicate the correct answer. This "off-brand" advice was used to control for the effects of advice when the influence of advice was the dependent variable.

Finally, participants performed a test block of 10 trials that offered them a choice on each trial of which of the two advisors they had encountered over the last two blocks would give them advice. No participants received feedback during the test phase, and all advisors gave on-brand advice according to their advice profile.

Advice profiles The High accuracy and High agreement advisor profiles defined marker placements based on the timeline based on the correct answer and the participant's initial estimate respectively. Both advisors used markers that spanned 7 years, and both placed the markers in a normal distribution around the target point with a standard deviation of 5 years. The target point for the High accuracy advisor was the correct answer, and the target point for the High agreement advisor was the participant's initial estimate. Neither advisor ever placed their marker exactly on the midpoint of the participant's marker (because doing so means the weight on advice statistic is undefined).

Note that just as the task is continuous rather than binary, so agreement is continuous rather than binary. There is no objective threshold at which to classify advice as 'agreement', although we can classify accuracy in a binary way as whether or not a marker includes the correct answer.

On off-brand advice trials, of which there were between 2 and 3 per Familiarization block, advisors neither indicated the correct answer nor agreed with the participant. This was achieved by picking a target point of the participant's answer reflected around the correct answer. A detailed example was given in Experiment 0§4.1.

Table 5.23: Participant exclusions for Dates task Accuracy vs Agreement experiment

Reason	Participants excluded
Too few trials	0
Insufficient advice taking	0
Too few choice trials	0
Wrong markers	2
Non-numeric advice	0
Total excluded	**2**
Total remaining	**33**

Results

Exclusions Individual trials were screened to remove those that took longer than 60s to complete. Participants were then excluded for having fewer than 11 trials remaining, fewer than 8 trials on which they had a choice of advisor, or for giving the same initial and final response on more than 90% of trials. Participants were also excluded for technical problems with the experiment and data: sometimes the widths of the markers placed by the participants had unrecognised values, and sometimes the values for the advisors' advice were corrupted. Overall, 2 participants were excluded, with the details shown in Table 5.23.

Task performance In this section, task performance is explored during the Familiarization phase of the experiment where participants received advice from a pre-specified advisor on each trial. There were an equal number of these trials for each participant for each advisor.

```
## Interaction calculation: Initial - Final
```

Participants generally improve their response accuracy following advice; they had lower error on final decisions than on their initial estimates ($F(1,32) = 69.02$, $p = <.001$; $M_{\text{Initial}} = 15.84$ [13.80, 17.89], $M_{\text{Final}} = 10.28$ [8.87, 11.69]). They also had lower error on their answers with the High accuracy advisor ($F(1,32) = 5.73$, $p = .023$; $M_{\text{HighAgreement}} = 14.28$ [11.86, 16.71], $M_{\text{HighAccuracy}} = 11.84$ [10.61, 13.07]). As expected, there was an interaction: participants reduced their error much more following advice from the High accuracy advisor ($F(1,32) = 60.26$, p

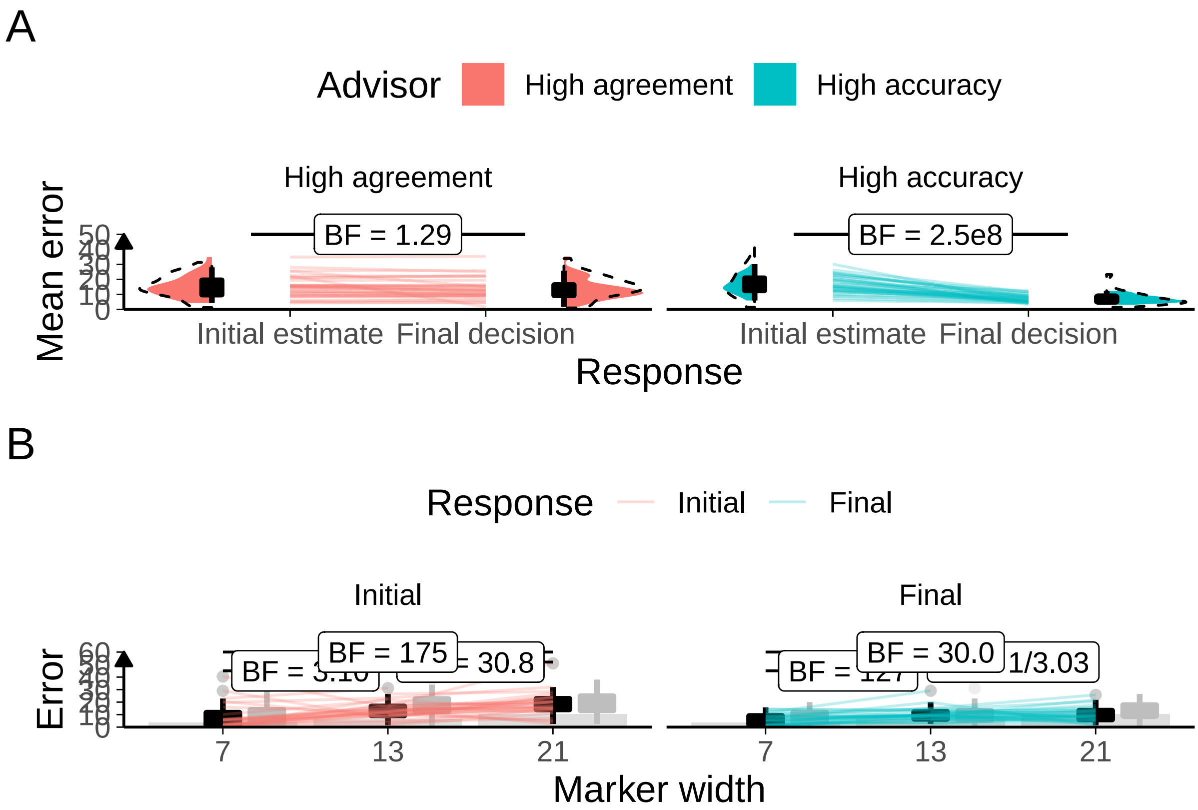


Figure 5.19: Task performance for the Dates task with accurate vs agreeing advisors.
A: Response error.
Faint lines show individual participant mean error (the absolute difference between the participant's response and the correct answer), for which the violin and box plots show the distributions. The dashed line indicates chance performance. Dotted violin outlines show the distribution of participant means on the original study which this is a replication. The dependent variable here is error, the distance between the correct answer and the participant's answer, and consequently lower values represent better performance. The theoretical limit for error is around 100.
B: Error by marker width.
Faint lines show individual participant mean error (distance from the centre of the participant's marker to the correct answer) for each width of marker used, and box plots show the distributions. Some participants did not use all markers, and thus not all lines connect to each point on the horizontal axis. The dashed box plots show the distributions of participant means in the original experiment of which this is a replication. The faint black points indicate outliers.
Grey bars show half of the marker width: mean error scores within this range mean the marker covers the correct answer.

= <.001; $M_{Reduction|HighAgreement}$ = 1.46 [0.05, 2.87], $M_{Reduction|HighAccuracy}$ = 9.67 [7.66, 11.68]; Figure 5.19).

Generally, we expect participants to be more confident on trials on which they are correct compared to trials on which they are incorrect. Confidence can be measured by the width of the marker selected by the participant. Where participants are more confident in their response, they can maximise the points they receive by selecting a thinner marker. Where participants are unsure, they can maximise their chance of getting the answer correct by selecting a wider marker. Participants' error was lower for each marker width in final decisions than initial estimates (Figure **??**). For both initial estimates and final decisions, error was higher for wider markers than for narrower ones.

Advisor performance The advice is generated probabilistically so it is important to check that the advice experienced by the participants matched the experience we designed. On average, the High accuracy advisor had lower error than the High agreement advisor ($t(32)$ = -8.75, p < .001, d = 2.22, BF = 2.0e7; $M_{HighAccuracy}$ = 6.02 [5.55, 6.48], $M_{HighAgreement}$ = 15.30 [13.25, 17.35]), and their advice was further away from the participants' initial estimates than the High accuracy advisor's ($t(32)$ = 12.47, p < .001, d = 2.24, BF = 9.0e10; $M_{HighAccuracy}$ = 20.28 [18.21, 22.35], $M_{HighAgreement}$ = 9.59 [8.39, 10.80]). 32/33 (96.97%) participants experienced the High accuracy advisor as having lower average error than the High agreement advisor, and 33/33 (100.00%) participants experienced the High agreement advisor as offering advice closer to their initial estimates than the High accuracy advisor. Overall, this indicates that the manipulation was implemented as planned.

Hypothesis test

```
## Warning: data coerced from tibble to data frame
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Consistent with the result from the Dots task (Experiment 3A§5.3.1), in the No feedback condition participants' preferences for receiving advice from the High accuracy advisor were not different from chance () on average, and varied widely

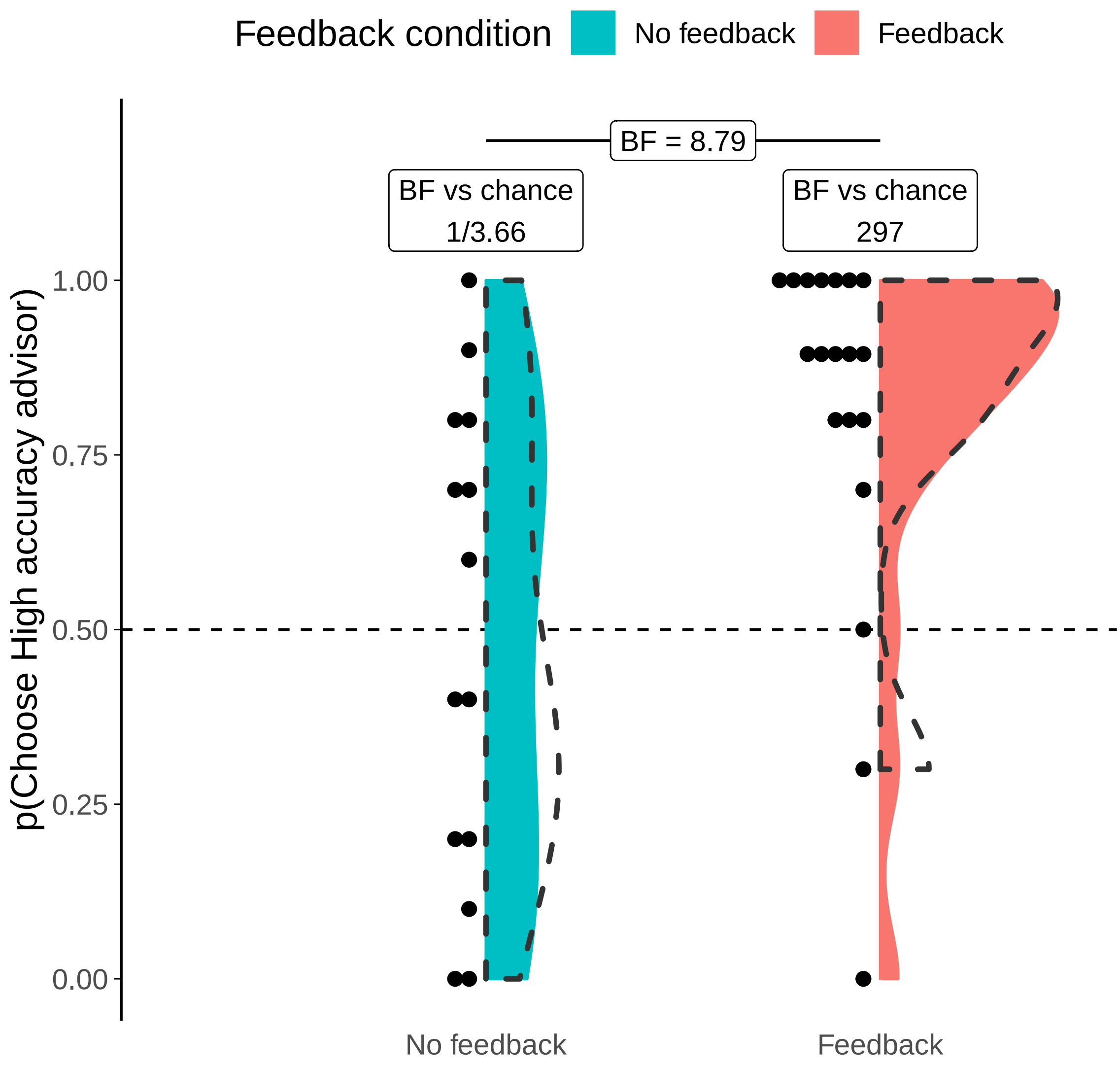

Figure 5.20: Dates task advisor choice for high accuracy/agreement advisors. Participants' pick rate for the advisors in the Choice phase of the experiment. The violin area shows a density plot of the individual participants' pick rates, shown by dots. The chance pick rate is shown by a dashed line. Participants in the Feedback condition received feedback during the Familiarization phase, but not during the Choice phase. The dotted outline indicates the distribution of participant means in the original experiment of which this experiment is a replication.

across individual participants: participant preferences in the No feedback condition were almost perfectly evenly distributed, both in terms of which advisor was preferred and the strength of that preference, in both the original study and the replication (Figure 5.20).

In the Feedback condition, in contrast, the mean of the participants' selection rates clearly favoured the High accuracy advisor (). This is consistent with a strategy which attempts to maximise the accuracy of final decisions. This qualitative difference from the No feedback condition also translated into a statistical difference: the two preference distributions were clearly different from one another ().

It was discovered after the completion of experiments that the advisor position (whether the advisor appears on the top or bottom of the advisor choice panel) was not counterbalanced between advisors. It had been decided during development of the tests to keep the advisors in the same position for every trial so that participants did not get mixed up between them. Together, this meant that the High accuracy advisor always appeared at the top and the Low accuracy advisor always appeared at the bottom, for every trial for every participant. We are thus unable to confirm that pick rate differences, or the absence of those differences, are caused by participants' preferences for the advice the advisor would provide or the position the advisor was in on the screen. Furthermore, it is possible that any genuine preference for one advisor over the other was *induced* by the position, rather than independent of it (Zajkowski and Zhang 2021).

Advisor influence

```
## Warning: Data is unbalanced (unequal N per group). Make sure you specified a
## well-considered value for the type argument to ezANOVA().
## Interaction calculation: High accuracy - High agreement
```

We included in our design a subset of trials on which advisors offered the same kind of advice.[5] This meant that we could investigate the influence of the advisors

[5] 1 participant was dropped from this analysis because their data did not have any remaining off-brand trials for one advisor.

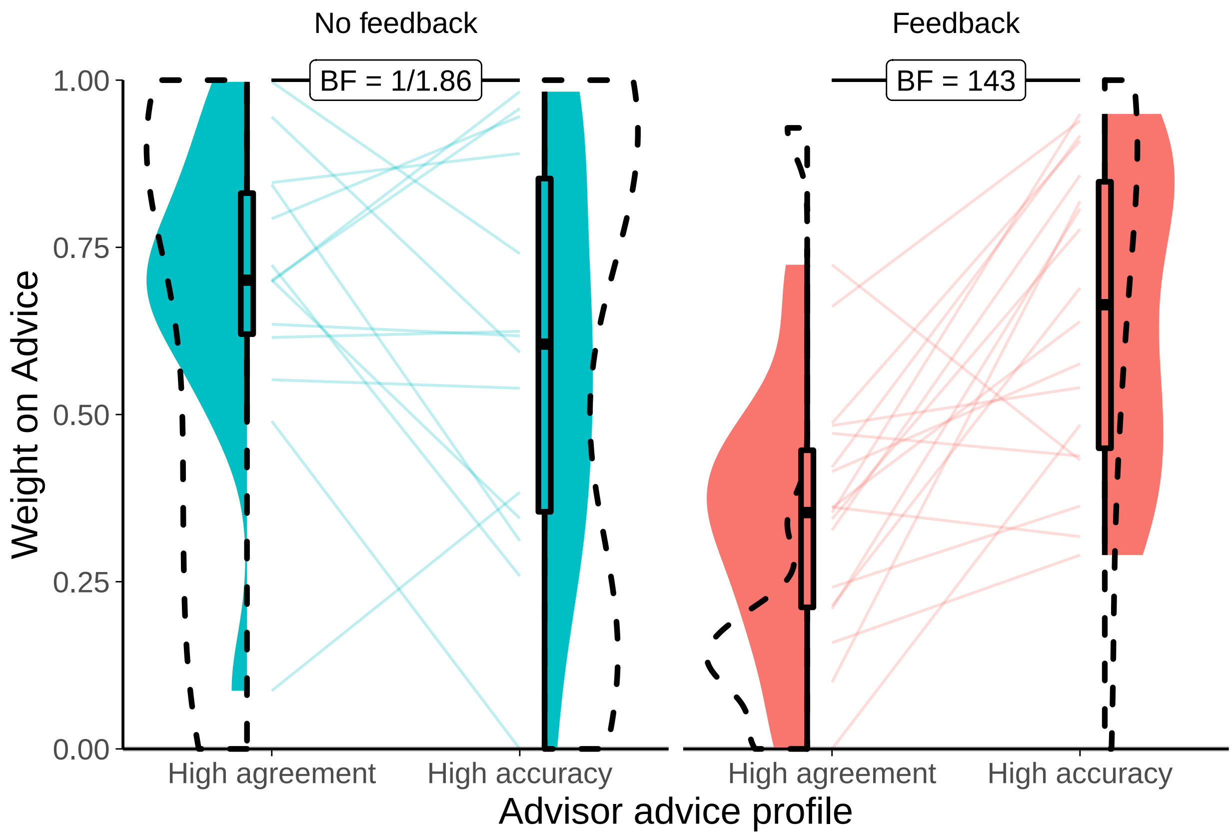

Figure 5.21: Date task advisor WoA for high accuracy/agreement advisors. Participants' weight on the advice for advisors in the Familiarization phase of the experiment. The shaded area and boxplots indicate the distribution of the individual participants' mean influence of advice. Individual means for each participant are shown with lines in the centre of the graph. The dotted outline indicates the distribution of participant means in the original experiment of which this experiment is a replication.

while controlling for differences in their advice. Examining the influence of these off-brand trials indicated that the High accuracy advisor was more influential than the High agreement advisor ($F(1,30) = 6.08$, $p = .020$; $M_{HighAccuracy} = 0.62$ [0.53, 0.72], $M_{HighAgreement} = 0.50$ [0.40, 0.59]). This difference was more pronounced in the Feedback condition ($F(1,30) = 15.82$, $p = <.001$; $M_{Accuracy\text{-}Agreement|Feedback} = 0.30$ [0.17, 0.44], $M_{Accuracy\text{-}Agreement|NoFeedback} = -0.10$ [-0.28, 0.07]). There was no evidence of a main effect of condition, however ($F(1,30) = 4.12$, $p = .051$; $M_{Feedback} = 0.50$ [0.42, 0.58], $M_{NoFeedback} = 0.64$ [0.51, 0.76]).

5.3.3 Discussion

Whereas previous experiments assessed the separate effects of advisor accuracy (1A§5.1.1 and 1B§5.1.2) and agreement (2A§5.2.1 and 2B§5.2.2), in these two experiments these factors were set in opposition. The results were clear when feedback was provided (in the Dates task), indicating that people are capable of attending to challenging but useful information provided they have a chance to learn that the information is actually useful. In contrast, when people are not given objective feedback against which to evaluate the advice they receive, there does not seem to be a systematic response: some people seek agreement while others seek alternate perspectives, and the extent to which each strategy is pursued to the exclusion of the other is also highly variable. It is an open question whether a person's strategy choice in the absence of useful cues as to the utility of the information they receive is due to random selection or related in a meaningful way to their personality or cognitive style.

The distributions of preferences were compatible with but slightly different to those seen in the previous experiments. In neither Experiment 1A§5.1.1 nor 2A§5.2.1 did we see a null effect of choice, so it is not clear whether that would look like random picking or, as it does here, a similar range of preference directions and strengths to those seen in the Dates task. The Dots task distribution seen here might be expected, given no systematic preferences, to cluster around the middle, representing the drives for fairness, novelty, etc. as discussed in previous chapters. The Dates task distribution is more in keeping with previous experiments: the expected range of preference directions and strengths is there in the absence of a systematic preference. In the Feedback condition where systematic differences are seen, we still see the entire range of preference directions and strengths, as we did before, although the clustering towards the High accuracy advisor is much more pronounced than in previous experiments. This clustering of the distribution suggests that the manipulation worked more effectively in this experiment, perhaps because of the use of the continuous Dates task: participants may have experienced the visual depiction of a distance between their estimate and the advice as a stronger signal than binary agreement or disagreement.

The results of the source selection behaviour in this experiment conceptually replicate the advisor influence results in the previous study (§4), and the influence results in the Dates task similarly replicate those findings. While this study was not set up to examine influence rigorously, the inclusion of trials where the advisors offered equivalent advice allowed us to explore advisor influence without the confound of the advice itself. We found, in two preregistered studies, that participants in the Feedback condition discriminated between their advisors, being more influenced by the accurate rather than agreeing advice. Contrary to our hypothesis, however, we found that participants in the No feedback condition either did not discriminate (original) or did not provide evidence of discrimination (replication).

The similarity of these results to those of the previous study is encouraging, although once again the methodology does not entirely equate the advice between the advisors. Although advice on the off-brand trials is constructed using the same rules for each advisor, off-brand advice from the High agreement advisor will be much more surprising than off-brand advice from the High accuracy advisor, because participants will usually have experienced the High accuracy advisor offering advice that deviates dramatically from their initial estimates whereas they will not have experienced this from the High agreement advisor. Assuming that participants in the No feedback condition have no better insight into the correct answer than they offer with their estimate, advice from the High agreement advisor may be discounted simply because it deviates more from the perceived correct answer than usual. Participants in the No feedback condition may also feel reassured by the High agreement advisor that they are good at the task, and thus reduce their reliance on advice. It is possible, in sum, for participants in the No feedback condition to suffer suppression of influence on off-brand trials specifically for the High agreement advisor, which would lead to genuine influence effects being difficult to detect.

Another consideration is that the off-brand advice trials that the influence analysis is based on occur during the Familiarization phase when participants are learning about the advisors. Participants in the Feedback condition appear to have learned rapidly that the advice of the High accuracy advisor is worth following, and the advice of the High agreement advisor is not informative. There is a slight suggestion from the individual participant data that many of the participants in the No feedback condition may have been creeping towards this conclusion, but the statistics are uninformative on the question. It is highly plausible that learning with feedback is far more rapid than learning without feedback, especially in noisy and heterogeneous tasks.

Whereas the previous experiments demonstrated that participants denied feedback would prefer to pick the High agreement advisor, that did not happen in this experiment. Together with the above, this might indicate that even without feedback participants were sensitive to the accuracy of advice over and above agreement. The Pescetelli and Yeung (2021) theory of metacognitive advice evaluation supports this because confidence is used to assess advice plausibility. Where people receive agreeing advice on questions that they feel confident about, they will consider the advice as highly likely to be accurate, whereas when they themselves are very unsure of the answer they will not learn very much about the advisor from the advice whether it agrees or not.

It is this confidence-based assessment of advice that we attempted to test in the next experiments using advisors whose advice was generated depending on the participant's confidence in their initial estimate.

5.4 Effects of confidence-contingent advice

The results of the experiments described previously (1A§5.1.1, 1B§5.1.2, 2A§5.2.1, 2B§5.2.2, 3A§5.3.1, 3B§5.3.2) were mixed, but indicated overall (and in combination with Pescetelli and Yeung 2021) that people prefer accurate over inaccuate advisors when they can assess the advisors based on feedback, but prefer agreeing over disagreeing advisors when feedback is absent. Here we test whether advice selection can depend on a more sophisticated mechanism when feedback is absent.

This mechanism, proposed by Pescetelli and Yeung (2021), weights the updating of trust in an agreeing advisor using the confidence of the judge's initial estimate. Where the initial estimate is made with high confidence, agreement is taken to indicate that the advisor has a high probability of being correct, while disagreement is taken to indicate that the advisor has a low probability of being correct. Where the initial estimate is low confidence, however, neither agreement nor disagreement is a good indicator of correctness. This intuition accords neatly with a Bayesian approach: the greater the uncertainty around whether or not the advice is correct, the lower the strength of the updating of trust in the advisor.

This mechanism can be directly tested by using advisors who have different agreement rates contingent upon the judge's confidence in their initial estimate, and Pescetelli and Yeung (2021) used just such an approach to provided evidence for the mechanism in the domain of advisor influence. Here, a very similar design is used to explore this effect in the domain of advisor choice.

Unlike the previous experiments, in which the same approach was implemented in both the Dots and the Dates tasks, this experiment uses only the Dots task. To balance the advisors' accuracy and agreement rates (such that only their *confidence-contingent agreement* is varied), precise control is required over the participants' initial estimate accuracy. Such control is achieved using a staircasing procedure in the Dots task, and cannot be done in the Dates task (because there are too few questions to choose from, the questions have too wide a range of difficulty, and a question's difficulty varies dramatically and unpredictably between participants).

Instead of a Dots and a Dates task, this experiment is repeated in two versions of the Dots task. The first version, directly adapted from Pescetelli and Yeung (2021), was performed in the lab with a correspondingly low sample size and high number of trials per participant. The second version was a replication run online and with a larger sample size and lower number of trials per participant. For ease of reference, the lab study is referred to as 'Lab study', while the online study retains the numbering and lettering used previously, meaning it is designated

Experiment 4A. There is no Experiment 4B, because it was not possible to use the Dates task with this design of advisors.[6]

5.4.1 Lab study of confidence-contingent advice

Pescetelli and Yeung (2021) used a judge-advisor system to demonstrate that judges are influenced to a greater extent by advisers who share their biases. Participants played the role of judge in a judge-advisor system, while the advisers were virtual agents whose advice-giving was dependent upon the confidence and correctness of the judges' initial decisions. The advisers were balanced for both overall agreement with the judge and objective correctness of advice. This was achieved by varying the agreement rates for the advisors contingent on the confidence of the judge's initial estimate. The Bias-sharing advisor would agree more frequently when the participant expressed high confidence in their initial estimate and less frequently when the participant expressed low confidence in their initial decision. The Anti-bias advisor did the opposite. Crucially, advisors performed identically on the largest minority of trials (when judges were moderately confident in their initial estimates), allowing the advisors to be compared directly on these trials (Pescetelli and Yeung 2021). We place participants in a similar paradigm in which they are given a choice between advisers, and hypothesise that they will more frequently seek advice from the *Bias Sharing* advisor than from the *Anti Bias* advisor. We predict that, given a choice, judges will prefer to receive advice from a *Bias Sharing* advisor over receiving advice from an advisor who does not share the judge's bias.

Open scholarship practices

This experiment was preregistered at `https://aspredicted.org/ze3tn.pdf`. One analysis in the preregistration is not reported here because the results are non-significant and they represented a branch of analysis that we sidelined in

[6] The lab version of this study was the first one performed in chronological order. The results were not as anticipated, so a replication was run online (Experiment 4A). These results were also odd, so other experiments were run to investigate the stability of the basic effects (Experiments 1A, 2A, and 3A).

other experiments. This analysis explored participants' subjective assessments of advisors. The experiment data are available in the `esmData::` package for R from !TODO[Zenodo esmData], and also directly from https://osf.io/vgcnb/. The code for running the experiment can be obtained from https://github.com/mjaquiery/nofeedback_trust.

Method

The method for the lab study was different in many small ways from the general method used for the Dots task online. The basic trial experience was the same: participants saw two boxes of dots flashed breifly on the screen, and were asked to indicate which box had more dots, along with how confident they were in this initial estimate. Participants then received advice and provided a final decision. On some trials participants were able to choose which of their advisors would provide the advice (Figure 5.22).

Overall, 26 participants were recruited from the University of Oxford participant recruitment platforms took part in the experiment and attended experimental sessions. 1 participant was excluded because the preregistered sample size had already been met when their data were collected, and one participant's data was lost due to technical issues. The remaining participants had an average age of 21.75 (±SD 4.7). Their self-identified genders were give as 5 male, 19 female, 0 other. Participants were compensated for their time with either course credit for a psychology degree, or 10GBP.

Each participant completed 363 trials (51 practice trials over 2 blocks + 12 x 26-trial experimental blocks). Prior to the first experimental block, after the final experimental block, and after the 4th and 8th experimental blocks, participants were presented with a questionnaire (Figure 5.23). The questionnaire contained 4 questions for each advisor. The questions asked for the judge's assessment of the advisor's likeability, trustworthiness, influence, and ability to do the task. The questions presented before the first experimental block were worded prospectively (e.g. 'How much are you going to like this person?' as opposed to 'How much do

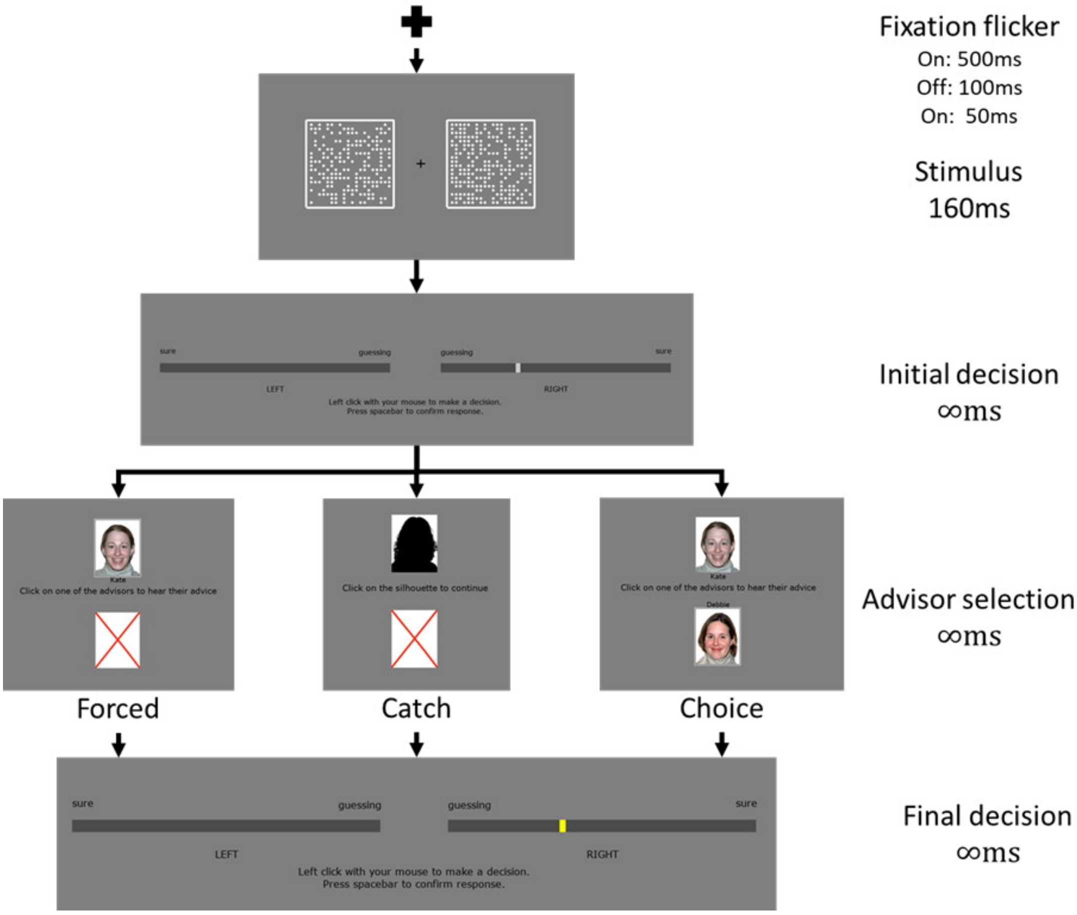

Figure 5.22: Experiment 1 procedure.
The task began with a blank screen containing only a fixation cross and progress bar. Momentarily prior to the onset of the stimuli the fixation cross flickered. The stimuli, two rectangles containing approximately 200 dots each, appeared for 0.16s, one on either side of the fixation cross. Once the stimuli disappeared, a response-collection screen appeared and prompted the participant to indicate their initial decision and its confidence by selecting a point within one of two regions. The left region indicated a decision that the target was on the left, and increasingly-leftwards points within that region indicated increasing confidence in that decision. The right region indicated a decision that the target was on the right, and increasingly-rightwards points within that region indicated increasing confidence in that decision.
Next, the participant was presented with a choice screen. The choice screen displayed two images, one at the top of the screen and one at the bottom. The images were one of the following: an advisor portrait, a silhouette, or a red cross. The red cross was not selectable, forcing participants to choose the other option. The silhouette offered no advice, and was only ever offered as a forced choice. Selecting an advisor image provided the participant with the opinion of that advisor on the trial.
Having heard the advice, the participant was again presented with the response-collection screen, with a yellow indicator marking their original response. A second (final) judgement was collected using this screen (except on catch trials), and the trial concluded.

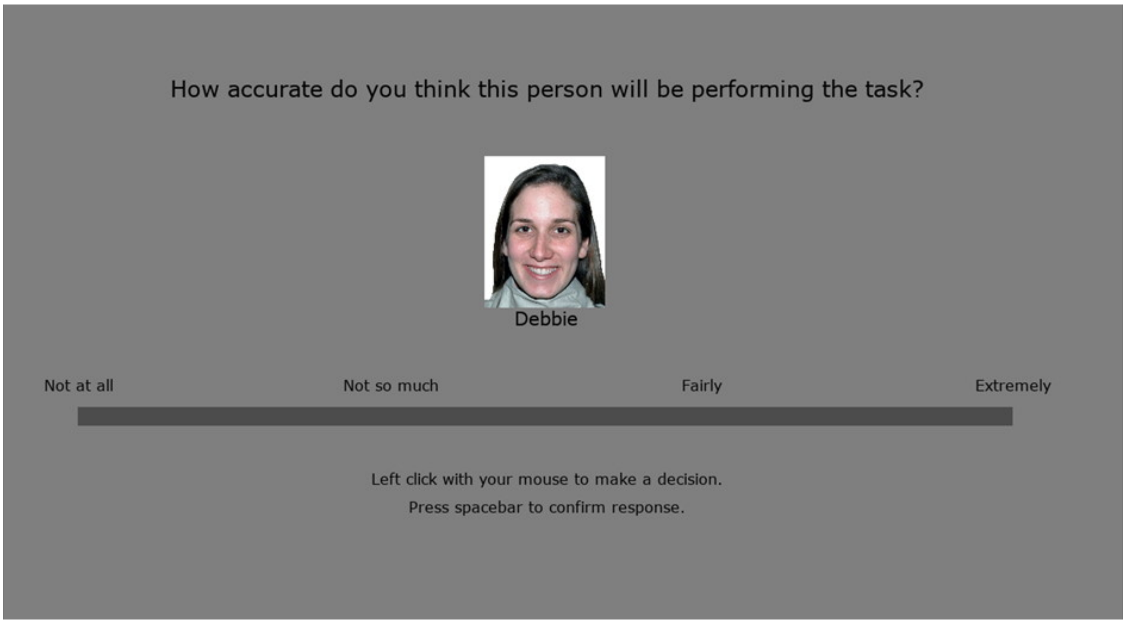

Figure 5.23: Experiment 1 advisor questionnaire.
Participants rated advisors on a number of different dimensions.

you like this person?'). Answers were provided by moving a sliding scale below the advisor's portrait towards the right for more favourable responses (marked 'extremely') or towards the left for less favourable responses (marked 'not at all').

Each participant attended the experiment individually, was welcomed and briefed on the experimental procedure, and had their informed consent recorded, before the experiment began. They were seated a comfortable distance in front of a 24' (1440x900 resolution) computer screen in a small, quiet, and dimly-lit room. The experiment took place wholly on the computer, and lasted around 45 minutes.

The experiment was programmed in MATLAB R2017b (*MATLAB* 2017) using the Psychtoolbox-3 package (Kleiner et al. 2007).

Key differences from online version There were several differences from the online version of the task that are worth mentioning. First, there were more trials and the experiment took longer to complete. Second, when the participant was forced to have advice from one or other advisor, they were presented with the advisor choice screen and one of the options was a blank advisor that they were unable to select. This meant that advisor influence on the forced trials could be

Table 5.24: Advisor advice profiles for Lab experiment

	Agreement				
	Initial estimate confidencetextsuperscripta			Subtotals by initial estimate correctne	
Advisor	Low[b]	Medium[c]	High[b]	Correct[d]	Incorre
Bias-sharing	60	70	80	70	3
Anti-bias	80	70	60	70	3

[a] Only correct trials received confidence-contingent advice
[b] 30\% of trials
[c] 40\% of trials
[d] Average correct agreement is the weigthed average of the previous three columns
[e] Overall agreement is p(correct) * p(agree|correct) + p(¬correct) * p(agree|¬correct)
[f] Overall accuracy is p(correct) * p(agree|correct) + p(¬correct) * p(¬agree|¬correct)

analysed as compared to choice trials. Third, 8% of trials were catch trials on which no advice was offered. Fourth, the questionnaires were more detailed and more numerous, and contained no free text fields. Fifth, participants had the opportunity to discuss the experiment with the experimenter after the experiment was complete, although none made use of this opportunity.

Advice profiles The two advisor profiles used in the experiment were Bias sharing and Anti-bias. The advisors are balanced for their overall accuracy and agreement rates, but the Bias sharing advisor agrees more frequently with participants when their initial estimate is correct and made with relatively high confidence. The Anti-bias advisor agrees more frequently with participants when their initial estimate is correct and made with relatively low confidence (Table 5.24). Note that the overall correctness and agreement rates of the advisers are equivalent. Importantly, on a large minority of trials, the middle 40%, the advisers are exactly equivalent, meaning these trials can be compared directly without confounds arising from agreement rates and initial confidence.

Results

Exclusions Participants could be excluded for having initial estimate accuracy below 60% or above 90%, or for performing the experiment after the stated sample

Table 5.25: Advisor agreement for Lab experiment

	Agreement					
	Initial estimate confidence			Subtotals by initial estimate correctness		
Advisor	Low	Medium	High	Correct	Incorrect	Ov
Anti-bias	78.7	70.3	63.6	71.2	31.6	
Anti-bias (Target)	*80*	*70*	*60*	*70*	*30*	
Bias-sharing	59.7	70.3	79.4	69.3	30.4	
Bias-sharing (Target)	*60*	*70*	*80*	*70*	*30*	

size had been reached. No participants were excluded for having accuracy out of range. 1 participant was excluded for being extraneous.

Advisor performance The advisors agreed with the participants' initial estimates at close to target rates in all confidence categories, and were as accurate on average as expected (Table 5.25).

Advisor selection We hypothesised that the participants would display different pick rates for the Bias Sharing advisor versus the Anti Bias advisor. This hypothesis was evaluated by calculating the proportion of choice trials on which each participant picked the Bias Sharing advisor, and then testing these values as a one-sample t-test against the null hypothesis that the pick rates would be 0.5. No support was found for this hypothesis (; Figure 5.24), although the Bayesian test indicated that the data were not sufficient to conclude that no effect was present. There was considerable variability across participants in the overall pick rate for the Bias Sharing advisor (range = [.10, .88]).

Unlike in other experiments, there was no systematic effect of the position of the advisors on the screen ($t(23) = 0.10$, $p = .920$, $d = 0.02$, BF $= 1/4.63$; $\mathrm{M}_{\mathrm{P(PickFirst)}} = 0.50$ [0.45, 0.55], $\mu = 0.5$). This placement is random, and has no relationship to the identity of the advisor (BF $= 1/2.43$; $\mathrm{M}_{\mathrm{P(BiasSharingFirst)}} = 0.51$ [0.49, 0.52], $\mu = 0.5$), so we expect that it would be random across participants.

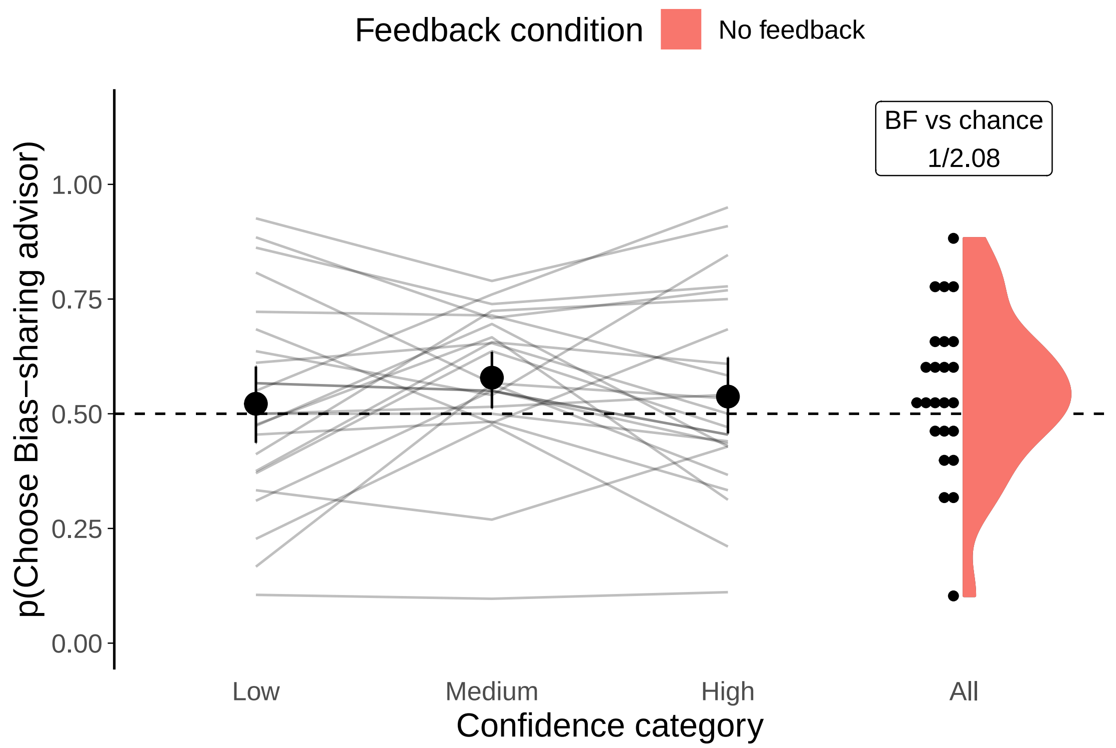

Figure 5.24: Advisor selection.
Proportion of the time each participant picked the Bias Sharing advisor. Faint lines and dots indicate data from individual participants, while the large dot indicates the mean proportion across all participants. The dashed reference line indicates picking both advisors equally, as would be expected by chance. Error bars give 95% confidence intervals.

Advisor selection on medium-confidence trials The advisers differed in their advice-giving as a function of the judge's initial confidence. In trials where the judge's initial decision was made with medium confidence, however, the advisers were equal on judge confidence and agreement rate. Comparing selection rates for these trials alone revealed a clear preference for the Bias Sharing advisor (; Figure **??** "Medium" confidence category), although the Bayesian analysis again indicated an insensitive result, albeit in the hypothesised direction.

Advisor influence Previous work in our lab demonstrated that the agree-in-confidence advisor exerted greater influence on the judges' final decisions than the agree-in-uncertainty advisor (Pescetelli and Yeung 2021). Influence was examined with a 2x2x2 (Bias-sharing versus Anti-bias advisor; choice versus forced trials;

Table 5.26: ANOVA of Advisor influence in Lab experiment

Effect	$F(1, 23)$	p		η^2
Advisor	0.28	.602		.001
Trial type	4.23	.051		.001
Advisor agreement	13.88	.001	*	.175
Advisor:Trial type	0.04	.842		< .001
Advisor:Advisor agreement	0.75	.395		.001
Trial type:Advisor agreement	1.99	.172		< .001
Advisor:Trial type:Advisor agreement	0.01	.935		< .001

Degrees of freedom: 1, 23

agreement versus disagreement trials) ANOVA (Figure 5.25). This was chronologically the first experiment we ran, and the analysis was preregistered prior to the development of the capped influence measure designed to allow agreement and disagreement to be compared more fairly. No main effect was found for advisor (F(1,23)=0.28, p=.602, $M_{\text{Bias-sharing}}$=0.09 [95%CI 0.06, 0.11], $M_{\text{Anti-bias}}$=0.08 [95%CI 0.06, 0.11]), meaning that the previous finding was not replicated. As shown in Table 5.26, the only statistically significant effect was the main effect of agreement, with disagreement producing higher influence than agreement (F(1,23)=0.04, p=.842, M_{Agree}=0.06 [95%CI 0.03, 0.08], M_{Disagree}=0.13 [95%CI 0.09, 0.17]).

When this analysis was repeated with the capped influence values (General Method - Capped influence§3.1.4) the effect of agreement was still significant, but less so (F(1,23)=1.28, p=.269, $M_{\text{Bias-sharing}}$=0.08 [0.06, 0.10], $M_{\text{Anti-bias}}$=0.08 [0.05, 0.10]), and a main effect of trial type also emerged (raw influence: F(1,23)=4.23, p=.051, M_{Choice}=0.09 [0.06, 0.12], M_{Force}=0.08 [0.06, 0.11]; capped influence: F(1,23)=5.38, p=.030, M_{Choice}=0.08 [95%CI 0.06, 0.11], M_{Force}=0.08 [95%CI 0.06, 0.10]).

Advisor influence on medium confidence trials

```
## Interaction calculation: Disagree - Agree
```

```
## Interaction calculation: Disagree - Agree
```

The agree-in-confidence and agree-in-uncertainty advisers differed by design in the frequency with which they agree with the participant as a function of the participant's confidence in their initial estimate. To control for the effects of initial

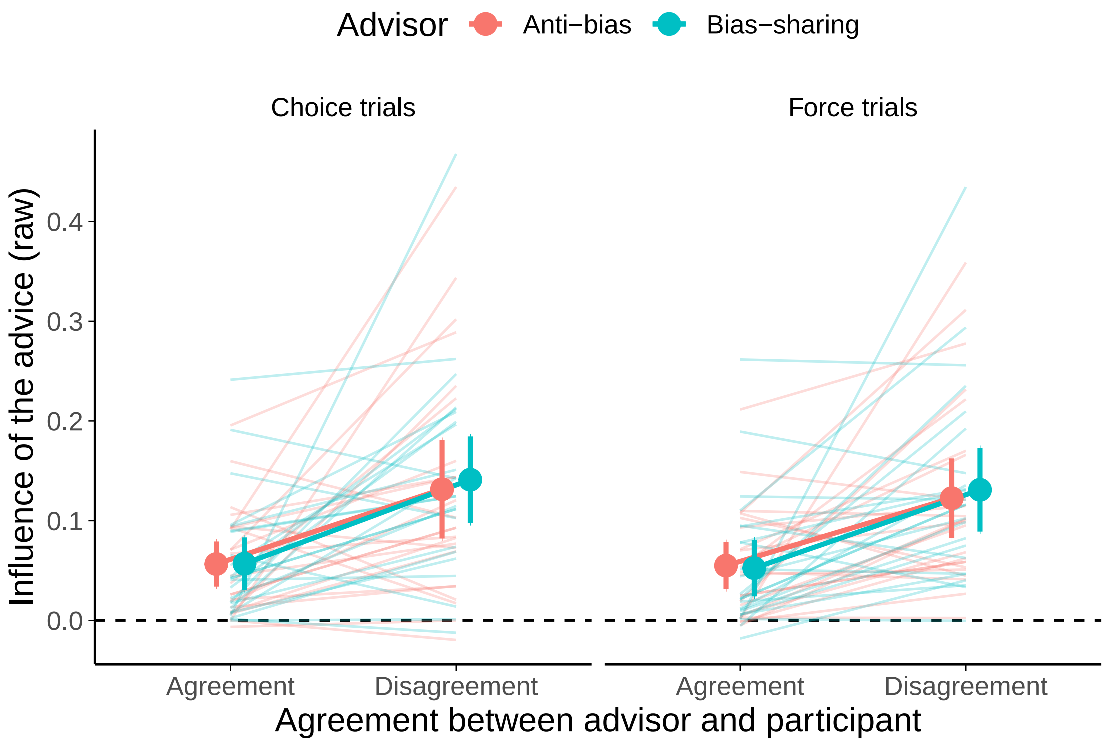

Figure 5.25: Advisor influence.
Influence of advice from each advisor by advisor, agreement, and trial type. Faint lines and indicate data from individual participants, while the dots indicate the mean proportion across all participants. Error bars give 95% confidence intervals.
Note: vertical axis is truncated to show group differences more clearly, the theoretical maximum influence given the scale is 110. The minimum is 0 as shown.

confidence on influence, the above analysis was repeated using only those trials on which the initial estimate was correct and given with medium confidence. In a deviation from preregistration, this analysis was constrained to only forced trials because some participants had missing data for some advisor-trial type-agreement contingencies in the Medium confidence trials.

The results were qualitatively identical to those for the trials at all confidence levels: a main effect of agreement (raw influence: ; capped influence: $F(1,23) = 8.70$, $p = .007$; $M_{\text{Disagree}} = 0.10$ [0.07, 0.13], $M_{\text{Agree}} = 0.05$ [0.02, 0.08]); and no significant effect of advisor (raw influence: $F(1,23) = 0.10$, $p = .759$; $M_{\text{Anti-Bias}} = 0.08$ [0.06, 0.11], $M_{\text{Bias-Sharing}} = 0.09$ [0.06, 0.12]; capped influence: $F(1,23) = 0.00$, $p = .989$; $M_{\text{Anti-Bias}} = 0.08$ [0.05, 0.10], $M_{\text{Bias-Sharing}} = 0.08$ [0.05, 0.10]) or interaction (raw influence: $F(1,23) = 1.36$, $p = .255$; $M_{\text{Disagree-Agree|Anti-Bias}} = 0.05$

Table 5.27: Linear regression of pick rate in later blocks by initial agreement difference

Effect	Estimate	SE	t	p	
(Intercept)	0.51	0.01	48.10	< .001	*
firstBlockAgreeDifference	0.12	0.06	2.27	.034	*
Model fit: F(5.1, 1) = 22; p = .034; R^2_{adj} = .152					

[0.00, 0.10], $M_{\text{Disagree-Agree|Bias-Sharing}}$ = 0.09 [0.04, 0.14]; capped influence: F(1,23) = 1.91, p = .180; $M_{\text{Disagree-Agree|Anti-Bias}}$ = 0.04 [0.00, 0.08], $M_{\text{Disagree-Agree|Bias-Sharing}}$ = 0.06 [0.03, 0.10]).

Sensitivity to the manipulation Finally, we planned to investigate the hypothesis that participants' choice of advisor would be sensitive to the differential agreement strategies of the advisers, e.g. participants might preferentially select the advisor with the greater likelihood of agreement given their initial confidence. This was investigated by testing the participants' mean Bias-sharing advisor pick rate in high- versus low-confidence trials. Pick rates did not differ (); Figure **??**).

Follow-up tests

Effect of initial agreement This experiment was run in the lab, and was longer than the online experiments to help justify participants' travelling time. This meant that the experiment contained multiple blocks, and consequently that we could explore the effects of early experience of advice on subsequent advisor preferences. We hypothesised that one reason we found no weak evidence for different pick rates for different advisors was that initial exposure to the advisers may have overshadowed information in subsequent blocks. To investigate this possibility, we examined the effect that agreement in the first experimental block had upon choices throughout the rest of the experiment (Figure 5.26). A simple regression (Table 5.27) indicated that the extent to which the Bias-sharing advisor agreed with the participant more frequently than the Anti-bias advisor in the first experimental block (Block 3) was a significant predictor of the preference for picking the Bias-sharing advisor in subsequent blocks.

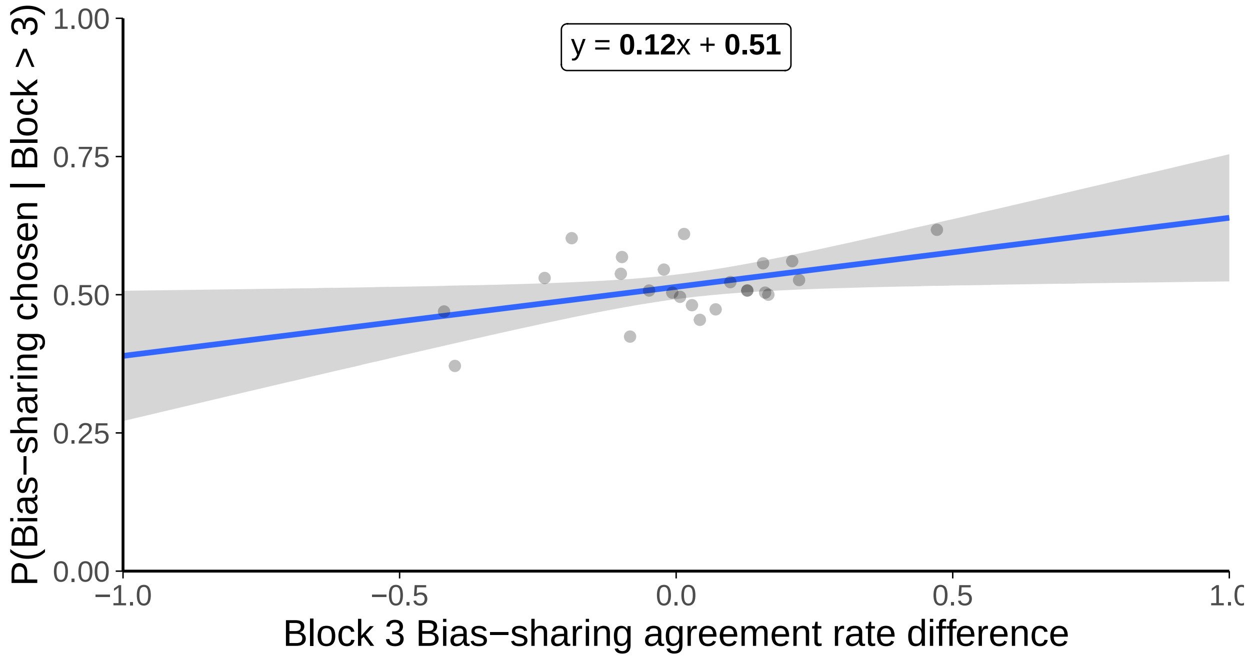

Figure 5.26: Initial agreement and subsequent preference.
figure shows the relationship between the agreement rate of the Bias Sharing advisor in the first experimental block (relative to the Anti Bias advisor) and the proportion of the time the participant picked the Bias Sharing advisor in later blocks. The dashed line shows the best-fit regression line, with shaded 99% confidence intervals.

Subjective assessment and choice of advisors Participants were asked to complete short questionnaires at several points in the experiment. These questionnaires collected ratings on a 100 point scale for each advisor across four dimensions: three loosely inspired by Mayer et al. (1995) accuracy, likeability, and influence; along with a general trustworthiness dimension. We did not observe significant differences in subjective ratings of the advisors at any time point. While overall differences might not show up, it is plausible that participants who had stronger preferences for one or other advisor would pick that advisor more frequently. We tested this using linear regression to predict the extent to which participants picked the Bias-sharing advisor from the differences in subjective advisor ratings. The questionnaire answers did not evolve systematically over the duration of the

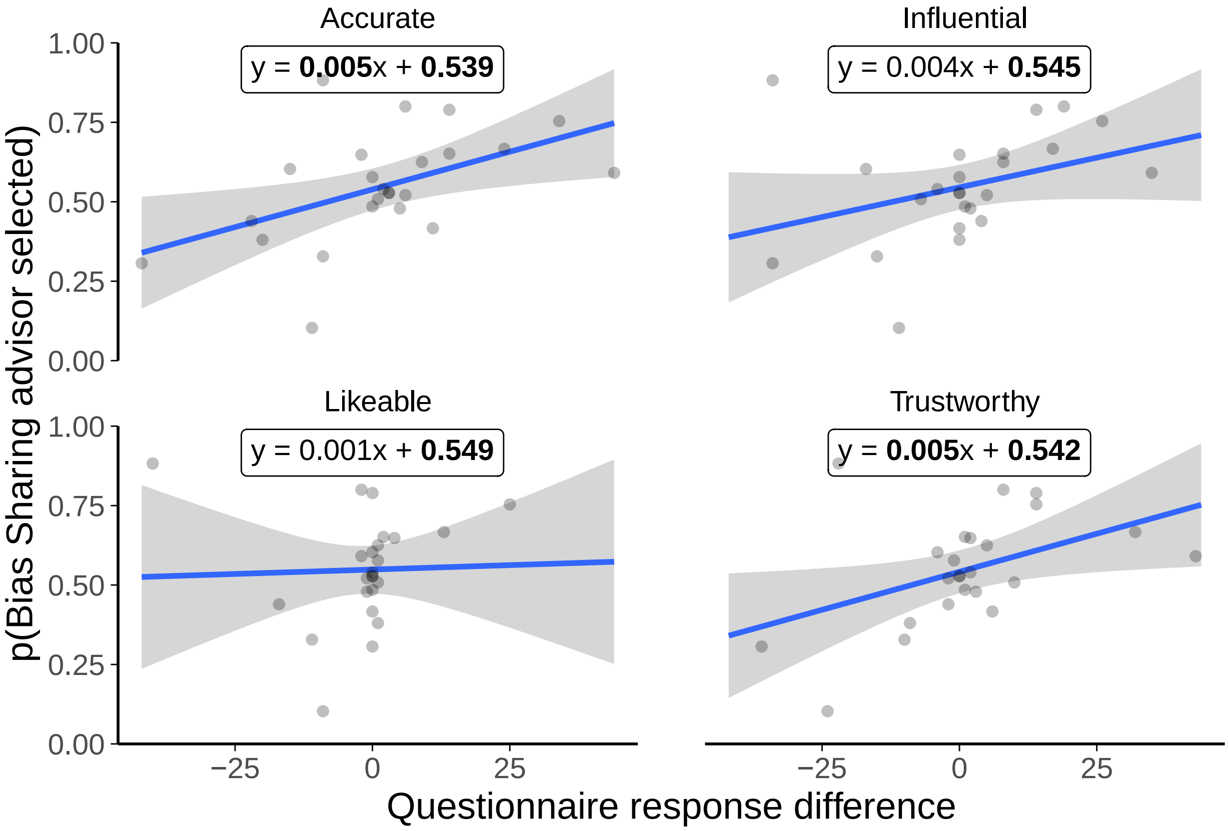

Figure 5.27: Behavioural and self-report consistency.
Relationship between questionnaire response scores on each item for the Bias-sharing advisor minus scores for the Anti-bias advisor and overall pick rate for the Bias-sharing advisor. Lines show best-fit linear model and shaded areas give 95% confidence intervals for the parameters. The theoretical limits on the horizontal axis are [-100, 100].

experiment, so the analysis was conducted on the final (post-experiment) answers.

This analysis (Figure 5.27) indicated that the association between subjective ratings and behavioural performance was present for the questionnaire dimensions of accuracy and trustworthiness, but not for influence or likeability.

There was a strong correlation between trustworthiness and accuracy (r=.866, p=< .001), and thus it was probable that the variance in the questionnaire responses explained by advisor preference was principally driven by this shared component. This was borne out by nested multiple regression Table 5.28, which demonstrated that a model using accuracy to predict advisor preference was not significantly improved by including trustworthiness.

Table 5.28: Iterative model comparison predicting advisor choice from questionnaire scores

		Model			
		1	2	3	4
(Intercept)	β	0.54	0.55	0.54	0.55
	p	$<$ .001	$<$ .001	$<$ .001	$<$ .001
Difference	β	0.00	0.00	0.00	0.00
	p	.015	.869	.029	.099
F	df	1 / 22	1 / 22	1 / 22	1 / 22
	F	6.99	0.03	5.46	2.96
	p	.015	.869	.029	.099
R^2_{adj}	R^2_{adj}	.207	-.044	.162	.079
	Δ		-.251	.206	-.084
	F				
	p				

Discussion

The lab experiment, chronologically the first of all the experiments,[7] produced equivocal results. While many of the expected relationships were found between participants' subjective perceptions of the advisors and their behaviour towards them, these did not reliably translate into differential picking rates for the Bias-sharing and Anti-bias advisors. There was a significant difference in pick rates in medium confidence trials (where the advisors were equivalent to one another), but the Bayesian test indicated that the evidence in favour of differential pick rates was weak. Furthermore, there was no convincing reason we could determine why the preference (if it were a real effect) would not show up at other confidence levels. The early experience of advisors did appear to predict their relative pick rates, with the advisor who agreed more with the participant in the initial experimental block being more likely to be picked more frequently on subsequent blocks.

Overall, these results were underwhelming and difficult to interpret. We conducted an online replication of this study so that we could collect data from more participants and hopefully determine more accurately whether or not participants were sensitive to the differences in the advisors.

[7] Later experiments were conducted to investigate why the hypothesised effects were not found in this experiment.

5.4.2 Experiment 4A: Confidence-contingent advice effects in the Dots Task

The results of the previous studies indicated that people update their preferences for advisors using agreement in place of feedback where objective feedback is unavailable. Previous results from our lab (Pescetelli and Yeung 2021) suggested that this capacity is heuristic is modified by confidence. The results of the previous experiment, however, failed to demonstrate these effects with advisor selection as the outcome.

The previous experiment was tightly controlled but the sample size was small. Using insights from its data, we refined the design and recruited a larger number of participants for the replication. Refinements included shortening the experiment, making the advisor profiles more extreme, reducing the questionnaires, and changing advisor representations from real faces and names to colours and numbers.

Open scholarship practices

This experiment was preregistered at `https://osf.io/h6yb5`. The experiment data are available in the `esmData::` package for R from !TODO[Zenodo esmData], and also directly from `https://osf.io/xb4kh/`. A snapshot of the state of the code for running the experiment at the time the experiment was run can be obtained from `https://github.com/oxacclab/ExploringSocialMetacognition/blob/90c04ff21d3a2876beadd` `AdvisorChoice/index.html`.

Method

54 participants each completed 368 trials over 7 blocks of a perceptual decision-making task. Each trial consisted of three phases: participants gave an initial estimate (with confidence) of which of two briefly presented boxes contained more dots; received advice on their decision from an advisor; and made a final decision (again, with confidence).

Participants started with 2 blocks of 60 trials that contained no advice to allow them to familiarise themselves with the task; to allow the staircasing process to titrate the difficulty to their ability in order to maintain approximately 71% initial

Table 5.29: Confidence-contingent advisor advice profiles

	Agreement				
	Initial estimate confidencetextsuperscripta			Subtotals by initial estimate correctne	
Advisor	Low[b]	Medium[c]	High[b]	Correct[d]	Incorre
Bias-sharing	50	70	90	70	3
Anti-bias	90	70	50	70	3

[a] Only correct trials received confidence-contingent advice
[b] 30\% of trials
[c] 40\% of trials
[d] Average correct agreement is the weigthed average of the previous three columns
[e] Overall agreement is p(correct) * p(agree|correct) + p(¬correct) * p(agree|¬correct)
[f] Overall accuracy is p(correct) * p(agree|correct) + p(¬correct) * p(¬agree|¬correct)

estimate accuracy; and to allow estimating of participants' idiosyncratic confidence reporting style. The first three trials were introductory trials that explained the task. All trials in this section included feedback indicating whether or not the participant's response was correct.

Participants then did 5 trials with a practice advisor to get used to receiving advice. They were informed that they would "get **advice** from an advisor to help you make your decision", and that "advice is not always correct, but it is there to help you: if you use the advice you will perform better on the task."

Participants then performed 2 sets of 2 blocks each. These sets consisted of 1 familiarisation block of 60 trials in which participants were assigned one of two advisors. The familiarisation block was followed with a test block of 60 trials in which participants could choose between the advisors they encountered in the familiarisation block. The participants saw different pairs of advisors in each set, with each pair consisting of one advisor with each of the advice profiles.

Advice profiles The two advisor profiles used in the experiment were Bias sharing and Anti-bias. The advisors are balanced for their overall accuracy and agreement rates, but the Bias sharing advisor agrees more frequently with participants when their initial estimate is correct and made with relatively high confidence. The Anti-bias advisor agrees more frequently with participants when their initial estimate is correct and made with relatively low confidence (Table 5.29).

Table 5.30: Participant exclusions for Dots task Confidence-contingent agreement experiment

Reason	Participants excluded
Accuracy too low	0
Accuracy too high	0
Missing confidence categories	3
Skewed confidence categories	1
Too many participants	0
Total excluded	**4**
Total remaining	**50**

Results

Exclusions In line with the pre-registration, participants' data were excluded from analysis where they had an average accuracy below 0.6 or above 0.85, did not have choice trials in all confidence categories (bottom 30%, middle 40%, and top 30% of prior confidence responses), had fewer than 12 trials in each confidence category, or finish the experiment after 50 participants have already submitted data which passed the other exclusion tests. Overall, 4 participants were excluded, with the details shown in Table 5.30.

Task performance

```
## Interaction calculation: Final - Initial
```

```
## Interaction calculation: Final - Initial
```

Basic behavioural performance was similar to that observed with the same Dots task in Experiments 1A§?? and 2A§??. Initial decision accuracy converged on the target 71%, and, as shown in Figure 5.28A, participants benefited from advice in terms of their final decisions being more accurate than their initial decisions (ANOVA main effect of Time: F(1,49) = 5.78, p = .020; M_{Final} = 0.72 [0.71, 0.73], $M_{Initial}$ = 0.71 [0.71, 0.72]), predominantly driven by following advice from the Anti-bias advisor (interaction of Time and Advisor: F(1,49) = 5.85, p = .019; $M_{Improvement|Anti\text{-}Bias}$ = 0.02 [0.01, 0.03], $M_{Improvement|Bias\text{-}Sharing}$ = 0.00 [-0.01, 0.01]). There was no main effect of Advisor (F(1,49) = 1.95, p = .169; $M_{Anti\text{-}Bias}$ = 0.73 [0.71, 0.74], $M_{Bias\text{-}Sharing}$ = 0.71 [0.70, 0.72]).

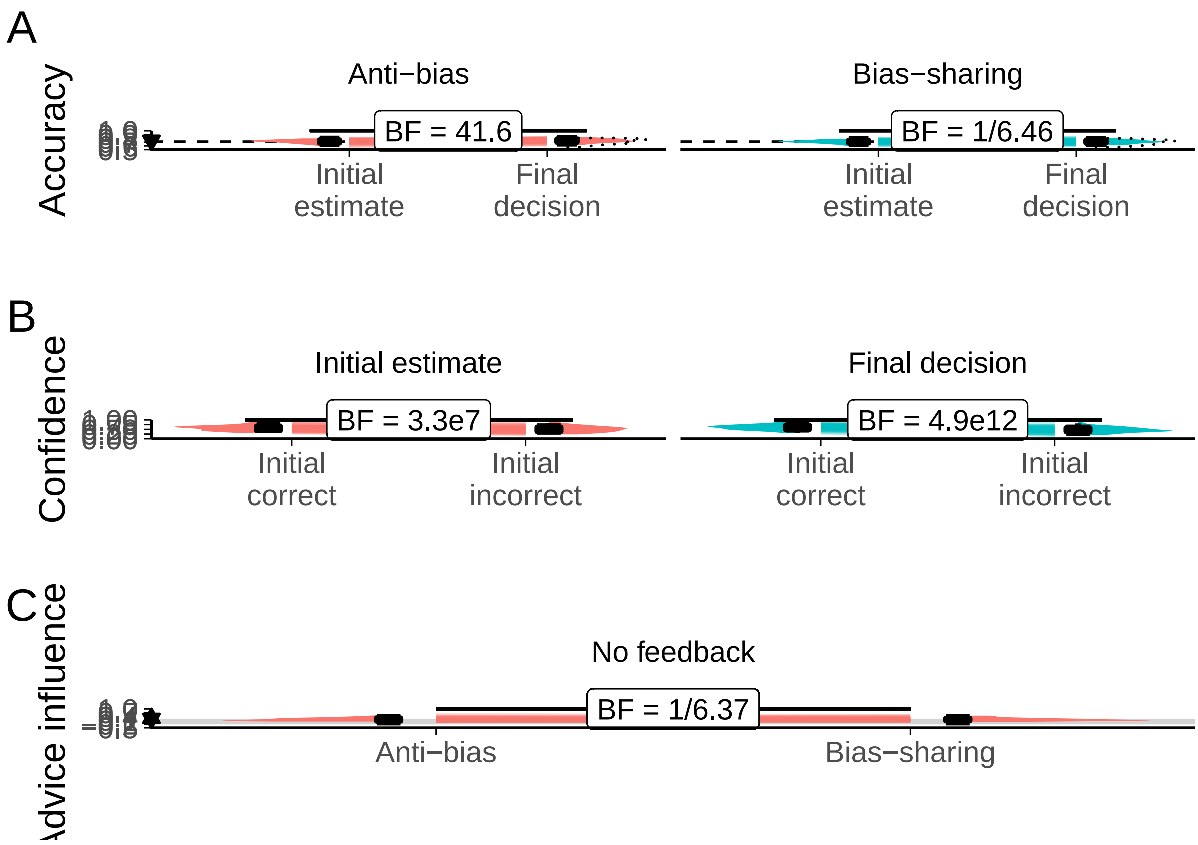

Figure 5.28: Task performance for the Dots task with confidence-contingent advisors.
A: Response accuracy. Faint lines show individual participant means, for which the violin and box plots show the distributions. The half-width horizontal dashed lines show the level of accuracy which the staircasing procedure targeted, while the full width dashed line indicates chance performance. Dotted violin outlines show the distribution of actual advisor accuracy.
B: Confidence. Faint lines show individual participant means, for which the violin and box plots show the distributions.
C: Influence. Participants' weight on the advice for advisors in the Familiarization stage of the experiment. The shaded area and boxplots indicate the distribution of the individual participants' mean influence of advice. Individual means for each participant are shown with lines in the centre of the graph. The theoretical range for influence values is [-2, 2].

Table 5.31: Advisor performance in Dots task with confidence-contingent advisors

	Agreement					
	Initial estimate confidence			Subtotals by initial estimate correctness		
Advisor	Low	Medium	High	Correct	Incorrect	Ov
Anti-bias	90.3	71.7	49.6	72.0	28.3	
Anti-bias (Target)	*90*	*70*	*50*	*70*	*30*	
Bias-sharing	49.0	69.0	90.3	67.8	28.8	
Bias-sharing (Target)	*50*	*70*	*90*	*70*	*30*	

Figure 5.28B and ANOVA indicated that participants were more confident in their answers when they were correct compared to incorrect (F(1,49) = 108.05, p = <.001; M_{Correct} = 29.59 [27.50, 31.68], $M_{\text{Incorrect}}$ = 25.02 [22.76, 27.28]), and that this difference was larger for final decisions than for initial estimates (F(1,49) = 80.93, p = <.001; $M_{\text{Increase|Correct}}$ = 1.83 [1.01, 2.66], $M_{\text{Increase|Incorrect}}$ = -2.34 [-3.34, -1.34]). There was no main effect of time independent of correctness (F(1,49) = 0.42, p = .520; M_{Final} = 27.18 [25.06, 29.29], M_{Initial} = 27.43 [25.21, 29.66]).

Figure 5.28C shows that there was evidence of equivalence in the influence of the two advisors. There was no evidence of a correlation between initial estimate accuracy and confidence (1.07), and no evidence for a correlation between final decision accuracy and confidence (1/1.62).

Advisor performance The advice is generated probabilistically from the rules described previously in Table ??. It is thus important to get a sense of the actual advice experienced by the participants.

Table 5.31 shows that advisors agreed with the participants' initial estimates at close to target rates in all confidence categories. There was a larger than expected difference in agreement on correct answers overall, but this was not a significant difference (). The advisors were also further apart in terms of overall accuracy than desired, because of the different agreement rates for correct answers, although once again this difference was not significant ().

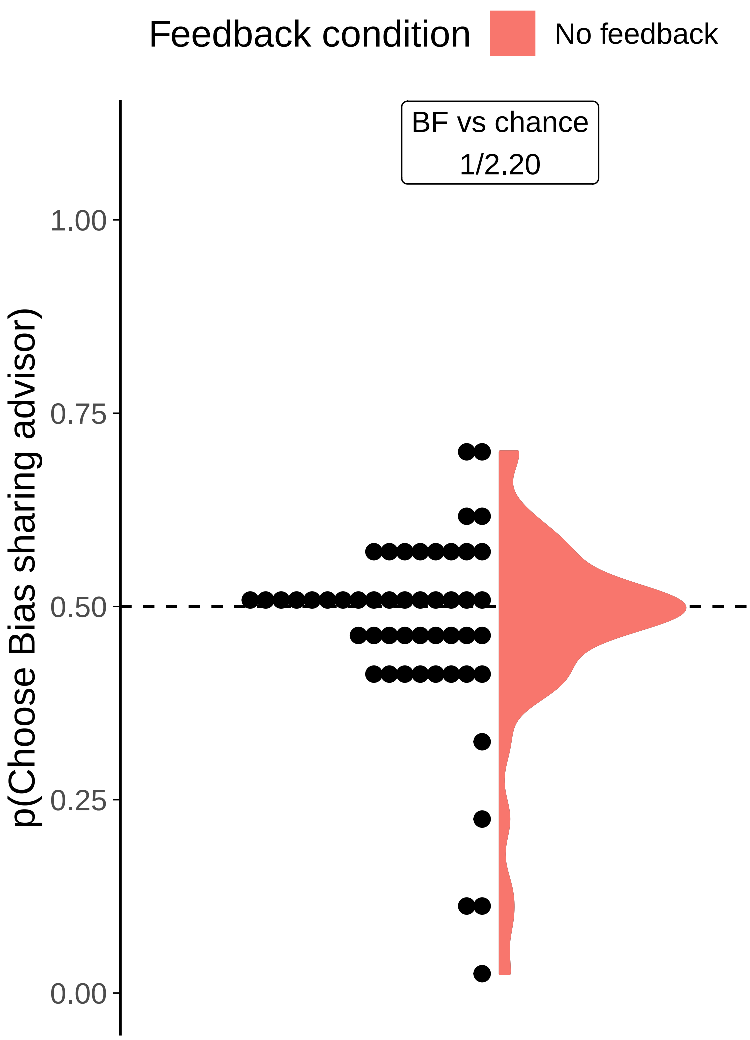

Figure 5.29: Dot task advisor choice for confidence-contingent advisors.
Participants' pick rate for the advisors in the Choice phase of the experiment. The violin area shows a density plot of the individual participants' pick rates, shown by dots. The chance pick rate is shown by a dashed line.

Hypothesis test There was a strong tendency for participants to express no, or slight preferences between advisors ($t(49)$ = -1.52, p = .134, d = 0.22, BF = 1/2.20; M = 0.47 [0.44, 0.51], μ = 0.5). Intriguingly, almost all participants who expressed a stronger preference expressed it towards the Anti-bias advisor: in the direction counter to that hypothesised (Figure 5.29).

In the Medium confidence trials, where the Lab experiment showed a glimmer of a difference, there was evidence against the existence of a difference ().

In this experiment we saw an extremely strong effect of picking the advisor in the first position on the screen ($t(49) = 5.00$, $p < .001$, $d = 0.71$, BF = 2.4e3; $\mathrm{M}_{\mathrm{P(PickFirst)}}$ = 0.65 [0.59, 0.71], $\mu = 0.5$), an effect that we would hope would be random and even out across participants. In this experiment, as a function of chance, the Bias-sharing advisor appeared in the favoured top position more frequently than we would expect (BF = 18.3; $\mathrm{M}_{\mathrm{P(BiasSharingFirst)}}$ = 0.48 [0.47, 0.49], μ = 0.5). The effect of preferring to pick the advisor in the first position, therefore, would enhance the pick rate of the Bias-sharing advisor. Despite this, we saw that the advisors were picked at equivalent rates, allowing us to be even more sure that there is no overall preference for the Bias-sharing advisor driven by that advisor's method of providing advice.

Discussion

While source selection and advice-taking are different domains, the previous experiments have shown strong similarities in the tendencies of participants: participants tend to be more influenced by the same kinds of advisors that they are more willing to hear from. On this basis, following Pescetelli and Yeung (2021), we would expect to see a preference for picking the Bias sharing advisor. We do not see this preference, and, insofar as we see any preference at all, we see the opposite.

5.4.3 Confidence-contingent advice effects in the Dates task

The Dates task was not used to study confidence-contingent advice because such advice requires both a precise control over the relative agreement and accuracy rates of the advisors and the ability to estimate confidence in responses. The advisors' agreement (and hence accuracy) profiles depend on the participant's performance, and this is unknown a priori in the Dates task whereas it is controlled in the Dots task using a staircase procedure. Different approaches to estimating participants' confidence were trialled, including a pilot experiment in which the width of marker used by participants was used as a proxy for confidence, but none of the approaches produced any discernible effect of confidence on advisor agreement.

We did attempt to design a version of the Dates task that would allow confidence-contingent advice, but pilot studies were largely unsuccessful and the time, cost, and effort required to refine the study were not deemed worthwhile in light of the null results from the Dots task studies.

5.4.4 Discussion

Across two tasks investigating whether people preferentially selected advisors who shared their biases we found scant evidence in favour of the effect. The two advisors, balanced for overall agreement and accuracy to eliminate the effects seen in previous experiments, differed in their likelihood of agreement based on the participants' initial estimate accuracy (following Pescetelli and Yeung 2021). The Bias-sharing advisor agreed more frequently with a participant when the initial estimate was made with high confidence, and less frequently when the initial estimate was made with low confidence, and the Anti-bias did the opposite. Overall, there was no evidence that participants picked these advisors at different rates. Looking just at the medium confidence trials, where the advisors were exactly equivalent, there was very weak evidence that the Bias-sharing advisor was picked more frequently in the Lab experiment and reasonable evidence against any difference at all in the online replication.

The data from these studies is not conclusive against the existence of this effect, especially when integrated with the findings of Pescetelli and Yeung (2021). It is plausible, for instance, that metacognitive moderation does happen in advice evaluation, but that these effects are more subtle than the broader effects of accuracy and agreement, and that the studies here were too underpowered to detect these effects. It seems unlikely that metacognitive moderation effects would exist in the advisor evaluation domain (as showing in Pescetelli and Yeung (2021)) but not in the advisor choice domain, given the rough parity demonstrated between these domains across the dimensions of accuracy and agreement in previous experiments.

5.5 General discussion

Previous work in our lab indicated that people are able to evaluate advice in the absence of feedback using agreement as a proxy (Pescetelli and Yeung 2021). We performed a series of experiments investigating whether the patterns observed for advice taking are also evident in advisor selection. Our experiments exercised tight control over the advisors' answers in the domains of agreement and accuracy, and allowed us to explore their relative contributions.

Modelling and theoretical work indicates that biased source selection can dramatically reshape communication networks and create echo chamber effects where accurate but unpalatable information is ignored (Sunstein 2002; Madsen et al. 2018). Empirical research on source selection behaviour has found relatively little indication that people behave this way in the real world (Marquart 2016; Sears and Freedman 1967; Nelson and Webster 2017), particularly in terms of avoiding exposure to unpalatable information (**jangChallengesSelectiveExposure2014**; Weeks et al. 2016). If the effects seen by Pescetelli and Yeung (2021) in the domain of advisor choice also occur in the domain of advisor selection, they may demonstrate in principle a psychological mechanism which could drive biased source selection that are rational and appropriate given the information available.

Advisor choice results

The preferences for advisors were broadly consistent with the pattern expected from previous work on advisor influence (Pescetelli and Yeung 2021). Experiments 1B and 3B showed that, where objective feedback could be used to calculate advisor performance, participants showed a systematic preference for picking the advisor who would provide the most accurate advice. Experiments 1A showed that this preference endured even when feedback was removed, although the result was not replicated in Experiment 1B.

Experiments 2A and 2B showed that participants who did not receive feedback systematically preferred High agreement advisors over Low agreement advisors. Experiments 3A and 3B indicated that this preference for agreement in the absence of feedback did not extend to a preference for agreement over accuracy. These results are consistent with an account of advisor trust updating which uses agreement as a proxy for advisor accuracy when more reliable information is not available, although some additional mechanism is required to explain how accuracy can with agreement in the absence of feedback. Pescetelli and Yeung (2021) offer confidence as an additional mechanism, but attempts to replicate their advisor influence experiment in the domain of advisor choice (Experiments 4A and the Lab experiment) did not provide empirical support for this.

NICK: Highlight key themes as they relate to your results specifically. The themes seem to be: 1. preference for accurate advice when this can be inferred from feedback, but not otherwise. 2. Preference for agreeing advice when feedback is absent, but not when it's present. Relate back to previous literature -> mixed evidence on how people seek info. May depend on domain? (Cf. our conversation all those years ago with Jonathan Bright about different knowledge/info domains.) + the more detailed points you make below (e.g., about similarities and differences between the tasks)

Accuracy of advice

Our experiments showed that people will prefer to seek advice from an advisor who is more accurate, provided that they can identify that advisor. This coheres well with results from Experiment 0§4.1 and other literature in the advice-taking domain where greater task accuracy is referred to as 'expertise' (Pescetelli and Yeung 2021; Yaniv and Kleinberger 2000; Gino, Brooks, et al. 2012; Rakoczy et al. 2015; Sniezek, Schrah, et al. 2004; Soll and Larrick 2009; Tost et al. 2012; Schultze, Mojzisch, et al. 2017; Wang and Du 2018; Önkal, Gönül, et al. 2017). A review of this advice-taking literature can be found in another chapter§7.2.3.

Like Pescetelli and Yeung (2021), we were interested in whether people are sensitive to accuracy for decisions they make where feedback is not provided. The majority of decisions on which we seek advice in everyday life do not come with feedback, or come with only infrequent and often delayed feedback. Although the perceptual decision-making task and the date estimation task used here are highly stylized, they can mimic this feedback regime. When feedback was denied to participants, leaving them unable to use the feedback to determine the accuracy of the advisors, their preferences only favoured the more accurate advisor in the Dots task (Experiment 1A§5.1.1), with no systematic preference in the Dates task (Experiment 1B§5.1.2). This difference might be explained by the relative difficulty of the tasks.

Confidence-weighted agreement as a proxy for accuracy

Where people cannot use objective feedback to determine the quality of advice, they use whether or not the advice agrees with them as a proxy for accuracy. This is shown by Experiments 2A§**??** and 2B§5.2.2, and is consistent with Experiment 1A§5.1.1 (although not with Experiment 1B§5.1.2 given Experiment 3B§5.3.2). It also reflects, in the advisor choice domain, the findings of Pescetelli and Yeung (2021) in the advice-taking domain.

Pescetelli and Yeung (2021) theorised, on the basis of their experiments, that people assess advice on the basis of confidence-weighted agreement. This means that, where a judge is confident in their own opinion, offering agreement or disagreement results in a large increase or decrease in trust in the advisor, respectively. Where a judge is very unsure of the accuracy of their own opinion, however, neither agreement nor disagreement affect trust in the advisor very much – it is as if the judge has no frame of reference by which to assess the advice they have been given.

Assuming that trust in an advisor translates directly into a preference for hearing advice from that advisor, this theory accounts for some, but not all of our experimental results. It provides a straightforward account of the results of Experiment 2A§**??** and 2B§5.2.2, where participants who did not receive feedback preferred to hear advice from the advisor more likely to agree with them (while overall advisor accuracy was held constant). It can also explain of the results of Experiment 1A§5.1.1, because the probability that two independent binary judgements agree is related to the probabilities that they are greater than chance (Soll and Larrick 2009). Given participants were above chance in their perceptual decisions, the more accurate advisor would agree more, as indicated in Table 5.1. Furthermore, if agreement is weighted by confidence, this discrepancy would be greater where the weighting is higher to the extent that participants were well calibrated in their confidence: where they were most confident they were most likely to be correct, and so they were proportionally more likely to be agreed with by the more accurate advisor.

Experiment 1B§5.2.2 is harder to explain under the Pescetelli and Yeung (2021) theory. Participants should be sensitive to differential agreement rates as in Experiment 1A§**??**, and this should be enhanced to the extent that participants' confidence is well calibrated. The results were doubtful that participants with greater accuracy or confidence calibration expressed a greater preference for the more accurate advisor. It could have been the case that individual high-impact trials where a participant was extremely confident of the correct answer happened to systematically coincide with the less accurate advisor agreeing with the participant, but this seems very unlikely indeed. All in all, the results from Experiment 1B do not cohere well with the theory.

Experiments 3A§5.3.1 and 3B§5.3.2 are also consistent with the theory. Participants denied feedback did not prefer to see advice from the agreeing advisor over the accurate one. This result would make sense if participants were able to detect the accuracy advantage of the accurate advisor: they might well have noticed that that advisor tended to agree with them where they were confident of the correct answer, and to disagree otherwise. The data did not support that conclusion. In the continuous Dates task used for Experiment 3B, the difference between agreement and accuracy is much clearer to participants because the advisors place markers on a timeline that also shows the participant's initial estimate. Participants in

Experiment 3B may have been able to detect the redundancy of the agreeing advisor's advice and preferred the accurate advisor for that reason. We included a debrief questionnaire asking what participants though was the difference between the advisors, and several participants indicated in their responses that they had identified that the agreeing advisor tended to reflect their own answer.

The most clear failure of the theory in accounting for the present results is in the experiments using confidence-contingent advice, the Lab experiment and Experiment 4A§5.4.2. In these experiments, the advisors were specifically constructed such that they would be balanced for overall agreement and accuracy, but differentiated at the extremes of participants' subjective confidence. One specific test indicated that there may be an effect – a frequentist test on the moderate-confidence Lab experiment trials where advisors' agreement rates were balanced. This test, and this experiment, are the closest to those reported by Pescetelli and Yeung (2021). The effect, if there is one, does not seem to generalise to advisor choice behaviour in shorter versions of the task performed by participants outside of a laboratory setting.

Overall, the results of this suite of experiments suggest that trust in advisors, as evaluated by advisor choice preference, is determined by a variety of factors, even in highly constrained perceptual and general knowledge estimation tasks. One of these factors may be confidence-contingent agreement, or agreement more generally. We have offered some speculation as to properties of the task and context that might alter the extent to which confidence-contingent agreement determines trust in advisors. Whether there is any dominant mechanism in play that accounts for a sizeable amount of the variation in trust remains unknown, but it appears unlikely that confidence-contingent agreement is that mechanism.

NICK: Highlight how results relate to the modelling to come: preference for agreeing advice when feedback is absent, but also wide variation in this preference. When preferences are not strong at the population level, we see almost all selection behaviours from always choosing A to always choosing B.

Variability between participants

The experiments produced a range of results, from clear evidence of systematic preferences to clear evidence of the lack of any such systematicity in preferences. Interestingly, participants displayed a range of preference patterns. Most participants in most conditions in most studies demonstrated fairly balanced picking behaviour; perhaps motivated by a sense of fairness or novelty, or perhaps as a result of random selection or attention to other factors such as the position of the advisor on the screen. Where effects appear in the data, they show up as a fat tail on a skewed distribution: a sizeable minority select a given advisor on most trials, and almost no one selects the other advisor on most of the trials.

The advisor preference distributions differ somewhat between tasks. Although advisor choice distributions in both tasks are roughly normally distributed, those in the Dots task results are sharper. This sharpness may be a result of many participants selecting advisors at approximately equal rates. If participants become bored or fatigued by the experiment they may disengage and select advisors in a random manner.

Where manipulations are effective in the Dots task (e.g. Experiment 3B§5.3.1) they change the direction of preferences: a good many participants continue to pick advisors at approximately equal rates, but preferences in those who do express a preference systematically favour one particular advisor. Systematic differences in the Dates task are signified by distributions in which the modal preference moves to an extreme preference for the relevant advisor while the tails of the distribution continue to cover the whole range. These differences are likely a consequence of the difference in the number of Choice trials in each Task. The Dates task only has around 10 Choice trials, and it is relatively common for participants to select a single preferred advisor repeatedly. The Dots task, however, has 30-60 Choice trials, meaning participants may become tired of seeing advice from the same advisor, increasing the novelty value of advice from the non-preferred advisor and thus reducing the apparent strength of preference. In support of this idea, the distribution of preferences in Experiment 3A§5.3.1), which had 30 Choice trials, was much more similar to those from the Dates task than the other Dots task experiments, which had 60 Choice trials. The similarity may have reflected the absence of any effect rather than the number of Choice trials, but it also more closely resembles the distributions of the Dates task where effects are see (1B§5.1.2, 2B§5.2.2, 3B§5.3.2) than the other Dots task distribution where they are not (4A§5.4.2).

The kind of heterogeneity revealed in the experiments' results is seldom included in population models of influence dynamics. In the following chapter§6, we explore the effects of including this kind of heterogeneity in agent-based models of advice giving.

Social and Personality perspectives

Source selection, most robustly selective exposure, is argued in the Social and Personality Psychology domains to be a product of a drive for cognitive consonance (Festinger 1957) or protecting a positive self-concept (Knobloch-Westerwick 2015) rather than an heuristic which is expected to improve the accuracy of decision-making under common real-world conditions. While these experiments do not directly contradict that literature, they do indicate that source selection can occur in the absence of any clear implications of the choice for a person's view of themselves. It may be argued that the effects in the absence of meaningful moral or political contexts reflects a bleed over from processes that perform useful work in vigilantly guarding a person's self-image, and that are always active when choices over information sources are made. Conversely, however, it could be argued that much of the selective exposure evidence represents a drive for accurate information: if I strongly believe in a liberal world-view I may judge the reporting of a liberal media outlet to be more objectively accurate, and be more willing to view its content on that basis.

Limitations

As noted above, there were several issues with the experiments that were only noticed after data had been collected. The most serious of these was that the

Dates task did not counterbalance the advisors' positions, and post hoc analysis of the Dots task data suggested that advisor position may have an unexpectedly important role in advisor choice behaviour. Another concern is that the difficulty of the Dates task was high, and also highly variable between participants. In some ways this is a strength, because it could theoretically allow us to detect variations in advisor choice behaviour as a function of participants' ability, but in practice we did not find these. We were left only with the drawbacks, therefore, of higher noise in the data and the possibility of strong effects of choice being driven by difficult-to-identify high-impact individual trials.

A related concern is the frequently large and significant effects of advisor position in the Dots task. Participants frequently picked the advisor at the top of the screen, regardless of that advisor's identity. This suggests disengagement with the task. Interestingly, the advisor at the top of the screen was further away from the answer bar (where participants input their initial estimates and final decisions) than the advisor at the bottom. We would expect lazy participants using a mouse to predominantly select the closer advisor. For participants using mobile phones, it would be equally easy to select either advisor.

Ecological validity is a concern more generally, too. While these studies were appropriate for capturing the behaviour of interest, they are also highly stylized in both their decision-making tasks and in the relationship between the participant and the advisor. The tasks are somewhat unusual in that people are seldom required to compare barely-seen visual scenes or estimate historical dates in their everyday lives. When they do, they are unlikely to seek advice on them. This is a limitation general to most psychology experiments in one way or another, but no less important for that.

The relationship between the participants and the advisors was also unusual. While some researchers have suggested that social influence phenomena can be explained by reinforcement learning from repeated interactions (Behrens et al. 2008; FeldmanHall and Dunsmoor 2019; Heyes et al. 2020), the kind of repetition presented in these tasks is not a typical feature of human relationships. In some particular cases, for example working with a colleague at a job that requires rapid and repeated joint decision-making, behaviour might approximate that of our tasks, although for our domain of advisor choice to be of particular interest these people would also have to have the power to select their partner.

Conclusion

These experiments paint a rather equivocal picture, without being able to offer the kind of strong evidence we would like concerning the validity of the Pescetelli and Yeung (2021) theory of confidence-weighted advice evaluation. One major observation revealed in the experiments was the extent to which participants exhibited a wide range of preference strengths and directions, particularly where systematic effects of advisor preference were absent. In the following chapter§6 we use agent-based computational modelling to investigate the role of this kind of heterogeneity in the formation of echo chambers in information exchange networks.

6

Network effects of advisor choice

6.1 Introduction

The behavioural experiments in the previous chapter, along with predecessors from others in our lab (Pescetelli and Yeung 2021), indicate that people show preferences for selecting advisors who agree with them and treat advice which agrees with their initial judgements as more valuable. Previous modelling work has shown that either of these properties is sufficient to produce echo-chamber-like structures in networks wherein individuals increasingly disregard information from those likely to disagree with them (Pescetelli and Yeung 2021; Madsen et al. 2018). In this chapter, similar models are constructed, with the parameters controlling the agents' trust updating and advice seeking behaviour based on values fitted to the participants in the behavioural experiments. The models indicate that heterogeneity within the population may act as a brake on echo chamber formation, with individuals who are slow to trust or balanced in their advice-seeking behaviour exhibiting less pull towards extremes.

6.1.1 Agent-based models

Agent-based models are models in which numerous discrete decision-making agents interact to produce a model state, typically used to investigate emergent phenomena

(Bonabeau 2002; Smith and Conrey 2007). Unlike typical models, which describe the overall behaviour of the system using equations, in an agent-based model each component agent (taken to represent an atom, a neuron, a starling, a person, etc.) has equations describing its behaviour at each time step. Crucially, the equations governing agents' behaviour will have terms relating to other agents, and the outcome of the equations will produce changes in those properties. For example, a neuron might have inputs indicating whether or not a neighbouring neuron is in action potential, and the equations will determine whether or not the neuron itself triggers an action potential.

Agent-based modelling is a powerful approach, because it can connect system-level phenomena to individuals' decision rules and incorporate individuals with outlying or different decision rules (Bonabeau 2002). The power of agent-based modelling also comes with drawbacks (Jones 2007). The intuitive nature of its explanations means that it is easy to lose touch with ecological validity (because even non-ecologically valid implementations may seem valid when expressed in agent-based terms) and easy to overclaim on the basis of results. The interactivity and emergent behaviours that it produces can mean that some models are unstable: it is important to demonstrate robustness of emergent behaviours by running repeated tests on families of models. Furthermore, the dispersed nature of the models means that there are more features to describe; modellers must make sure that the models are precisely specified and that those specifications are shared accurately so that others are able to reproduce and explore the models (Jones 2007; Bruch and Atwell 2015).

In social science research, agent-based models are best used to investigate "the implications for social dynamics of one or more empirical observations or stylized facts" (Bruch and Atwell 2015). The models used here are employed to investigate the effects of introducing the advisor selection phenomena documented in the previous chapter into network models of social influence. Bruch and Atwell (2015) argue strongly that agent-based models would benefit from greater empirical realism, and others, including Bonabeau (2002), have called for greater ecological validity.

This chapter provides some evidence for this discussion by demonstrating the effects of implementing agents which copy individual participants versus drawing agents' coefficients from an empirically-defined distribution.

6.1.2 Similar models in the literature

The use of agent-based models has increased substantially in many fields in recent decades. Rather than attempt to present a full picture of agent-based models in which agents advise one another, this section briefly covers models that include a tendency for agents to seek out advice from likeminded others and compare the effects of varying this property. Models such as Song and Boomgaarden (2017) that include inter-agent influence but no variation in this propensity are not discussed.

Pescetelli and Yeung (2021) present a model which is a conceptual predecessor for the current model. In this model, agents make a single binary decision at each step on the basis of a perception of the true value and an idiosyncratic (static) bias. Agents then consult another agent for advice, and reach a final decision by integrating that advice with their initial estimate, weighted by the trust they have in their advisor. Model parameters varied according to whether or not agents received feedback, and whether or not agents sought out other agents in whom they had higher trust (similar to weighted sampling in our model). They showed a cluster of interesting results, starting with the demonstration that using agreement as a proxy for accuracy in the absence of feedback is beneficial provided biases are uncorrelated. Secondly, they found that agents' trust weights aggregated into separate camps defined by shared biases where feedback was infrequent. Lastly, they observed that agents' biases tended to polarise in the absence of feedback when biased agents preferentially sought likeminded advice. Key differences between this model and the current model are that the current model allows for more heterogeneity in agents' properties and draws all agents' biases from a normal rather than uniform or point distribution.

In a similar vein, Madsen et al. (2018) demonstrated that echo chambers could form within large networks of rational agents. Their models used Bayesian agents whose likelihood of accepting advice was dependent upon the similarity of the advice to their own prior beliefs. Under a range of parameters, at at a range of network sizes from 50 to 1000 agents, they showed that echo chambers formed consistently in most cases.

Gao et al. (2015) report that agents preferring to hear from likeminded others allowed stable minority opinion to persist where agents had to subscribe to one or other opinion.[1]

Van Overwalle and Heylighen (2006) replicated a several social psychology results using small networks of connectionist neural networks. These results included a demonstration of majority influence (the authors describe it as 'polarisation'), in which repeated interactions bring about a consensus opinion in a small network of 11 agents.

!TODO[Maybe discuss dugginsPsychologicallyMotivatedModelOpinion2017]

6.1.3 Current model

In the current model, the agents face a perceptual decision in which they integrate sensory evidence with prior expectations in a Bayesian manner to produce an initial estimate The agents then receive advice from another agent, whom they choose on the basis of their trust in that agent compared to all the other agents combined with their preference for receiving advice from more trusted advisors. 'Trust' as encapsulated in the model is double-ended: very untrustworthy advisors can give reliable information because they are firmly expected to give the wrong answer (and therefore contain information about the correct answer because the decision is a binary one). The advice is integrated with the initial estimate, again in a Bayesian manner, to produce a final decision. The final decision is used to update the agent's perception of the world and their advisor. The agent's bias drifts towards the final decision answer by weighted averaging, while the trust they

[1] This paper is extremely brief in its reporting of the agent-based model, and it is difficult to determine precisely what was done.

have in their advisor increases (or decreases) depending upon whether (or not) the advice is in agreement with the final decision.

The estimate-advice-decision-update cycle is repeated thousands of times. The models illustrate the dynamics of advice interactions: how prior expectations change over time and whether and how polarisation or consensuses emerge; and how agents' trust in one another changes as they gain experience with one another.

6.1.4 Goals of this approach

There are two parts to the modelling work in this chapter: exploring the characteristics of participants' recovered model parameters; and exploring the effects of those parameters on the dynamics of the agent-based models.

In the first part, models parameters are recovered from the behavioural data for each participant using a gradient descent algorithm. The two free parameters estimated during the parameter recovery process are: how rapidly trust updates (*trust volatility*); and the extent to which more trusted advisors are preferred (*weighted selection*).

The model and parameter recovery process are both tested using simulation. The parameter recovery process is tested by attempting to recover data from simulated data (where the parameters are know), and shows a reasonable level of accuracy in recovering parameters from data, suggesting that the recovered values for participants are likely to capture some aspects of participants' behaviour on the task. The model itself is tested by comparing errors for fitted parameters to errors for those same parameters when fitted to data that has had the advisor agreement information scrambled. Given the models are driven by the advisors' agreement with the judge's initial decision, scrambling this column should break the temporal relationship between experience and trust updating, leading to worse parameter fits. This is indeed the case.

The parameter recovery results, distributions and correlations of the coefficients, illustrate the heterogeneity of the participants. Some participants' data does not fit well to the model, indicating that these participants may have used strategies which bear little resemblance to the model the data are fit to.

We then explore the behaviour of the agent-based models (within a narrow area of their parameter space), varying the way agents' preferences for more trusted advisors are determined. The baseline model assigns all agents no preference for more trusted advisors, meaning that they sample from all possible advisors at random. The Homogenous model applies the mean of the recovered parameter distribution to all agents. The Heterogeneous model draws values from a distribution defined by the mean and standard deviation of the recovered parameter distribution. The Empirical model selects values from those observed in the recovered parameter data (with replacement). For the first three models, trust update values are sampled from a distribution with the mean and standard deviation of the recovered parameters, while in the Empirical model the agents receive the trust update value for the agent whose preference strength coefficient they received. Two versions of each model are run, one which starts with a random trust network and one which starts with a homophilic trust network.

6.2 Method

Agent-based modelling is used to simulate the interactive effects of repeated paired decision-making. The agents in the model perform a cycle of making an initial decision, selecting one of the other agents as an advisor, making a final decision, and updating their trust in their advisor and their internal beliefs about the world. The task the agents face is roughly analogous to the task faced by human participants in the Dots Task experiments (§Perceptual decision (Dots Task)), where the participants make a decision about which of two rapidly and simultaneously presented grids contained more dots.

Each model consists of a population of agents implementing the same mathematical model of decision-making, trust-updating, and advisor selection, with different coefficients for parameters of that mathematical model (Table 6.1) drawn from appropriate distributions.

6.2.1 Open code

The agent-based modelling is performed using a custom-written R package. Zenodo The functions which implement each of the steps below are listed in the (non-exported) simulation loop function `simulationStep`. Each of these sub-functions includes unit tests to verify its behaviour. For readers who prefer to follow along with code rather than maths, each section below includes details of the function implementing the equations. These small functions are not exported by the package, so links are to source code rather than package documentation.

6.2.2 Model details

Each model is defined by a set of model parameters (6.2), which are assigned capital letters in the equations below. These parameters are used to generate further parameters which govern agents' tendencies, and these are assigned lower-case letters and superscripted with the agent to whom they belong. Variables which change over the course of the model have a subscript indicating which step of the model they belong to.

This is best illustrated with an example. Each agent has a sensitivity parameter, s^a which governs the amount of random noise which is attached to their perception of the world. On any given decision, the amount of this noise ϵ_t^a is determined by drawing from a normal distribution with standard deviation defined by the agent's sensitivity ($N(0, \frac{1}{s^a})$). When the agent is created, the parameter defining its sensitivity distribution (s^a) is itself drawn from a normal distribution with mean and standard deviation defined by the model settings ($N(S^\mu, S^\sigma)$).

Table 6.1: Agent properties

Property	Description	Updates each step
a	Agent unique identfier	No
s^a	Accuracy of agent's perceptions	No
c^a	Agent's subjective confidence scaling	No
τ^a	Size of agent's trust updates	No
λ^a	Size of agent's bias updates	No
w^a	Extent of agent's preference for trusted advisors	No
b^a_t	Agent's prior expectation about the task answer	Yes

Creating agents

Each agent has the properties listed in Table 6.1. An agent's values for each of these properties (except id) are created by drawing values from normal distributions defined by parameters in the model settings (Table 6.2).

Additionally, each agent has a 'trust' in each other agent: an estimate of that agent's reliability ranging from being convinced that the other agent is always correct to being convinced that the other agent is always wrong. These values are updated at each step and initialised by drawing from a uniform random distribution with limits [0.5, 1].

There is an ongoing debate about whether these kinds of advice models should support expectations of lying. In some models, an agent's lowest trust value for another agent indicates that they expect that agent's advice will be useless (no better than random chance), while others use a floor which indicates the agent believes the advisor to be deliberately misleading (providing information in the strict sense that advice is predictive of correct answers). Collins et al. (2018) provide a discussion of this issue along with initial evidence that advice can be considered misleading in some cases. The current model treads a hybrid path between these positions, by initialising the trust weights such that the lowest level of starting trust is equivalent to considering the advice as random (containing no information about the truth), but allowing trust weights to decrease following interaction so that advice is considered as pointing away from the truth.

Agents are created in the `make_agents` function.

Model step

Establishing the stimulus The same stimulus is presented to each agent, in the form of a value (v_t) drawn from a normal distribution with mean (V^μ) and standard deviation (V^σ) defined in the model settings:

$$v_t \sim N(V^\mu, V^\sigma)$$

The agents' task is to determine whether v_t is greater than or less than zero.

The establishing of true values is achieved in the code within the `simulationStep` function, and executes a function supplied by the user. The default function, used in all models described below, is drawing from the normal distribution. This default can be seen in the `truth_fun` parameter for the `runSimulation` function.

Initial decision-making Agents perform a noisy perception of the stimulus, by combining the true stimulus value with random noise (ϵ_t^a), to produce a percept q_t^a.

$$q_t^a = v_t + \epsilon_t^a$$

Where: $\epsilon_t^a \sim N(0, \frac{1}{s^a})$

This sensory percept is then converted into a subjective probability that the stimulus was greater than zero (p_t^a) using a sigmoid function with a slope defined by the agent's subjective confidence scaling parameter.

$$p_t^a = \varsigma(q_t^a, c^a)$$

Where: $\varsigma(x, y) = \frac{1}{1+e^{-xy}}$

The conversion of the percept (q_t^a) to the subjective probability (p_t^a) changes the representation from a theoretically-unbounded normal distribution centred around 0 to a probability in the interval [0,1] centred around 0.5. The subjective probability expresses a judgement about whether or not v_t is greater than or less than zero, but does not contain an estimate of v_t itself.

In the code, the percept is calculated using `getPercept`. The inclusion of the agent's confidence scaling is done as part of the second step in the determination of the initial decision.

This subjective probability is then integrated with the agent's prior expectation about whether stimuli are generally less than or greater than 0 (b_t^a), or 'bias', to produce an initial decision, i_t^a. This integration is performed using Bayes' rule.

$$i_t^a = \frac{b_t^a \cdot P(p_t^a | v_t > 0)}{b_t^a \cdot P(p_t^a | v_t > 0) + (1 - b_t^a) \cdot P(p_t^a | v_t \leq 0)}$$

Where: $P(p_t^a | v_t > 0) = z(p_t^a, 1, V^\sigma)$;
and: $P(p_t^a | v_t \leq 0) = z(p_t^a, 0, V^\sigma)$;
with: $z(x, \mu, \sigma)$ giving the density of $N(\mu, \sigma)$ at x.

The prior, b_t^a, is equal the prior probability that the value is greater than 0: $b_t^a = P(v_t > 0)$

In the code, the calculation of initial decision from the percept (including the scaling of the percept according to the agent's confidence scaling parameter) is performed in `getConfidence`.

Note that in these initial decision equations the agents have direct access to a model property, V^σ, the standard deviation of the true values of the stimuli. Ideally, in an agent-based model, the agents would not have such direct access to non-observable properties, and would instead build up specific expectations about the variability of stimuli from observation, perhaps seeded with a loosely-informative prior. The agents are allowed to know the value here as a shorthand for such exploration. This makes the agents' task analogous to well-practised real-world tasks such as perceptual decision-making. It is unlikely to seriously affect any conclusions or illustrations drawn from the models.

The initial decision contains both a discrete decision (whether the stimulus value was more likely to be less than zero, $i_t^a < 0.5$, or greater than zero $i_t^a \geq 0.5$), and how much so ($|i_t^a - 0.5|$). It thus represents both the agent's decision and the confidence in that decision.

Advisor selection Having made an initial decision, each agent selects a single advisor from whom to receive advice. This choice is made based on the trust in each potential advisor scaled by the strength of the agent's preference for receiving more-trusted advice.

The identity of an agent's advisor on a given trial is designated by a', and the weight assigned to advisor a' by agent a at step t by $\omega_t^{a,a'}$. Each agent's trust value is adjusted to be relative to the most (or least) trusted advisor, depending upon whether the agent prefers trusted ($\mathrm{w}^a > 0$) or untrusted ($\mathrm{w}^a \leq 0$) advisors.

$$\omega'^{\ a,a'} = \begin{cases} \mathrm{w}^a > 0, \omega_t^{a,a'} - \max(\Omega_t^a) \\ \mathrm{w}^a \leq 0, \omega_t^{a,a'} - \min(\Omega_t^a) \end{cases}$$

Where Ω_t^a is the set of trust weights in all potential advisors for agent a at step t.

These relative trust values are then assigned probability weightings for selection based on a sigmoid function. Because of the relative scaling, each probability is in fact drawn from a half sigmoid, where the probability weight assigned to the most likely candidate is 1 and other candidates' probability weights between 1 and 0.

The identity of advisor $_t^{a,a'}$ is determined by sampling at random from the advisors, weighted by the probability weights assigned. In the R code the advisor selection procedure occurs in `selectAdvisorSimple`.

Differences from parameter estimation The approach used for fitting participant data from the behavioural experiments is subtly different from that used for advisor selection in the agent-based models. In the behavioural experiments, rather than being presented with a choice of many different advisors whose appeals all had to be considered, participants were presented with a choice of two advisors only. The parameter recovery process estimated a trust update rate (τ^a in the agent-based model) which tracked the trust in each advisor, and the difference between these trust values was fed into a sigmoid governing selection. The slope of that best-fitting sigmoid function was taken as equivalent to w^a in the agent-based models. This approach is reasonable given the differences in the advisor selection task facing the human participants and model agents, but should be noted as a caveat for drawing interpretations concerning the role of weighted selection values based on human participants' performance.

Code for the implementation of advisor choice fitting can be found in the function `advisor_pick_probability`.

Final decision-making Final decisions are made by Bayesian integration of the initial decision and advice. Advice takes the form of a binary recommendation, and is weighted by the trust the agent has in their advisor.

First, initial decisions and advice are reoriented to the direction of the initial decision, such that initial decisions represent confidence in the initial decision and advice represents agreement with that decision.

$$i'\,{}^{a}_{t} = \begin{cases} i < 0.5, 1 - i^{a}_{t} \\ i \geq 0.5, i^{a}_{t} \end{cases}$$

$$i'\,{}^{a,a'}_{t} = \begin{cases} i < 0.5, \mathrm{round}(1 - i^{a,a'}_{t}) \\ i \geq 0.5, \mathrm{round}(i^{a,a'}_{t}) \end{cases}$$

The trust weight is slightly truncated to avoid very extreme values, and oriented based on whether the advice agrees, giving the expectedness of the advice provided the initial answer was correct:

$$\omega'\,{}^{a,a'}_{t} = \begin{cases} i'\,{}^{a,a'}_{t} = 0, 1 - \min(0.95, \max(0.05, \omega^{a,a'}_{t})) \\ i'\,{}^{a,a'}_{t} = 1, \min(0.95, \max(0.05, \omega^{a,a'}_{t})) \end{cases}$$

The final decision is obtained by performing Bayesian integration. In Bayesian terms, agents are trying to discover the probability of their final answer being correct given the agreement (or disagreement) observed from their advisor. Thus they multiply the initial probability of being correct (subjective confidence, $i'\,{}^{a}_{t}$) by the probability of the advisor agreeing if they are correct ($w'\,{}^{a,a'}_{t}$). This is divided by all of the options that could have led to the observed advice: the probability that they are correct multiplied by the probability of the advice if they are (the numerator), plus the probability that they are incorrect multiplied by the probability of the advice if they are incorrect. Because correctness and advice agreement probability are both mutually exclusive binaries, the probability of being incorrect is 1 - the probability of being correct, and the probability of the advice if they are incorrect is 1 - the probability of the advice if they are correct.

$$f'^{\,a}_{\,t} = \frac{i'\ a_t \cdot \omega'^{\,a,a'}_{\,t}}{i'\ a_t \cdot \omega'^{\,a,a'}_{\,t} + (1 - i'\ a_t)(1 - \omega'^{\,a,a'}_{\,t})}$$

The final decision (with confidence) is acquired by reversing the transformation applied earlier:

$$f^a_t = \begin{cases} i < 0.5, 1 - f'^{\,a}_{\,t} \\ i \geq 0.5, f'^{\,a}_{\,t} \end{cases}$$

Final decisions are calculated in the code in the `bayes` function.

Feedback On a proportion of trials a proportion of the agents are randomly selected to receive feedback. The feedback is always accurate, and implicitly trusted by all the agents. When they receive feedback, the agents use that feedback rather than their own final decisions to update their trust and bias. The R implementation is part of the `simulationStep` function.

Bias updating The agents update their bias after each final decision, taking an average of their current bias and the feedback or their final decision, weighted by the agent's bias volatility (λ^a). They therefore adjust their prior expectations on the basis of their experience (and advice). Participants in perceptual decision experiments can update their prior expectations in response to base rates, even when feedback is not provided (Zylberberg et al. 2017). Indeed, (Haddara and Rahnev 2020) suggest part of the effect of feedback is to reduce bias.

$$b'^{\,a}_{\,t+1} = b^a \cdot (1 - \lambda^a) + f^a_t \cdot \lambda^a$$

The bias is clamped to within 0.05 and 0.95 to keep agents at least a little open-minded.

$$b^a_{t+1} = \min(0.95, \max(0.05, {b'}^{\,a}_{t+1}))$$

This is implemented using the `getUpdatedBias` function.

Trust updating The agents update their advice by taking an average of their current trust in their advisor and the advisor's agreement, weighted by their trust volatility (τ^a).

$$x = w^{a,a'}_t \cdot (1 - \tau^a) + {i'}^{\,a,a'}_{t} \cdot \tau^a$$

A tiny cap is used for truncation to prevent agents wholly disregarding or blindly trusting others:

$$w^{a,a'}_{t+1} = \min(0.9999, \max(0.0001, x))$$

Trust updates are accomplished using the `newWeightsByDrift` function.

6.2.3 Model parameters

The models have a large number of parameters. These can be broadly divided into parameters which govern the distributions from which the agents' parameters are drawn, and parameters which define the operation of the model.

These parameters can be seen in full by inspecting the parameters for the `runSimulation` function.

Parameters varied between model runs

There are three parameters which are varied between runs:

- `decision_flags`, whether or not there is time for the random network weights to update before agents' biases begin to update
- W^μ, W^σ, the agents' weighted selection values

The exact specification of how these parameters are varied can be seen in the source code for this document Zenodo thesis code when it's all working.

Table 6.2: Model properties

Property	Description
`n_agents`	Number of agents to simulate
`n_steps`	Number of decisions to simulate
`decision_flags`	Binary flags indicating whether trust and/or bias update at each step
`feedback_probability`	Probability feedback is provided each step
`feedback_proportion`	Proportion of the population receiving feedback when provided
`random_seed`	Seed used for the pseudorandom number generator to allow repetition o
`truth_sd`	Standard deviation for true world values
`confidence_weighted`	Whether agents use their own confidence to modify their trust updates
B^μ, B^σ	Mean, standard deviation of distribution of agents' biases
S^μ, S^σ	Mean, standard deviation of distribution of agents' sensitivities
T^μ, T^σ	Mean, standard deviation of distribution of agents' trust volatilities
$\Lambda^\mu, \Lambda^\sigma$	Mean, standard deviation of distribution of agents' bias volatilities
C^μ, C^σ	Mean, standard deviation of distribution of agents' confidence scaling
W^μ, W^σ	Mean, standard deviation of distribution of agents' trusted advisor pref

Constant parameters

Many parameters are held constant across runs. Within reasonable tolerances, these parameters do not have substantial effects upon the model dynamics. An detailed exploration of the model space is beyond the scope of this work, but all parameters were varied in some way during model development. The main parameters which might vary but are held constant are:

- `feedback_probability` and `feedback_proportion`, which force the agents' biases to remain closer to the optimal value of 0.5. The bias reduction effect is as expected from Haddara and Rahnev (2020).
- B^μ, which increases polarisation and homophily by separating the mean of the agent groups
- B^σ, which has little effect independent of B^μ, but can increase the frequency of extreme and moderate biases
- S^μ and S^σ, which interact with `truth_sd` and C^μ, C^σ to determine agreement rates given a constant bias, increasing or decreasing the power of weighted sampling to shape network dynamics
- Λ^μ and Λ^σ, which increase the speed of the network dynamics (so that similar trajectories occur over fewer generations)

- and `confidence_weighted`, which decreases the speed of the network dynamics.

It is plausible that there are interactions between these and other model parameters that were not detected during model development. The model code is made available to allow others who may be curious about aspects of the model beyond the scope of this thesis to investigate behaviour in these regions of the parameter space. Formal runs illustrating the effects described above are not provided here because they would require a huge amount of computational time to generate for the full models described in this chapter.

Empirically estimated coefficients

The models which have their weighted_sampling value marked as 'emp' are constructed by drawing parameter values for trust volatility (τ^a) and weighted selection (w^a) directly from the parameter estimation coefficients. The weighted selection values in other models are taken from distributions defined by the estimated values for the population, which allows for the generation of an unlimited number of unique participants but means that the agents produced will be more homogeneous in their overall strategies. For example, if a minority strategy exists within the population, the parameter estimate distributions reflective of the dominant strategy would be slightly modified through averaging towards the minority strategy, which may in turn produce agents whose behaviour is not reflective of any plausible real individuals.

Using parameters estimated from actual individuals instead of drawing from a distribution derived form those values has both strengths and weaknesses. The strengths are that the values represent genuine best-estimates of both trust volatility and weighted selection. Given that these two parameters are related to one another, with weighted selection being dependent on trust volatility, it may be important to use observations of both simultaneously to appropriately model individuals' behaviour. The weaknesses are that the task given to human participants differed in potentially important ways from the task modelled in the agent-based model. The most potentially important difference in this respect is the choice of advisor: in the behavioural experiments the human participants were familiarised with two advisors and then given the choice between them; whereas in the model the agents are picking from a large number of potential advisors. The parameter is estimated on the basis of a sigmoid function applied to the trust difference between advisors, whereas it is used in the agent-based models in a half-sigmoid applied to the difference between each advisor and the most trusted advisor.

These considerations mean that the model dynamics arising from using estimated coefficients may be informative, but only in an illustrative capacity. Too much differs between the behavioural and simulated situations to draw strong conclusions.

6.3 Results

6.3.1 Model checks

Verification of this model fitting approach was conducted on the Advanced Research Cluster (Richards 2015) in three ways.

Firstly, simulated participant datasets were generated by using the recovery model in reverse with know coefficients, and then these coefficients were recovered using the recovery procedure. This illustrated that known coefficients could be recovered.

!TODO[Graph with parameter recovery accuracy stuff] !MAYBE[Figure showing the shuffling process?]

Secondly, error scores were compared for coefficients fitted to each participants data and coefficients fitted to versions of that participant's data where the advisor agreement column values had been shuffled. Overwhelmingly, the fitting error on the shuffled data tended to be higher than the fit to the original data, indicating that the models were sensitive to participant patterns related to advisor agreement.

Finally, the error for coefficients fitted to the participants' data were compared to errors for the same coefficients fitted to shuffled data. Error values for the

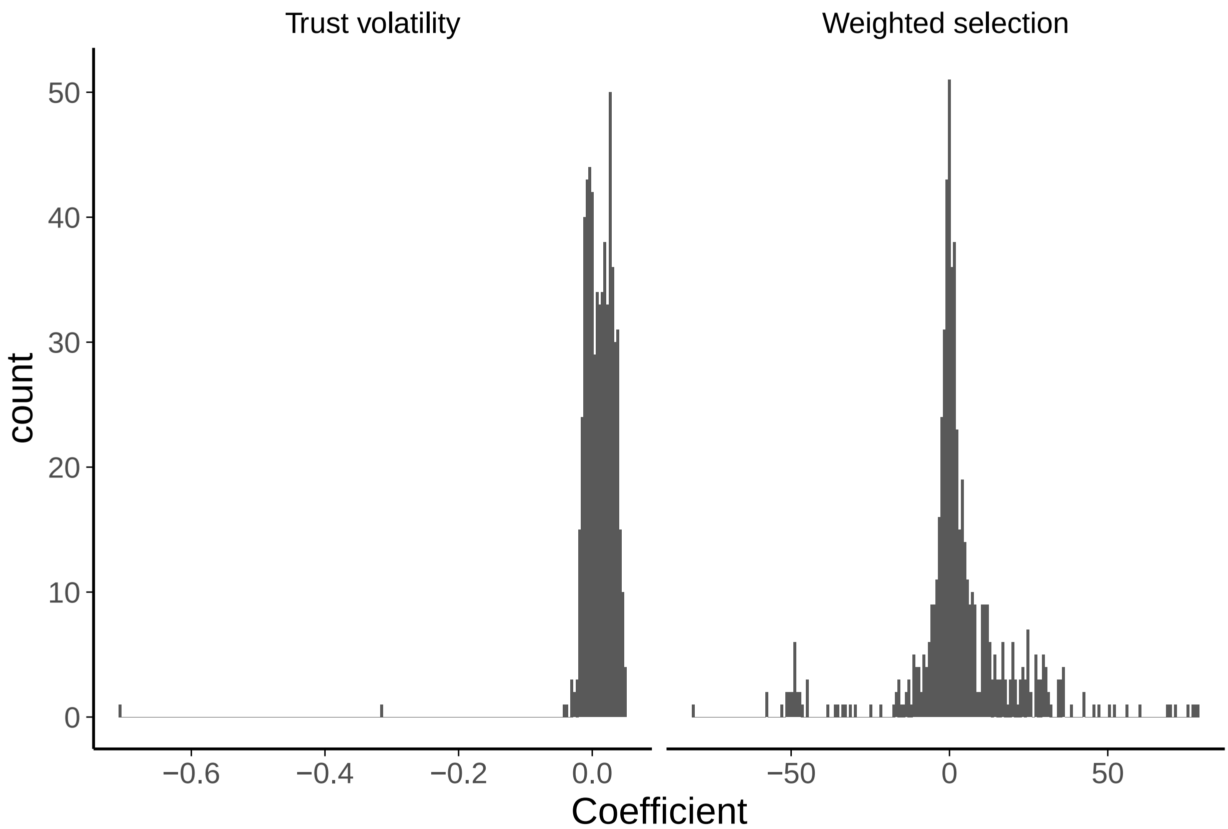

Figure 6.1: Histograms of recovered parameters.

shuffled data tended to be higher than values for the original data, suggesting that participants' behaviour shared some features with the model's expectations.

!TODO[Error distributions for real/shuffled data and shuffle positions]

6.3.2 Participants' coefficients

The mathematical model was fit to participant each participant's data using a gradient descent algorithm. The resulting coefficients were clustered close to zero for most participants, with a roughly gaussian distribution, as shown in Figure 6.1. There was a slight correlation between trust volatility and weighted selection, though this was only significant when excluding two participants with outlying trust volatility coefficients (Figure 6.2).

!MAYBE[Other stuff of interest – maybe we want to relate the variables to participants' ability to do the task or propensity to pick one advisor over another?]

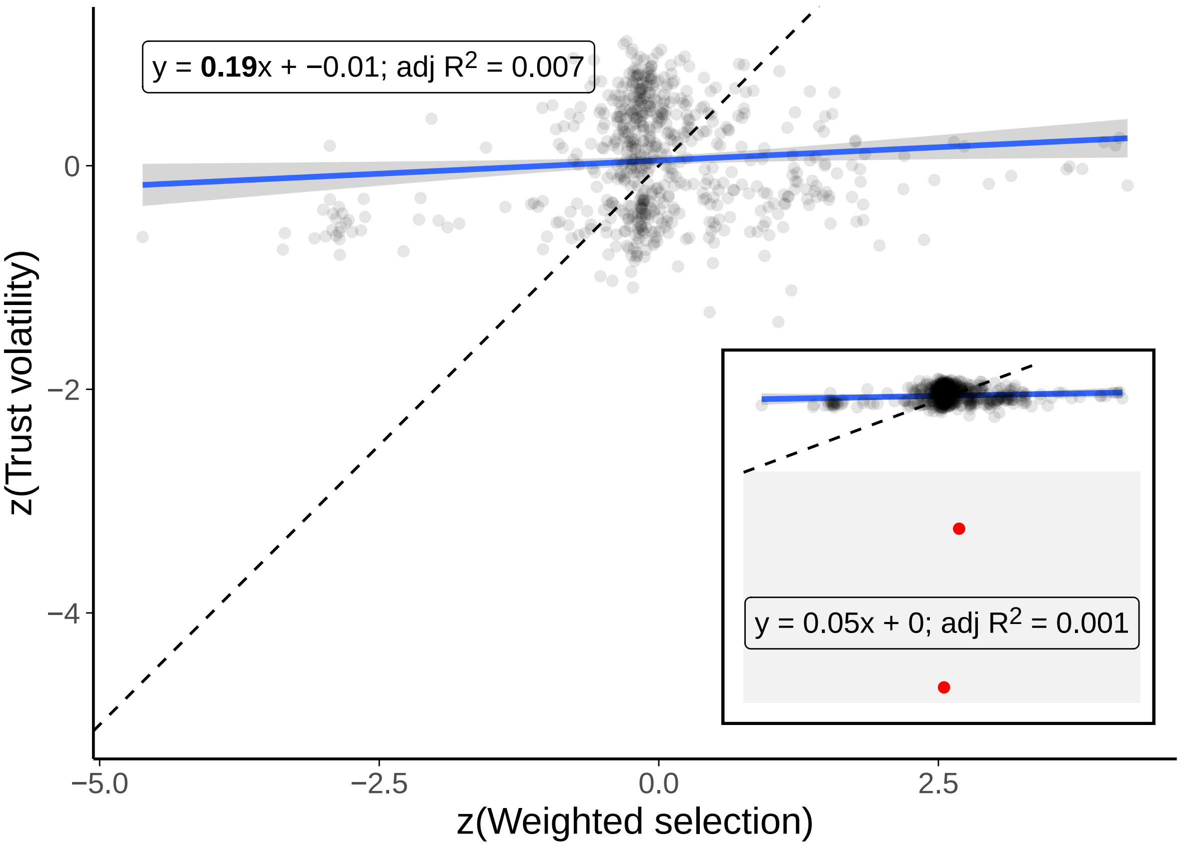

Figure 6.2: Correlation of recovered parameters.
Two outliers with extremely low trust volatility values (z < -5) have been dropped, and are shown on the inset plot for context. Each point is a participant, and the solid blue line shows the overall relationship (with 95% confidence intervals shaded in grey). The formula for the line is given in the top-left. Dashed line indicates a 1:1 correlation.

6.3.3 Validity of the model

The model reproduces key effects of interest. Firstly, agents reinforce one another's biases, leading to the emergence of extreme biases and echoing the effects in the literature. This can be seen in Figure 6.3 where the biases of all but a handful of agents reach and remain at extreme values. Secondly, agents develop greater trust in agents who share their bias, as shown by Pescetelli and Yeung (2021). This can be seen in Figure 6.4 where the correlations between agents' shared bias with an advisor and their trust in that advisor rises over time.

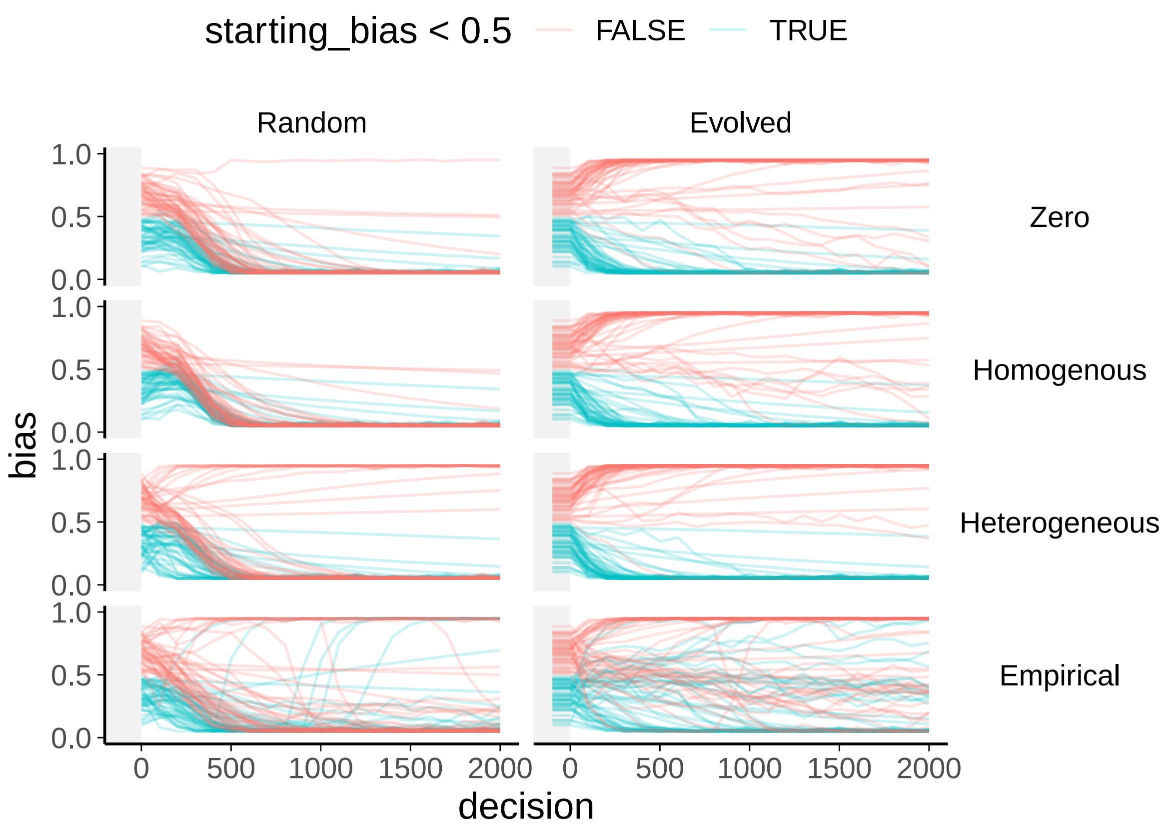

Figure 6.3: Bias evolution within each model.
Each plot shows a model run with a different combination of starting trust weights and weighted selection values. Each line shows the bias of a single agent, coloured according to whether the agent's starting bias was more or less than 0.5. In the Evolved weights graphs the evolution period prior to biases being allowed to shift is truncated (and marked with grey shading).

Random vs evolved trust networks

The model shows a substantial effect of whether or not the agents' trust weights have a chance to evolve prior to biases being allowed to shift. The models in Pescetelli and Yeung (2021) implemented this approach, and, like the current model, demonstrated group polarisation where agents selectively sampled from their own groups and thereby increased their bias.

Where the models are started with random trust weights, agents also tend towards extremes, but there is a single consensus for all agents in the model. When this happens, the population is no more accurate than in other models, but *is* more unified.

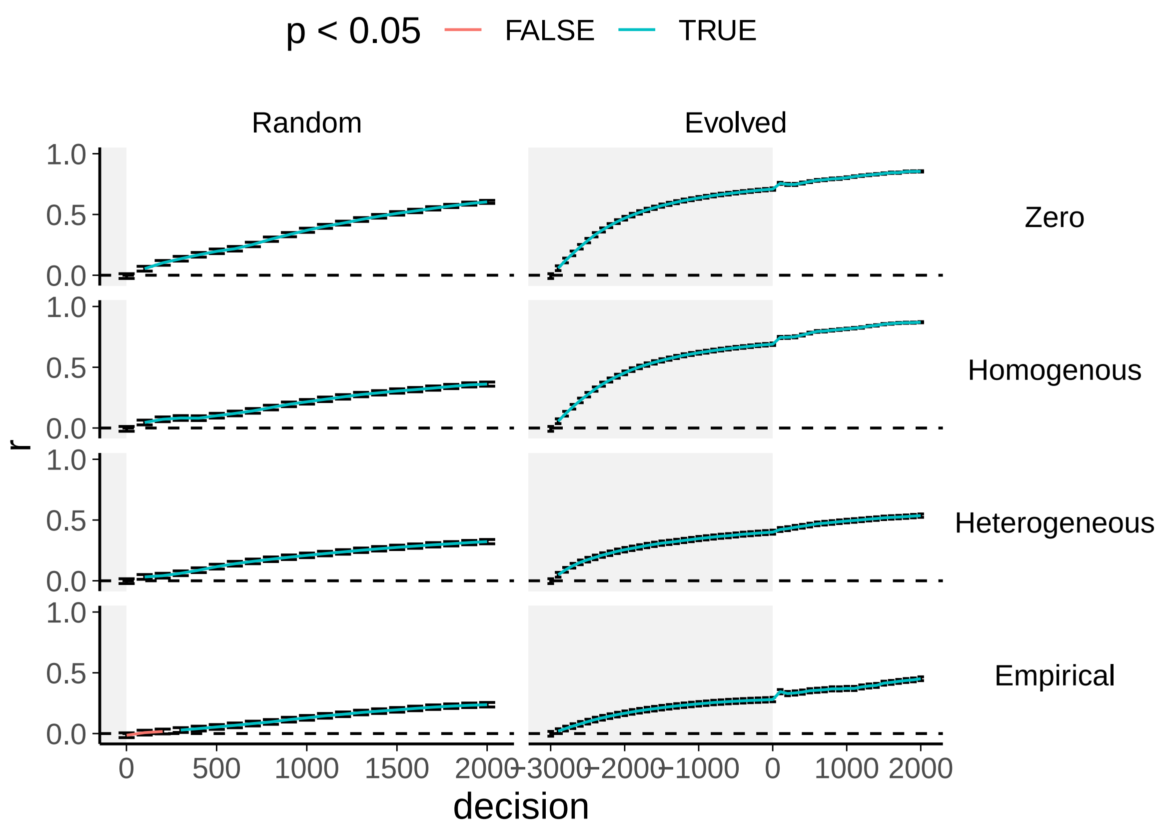

Figure 6.4: Shared bias-weight correlation evolution for each model.
Each plot shows a model run with a different combination of starting trust weights and weighted selection values. Each graph shows the mean correlation between shared weight between agents and trust weights at each decision.

Figure 6.6 shows an overview of this pattern for multiple runs with different random seeds: the evolved trust weight runs tend to show higher group ratios, indicating greater polarisation.

Effect of weighted selection

Providing the agents with weighted selection has little effect, likely because the mean value estimated from the Dots task participants was small. There is virtually no effect on bias trajectories, and only a slight one on the distribution of trust weights (Figure 6.5), with fewer very low weights in the weighted selection runs. The most notable effect was in the correlation between shared bias and trust weight where trust weights are random. In this case, the correlation between shared bias and trust weight does not increase as fast or get as high when weighted

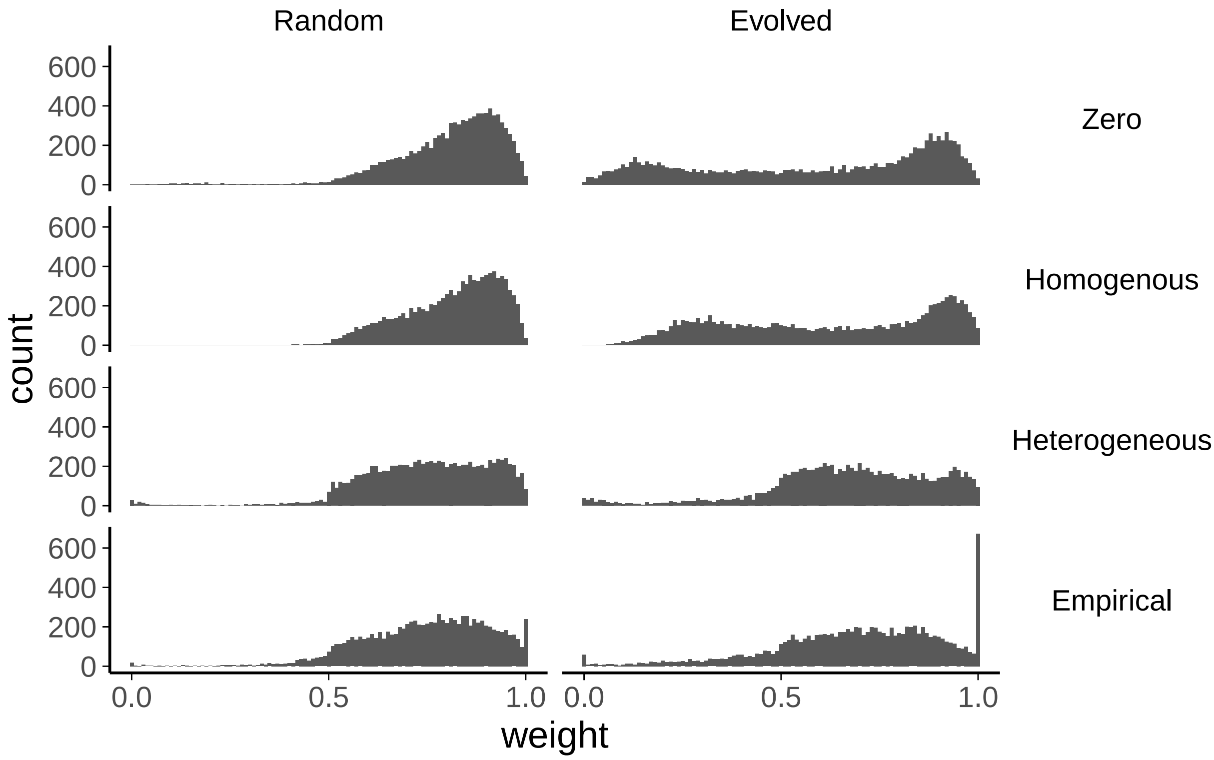

Figure 6.5: Distribution of final weights for each model.
Each plot shows a model run with a different combination of starting trust weights and weighted selection values.

selection is introduced. The group ratio, which is the ratio of agents' weights for their ingroup members to their weights for their outgroup members, is nearly always lower when weighted selection is introduced.

These effects are due to agents becoming increasingly unlikely to sample advice from advisors as the trust in those advisors decreases, which in turn means that only the higher trust weights are likely to be adjusted. In effect, when weighted selection occurs the lower tail is ignored, meaning that trust remains unchanged whether the shared bias is moderate or low, and thus reducing the correlation between them. In this sense, the effects of weighted selection itself are more a feature of the operationalisation than the underlying model.

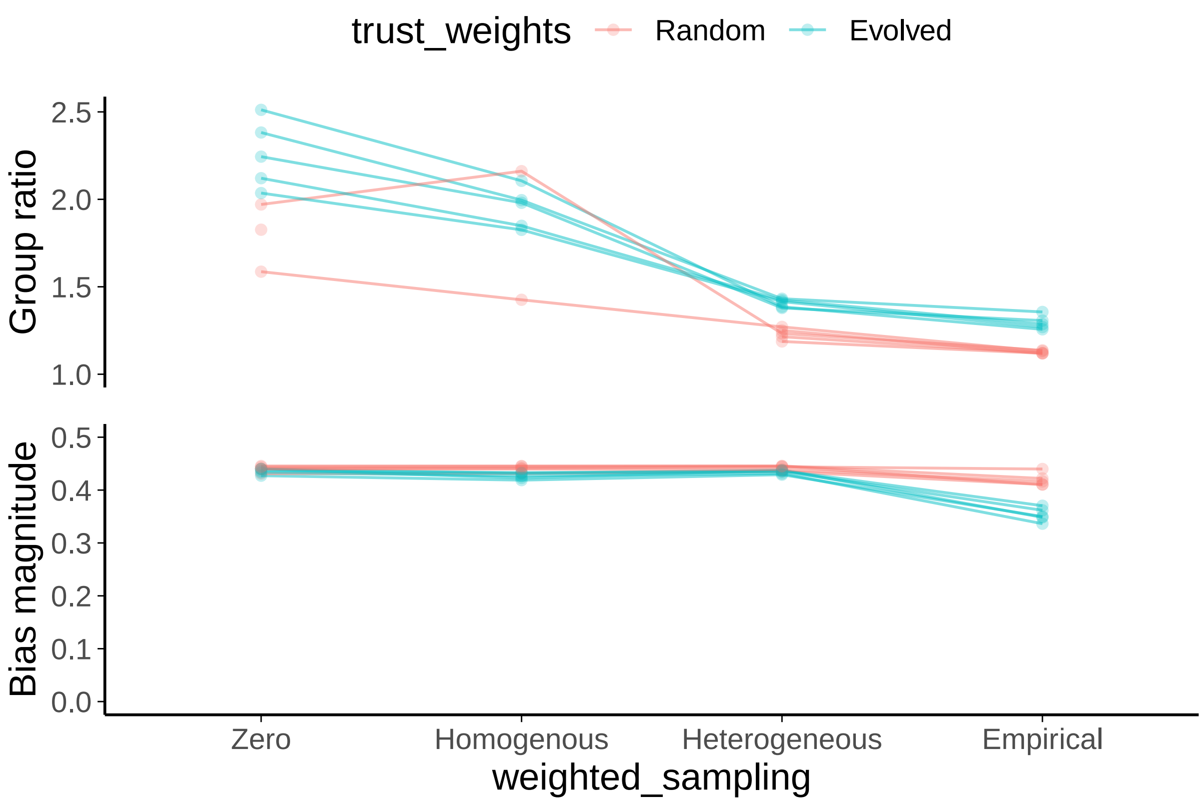

Figure 6.6: Model final decision group ratio and mean final bias magnitude.
Each of the trust weights and weighted selection sampling combinations in the figures above was run multiple times with different random seeds. Each of these runs is represented by a single line. Some models have no group ratio because there is only one group at the final decision (because there is a consensus wherein all agents have the same direction of bias), which means that the ratio of weights is undefined and consequently the line is broken.

Heterogeneity

Allowing heterogeneity in weighted selection coefficients has a more pronounced effect. Whether the heterogeneity comes from sampling from the distribution of estimated participant coefficients (Heterogeneous) or sampling the values themselves (Empirical), varied weighted selection values create subsets of agents who polarise away from the consensus value formed when starting trust weights are random.

There are differences, too, between the Heterogeneous and Empirical sampling approaches. Compared to the Heterogeneous sampling, the model with Empirical sampling shows more moderate biases and more extreme weights (Figure 6.6). This apparently contradictory finding is explicable because agents in the Empirical runs include both positive and negative weighted sampling coefficients. This leads to the peaks of weights at both the 0 and 1 end of the histogram for evolved weights. The negative weighted selection values can also drive agents to sample advice that is likely to disagree with them, moderating their biases as seen by a larger minority of agents with moderate biases in the Empirical run than in other runs. Madsen et al. (2018) termed agents with this advice-seeking strategy 'Socratic' because they aimed to engage with others of a different opinion.

6.3.4 Friends in strange places

Combining the results above, the models raise the interesting observation that consensuses form much more readily where issues arise on which people are likely to encounter different opinions within their circle of trusted informants. It is important to note that, at least in these models, the consensus is not more accurate than the alternative (and is less accurate than the starting positions), but it is nevertheless intriguing to see dynamics that resemble broad versus insular consultation.

Heterogeneous weighted selection can prevent a full consensus forming where individuals have access to trusted advisors of a variety of different opinions by enabling those who have a strong preference for agreement to rapidly disregard the advice of their previously-trusted advisors who offer a different opinion. These agents form their own polarised minority, consulting and advising one another consistent with their bias (as the majority are also doing, just with the other bias).

6.3.5 Consistency of the models

Each model was run 5 times with different random seeds to check which features were consistent across runs, indicating the level of stochastic uncertainty (Bruch and Atwell 2015). The features described above were all consistent across runs, as can be seen by observing the variability in Figure 6.6.

For parameter sets in which all agents' biases tend towards the same extreme, which extreme was favoured varied according to the random seed used. This stochasticity is akin to placing a ball on a gabled roof: tiny variations in the initial conditions affect which way it will roll, but it will always roll down one pitch or the other.

Similarly, in some of these models, adding in weighted selection switches which bias extreme is adopted. This effect is not consistent between runs with different random seeds, and is an outsized effect of minor differences to early states, analogous to a breath of wind nudging the ball in the previous example one way or another.

Output in the style of Figures 6.3, 6.4, and 6.5 for different random seeds can be visualised by tweaking the source code for this chapter.

6.4 Discussion

As with the other simulation-based approaches (§Similar models in the literature), this work showed that echo chambers and polarisation can occur in networks of agents using heuristics that appear from the agent's perspective to be rational. This emergence of collective irrationality from the interaction of rational individuals is similar to the well-known phenomenon of the Tragedy of the Commons, in which individual rewards purchased with distributed costs results in everyone becoming poorer.

6.4.1 Weighted selection

The models indicate that weighted selection, the extent to which agents prefer to receive advice from those whom they consider most trustworthy, has relatively little effect on model dynamics, at least at the level estimated from participant behaviour. The results from more heterogeneous runs discussed below indicate that agents with high weighted selection values do behave in notably different ways. During development we observed that higher weighted selection values (around 10) tended to accelerate the polarisation of the model.

6.4.2 Heterogeneity

Beyond the basic effects of including weighted selection, some models used agents whose weighted selection coefficients were heterogeneous. In both the Heterogeneous and Empirical models, heterogeneous weighted selection coefficients led to the emergence of hold-out groups – small subsets of the population who formed their own minority consensus at the opposite extreme to the majority consensus.

Where the models had evolved trust networks prior to beginning their bias updates, polarisation was common, and heterogeneous weighted selection coefficients sped up this process. They also appeared to make it harder for agents to maintain low biases, although many of the agents in the Empirical model managed to maintain these moderate biases for the duration.

These effects, the presence of hold-out groups and the speeding up of polarisation, suggest that other models of advice dynamics may underestimate the speed with which polarisation occurs while overestimating the degree to which it takes over.

The hold-out groups captured in the heterogeneity models is an observable feature of opinion networks in the real world. For each consensus opinion there is likely to be an outspoken minority who cling resolutely to the opposite opinion. The models presented here cannot tell us how such heterogeneity comes about, but they do underscore its importance in the wider ecosystem of opinion interactions, and invite questions concerning its origins. One suggestion we raise here is that being contrarian may be a stable sub-strategy within an environment where consensus dominates. !TODO[what do we mean by this, exactly? Can we offer an example? Potentially having diverse needs/wants/opinions/tolerances means you can exploit underexploited resources/explore new things. But this seems a bit hand-wavy and not quite what we're modelling...]

!MAYBE[Are these contrarians related to the people who don't take advice in our studies - can we do stats to find out (relate low trust update and/or weighted sampling to low influence? Ideally among people who show a strong preference for one or another advisor to avoid people who just aren't engaged skewing the analysis)]

6.4.3 Considerations

There are several considerations which should be borne in mind while interpreting the results of the models. These considerations fall into several different categories, from technical constraints to limitations on generalisability.

Context

Polarisation and homophily are not inherently bad features. In some discussions (e.g. debates about whether slave ownership is morally permissable) we can be glad that near-universal opinion reigns on the issue, despite its once being a contested issue. In others (e.g. discussion concerning the extent to which the state should be involved in individuals' lives) it is helpful to have a range of opinion to maintain the diversity of approaches (Diamond 1998, p482).

This means that the context must be considered when evaluating the implications of these models. The mathematics and computational results will remain the same, but the real-life implications would change dramatically depending upon the context which the models were chosen to represent[2]. The model presented here could be considered a model of many different kinds of social interactions, from those where consensus is desirable such as norm formation to those where consensus is subordinate to accuracy, such as group and individual decision-making. The implications of polarisation and consensus formation appear beneficial in some cases and harmful in others. The model is sufficiently specified that it is unlikely to fit many interpretations outside of the domain of social influence, but there are plausibly areas in other disciplines which may have a different gloss on this or a very similar model implementing a combination of Bayesian and weighted average integrations and updates.

[2]In one exchange in the literature, Jones (2007) view a model as showing drinking behaviour over a day, while the authors, Mezic et al. (2007), see each tick in the model as representing a day!

Generalisabilty

There are two major question marks concerning the generalisability of the model results. The first is to do with the assumptions of the model, specifically whether the recovered parameters are features of individuals that remain stable across contexts; and the second to do with the results, specifically whether people polarise to the extent indicated by the model.

The first of these generalisability concerns is part of what Bruch and Atwell (2015) term 'input uncertainty'. This is uncertainty over the inputs of a model, in the present case the coefficients recovered from participant data. For each of our participants, we fitted our underlying mathematical model to their behavioural data so that we obtained coefficients that minimised the fitting error across both parameters simultaneously. The existence of such minima, though not unique for every participant, is an observed fact. What is less clear, however, is whether these coefficients represent a stable and enduring feature of the participants or whether they are largely arbitrary – an ephemeral product of time, psychological context, and task. It is a basic assertion of the model that the coefficients observed for an individual do not change during the model: the size of the trust update is the same at decision 1000 as it is at decision 1, and likewise the strength of preference for hearing from an advisor who is more likely to agree. The results of the model can inform us about what might happen if these properties are stable, although even then it is unlikely that the values themselves are exactly accurate to some posited underlying propensity. The constant model parameters were tuned during development to give sensible results given the mean value of weighted selection and the mean and standard deviations of trust volatility, so the performance of the model should not be taken as strong evidence for the existence of the posited psychological features. !MAYBE[can we do a comparison of variability for participants who saw multiple pairs of advisors?]

The second generalisability concern is in the conflict between the model behaviour and the results of research into the level of polarisation and consensus in society. Research consistently demonstrates that most people occupy central positions on political issues. !TODO[Pretty sure this is true, Pew and other surveys usually show bell curve stuff. Sunder Katwala has things on this, but are there non-think-tank publications? Interpretation slightly depends on what we think models are showing - are they showing the drift leftwards of political opinion over time (consensus forming) or giving us misleading ideas about the uninhabitability of moderate areas? This is all pretty complicated because whether people polarise or not and what changes that is a massive open question and partly what my thesis is about. This would be better framed as the debate into which my work feeds in a broader discussion chapter.]

Known psychological aspects which are left out

The modelling work in this chapter follows the recommendations of Bruch and Atwell (2015), who suggest that modelling is best used to characterise the effects on system dynamics of a particular effect derived from empirical observation, and Jackson et al. (2017), who advocate for simplicity where possible. Consequently, many decision-making and advice-taking phenomena reported in the literature are left out. East et al. (2016) argue that an agent-based model only achieves sufficiency by including "all relevant processes and conditions". To allow the reader to determine whether the omissions are important, this section briefly covers those aspects which are left out of the models.

Advice in the models, as in the experiments, is a one-way transaction. The agents giving advice are unaffected by having given the advice, except by the roundabout route of potentially causing other agents to give advice in the future which is likely to make them appear trustworthy to the agent. This means that there is no deliberate reciprocity to advice-seeking behaviours (Mahmoodi et al. 2015), and agents do not alter their advice to appear more paletable to others or so that they are selected as advisors more frequently (Hertz, Palminteri, et al. 2017; Hertz and Bahrami 2018).

Advice is given as an absolute recommendation, as in the Dot task experiments for which the model task in an analogue. This means that !QUESTION[does this actually have a relevant implication?]

The agents use advice rationally, given their estimation of the trustworthiness of their advisor. A plethora of research has shown that people do not do this in practice. Most research has documented processes which result in the underweighting of advice (e.g. Yaniv and Kleinberger 2000, and many others), while some has indicated conditions under which people are over-reliant on advice (Schultze, Mojzisch, et al. 2017; Dietvorst and Simonsohn 2019). This literature is reviewed in the chapter on the Context of Advice.

- Agents don't have a drive for novelty !TODO[Cite something about the attractiveness of novelty for humans; talk about people perhaps picking randomly where they don't have a preference?]
- Agents don't get bored with the task !TODO[Something on attentional effects in psychophysics/cognitive tasks]

Paths not taken

There are several other implementation questions for which empirical evidence is slight or absent, but which represent potentially important decisions and which further extensions may wish to implement.

It is common for models of social interactions to use connectivity graphs that reflect structural features common to social networks. The most common of these structures for the kinds of social interactions modelled here is the small-world network (Watts and Strogatz 1998), in which each agent is connected to several others such that the network is constructed of small clusters with a low average path length (the minimum number of steps required to connect two nodes, for all pairs of nodes in the network). In some cases, the small-world networks also implement a scale-free power law structure, in which a few nodes are highly connected while most are less well connected (Humphries and Gurney 2008). The networks used here do not have these properties, and are instead fully-connected networks, wherein each agent is connected to each other agent, and the likelihood of two agents interacting is goverened by their trust weights for one another. These implementations are not mutally exclusive, but there are several reasons why fully-connected networks are used here rather than small-world or small-world scale-free networks.

Firstly, the impetus for this work arose from Pescetelli and Yeung (2021), in which fully-connected networks were used. Secondly, small-world scale-free networks are not necessary for exploring the role of weighted sampling in agent interactions. Thirdly, using non-fully-connected structures with trust weight updating would mean that agents ought to be able to drop and forge new connections, transforming the conceptualisation about what is being modelled and inviting open-ended questions of how this is implemented and how it relates to trust weight updating. These issues, and similar questions about whether new connections and/or dropped connections ought to prompt reciprocal connections create a fascinating but extremely large problem space. Lastly, using non-fully-connected networks means that it becomes important to understand how an agent's position within the network affects their behaviour (McClain 2016), which requires a full investigation beyond the scope of this thesis.

Another implementation question concerns the behaviour of trust weights for non-selected advisors. In the current model, trust weights for an advisor do not change unless advice is received from that advisor. This is not the only approach. An alternate valid approach would be implementing a small decay of trust over time towards a default value, although because only one advisor is selected at each step this can quite rapidly remove most trust information from the model. A different alternative would be normalisation of trust weights such that the sum of an agent's trust weights for all their advisors remains constant. This would mean that as an advisor becomes more trustworthy they slightly decrease the trust weights for other advisors (either proportionally or equally). In practice this happens in the current model for the likelihood that advisors are selected (based on trust weights and preference strength for more trusted advisors), but not with the trust weights themselves. Whether or not increased trust in one advisor diminishes trust in other advisors is an empirical question (and may change based on context), but we do not expect the results of the model would be substantially different if trust weights were updated in this way.

Differences between experiments and model

For the most part, the task in the model was analogous to that facing the participants in the Dots task experiments. The key difference in task terms was in the choice of advisor. Participants in the experiments were offered a choice of two advisors (with whom they had recently had blocked practice), whereas agents in the model chose freely from all other agents in the model and begin with a knowledge of how trustworthy they consider each of those potential advisors.

Technical constraints

Agent-based models are computationally expensive to run because each step in the model depends upon the step before. This poses some challenges for sources of uncertainty Bruch and Atwell (2015) term 'stochastic uncertainty' and 'model uncertainty'. Stochastic uncertainty refers to the consistency of a model's dynamics when different random numbers are selected in places where they are required. Model uncertainty refers to the consistency of a model's dynamics to variation in parameters other than those selected for exploration. Stochastic uncertainty has been addressed to the extent it is possible given temporal and computational constraints by running each model 5 times with different random seeds.

Model uncertainty is more difficult to address. There are certainly values for model parameters which have substantial effects on dynamics. A trivial example is that when agents are given feedback on every decision they tend towards very low bias magnitudes. Feedback and other parameters were explored to some extent during the development of the model, as documented in §Constant parameters. Despite this, readers should be aware that the parameter space is very large, and there are likely to be substantial interaction effects which were not found during development.

Many models in the literature use very large populations. While the models discussed in §Similar models in the literature had smaller sizes, the largest being

1000 agents, it is not uncommon to see models with 2-3000 agents (Nowak et al. 1990; Duggins 2017; Song and Boomgaarden 2017). The size of the current model runs seems sufficient to illustrate the effects of interest, but it is possible that different effects may emerge in larger populations. Smaller populations examined during testing showed less consistent effects. For very large models the underlying package code would have to be adjusted and the data extracted from model runs sampled much more sparsely.

6.5 How to choose an examiner

- You want them to read your thesis because…
  - You want their insight about some aspect
  - You want to impress them

6.5.1 Internal

- From the department/a closely allied one
- i.e. a psychologist

6.5.2 External

- Fewer constraints
- A couple of choices would be nice

7 Context of Advice

7.1 Egocentric discounting

Egocentric discounting (also known as egocentric 'advice discounting') is a phenomenon wherein advice is under-weighted during integration with the advice-taker's existing opinion, relative to a normative expectation. Most experiments that explore egocentric discounting use the Judge-Advisor System. The Judge-Advisor System has roles for a judge, usually the participant, and one or more advisors, often other participants. In a typical design, the judge offers an initial estimate for some decision, e.g. the total value of coins in a jar of change, then receives advice from the advisor, and then makes a final decision. The difference between the initial estimate and the final decision is taken as measure of how influential the advice was, typically expressed in terms of the contributions of the initial estimate and the advice to the final decision.

In these experiments, the task performance for the advisor is usually as good or better than that of the judge. This performance structure can be well captured for most tasks with Gaussian answer + error distribution where the answer supplies the mean and the error supplies the variance. When combining individual estimates from multiple distributions, optimal results are obtained by weighting the estimates according to the relative precision of their parent distributions (Soll and Larrick

2009; Bahrami et al. 2010). This is analogous to multisensory integration (Ernst and Banks 2002; Körding et al. 2007), and many other cognitive processes argued to be modelled on Bayesian integration, such as those listed in Section 2 of Colombo and Hartmann (2015). Where the performance of the advisor is higher than the judge, the advisor's error will be lower than the judge's, and thus the variance of the advisor's distribution will be narrower and therefore the precision of the advisor's distribution will be higher. When a judge combines their own estimate with that of an advisor who is at least as good, an optimal judge will weight the advisor's opinion at least as highly as their own. !TODO[We'll have got a whole section on models of advice integration stuff, probably, so we can point to that here.] The classic presentation of egocentric discounting is when, in these scenarios, the weight applied to the advice is lower than the optimal weight.

Egocentric discounting is a robust phenomenon in advice-taking. It is not a generic inability to combine estimates: people can accurately combine estimates that do not include their own opinion (Yaniv and Choshen-Hillel 2012), even adjusting for differences in ability between advisors (Soll and Mannes 2011). Similarly, Trouche et al. (2018) showed that when advice and initial estimates were surreptitiously switched, participants discounted their own initial estimates in favour of advice, suggesting that a person's own opinion has a privileged status.

In this chapter, I review the literature on egocentric discounting, with particular attention to manipulations that have been used and explanations that have been offered. I conclude the chapter by expounding an alternative perspective from which to view the phenomenon which, in my view, clarifies the phenomenon and opens the field for a wider range of effective explanations.

7.2 Manipulations affecting egocentric discounting

A sizeable body of research has been conducted into egocentric discounting, using a wide variety of manipulations. These manipulations fit roughly into four categories:

properties of the task, properties of the advice, properties of the advisor, and wider social factors.

Many of the experiments detailed below have not looked at egocentric discounting itself, but have instead looked at changes to the weight given to advice in integrated decisions. While egocentric discounting and advice weighting are inversely related, establishing the degree to which advice has been discounted requires a reference value from which its weighting has deviated. Many experiments do not calculate this normative value, nor even give sufficient details to establish a reference value, either an objective normative value or an expected value from the perspective of the participant.

One obvious reference value is the case where equal weight is given to one's own opinion and the advice received. This is often the value implicitly or explicitly stated. However, equal weighting is only normatively prescribed if advice is exactly as reliable as one's own initial estimate.

Nevertheless, it should be borne in mind, therefore, that depending on the circumstances, very high or low levels of advice weighting might not correspond to very low or high levels of discounting. When Schultze, Mojzisch, et al. (2017) provide participants with advisory estimates from a random number generator, ascribing any weight at all to the advice is suboptimal. Conversely there are many experiments in which expert advice is provided, and in these experiments weighting advice evenly corresponds to egocentric discounting because the advisor's estimates are more accurate on average.

As a consequence of the uncertainty about which normative weighting strategy is required by each experiment (and sometimes this strategy is different from the perspectives of the researcher and the participant), egocentric discounting is primarily examined here in terms of changes in advice weighting. Where advice weighting diminishes, egocentric discounting is said to increase, without specific comment being possible on the exact level of discounting on display. In studies where normative strategies can be determined (e.g. Yaniv and Kleinberger 2000), advice weighting is below that predicted by the normative strategy, indicating egocentric discounting.

7.2.1 Task properties

The properties of the task chosen can affect the levels of egocentric discounting. Task difficulty is a major factor, perhaps mediated by the judge's confidence, but broader features also play a role, including how advice is provided and whether unified estimates are required at any point.

Task difficulty

The most prominent feature of the task which affects egocentric discounting is difficulty. Gino and Moore (2007) asked participants to estimate a person's weight from a clear (easy condition) or blurry (hard condition) picture, and saw less discounting on the hard task. Likewise, Wang and Du (2018) used blurring to increase the difficulty of estimating the number of coins in a photograph of a jar and found that participants discounted less in the blurry compared to the clear condition. Yonah and Kessler (2021) included a condition where there was no objectively correct answer in a random dot motion display, and saw increased willingness to take advice compared to conditions where participants were on average 67%. They also found that participants were less likely to seek advice where they rated their performance on the task as more competent, i.e. where they experienced the task as easier.

Judge's confidence

Wang & Du saw full mediation of their difficulty manipulation by participants' confidence on the task, while Gino and Moore saw only partial mediation. Other studies have manipulated the judge's confidence through other mechanisms. See et al. (2011) used a power manipulation which was effective in part through raising judges' confidence; and Gino, Brooks, et al. (2012) used anxiety manipulations to decrease judges' confidence. In both cases, partial or full mediation through confidence occurred such that participants' higher confidence in themselves and their decisions was associated with greater egocentric discounting. In many other experiments, including other experiments in Wang and Du (2018), confidence is not manipulated but is still associated with greater egocentric discounting. Using a similar methodology, but a different analytical approach, Moussaïd et al. (2013) observed that highly confident participants rarely updated their views following advice.

Judge-Advisor System structure

More complex task designs, in which reflection and discussion are encouraged, can reduce discounting. Minson, Liberman, et al. (2011) and Liberman et al. (2012) asked dyads to take simultaneous roles as judge and advisor, providing initial estimates, exchanging advice during a discussion, and then providing final decisions on estimation tasks. Discounting was reduced, but still evident in this process, as it was in Van Swol (2011), which used a traditional Judge-Advisor System paradigm where advice was delivered face-to-face. Liberman and colleagues did manage to eliminate discounting where, between exchanging advice and providing a final decision, participants produced a single mutually satisfactory collaborative judgement, and showed that the value of this collaborative judgement was itself improved by open-minded discussion over justifying estimates or exchanging bids. Schultze, Mojzisch, et al. (2017) demonstrated that asking judges to selectively generate reasons why the advice might be correct or incorrect led to lower and higher levels of egocentric discounting respectively.

7.2.2 Advice properties

Several features of advice itself have been explored: the confidence of advice, its similarity to the initial decision, whether it is solicited, and the amount of it provided.

Confidence of the advice

When judges are more confident, they tend to be less influenced by advice, and the expected corollary of this is that when advice is expressed more confidently the advice will be more influential. Soll and Larrick (2009) measured the confidence of advice and saw that higher advice confidence was associated with higher influence of advice. Moussaïd et al. (2013) also found that differences in confidence between judges' and advisors' estimates were useful in producing a decision tree determining the extent to which advice was taken.

Similarity of advice to the initial estimate

A frequently-manipulated property of advice, and the most interesting in the context of the first part of this thesis, is the similarity of the advice to the initial estimate. This is sometimes expressed as agreement or reasonableness of advice.[^Note that the influence of advice that is perfectly in accord with an initial estimate is undefined when using §Weight on Advice measures.] The evidence on the effects of advice distance on advice influence is equivocal. Some studies show that advice is less influential the further it is from the initial estimate, while other studies show a greater influence of more distant advice. Other studies have indicated that the relationship is quadratic: low weight is assigned to advice which is too near or far from the initial estimate, and a greater weight assigned to advice which is in the moderately distant.

Several studies have provided evidence for a simple agreement effect whereby advice that is nearer to the initial estimate is more influential. Yaniv (2004) manipulated advice to be nearer to or further away from the initial estimate and saw that the influence of advice decreased as the advice was further from the initial estimate (although this pattern did not hold for low-expertise judges in one experiment). Minson, Liberman, et al. (2011) found that more distant advice was associated with less advice-taking behaviour once average distance between dyad members was controlled for, although their results were not expressed using standard advice-taking metrics. Yaniv and Milyavsky (2007) observed that advice closer to the initial estimate was also more influential when combining multiple pieces of advice simultaneously.

The opposite effect was demonstrated by Hütter and Ache (2016). They found consistently higher influence for advice that was further from the initial estimate, both for single pieces of advice and for integrating multiple pieces of advice.

A non-linear, U-shaped relationship between advice distance and advice influence has been shown in other studies. Moussaïd et al. (2013) asked participants to estimate answers to a variety of questions and give confidence ratings, both before and after receiving another person's initial estimate as advice. They identified a three-zone structure to the influence of advice according to the distance between the initial estimate and the advice. Similar advice fell into the 'confirmation zone', where opinion was unchanged but confidence increased; moderately distant advice fell into an 'influence zone' where opinion changed to accommodate the advice; and distant advice was generally ignored. Likewise, Schultze, Rakotoarisoa, et al. (2015) showed in an elegant series of Judge-Advisor System experiments that relationships between egocentric discounting and advice distance were U-shaped. Advice was given very little weight where it was close to the initial estimate, with the weight then increasing precipitously to a peak for middle-distance advice, after which it either remained stable or decreased slightly as the distance increased further. Schultze et al. also showed that confidence in final decisions was dramatically boosted by near advice, and that confidence gains decreased sharply with distance, consistent with Moussaïd et al.'s account.

Hütter & Ache used two discrete advice distances in each of their experiments, and thus could not demonstrate a U-shaped relationship with only two points. Hütter & Ache's results can be considered consistent with the studies demonstrating a U-shaped relationship, although there still remains a difficulty in integrating the results of these studies with those showing a monotonic decrease in advice weight by distance. This is consistent with the argument that high-confidence decisions reduce the processing of disconfirmatory evidence (Rollwage et al. 2020). This mechanism will function more effectively if there is a bias for processing nearby advice before more distant advice.

This mechanisms can be extended to explain the results in Yaniv (2004) (in which there is only a single piece of advice) by allowing the updating of confidence and the adjustment of the estimate to be separate processes. If updating is a two-stage process wherein confidence is updated first, and then the updated confidence value is used to determine the extent to which the estimate is adjusted, we can reproduce the U-shaped relationship from two monotonic relationships. First, confidence is increased by nearby advice (and perhaps decreased by distant advice). Second, advice is more influential the nearer it is to the initial estimate. This means that, for nearby advice, the judge's confidence is boosted by agreement, meaning that the ostensibly highly-influential nearby advice is subsequently assigned very little weight (because highly confident judges ascribe little weight to advice - §Judge's confidence). For advice that lies at a middle distance, there is no boost to confidence and advice is somewhat influential. At greater distances, while the judge's confidence in their own opinion might be reduced slightly, there is little potential influence in the advice.

This unifying explanation is consistent with Moussaïd et al.'s demonstration that the effects of advice on confidence and estimate were somewhat separable. Nevertheless, it is somewhat unintuitive that the metacognitive effects of advice on confidence would happen prior to the cognitive effects of advice on estimate updating. The Yaniv (2004) results are also not wholly explained by this kind of process because Schultze, Rakotoarisoa, et al. (2015) included an experiment designed to reproduce Yaniv's methodology as closely as possible, the results of which were inconsistent with Yaniv's.

Related to the distance of advice is the reasonableness of advice, because where the judge has a somewhat reasonable estimate the distance serves as a reliable proxy for reasonableness. Gino, Brooks, et al. (2012) included an experiment in which (non-anxious) participants heavily discounted unreasonably high and unreasonably low advice, while discounting reasonable advice at a rate typically seen in Judge-Advisor System experiments. Similarly, Schultze, Mojzisch, et al. (2017) saw judges discount wildly implausible advice more heavily, although it was still assigned some weight, even when labelled as coming from a random number generator.

The level of agreement may act as a cue to the plausibility of both the advice and the initial estimate. Somewhat counter-intuitively, this can lead to agreeing advice being assigned less weight than we might expect because of its bolstering of the confidence in the initial estimate.

Solicitation of advice

The extent to which advice is discounted may also be related to whether advice is wanted. This is hard to disentangle from the effects of task difficulty, because people are more likely to seek advice when they find the task more difficult to do (and hence their confidence in their response is lower).

Gino and Moore (2007) compared advice-taking behaviour across two experiments using a task in which participants had to estimate people's weight from photographs. In one of these experiments participants received advice automatically and in the other participants had the option of clicking a button to receive advice. Participants opted to receive advice on almost all trials, including in an easy condition, and no differences were found in the extent to which advice was used between the compulsory and optional advice experiments. The very high rates of advice seeking in the optional experiment suggest that participants in the compulsory advice experiment may have been very welcoming of the advice due to the difficulty of the task. Gino (2008) showed that more expensive advice was sought less frequently but used more heavily, that expensive advice was used more heavily than free advice even when both are compulsory, and that paid-for advice was used more heavily than when the same advice was given for free as the result of a coin-flip. This study packaged questions together in blocks, and participants purchased advice for a whole block at once. This procedure means that the solicitation of advice is decoupled from the difficulty of the question on a trial-by-trial basis, although it is still likely that those participants who found all the questions in a block more difficult were more likely to solicit advice. In the real world, price is often an indicator of quality, albeit an imperfect one, and solicitation is likely to act as a proxy for confidence, which is again related to task difficulty: participants in the experiments may have taken more advice because they were less sure on the blocks in which they sought advice. !TODO[Michael and/or Naomi worked on seeking advice more when less confident? That's relevant here] Hütter and Ache (2016) found that participants opted to see a large number of advisory estimates in a calorific content estimation task when allowed to sample ad lib, although the influence of the advice was low.

Number of advisory estimates

Hütter and Ache (2016) !TODO[There are definitely others! Yaniv for a start? Find them. E.g. @yonahTheyDonKnow2021] saw levels of advice usage for multiple pieces of advice which were relatively similar to levels of advice usage for a single piece of advice. In other words, people integrating their own opinion with two advisory estimates tend to integrate the advisory estimates and then treat them as a single piece of advice for integrating with their own opinion. This may be an artefact of presentation because all these studies presented multiple estimates simultaneously as a list. If this is a real phenomenon, however, it is a critical bias: while sensible motivations for favouring one's own opinion over another's will be posited below, it is far harder to argue that the weight assigned to one's own opinion should remain the same whether being integrated with one other estimate or ten other estimates.

7.2.3 Advisor properties

A common strategy in advice-taking experiments is to manipulate the properties of the advisors (either within- or between-participants). Expertise is often manipulated, but some research has investigated factors such as the pre-existing relationship between judge and advisor, whether advisors are human or algorithmic, and whether advisors have a clear conflict of interest.

Expertise of advisors

By far the most frequently manipulated property of advisors is their ability to perform the task in question, known as their expertise. Advisor expertise can be communicated to participants in various ways, which can be broadly categorised into ‘showing’ approaches in which participants build up a picture of advisors’ performance over a series of instances, and ‘telling’ approaches where participants are presented with a summary of an advisor’s performance. In approaches towards the middle of the spectrum participants are offered evidence of the advisor’s performance in one go, e.g. a scorecard showing an advisor’s performance on practice questions. In some experiments advisor expertise manipulations are genuine, whereas in others they are merely apparent and the participants receive equivalent advice regardless of the label or historical performance of the advisor. As noted above, the ability of the advisor to perform the task (relative to the judge) alters the optimal level of advice-taking according to the normative model. This means that discounting may still occur even when the advisor’s estimate is weighted more highly than the judge’s.

Yaniv and Kleinberger (2000) showed in a series of historical date estimation experiments that better-performing advisors were more influential than worse-performing advisors, especially where feedback was provided on judge’s final decisions but also where it was not. Gino, Brooks, et al. (2012) used a coin-jar estimation task where participants were shown the advisor’s past performance to demonstrate that people give the same advice more weight if it comes from an advisor with a history of good performance, although this effect was not visible in an anxiety arm of the experiment. Rakoczy et al. (2015) saw that 3-6 year-old children took advice from advisors who had named animals correctly more seriously than advice from advisors who acknowledged their own ignorance in a version of the Judge-Advisor System adapted for young children.

Sniezek, Schrah, et al. (2004) saw greater dependence on advice provided by specially-trained ‘expert’ peers, but only where the proportion of reward money for accurate judgement paid to the advisor had been decided *before* advice was provided. Soll and Larrick (2009) observed greater influence of more expert advisors across three experiments in which expertise was signalled by familiarity ratings with a university for which the graduate salary was being estimated, country of origin in a geography knowledge task, and confidence in a set of trivia questions. Tost et al. (2012) used a weight-estimation task and observed that greater weight was placed on advice from advisors labelled as experts compared to the same advice from advisors labelled as novices. Schultze, Mojzisch, et al. (2017) labelled advisors using a ranking system and saw that participants placed a higher weight on advice from highly-ranked advisors compared to low-ranked advisors, although the actual advice was (unbeknownest to the participants) the same. Wang and Du (2018) showed that participants placed greater weight on advice from expert as opposed to novice advice in a coin-jar estimation task using advice that was genuinely expert or novice.

Önkal, Gönül, et al. (2017) reported a series of experiments using a stock price estimation task in which advisor expertise was manipulated using both labels and experience. Where experience alone was provided there was no clear effect of expertise, while labelling had a substantial effect. Further experiments crossing labelling (high or low expertise) with experience (of high or low expertise) indicated that participants responded to both labelling and experience in a rational way: the most influential advisors were both labelled and experienced as experts, followed by those labelled as having low expertise but demonstrating high expertise. Both advisors whose expertise was experienced as low were weighted less heavily, with no clear differences between them. [1]

Familiarity of advisors

So far as I can ascertain, no one has reported on a Judge-Advisor system in which comparable advice is received from friends and non-friends. Sniezek and Van Swol (2001) and van Swol and Sniezek (2005) found a correlation between the amount classmates had interacted and ratings of trust in those classmates as advisors, but they did not use a measure of advice-taking which allows calculation of advice weighting. Minson, Liberman, et al. (2011) had long-term dance partners make estimates about their own dance performances in relation to professional

[1] Numerically, there was an indication of a backlash effect whereby the advisor labelled as high-expertise but demonstrating no expertise in practice was less influential than an advisor labelled as low-expertise and experienced as performing similarly.

assessment, and saw normal levels of egocentic discounting. [2]

It seems intuitive that advice will be weighted more highly when coming from people we know, but this does not appear to have been tested. With regard to the three-factor model of trust put forward in Mayer et al. (1995), knowing and trusting an advisor would be expected to increase the extent to which the advisor's advice is taken.

Humanity of the advisor

Some experimenters have examined advice weighting for non-human advisors. In a stock-market forecasting task, Önkal, Goodwin, et al. (2009) found that judges placed less weight on (identical) advice when it was labelled as coming from a statistical model versus a human expert forecaster. In their study on overweighting, Schultze, Mojzisch, et al. (2017) provided participants with advice labelled as coming from a random number generator and noted that its estimates were still assigned some weight by judges. The influence of this randomly-generated advice was roughly equivalent to that of human advisors not labelled as having high expertise.

Advisor conflict of interest

Where advisors have a conflict of interest, following the advice may benefit the advisor at the expense of the judge. Gino, Brooks, et al. (2012) observed that non-anxious judges assigned less weight to advice from advisors with a conflict of interest, while anxious judges assigned similar weight regardless of conflict of interest. Bonner and Cadman (2014) similarly saw less influence from advice from advisor with a conflict of interest in a CEO-remuneration task.

7.2.4 Wider social factors

The Judge-Advisor System in experiments is usually quite abstracted away from advice-taking in the real world, in which broader social factors are likely to offer a

[2] The partners' estimates in this study were highly correlated, meaning that they were not independent and thus did not necessarily bracket the correct answer. This means that both partners assigning positive weight to the other's advice is not likely to approach the correct answer.

highly influential context dictating or suggesting norms for advice-taking. Despite being somewhat shielded from these broader factors, they are nevertheless apparent in some Judge-Advisor System experiments.

Fairness may be a fairly universal value !TODO[Do we need to back this up? There'll be some cross-cultural studies to reference, and even de Waal's research on fairness in non-human primates.], and it is certainly a lauded value in the developed Western societies where the majority of Judge-Advisor System research takes place. Consistent with this, judges in the Sniezek, Schrah, et al. (2004) study tended to offer both novice and expert advisors equal shares in their reward money. This sense of fairness is seen to extend to advice-weighting, too. Harvey and Fischer (1997) saw expert judges consistently placed some weight on novice advice, and attributed this advice-taking behaviour to a social requirement for fairness. Likewise, Mahmoodi et al. (2015) showed that dyads making perceptual decisions would consistently over-weight the estimate of the less accurate dyad member despite evidence that this impaired performance and thus decreased the dyad's reward money.

Feeling powerful was observed to lead to decreases in advice-taking by See et al. (2011) and Tost et al. (2012), with the latter showing a partial mediation via the judge's confidence. Somewhat similarly, Gino, Brooks, et al. (2012) saw angry judges took less advice through being more self-confident, while anxious judges sought and took more advice through being less self-confident.

Finally, developmental maturity seems to be an important factor. Rakoczy et al. (2015) found that while 3-6 year old children did differentiate between knowledgeable and ignorant advisors, they nonetheless placed exceptionally high weight on advice from both advisors.

7.3 Purported explanations for egocentric discounting

Almost as numerous as the studies into egocentric discounting are the explanations offered to account for it. Despite this, no explanation has managed to withstand critical empirical scrutiny. Below, I offer a brief review of the explanations which have been put forward to date.

7.3.1 Egocentric bias

Among the earliest explanations for egocentric discounting was egocentric bias: the belief that one's judgement is superior to that of others. Harvey and Fischer (1997) attributed egocentric discounting to such self-serving estimates of ability, such as when car drivers report on average being better than average (Svenson 1981). The explanation also suggested a mediating role of overconfidence, tying it neatly into later similar explanations from See et al. (2011), Gino, Brooks, et al. (2012), and Tost et al. (2012). For all these authors, a judge's confidence in the initial estimate is the principal driving force behind the weighting of advice. This view is appreciable from a Bayesian perspective on advice integration (Bahrami et al. 2010): as with multi-sensory integration (Fetsch et al. 2012), informational cues coalesce optimally onto the correct answer where they are weighted according to their precision. In such a framework, there is an optimal (Bayesian) integration process which is being fed faulty inputs (because the judge's own estimate is believed to have erroneously high precision).

Overconfidence certainly seems to have a role, and accounts well for many of the nuances in advice-taking experiments including the findings that advice is taken more readily for difficult tasks (where judge confidence is lower) and that judges made to feel more powerful (and hence more confident) take less advice. It is unlikely to be the whole story, however: Trouche et al. (2018) note that Soll and Mannes (2011) were able to disentangle ratings of ability and advice-taking behaviour and saw that the egocentric discounting occurred even if the judge's assessment of relative ability were taken as true.

7.3.2 Access to reasons

One of the most influential explanations of egocentric discounting has been Yaniv's argument that judges have greater access to the reasons justifying their own decisions than to those justifying the decisions of others, due to the opaqueness of other minds (Yaniv and Kleinberger 2000). This differential access to reasons suggests, from the perspective of the judge, that there is greater evidence favouring the judge's own opinion than favouring the estimate of their advisor, and that, analogous to the confidence case above, the more well-supported opinion should be given a greater weight during integration.

Trouche et al. (2018) presented judges with their own estimates labelled as advice, and with advice labelled as their own estimates, and observed that judges persisted in placing greater weight on what they *believed* was their own initial estimate rather than what was *actually* their own initial estimate. This result is a serious problem for the access-to-reasons explanation, because there is no good reason why simply relabelling estimates should change the judge's internal census of evidence supporting the estimates.

7.3.3 Anchoring

Some researchers have argued that egocentric discounting can be explained by anchoring. Anchoring is a well-established phenomenon whereby numbers clearly unrelated to a numerical estimation task can nevertheless bias estimates, as when participants asked to estimate the height of Mount Everest in feet but first asked to say whether '2,000' is higher or lower than the correct value (12,000) give far lower estimates than participants asked to say whether '45,500' is higher or lower than the correct value (Jacowitz and Kahneman 1995). Bonner and Cadman (2014) suggested that anchoring was responsible for judges' over-use of outlandishly extravagant suggestions for CEO remuneration. In a more thorough set of studies, Schultze, Mojzisch, et al. (2017) observed a consistently greater-than-zero weight on transparently useless advice, which they ascribed to anchoring to the advice. While these cases for anchoring may be admitted, they are to do with over-weighting advice rather than the under-weighting advice which characterises egocentric discounting. This is because the putative anchor is the advisory estimate.

Historically, it has been suggested that the judge's initial decision acts as an anchor (Harvey and Fischer 1997), but this was later ruled out when Harvey and Harries (2004) demonstrated egocentric discounting persevered when the labels for the judge's initial estimate and the advisor's advice were switched. If anchoring to the initial estimate were responsible for egocentric discounting the relative weighting of initial estimate and advice would have followed the actual values rather than the labelled values, whereas the data showed that discounting occurred towards the *labelled* rather than *actual* initial estimate.

7.3.4 Sunk costs

Another general cognitive bias recruited as an explanation for egocentric discounting is the sunk costs fallacy, in which one perseveres with a poor strategy in order to justify the cost or effort that has already gone into pursuing it. Interestingly, sunk costs have been recruited to explain both egocentric discounting *and* following advice.

Gino (2008) had participants receive advice for free or for a fee depending upon the outcome of a coin flip. They found that the same advice was more influential where payment had been taken, and that the more expensive the more influential it was whether paid for or free. Assuming that participants viewed the greater cost as a marker of quality, the remaining effect contingent on whether or not payment had actually been taken can be explained by sunk costs. Similarly, Sniezek, Schrah, et al. (2004) found that, at least for expert advisors, judges who allocated a portion of their prospective reward money to the advisor *before* receiving the advice placed more weight on that advice.

Ronayne and Sgroi (2018) also invoked the sunk costs fallacy, but suggested that it could account for using less advice, i.e. discounting. These authors presented participants with the opportunity to use another participant's results rather than their own in a reward lottery, which they somewhat oddly labelled 'advice'. Despite this 'advice' being transparently better, participants frequently chose to keep their own results rather than switch, and this finding is explained on the basis that those participants were loath to forfeit the work they had done to obtain their own results by adopting another's. Extended to the Judge-Advisor System, this explanation would predict that more effortful tasks would lead to greater egocentric discounting. This is an intriguing prediction, but perhaps because effortfulness tends to covary with difficulty (which in turn decreases egocentric discounting), it does not appear to have been studied.

7.3.5 Naïve realism

A third cognitive bias invoked to explain egocentric discounting is naïve realism. Naïve realism occurs when people treat their own perceptions as reflective of shared underlying reality and others' perceptions as misguided or biased to the extent that they do not agree. Minson, Liberman, et al. (2011) argue strongly for a naïve realism explanation of egocentric discounting, although it is never fully explained why, on a naïve realism view, *any* adjustment to advice is warranted (because ex hypothesi the initial estimate reflects the true answer). The most creative element of the experiments involves funnelling integrated decisions (corresponding to judges' final decisions in the typical Judge-Advisor System) through a collaborative joint decision process before extracting a final individual decision. Naïve realism once again fails to provide a compelling explanation why the final individual decisions closely reflect the collaborative join decisions rather than the initial estimates: rather than continuing to endorse their own view of 'reality', participants appeared to be willing to accept the joint view once it had been established.

7.3.6 Responsibility / feeling of deserving outcomes

Some advice taking may be explicable on the basis of responsibility sharing. Harvey and Fischer (1997), Mahmoodi et al. (2015), and Ronayne and Sgroi (2018) have all suggested that taking advice can transfer some of the responsibility for the outcome of a decision onto the advisor. Where uncertainty is high, or where rewards are shared, this can be particularly useful.

While distribution of responsibility is more a reason for *reduced* rather than *increased* egocentric discounting, when combined with an account that predicts ascribing very low or no weight to advice by default (e.g. naïve realism), it can explain why advice weighting is higher than the zero that would be expected. It seems more plausible, however, that factors which promote advice-taking such as fairness, advisor expertise, and distribution of responsibility serve to place a limit on egocentric discounting, rather than that complete discounting is a default strategy from which these factors move judges.

7.3.7 Wariness

Trouche et al. (2018) designate the above explanations ‘proximal explanations’, because they offer a mechanistic account of how discounting occurs. Rather than partaking in this discussion, they instead provide an ‘ultimate explanation’, which may explain why discounting occurs. They suggest that discounting occurs because advisors’ interests do not always align with judges’, and thus some level of discounting offers protection from relying too heavily on advice that may be deliberately harmful. They suggest thus that discounting is an evolved response to misaligned incentives between judges and advisors. The account I offer below is in the mode of this explanation.

7.4 A wider view of egocentric discounting

The explanations outlined above are all explanations pitched at the level of are all offered as explanations for a deviation from normative optimality. I have chosen to take a different approach, asking instead under which circumstances the observed behaviour would be an optimal policy, and exploring the plausibility of those circumstances continuing to have an influence in experimental settings where the normative behaviour might be averaging one’s own opinion with advice.

As a starting point, I note that if one tells anyone who is not an advice-taking researcher that people do not take others’ opinions as seriously as they take their own when making decisions, the response is likely to be a flat “of course”, perhaps accompanied by a perplexity as to why such an obvious statement is being presented as a valuable insight. The approach taken here works to codify this intuition as a set of hyperpriors: expectations about the relative utility of advice as compared to one's own opinion. I argue that the normative model of advice-taking, and the pared-down experimental design with which is it entwined, seek to take the situation out of the evolutionary history of advice, but cannot take the evolutionary history of advice out of the situation.

According to this hypothesis, the hyperpriors on advice-taking are a consequence of the many opportunities for deception and misunderstanding which apply to advice but not to one's own opinion. The most obvious of these is the opportunity for deception. As Trouche et al. (2018) point out, advisors do not always have the judge's best interests at heart when supplying advice. Consider, for example, a situation where one coworker, Sally, asks another, Hanan, whether it is a good idea to apply for a promotion. Hanan may think it would be good for Sally to apply, because Sally is well-qualified and hard-working, but nevertheless discourage Sally from applying because Hanan herself is going to apply and wishes to reduce the competition.

It is not necessary, however, for there to be misaligned incentives of this kind. Advice may be less informative than one's own opinion where the advisor is less able to perform the task, or does not perform the task as effectively. Consider, for instance, Sally asking Hanan for advice on where to go on holiday. Hanan may wish to maximise the probability Sally has a wonderful holiday by offering the best advice possible, but, because Hanan does not have a perfect knowledge of Sally's preferences, nevertheless advice Sally to select a non-optimal destination. Likewise, Sally may well have spent considerable time researching and thinking about the question, and it is not reasonable to believe that Hanan would do likewise because it is, after all, Sally's holiday.

There is room for misunderstanding even where an advisor's interests are aligned with the judge's and the advisor's ability equals the judge's. Advice must be communicated to the judge, and communication in the real world is inherently noisy. Communication of advice requires something in the mind of the advisor to be encoded into a set of signals, transmitted to the judge, and then reencoded into something in the mind of the judge, at which point it can be integrated with what the judge already believes. Information can be degraded at any of these steps, resulting in advice that is less informative than the judge's own opinion. When Sally asks Hanan what to do about a work problem, and Hanan rapidly and confidently rattles off a suggestion, Sally may be forgiven for thinking the rapidity and confidence are a property of Hanan's confidence in the suggestion rather than an underlying characteristic of Hanan. If Sally does not adjust for the fact that Hanan is always more confident about things than Sally is, then Hanan's suggestion will be overly dominant relative to its informational value.

7.4.1 Compatability with existing explanations

As noted by Trouche et al. (2018), ultimate explanations of the kind offered here do not invalidate proximal explanations of the kind offered in the middle of this chapter (§Purported explanations for egocentric discounting). My view is most consistent with a Bayesian integration view in which advice is weighted by a range of features of advice, advisor, and context, with the further proviso that there are hyperpriors which govern the default level of discounting. I am thus content to observe the contest to provide proximal explanations for changes in the level of egocentric discounting, and only baulk at claims that egocentric discounting is 'irrational'. I suggest that, once egocentric discounting as a default is accepted, the adjustments in the level of discounting are almost all transparently rational.

7.4.2 Evidence

In the chapters that follow, I present evidence from computational agent-based evolutionary simulations and online human behavioural experiments to illustrate the plausibility of the claims made above. The evolutionary simulations demonstrate that an array of plausible factors affecting the relative utility of advice can create an environment in which egocentric discounting is adaptive, and the behavioural experiments demonstrate that some of these factors can be responded to by individual humans by adjustments in behaviour. I note that only the first of these is necessary for illustrating the plausibility of the theory: the behavioural adjustments serve more to support the extension of the argument, that changes in the level of discounting are rational adjustments.

8

Sensitivity of advice-taking to context

Advice-taking is often overly-conservative as compared to the normative level of advice-taking for a given experimental design. I argue that participants' performances in advice-taking experiments reflect both the specifics of the experimental design and prior expectations about advice-taking situations. These prior expectations may be both learned, as where individuals who grow up in less stable environments show lower propensity to trust !TODO[Find reference], and inherited. Most useful for the current argument would be a demonstration that conservatism can emerge within a population even where detrimental advice is rarely experienced, and that this can thus produce individuals who exhibit conservatism without ever experiencing detrimental advice. This demonstration is presented in the form of evolutionary modelling.

As discussed in the [previous chapter]}(#chapter-context), conservatism is optimal under some circumstances, and thus we expect that simulated agents allowed to evolve an advice-taking policy in those circumstances will evolve a conservative policy. I explored this tendency as a function of three plausible scenarios. The first scenario is one in which agents occasionally give deliberately poor advice to their advisee, which represents situations where advisors' interests may sometimes be contrary to judges' interests, unbeknownst to the judges. In the second scenario, advice is simply noisier than the judge's own initial estimate, either because the judge is less competent at the task, less willing to exert the required effort for the task, or because the advice is communicated imperfectly. In the third scenario, agents belong to either a 'cautious' or a 'confident' group in how they express and interpret advice, which is a simple analogue of the observation that people's expressions of confidence are idiosyncratic (Ais et al. 2016; Navajas et al. 2017). In each of these three scenarios, it is hypothesised that some level of egocentric discounting will emerge as the dominant strategy, i.e., the mean population weighting for initial estimates versus advice will be greater than .50.

8.1 General method

Agent-based computational models of an evolutionary process were programmed in R (R Core Team 2018) and run variously on a home computer and the Oxford Advanced Research Computing cluster (Richards 2015). The code is available at `https://github.com/oxacclab/EvoEgoBias`, and the specific data presented below are archived at !TODO[find somewhere...]

The models reported here use 1000 generations of 1000 agents which each make 30 decisions/generation on which they receive the advice of another agent. Decisions are either point estimation (Scenarios 1 and 2) or categorical decision with confidence (Scenario 3). Each agent combines their own initial estimate with the advice of another agent, with the relative weights of the initial estimate and advice set by the agent's egocentric bias parameter, to produce a final decision. Final decisions are evaluated by comparison with the objective answer, and an agent's fitness is the sum of its performance over the 30 decisions of its lifetime.

8.1.1 Initial estimates

The agents perform a value estimation (category estimation in Scenario 3) task. Agent i's initial estimate t is the true value (v_t), plus some noise drawn from a normal distribution with mean 0 and standard deviation equal to the agent's insensitivity parameter (s^i, which is itself drawn from a positive-clamped normal distribution with mean and standard deviation 10 when the agent is created).

An agent's initial estimate (e^i_t) is thus:

$$e^i_t = v_t + N(0, s^i) \tag{8.1}$$

8.1.2 Advice

Each agent receives advice from another agent which it combines with its initial estimate to reach a final decision. The advice has a probability of being mutated in some fashion. The mutation depends upon the scenario and is described separately for each.

8.1.3 Final decisions

In the basic model from which other models inherit their decision procedure, agent i produces a final decision t as the average of the agent's initial estimate (e^i_t) and another agent's advice (a^i_t), weighted by the agent's egocentric bias (b^i). The models typically change the value of a^i_t, which is typically a function of some other agent j's initial estimate e^j_t.

An agent's final decision (d^i_t) is thus:

$$d^i_t = \frac{e^i_t \cdot b^i + a^i_t \cdot (1 - b^i)}{2} \tag{8.2}$$

The final decisions in Scenario 3 are more complex, but follow a similar structure.

8.1.4 Reproduction

Roulette wheel selection is used to bias reproduction in favour of agents performing best on the decisions. Performance is determined by a fitness function which differs slightly between categorical and continuous decisions. For scenarios 1 and 2, which use continuous decisions, this fitness is obtained by subtracting the absolute difference between the final decision and the true value for each decision (mean absolute deviance):

$$u^i = -\sum_{t=1}^{30} |v_t - d_t^i| \tag{8.3}$$

Scenario 3 uses categorical decisions. These are processed using a fitness function that TODO: Categorical fitness function description

The selection algorithm proceeds as follows: The worst performance is subtracted from each agent's fitness and 1 added to put fitness scores in a positive range. These scores are then continually multiplied by 10 until the lowest score is at least 10 to improve resolution. QUESTION: Wouldn't the lowest score be 1 so the multiplication by 10 would only need to happen once? Each agent is then given a probability to reproduce equal to their share of the sum of all fitness scores:

$$r^i = \frac{u^i}{\sum_{j=1}^{n} u^j} \tag{8.4}$$

where n is the number of agents and u has undergone the transformations described above.

Reproducing agents pass on their egocentric bias to their offspring. Other agent features, e.g. decision-making accuracy, are randomised when they are created. In the present simulations, agents receive no feedback on decisions, and cannot learn about or discriminate between their advisors. The key outcome of interest in each simulation is whether the population evolves towards egocentric discounting as the dominant adaptive strategy.

8.2 Scenario 1: misleading advice

In scenario 1, agents sometimes choose to offer misleading advice to their advisee.

8.2.1 Method

The true value (v_t) is fixed at 50 in this scenario. The agents do not learn about the true value over time, so a fixed and arbitrary value does not alter the results of the simulation. Advice in this scenario is either the advising agent's initial estimate (e_t^j), or an extreme answer in the opposite direction to the advising agent's initial estimate (i.e. lower than 50 if e_t^j was above 50, and vice-versa). !TODO[describe what we actually do with this work; it has moved on a bit since transfer report]

8.2.2 Results

Where advice is always genuine, egocentric discounting does not emerge. Where advice is sometimes misleading, however, egocentric discounting emerges rapidly and remains stable throughout the rest of the stimulation.

8.2.3 Discussion

As one might expect, where there is the potential for exploiting trust it is safer to trust less, even if this reduces some of the benefits which would be gained from trusting. Egocentric discounting may be a viable strategy, even if there is no difference in ability between advice-seekers and advice-givers, given social contexts where the interests of advice-seekers and advice-givers are not perfectly aligned.

8.3 Scenario 2: noisy advice

In this scenario, agents offer advice which is of slightly lower quality on average than their own initial decisions. This could reflect a difference in effort or in expertise.

8.3.1 Method

As with Scenario 1, the true value (v_t) is fixed at 50.

Advice in this scenario has additional noise added:

$$a_t^i = e_t^j + N(0, x) \tag{8.5}$$

Where $x = 0$ in the no-noise condition and 5 in the noise condition.

Models were also run where a new decision was used as the basis for advice, rather than an initial estimate that some other agent would be using as a component in its own final decision:

$$a_t^i = v_t + N(0, s^j + x) \tag{8.6}$$

but the results were the same.

8.3.2 Results

As before, egocentric discounting emerges rapidly in the active condition and remains stable throughout the rest of the simulation.

8.3.3 Discussion

Where the advice is of worse quality on average, encoded here as additional variation, egocentric discounting tailors the relative weights of the estimates according to their average quality. As with situations in which an advice-seeker is more competent at making the relevant decision than their advisor, some measure of egocentric discounting is warranted in this scenario. It is worth noting that different competencies are not the only reason why advice may be systematically less valuable than initial decisions: difficulties in communicating the advice or different levels of conscientiousness in decision-making may also produce this effect.

8.4 Scenario 3: confidence confusion

While lackadaisical, incompetent, deliberately bad, or poorly communicated advice produces an obvious adaptive advantage for egocentric discounting, it is plausible that scenarios may exist where equally competent, wholly well-intentioned advice may still favour egocentric discounting. A common feature of advice is the communication of confidence, and this improves outcomes (Bahrami et al. 2010). Notably, there are large and consistent individual differences in people's expressions of confidence (Ais et al. 2016; Song, Kanai, et al. 2011). In this scenario agents are equally competent at the task, and do their best to assist one another, but may be hampered by expressing their estimates and advice with different confidences.

8.4.1 Method

The true value was drawn from a normal distribution around 50:

$$v_t = N(50, 1) \tag{8.7}$$

This allowed categorical answers which identified whether or not v_t was greater than 50. Agents were equiprobably assigned a personal confidence factor (c) of 1 or 10. This was used to scale the difference between the advising agent's initial estimate and the category boundary to produce the advice:

$$a_t^i = (e_t^j - 50 \cdot c^j) + 50 \tag{8.8}$$

Each agent then used the reciprocal of this process to translate advice back into its own confidence scale before integrating it with the initial estimate and arriving at a final decision:

$$d_t^i = \frac{e_t^i \cdot b^i + (\frac{1}{c^i}(a_t^i - 50) + 50) \cdot (1 - b^i)}{2} \tag{8.9}$$

This process amounts to the outgoing advice being translated into the advising agent's confidence language, and incoming advice being translated into the advised agent's confidence language. Where these languages are the same, the resulting advice is understood equivalently by both agents, but where there are differences the advice will be of greater or lesser magnitude compared to the initial estimate.

8.4.2 Results

Egocentric discounting again emerges in the condition where agents can have different confidence factors. The adaptiveness of egocentric discounting in this scenario arises because advice from a mismatched partner (e.g. one using a confidence scaling of 10 when the judge uses 1, or vice-versa) requires a different aggregation approach than advice from a matched partner. The application of an inappropriate aggregation approach results in misleading advice, so there is a pressure for a middle-of-the-road strategy whereby advice is counted, but not too much.

8.4.3 Discussion

Even where there is no intent to mislead, and no difference in basic ability, it is possible for egocentric discounting to emerge as an optimal strategy, purely because of differences in how agents communicate and understand estimates.

8.5 General discussion

The computational models show that egocentric discounting is an adaptive strategy in an array of plausible advice contexts: misleading advice, noisy advice, and different interpretations of confidence. While these models are necessarily limited in applicability to real life, they do demonstrate that egocentric discounting, while irrational for simple estimation problems with an objective answer and with advice that is not systematically better or worse than an individual's own judgement, may be beneficial for many of the kinds of decision for which we have sought and used advice in everyday life throughout our evolution. This argument invites attention to the advice-taking task as much as to the properties of the advisor: it predicts that egocentric discounting will be attenuated where the outcome of decisions affects judges as well as advisors (Gino 2008, observed this effect but attributed it to judges falling prey to the sunk costs fallacy); where decisions rely more on objective than on subjective criteria (Van Swol 2011); where advisors and judges have opportunities to calibrate their confidence judgements with one another by completing training trials where they have to produce a shared decision with a shared confidence; and where incentives for judges and advisors are more closely aligned (Gino, Brooks, et al. 2012; Sniezek, Schrah, et al. 2004).

Notably, the utility of these heuristics does not depend on malice, mistake, or miscommunication: inconsistency in the usage of confidence terminology can produce adaptive pressure for egocentric discounting. More generally, the results indicate that properties of the advice-giving milieu can influence advice-taking strategies.

These models establish that it is plausible that people have deeply ingrained hyperpriors towards discounting advice. It is also possible, however, that people can flexibly respond to contexts, modulating their advice-taking appropriately. The voluminous experiments in advice taking discussed in the introduction to this section can be seen as eliciting exactly this behaviour. In the next chapter I present new behavioural experiments which suggest that people do indeed modulate their advice-taking to the specifics of the context in which they are in.

9

Behavioural responses to advice contexts

We suggest that the long-standing view that egocentric discounting reflects sub-optimal information processing is a consequence of taking a narrow view of the problem being solved. While it is indeed demonstrable that people in Judge-Advisor System experiments would perform better and earn more reward if they took more advice CITE, the people who participate in those experiments also have to function in the real world where being too trusting can produce very negative outcomes. Furthermore, advice-taking and decision-making are, like the behaviour of other organisms, ultimately mechanisms for the more efficient propagation of genes. The task being solved by the genes in the cells of the participant in a Judge-Advisor System experiment is to effectively exploit a complex, often-unreliable, and frequently-changing world. We suggest that egocentric discounting may result from genetic and cultural evolution favouring the default assumption that an individual's own information is a more reliable basis for decision-making for that individual than another individual's information.

The evolutionary models offered a proof-of-concept illustration that hyperpriors may evolve under a variety of contextual features that are almost always at least partially true of advice-taking situations. In this chapter, we explore the flexibility of egocentric discounting in these contexts. We present three experiments that examine how advice-taking changes according to the benevolence and the identifiability of advisors. We also include a brief discussion of the literature on advisor expertise.

9.1 Experiment 5: Benevolence of the advisor population

The evolutionary models discussed in the last chapter demonstrated that optimal advice-taking strategies depend in part upon the advice one receives being a genuine effort to help. Difference in benevolence, or the extent to which the interests of the advisor and the judge overlap, is one of the three pillars of the **Mayer** model of advice-taking. Despite this, relatively little investigation has been made into the role of benevolence in trust, as discussed previously§7.2.3.

Advice-taking can be contingent on the properties of the advice, or on the properties of the advisor. In order to maximise the value of advice while minimising the potential exposure to exploitation, advice-taking should be contingent on a combination of these factors. Where advice is plausible it should be weighted relatively equally, whether it comes from an advisor who is sometimes misleading or not, but where advice is more implausible it should only be trusted when it comes from an advisor who is never misleading. To explore whether people's behaviour matches this pattern, participants were recruited for a series of behavioural experiments in which they were given advice on a date estimation task from advisors who were described as either always helpful or occasionally misleading.

We expected that advisor influence would be higher for advice that participants rated as 'honest' versus advice rated as 'deceptive'. Likewise, we expected that influence would be higher for advisors who were described as 'always honest', even for advice rated as 'honest'. In other words, we expect that participants' advice-taking depends upon both the plausibility of the advice and the benevolence of the advisor.

Open scholarship practices

This experiment was preregistered at `https://osf.io/tu3ev`. The experiment data are available in the `esmData::` package for R from !TODO[Zenodo esmData], and also directly from !TODO[OSFify Dates task data]. A snapshot of the state of the code for running the experiment at the time the experiment was run can be obtained from https://github.com/oxacclab/ExploringSocialMetacognition/blob/b4289fea196f71ccf0ba0b2ae8fde12139a16301/ACv2/db.html.

This is a replication of a study of identical design that produced the same results. The data for the original study can be obtained from the `esmData::` R package or from !TODO[OSFify].

There were a number of deviations from the preregistered analysis in this Experiment. Several participants used translation software to translate the experiment website. We could not guarantee that the questions were accurately translated and so these participants were excluded. Some participants never rated the Sometimes deceptive advisor's advice as 'honest', and so they were excluded in the t-test comparing advisors.

9.1.1 Method

Procedure

20 participants each completed 28 trials over 3 blocks of the continuous version of the Dates task§3.1.3. On each trial, participants were presented with an historical event that occurred on a specific year between 1900 and 2000. They were asked to drag a marker onto a timeline to indicate the date range within which they thought the event occurred. The markers covered 11 years, meaning that it would cover an entire decade inclusively, e.g. 1965-1975. Participants then received advice indicating a region of the timeline in which the advisor suggested the event occurred, and rated the honesty of the advice on a three-point scale (Figure 9.1). Participants could then mark a final response in the same manner as their original response, and could choose a different marker width if they wished.

Participants started with 1 block of 7 trials that contained no advice to allow them to familiarise themselves with the task. All trials in this section included feedback for all participants indicating whether or not the participant's response was correct.

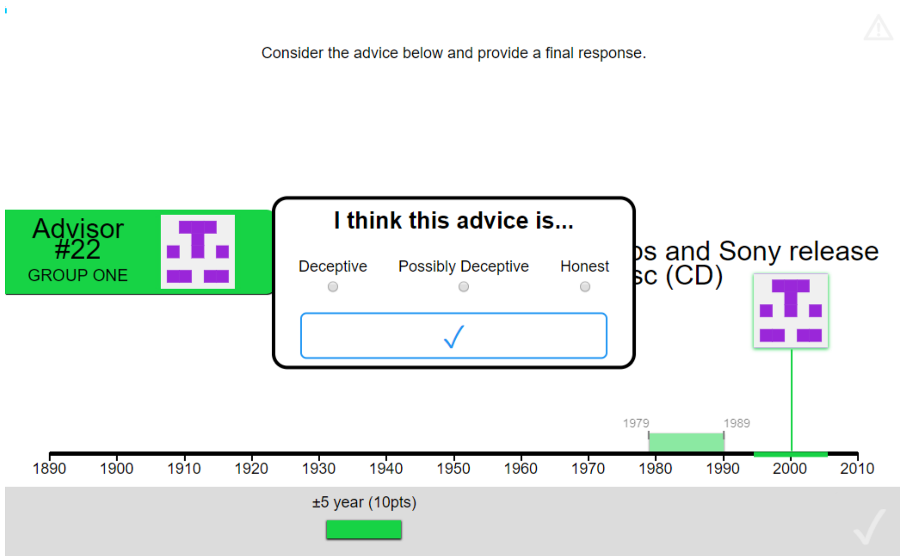

Figure 9.1: Advice honesty rating.
Participants rated advice on a three-point scale according to whether they thought the advice was deceptive or honest.

Participants then did 7 trials with a practice advisor to get used to receiving advice. They also received feedback on these trials. They were informed that they would "get advice on the answers you give" and that the feedback they received would "tell you about how well the advisor does, as well as how well you do". Before starting the main experiment they were told that they would receive advice from multiple advisors and that "advisors might behave in different ways, and it's up to you to decide how useful you think each advisor is, and to use their advice accordingly".

Participants then performed 2 blocks of trials that constituted the main experiment. In each of these blocks participants had a single advisor for 6 trials, plus 1 attention check. No feedback was given on answers in the main experiment blocks.

The two advisors were identical in how they generated advice, but they were labelled differently. The advisors had different colours, and participants were told that the advisor who matched the participant's colour "will give you the **best advice that they can**", while the advisor who did not match the participant's colour "might sometimes try to direct you **away from the correct answer**". Colours and the order in which the advisors were encountered were counterbalanced.

Table 9.1: Participant exclusions for Experiment 5

Reason	Participants excluded
Attention check	2
Multiple attempts	0
Missing advice rating	2
Odd advice rating labels	1
Not enough changes	2
Too many outlying trials	0
Total excluded	**3**
Total remaining	**17**

Advice profiles

Despite differences in labelling, the advisors were identical in terms of how they actually produced advice. The advisors offered advice by placing an 11-year wide marker on the timeline. The marker was placed with its centre on a point sampled from a roughly normal distribution around the correct answer with a standard deviation of 11 years.

9.1.2 Results

Exclusions Participants (total $n = 20$) could be excluded for a number of reasons: failing attention checks, having fewer than 11 trials which took less than 60s to complete, providing final decisions which were the same as the initial estimate on more than 90% of trials, or using non-English labels for the honesty questionnaire. The latter exclusion was added after data were collected because it was not anticipated that participants would use translation software in the task. The numbers of participants who failed the various checks are detailed in Table 9.1.

The final participant list consists of 17 participants who completed an average of 11.88 trials each.

Task performance

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## Interaction calculation: Initial - Final
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Participants performed as expected, decreasing the error between the midpoint of their answer and the true answer from the initial estimate to the final decision

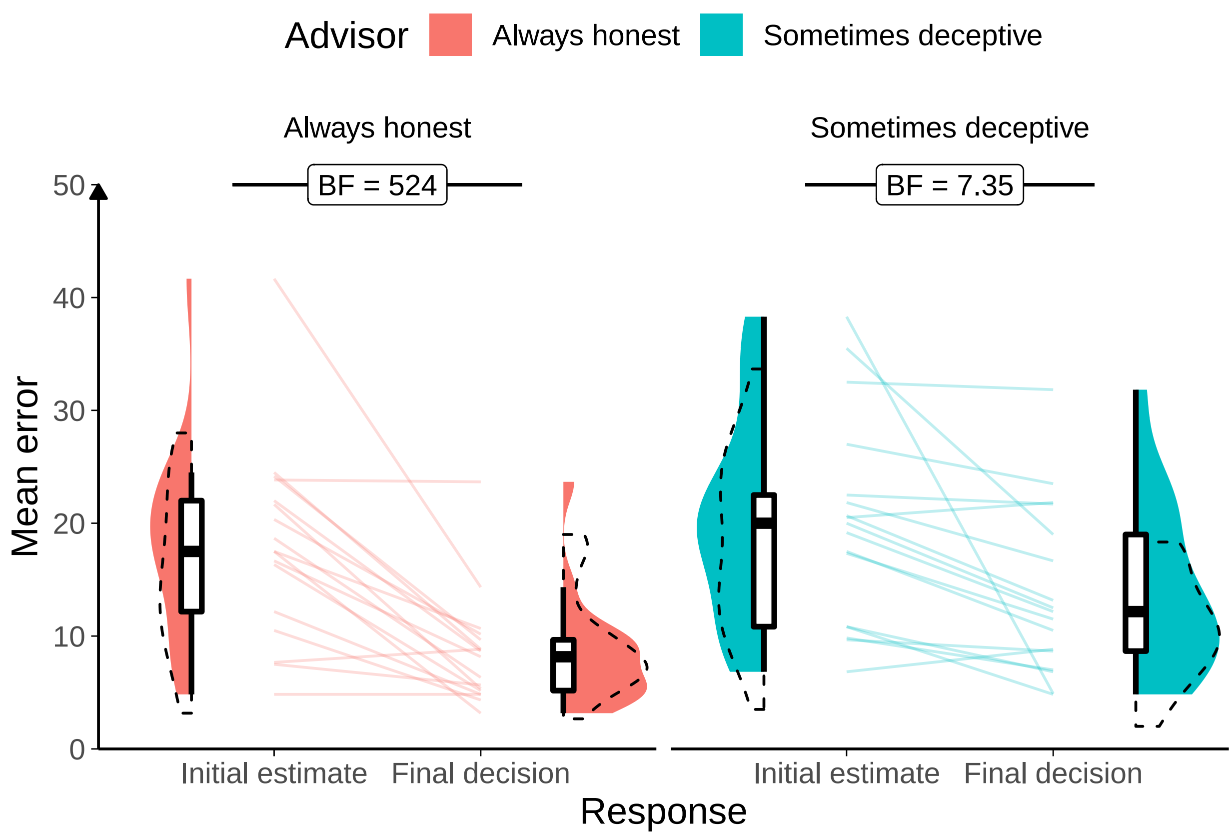

Figure 9.2: Task performance for Experiment 5.
A: Response error. Faint lines show individual participant mean error (the absolute difference between the participant's response and the correct answer), for which the violin and box plots show the distributions. The dashed line indicates chance performance. Dotted violin outlines show the distribution of participant means on the original study which this is a replication. The dependent variable here is error, the distance between the correct answer and the participant's answer, and consequently lower values represent better performance. The theoretical limit for error is around 100.

(F(1,16) = 31.85, p = <.001; M_{Initial} = 19.07 [14.98, 23.16], M_{Final} = 11.12 [8.27, 13.97]), which suggests that they incorporated the advice, which was indicative of the correct answer (Figure 9.2). The participants had less error on decisions made with the Always honest advisor than the Sometimes deceptive advisor (F(1,16) = 9.38, p = .007; $M_{\text{AlwaysHonest}}$ = 13.24 [10.16, 16.32], $M_{\text{SometimesDeceptive}}$ = 16.95 [13.18, 20.72]), although surprisingly there was no evidence that this was due to greater reduction in error scores over time for that advisor (F(1,16) = 1.94, p = .183; $M_{\text{Reduction|AlwaysHonest}}$ = 9.70 [5.95, 13.44], $M_{\text{Reduction|SometimesDeceptive}}$ = 6.20 [1.96, 10.43]).

Participants only had one marker they could place, and separate confidence

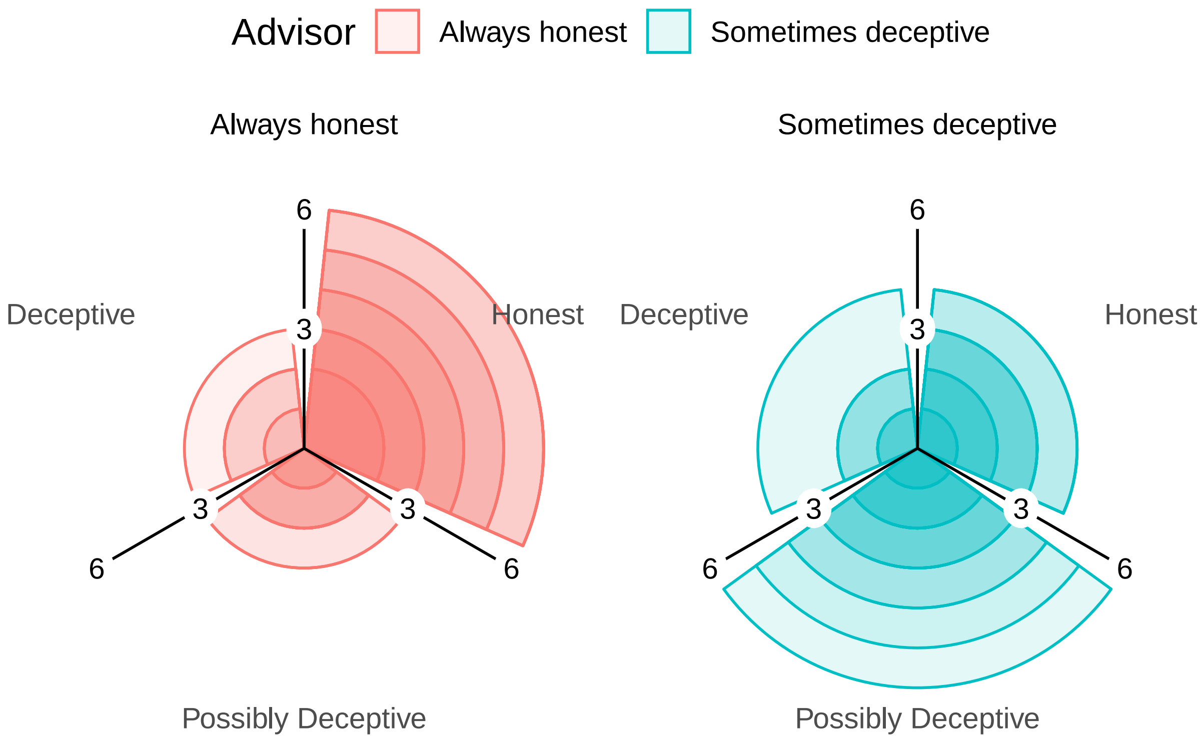

Figure 9.3: Advice rating in Experiment 5.
The polar plots show the number of times each participant gave the given rating to the advice of each advisor. The colour density illustrates the number of participants who gave at least that many ratings to the advice of an advisor.

judgements were not asked for, so we cannot directly assess confidence in these data.

Advisor performance The advice given by the advisors was generated stochastically from the same distribution. Any differences will be random. This was demonstrably the case in the domain of advice error (absolute distance between the centre of the advice marker and the correct year; BF = 1/3.92; $M_{AlwaysHonest}$ = 8.05 [6.81, 9.29], $M_{SometimesDeceptive}$ = 7.90 [6.84, 8.96]). The Bayes' Factor for the domain of agreement – the absolute distance between the centre of the advice marker and the centre of the participant's initial estimate marker – was essentially on the threshold (BF = 1/2.96; $M_{AlwaysHonest}$ = 20.14 [16.09, 24.19], $M_{SometimesDeceptive}$ = 21.85 [17.00, 26.70]).

Advice ratings The advisors' advice was rated differently by the participants, as shown in Figure 9.3 ($\chi^2(2)$=18.14, p=< .001, BF=335; AlwaysHonest:SometimesDeceptive ratio: Deceptive 0.65, Possibly Deceptive 0.49, Honest 1.82). This indicates that participants understood the task and that the manipulation worked as intended, with more suspicion applied to the advice from the Sometimes deceptive advisor.

Effect of advice Participants chose their own ratings for the advice, and it was common for participants not to use all ratings for all advisors (e.g. many participants never rated advice from the Always honest advisor as Deceptive). This meant that the statistical tests preregistered for this hypothesis were broken down into separate contrasts of advice and advisor. Using a more complete test, such as 2x3 ANOVA, would have suffered greatly from missing values.

To explore the effect of advice, a 1x3 ANOVA was run on weight on advice across ratings. In all, 11/17 (64.71%) participants had at least one trial rated with each of the three ratings. The weight on advice differed according to the rating assigned the advice (F(2,20) = 9.37, p = .001; $M_{\text{Deceptive}}$ = 0.18 [-0.04, 0.41], M_{Honest} = 0.60 [0.46, 0.74], $M_{\text{PossiblyDeceptive}}$ = 0.32 [0.09, 0.55]; Mauchly's test for Sphericity W=.935, p=.739). As expected, participants were more influenced by advice they rated as Honest compared to advice they rated as Deceptive.

Effect of advisor To distinguish the effects of the advisor from the effects of the advice, we compared weight on advice for only those trials where the participant rated the advice as Honest. This approach has the limitation that 2 (11.76%) participants had to be dropped due to never rating advice from the Sometimes deceptive advisor as Honest.

Comparing the two (Figure 9.4) showed that participants placed more weight on the Honest advice from the Always honest advisor ($t(14) = 2.67$, $p = .018$, $d = 0.78$, BF = 3.38; $M_{\text{AlwaysHonest}}$ = 0.67 [0.54, 0.80], $M_{\text{SometimesDeceptive}}$ = 0.48 [0.34, 0.62]).

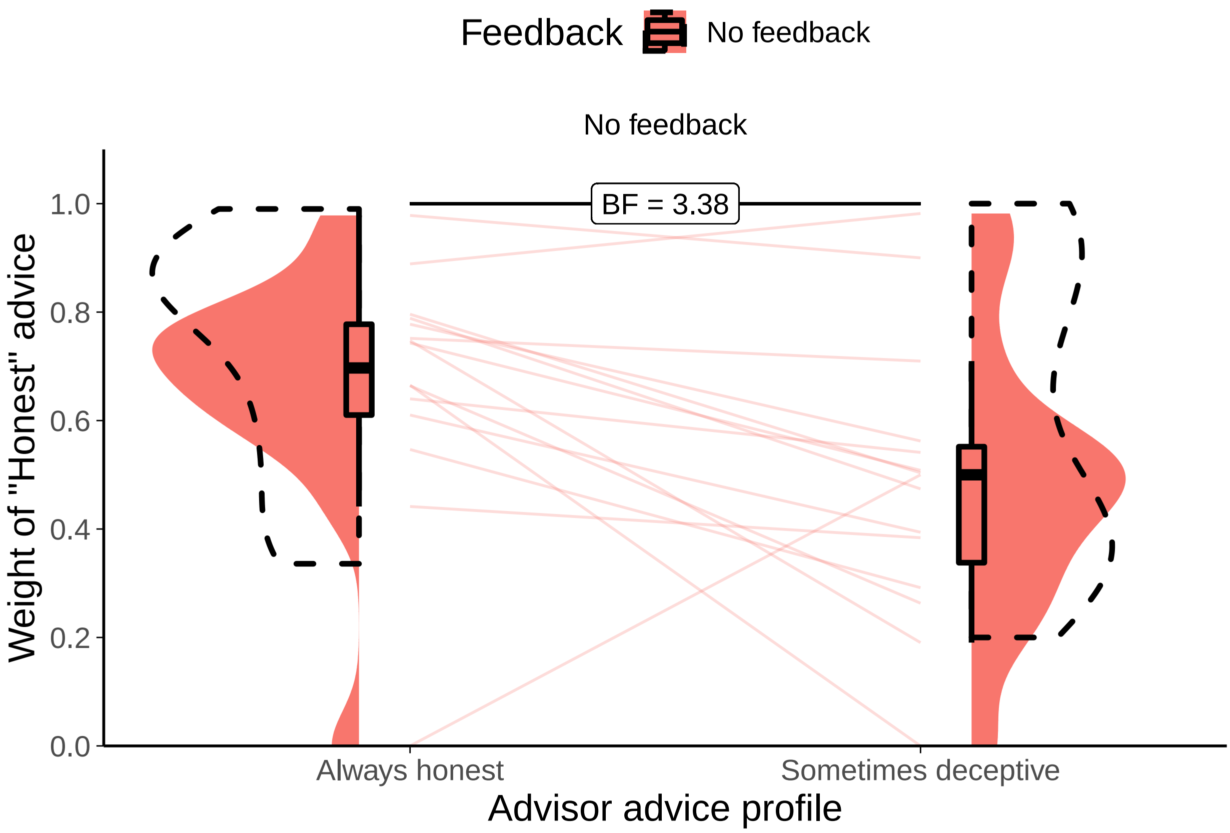

Figure 9.4: Weight on Advice in Experiment 5.
Shows the weight of the advice of the advisors. Only trials where the participant rated the advice as Honest are included. The shaded area and boxplots indicate the distribution of the individual participants' mean weight on advice. Individual means for each participant are shown with lines in the centre of the graph. The dashed outline shows the distribution of participant means in the original study of which this is a replication.

Exploration

Figure 9.5 shows the overall pattern of advice-taking in the experiment. The difference in the number of dots of each colour reflects the difference in participants' ratings of advice between advisors.

The lines illustrate the relationship between the distance of the advice from the participant's initial estimate and the amount the participant moved the marker towards the advisor's advice for their final decision. The dashed line indicates complete acceptance of the advice, while the bottom triangle below the dashed line indicates incomplete acceptance.

A linear mixed model was used to better understand the significance of the

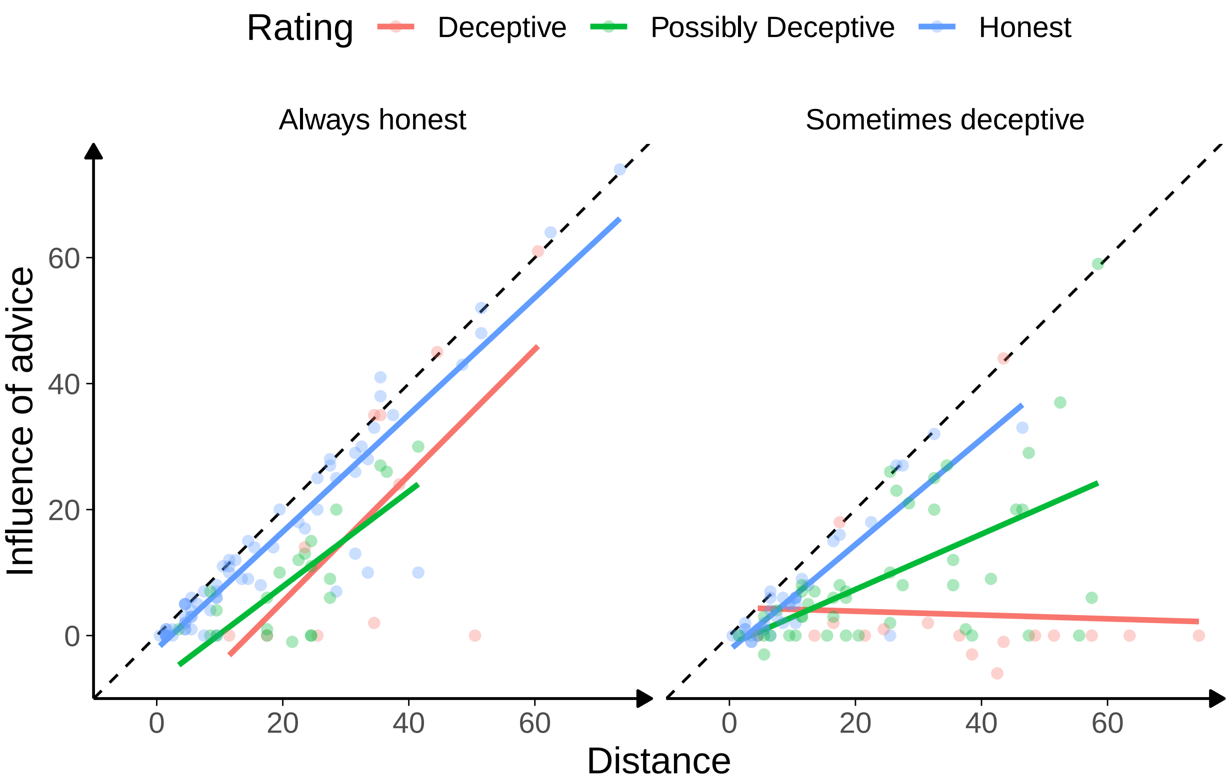

Figure 9.5: Advice response in Experiment 5.
Scatter plot of advice distance and advice influence. The dashed line shows x = y. Colours represent different advice ratings. Dots represent individual trials, and lines show the pattern within each rating.

relationships in the figure, predicting influence on a trial from the advisor, the distance of the advice, and rating (dummy-coded). Random intercepts were included for participant. A similar Bayesian model was included to produce Bayes factors.

Overall, the distance between the initial estimate and the advice was hugely influential, with an increase in distance leading to an approximately equal increase in influence (β=0.94 [0.83, 1.05], df=184.30, p=<.001, BF=4.1e27). This indicates that the advice rated Honest from the Always honest advisor was generally followed, as is shown clearly in Figure 9.5. This pattern indicates that the participants see the difference between their own estimates and the advisor's as indicating that they themselves are mistaken.

Advice rated as Deceptive from the Sometimes deceptive advisor was substantially less influential (β=16.24 [1.51, 31.07], df=175.94, p=.037, BF=1/6.71), and this was complemented by the relationship between distance and influence being substantially reduced in this case (β=-0.78 [-1.25, -0.33], df=177.51, p=.001, BF=23.0).

There are other patterns in Figure 9.5 that do not come out in the statistics, possibly due to the very low trial and participant numbers, but are suggestive of a pattern we might expect. The relationship of advice rated Honest is similar between the advisors: as the advice gets further away it retains its level of influence and the participants are willing to go further to match it. This is not tested directly in the stats, but there is no significant effect of advisor alone (β=0.87 [-3.57, 5.28], df=180.44, p=.705, BF=1/2.53), the presence of which would indicate that Honest-rated advice differed between advisors. The Bayesian analogue of the test indicated that there was not enough evidence to decide whether Advisor alone was an important factor.

Similarly, there may be a difference in the way the Possibly Deceptive advice was received from the Sometimes deceptive advisor as compared to the Always honest advisor (7.57 [-2.34, 17.43], df=181.63, p=.144, BF=1/5.67), or in how the influence of this advice related to the distance between the advice and initial estimate markers (-0.29 [-0.74, 0.17], df=182.50, p=.225, BF=1/3.05). The statistics did not support this, and the Bayesian statistics suggested that there was sufficient evidence against either effect being present.

9.1.3 Discussion

Both the source and the plausibility of the advice matter, and in this task they interacted such that discounting primarily occurred where advice was offered that differed substantially from the participant's initial estimate and came from the advisor the participants were told might mislead them. Participants adapted to the context of the advice, both in terms of how readily they were to categorise the advice as dishonest and in terms of how much they were influenced by the advice. As advice got more distant from the initial estimate, i.e. decreased on our measure of plausibility, participants had to decide whether the discrepancy was due to their own error or their advisor's. Where they were in a context that suggested the advisor was trustworthy, they were more likely to ascribe the error to themselves, rating the advice as honest and adopting it for their final decision. Where they were in a context that suggested the advisor was not trustworthy, they were more likely to label the advice as misleading and to retain their initial estimate as their final decision.

These observations complement the evolutionary simulations and the benevolence component of the three-factor model of trust described by Cite Mayer et al 1995. Nevertheless, we have demonstrated sensitivity to context, but not universal discounting due to hyperpriors. It remains a matter of speculation that people exercise epistemic hygiene by ensuring that information comes from trusted sources before integrating it, and that no source is as trusted as one's own mind. This result is in keeping with evidence from experiments where initial estimates are labelled as advice (and vice-versa), in which participants frequently show an unusual level of acceptance of the advice suggestive of their performing egocentric discounting of their own initial estimate because they have accepted the labels and adopted the advice as their own initial estimate (Soll and Mannes 2011).

Limitations

Even compared to the other experiments in this thesis, these experiments had a low participant count. As with other Dates task studies, data collection stopped when the Bayes Factor for the main experimental hypothesis reached one of the two thresholds ($1/3 > \text{BF} > 3$). This, combined with the low trial count for each participant, meant that there were interesting follow-up questions that the data were unable to address. In keeping with the open science approach, we suggest that future investigations exploring those questions take them as preregistered hypotheses.

The Dates task is one that participants find challenging, leading to generally high levels of advice taking in the absence of other effects (Gino and Moore 2007; Yonah and Kessler 2021). This means that the Honest advisor seemed to be entirely trusted. As discussed in the introduction to this section§7, however, even greatly trusted advisors' advice is usually subject to some egocentric discounting.

9.2 Noise in the advice

The second scenario explored in the evolutionary models added noise to the advice agents received and demonstrated that this provided an evolutionary pressure towards egocentric discounting. The addition of noise in a point-value estimation task lowers the relative performance, and thus this scenario was essentially a manipulation of advisor expertise. This scenario is not explored in behavioural experiments because its conclusions are well supported by existing literature. Specifically, the literature demonstrates the normativity of discounting where the judge outperforms the advisor, that advice-taking is sensitive to advisor expertise, and that people are likely to consider themselves superior to the average advisor. This demonstration is supplemented with a more comprehensive model of the utility of advice according to the ability to identify the more expert opinion, the relative accuracy of the advisor and judge, and the independence of the opinions (Soll and Larrick 2009). A more complete account of the effects of advisor expertise was provided earlier§7.2.3.

Views of relative expertise may also be affected by self-enhancement bias (Brown 1986). We expect that this is somewhat dependent upon the difficulty of the task presented; people faced with a difficult task will under- rather than overestimate their ability relative to others (Gino and Moore 2007). Taken together, then people are likely to consider themselves more able on a given task than an arbitrary advisor, and consequently that they are likely to downweight advice relative to their own initial estimate. This behaviour is supported by normative models which show biasing towards the better estimator (in this case the judge) is the optimal strategy (Soll and Larrick 2009; Mahmoodi et al. 2015). As discussed in the scenario, the belief that one is better at a task than the average advisor may not be misguided: advisors may not dedicate the same amount of time, concentration, or thought to producing advice as judges do for initial estimates. Judges have to live with the consequences of their decisions, while advisors do not.

9.3 Confidence mapping

The third scenario explored in the evolutionary models assigned each agent a confidence mapping, and demonstrated that discounting emerged as an appropriate response where the advisor's confidence mapping was unknown. The key difficulty in conducting behavioural experiments to test the effects of known vs unknown confidence mapping is finding a manipulation of confidence mapping knowledge which is not confounded by familiarity with an advisor or the amount of information provided by an advisor.

We attempted to investigate this in two ways. Firstly, we performed an experiment where participants received advice over blocks of trials from either the same advisor on each trial or a different advisor on each trial. Secondly, we familiarised participants with two advisors who differed in their use of the confidence scale, and then tested whether participants responded differently based on whether they could identify which of those two advisors was providing advice.

Neither experiment provided evidence consistent with the hypothesis that familiarity with an advisor's confidence calibration leads to greater trust in that advisor's advice. We suggest that this may be a failure in experimental design rather than reliable evidence of the absence of this feature of advice-taking behaviour.

9.3.1 Experiment 6: Individuality as a cue to confidence

The expression of confidence is highly idiosyncratic (Ais et al. 2016; Navajas et al. 2017). Given this, we hypothesised that the same advice may be treated differently depending upon whether or not it was labelled as coming from repeated interactions with a single individual or from a series of one-off interactions with different individuals. Where the advice comes from the same individual over and over again, participants should become more aware of that individual's confidence calibration (the relationship between their confidence and the probability they are correct), and should thus be able to better discriminate higher-quality (more confident) advice.

In this experiment the effect of individuality is confounded with familiarity, a feature that we expect will increase the palatability of advice. While participants are learning about the advisors, and learning more about the confidence calibration of the individual advisor than that of the group members, they are also becoming more familiar with the individual advisor. Nevertheless, this is a useful experiment because it is capable of indicating the absence of an effect: if the manipulation proves unable to produce greater influence from the individual advisor it will do so without a confound. If the experiment demonstrates higher influence for the individual advisor then further experiments can be run to attempt to isolate confidence calibration knowledge from mere exposure.

Open scholarship practices

This experiment was not preregistered because it was highly exploratory, and we were doubtful that the manipulation would work. The experiment data are available in the `esmData::` package for R from !TODO[Zenodo esmData], and also directly from !TODO[OSFify Dates task data]. A snapshot of the state of the code for running the experiment at the time the experiment was run can be obtained from `https://github.com/oxacclab/ExploringSocialMetacognition/blob/4d8aace6b864c4465cacd8579ec1a` `ACBin/ce.html`.

Method

Procedure 33 participants each completed 40 trials over 5 blocks of the binary version of the Dates task§3.1.3. On each trial, participants were presented with an historical event that occurred on a specific year between 1900 and 2000. They were then given an anchor year within the 1900-2000 range. The task for the participant was to indicate whether the event occurred before or after the anchor year. To answer 'before' they selected a point on the lefthand column, and to answer 'after' they selected a point on the righthand column. They indicated their confidence by the point they selected on the chosen column: the higher up the column they chose the greater confidence the signalled in their answer.

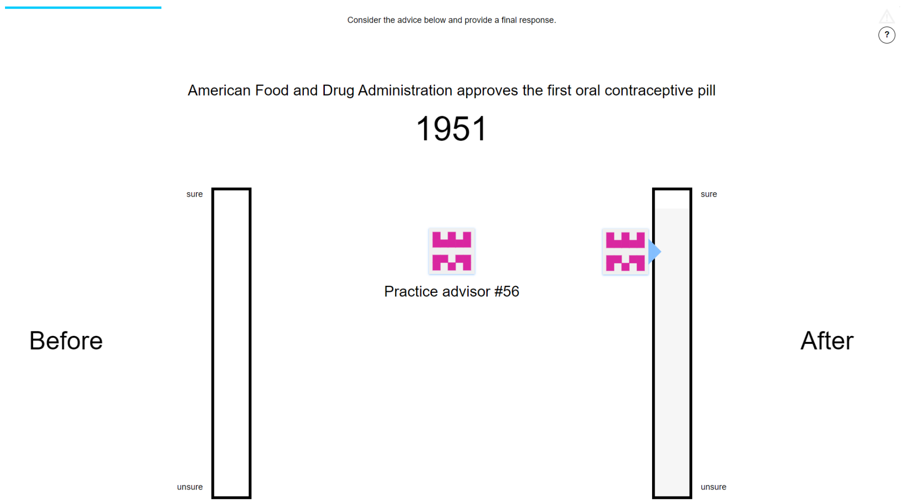

Figure 9.6: Advice confidence.
Advisors' markers shifted position vertically to indicate the confidence with which the advice was give. As with the participants' own responses, higher placement on the column indicated higher confidence.

Participants then received advice from an advisor indicating which of the columns the advisor considered the correct answer. The advisor also indicated a confidence in their response, indicated by the height of the advisor's avatar on the indicated bar (Figure 9.6).

Participants started with 1 block of 7 trials that contained no advice to allow them to familiarise themselves with the task. All trials in this section included feedback for all participants indicating whether or not the participant's response was correct.

Participants then did 3 trials with a practice advisor to get used to receiving advice. They also received feedback on these trials. They were informed that they would "get advice about the correct answer" and that the "advisors indicate their confidence in a similar way to you, by placing their marker higher on the scale if they are more sure".

Participants then performed 4 blocks of trials that constituted the main experiment. These blocks came in pairs, with each pair containing one block of 10 trials that included feedback followed by one block of 5 trials that did not include feedback. 1 trial in the first block in each pair was an attention check trial. Each pair of blocks had a different advisor presentation.

Table 9.2: Participant exclusions for Experiment 5

Reason	Participants excluded
Multiple attempts	0
Not enough changes	3
Too many outlying trials	3
Total excluded	**6**
Total remaining	**27**

In one of the pairs of blocks, the advice was always labelled as coming from the same advisor. In the other pair of blocks, the advice on each trial was labelled as coming from a different advisor. The order that the participants encountered these presentations in was counterbalanced.

Advice profiles Despite differences in labelling, the advisors were identical in terms of how they actually produced advice. The advice was determined on each trial by selecting a point drawn from a normal distribution around the correct answer with standard deviation of 8 years. This point served as the advisor's internal estimate of the correct answer. The advisor then gave the appropriate answer according to whether or not the internal estimate was before or after the anchor year, and scaled their confidence based on the distance between those two years. The advisor was maximally confident where the difference between the internal estimate and the anchor year was at least 20 years.

Results

Exclusions Participants (total $n = 33$) were excluded for a number of reasons: having fewer than 10 no-feedback trials which took less than 60s to complete, or providing final decisions which were the same as the initial estimate on more than 90% of trials. Table 9.2 shows the number of participants excluded and the reasons why. Participants who failed attention checks would have been immediately dropped, but no participants did so.

The final participant list consists of 27 participants who completed an average of 28.00 trials each.

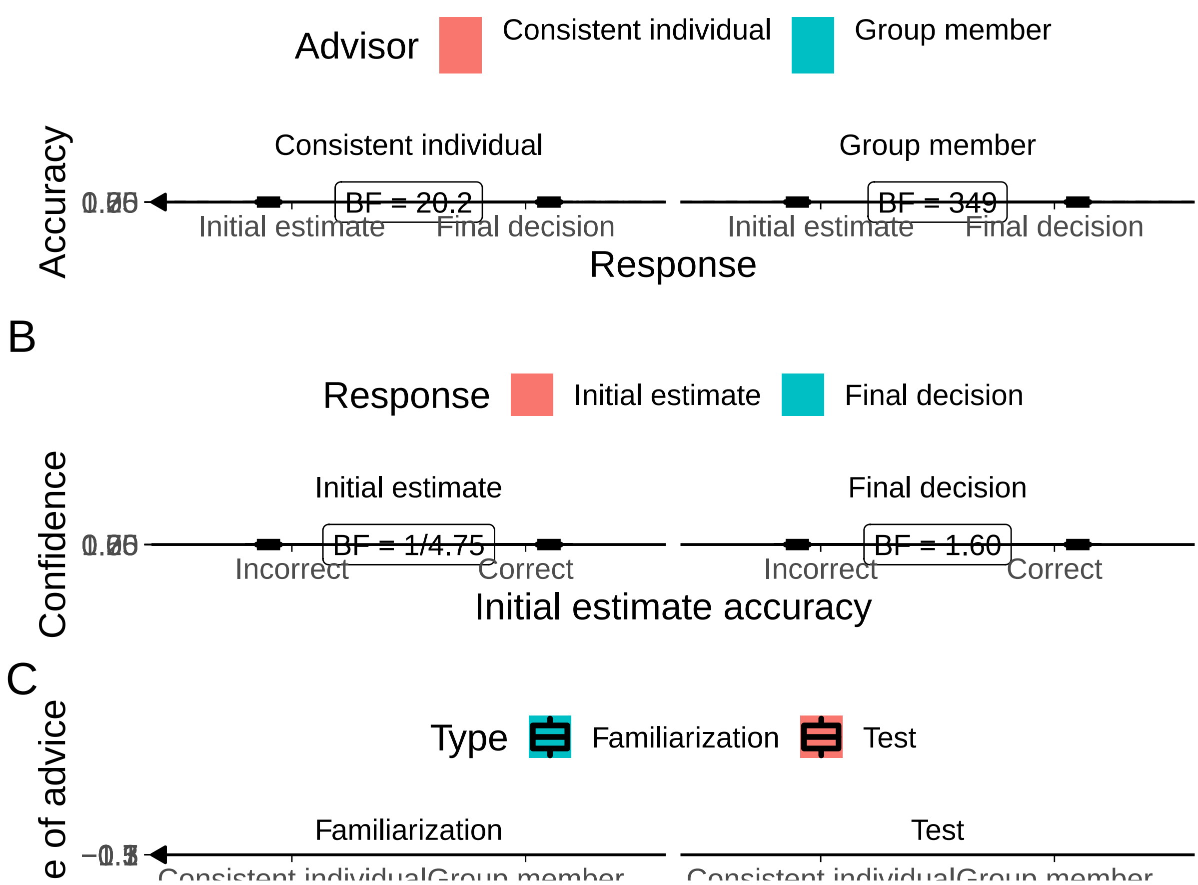

Figure 9.7: Task performance for Experiment 6.
A: Response accuracy. Faint lines show individual participant means, for which the violin and box plots show the distributions. The dashed line indicates chance performance. Dotted violin outlines show the distribution of actual advisor accuracy.
Because there were relatively few trials, the proportion of correct trials for a participant generally falls on one of a few specific values. This produces the lattice-like effect seen in the graph. Some participants had individual trials excluded for over-long response times, meaning that the denominator in the accuracy calculations is different, and thus producing accuracy values which are slightly offset from others'.
B: Confidence. Faint lines show individual participant means, for which the violin and box plots show the distributions.
C: Influence. Participants' weight on the advice for advisors in the Familiarization and Test stages of the experiment. The shaded area and boxplots indicate the distribution of the individual participants' mean influence of advice. Individual means for each participant are shown with lines in the centre of the graph. The theoretical range for influence values is [-2, 2].

Task performance

```
## Interaction calculation: Final decision - Initial estimate
```

```
## Interaction calculation: Final decision - Initial estimate
```

```
## Interaction calculation: Consistent individual - Group member
```

Accuracy during the Familiarization trials (Figure 9.7A) was higher for final decisions than for initial estimates ($F(1,26) = 31.69$, $p = <.001$; $M_{FinalDecision} = 0.72$ [0.68, 0.76], $M_{InitialEstimate} = 0.58$ [0.51, 0.64]), and higher for the Consistent individual advisor than for the group ($F(1,26) = 6.35$, $p = .018$; $M_{ConsistentIndividual} = 0.69$ [0.63, 0.75], $M_{GroupMember} = 0.60$ [0.55, 0.66]). There was no significant effect of advisor presentation on improvement ($F(1,26) = 2.81$, $p = .106$; $M_{Improvement|ConsistentIndividual} = 0.10$ [0.04, 0.16], $M_{Improvement|GroupMember} = 0.18$ [0.10, 0.26]).

There were no significant effects for the ANOVA on confidence (Figure 9.7B). Neither the correctness of the initial estimate ($F(1,26) = 3.60$, $p = .069$; $M_{Correct} = 0.57$ [0.51, 0.63], $M_{Incorrect} = 0.53$ [0.46, 0.60]), nor whether the response was the initial estimate or final decision ($F(1,26) = 0.14$, $p = .709$; $M_{FinalDecision} = 0.55$ [0.48, 0.61], $M_{InitialEstimate} = 0.56$ [0.49, 0.62]) appeared to systematically affect confidence. There was also no significant interaction to indicate that confidence changed over time differently for correct versus incorrect initial estimates ($F(1,26) = 3.78$, $p = .063$; $M_{Increase|Correct} = 0.03$ [-0.02, 0.08], $M_{Increase|Incorrect} = -0.04$ [-0.11, 0.02]).

Likewise, no significant effects were found for the ANOVA on influence (Figure 9.7C). Neither the advisor presentation ($F(1,26) = 0.61$, $p = .442$; $M_{ConsistentIndividual} = 0.23$ [0.16, 0.29], $M_{GroupMember} = 0.26$ [0.17, 0.35]), nor whether the trial was in the Familiarization stage (with feedback) or the test stage (without feedback) appeared to be systematically linked to the influence of the advice ($F(1,26) = 3.62$, $p = .068$; $M_{Familiarization} = 0.27$ [0.19, 0.35], $M_{Test} = 0.21$ [0.15, 0.27]). There was also no indication that the influence of the advisors' advice changed differently between stages ($F(1,26) = 0.64$, $p = .432$; $M_{Individual\text{-}Group|Familiarization} = -0.06$ [-0.20, 0.07], $M_{Individual\text{-}Group|Test} = -0.01$ [-0.10, 0.09]).

Advisor performance The advice is generated probabilistically using the same rules for both advisor presentations. The advisors' advice is not dependent on the participants' initial estimates, but it is useful nevertheless to compare the advisors' agreement and accuracy rates. The advisors were demonstrably similar in their agreement rates (BF = 1/4.74; $M_{\text{ConsistentIndividual}}$ = 0.59 [0.52, 0.66], $M_{\text{GroupMember}}$ = 0.60 [0.53, 0.68]), but not so in their accuracy rates (BF = 1/2.24; $M_{\text{ConsistentIndividual}}$ = 0.78 [0.74, 0.82], $M_{\text{GroupMember}}$ = 0.74 [0.69, 0.79]), driven by a difference in the likelihood of agreeing with incorrect initial estimates. It is unlikely, but plausible, that the Consistent individual advisor may have been slightly more influential because of their higher agreement rate, a systematic effect that would go in the same direction as our hypothesis and that would therefore require a caveat during interpretation. We note here, however, that participants received feedback during this Familiarization phase, and that Experiment 3B§5.3.2 demonstrated that accuracy is an important feature, and agreement an unimportant feature, in the Dates task where feedback is available.

Effect of advisor presentation The hypothesis for this study was that, in the key Test trials where participants did not receive feedback, the advisor presented as a consistent individual would be more influential than the advisor presented as different members of a group on each trial. As illustrated in the righthand panel of Figure 9.7C, influence was not systematically different between the advisor presentations on the Test trials ($t(26)$ = -0.14, p = .886, d = 0.03, BF = 1/4.86; $M_{\text{ConsistentIndividual}}$ = 0.21 [0.14, 0.28], $M_{\text{GroupMember}}$ = 0.22 [0.13, 0.30]).

From this we can conclude that the manipulation was not effective in differentiating the advisors or that such differences as were produced do not affect the influence of advice.

Simulation

We conducted a simulation in which we modified participants' actual data by substituting the final confidence values for values calculated by a model that reduces

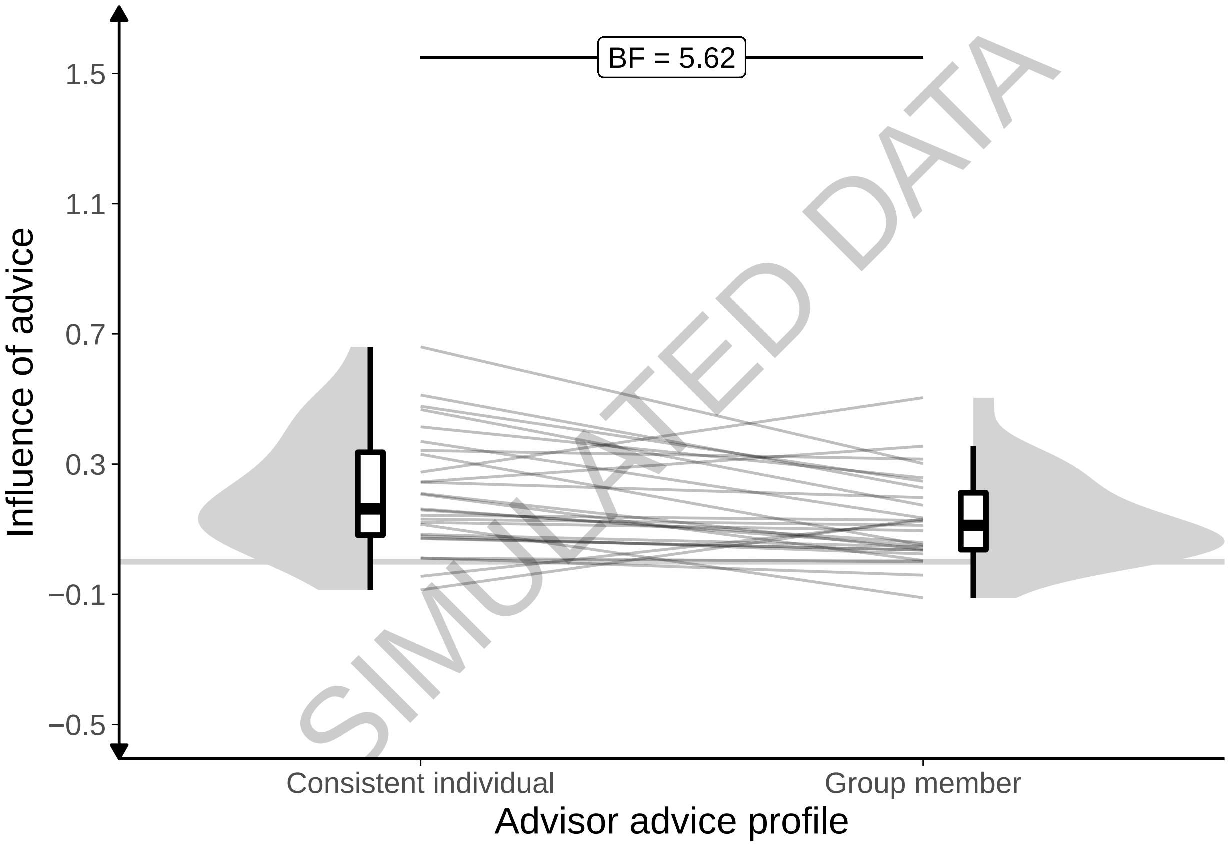

Figure 9.8: Simulated data for Experiment 6.
Simulated participants' weight on the advice for advisors in the Test stage of the experiment. The shaded area and boxplots indicate the distribution of the individual participants' mean influence of advice. Individual means for each participant are shown with lines in the centre of the graph. The theoretical range for influence values is [-2, 2].

egocentric discounting for the Consistent individual advisor by a set effect size. The simulations indicated that, even where we directly simulated differences in egocentric discounting for each advisor, we were unable to detect the effects we were looking for (Figure 9.8; $t(26) = 2.87$, $p = .008$, $d = 0.53$, BF = 5.62; $M_{\text{ConsistentIndividual}} = 0.21$ [0.14, 0.29], $M_{\text{GroupMember}} = 0.13$ [0.08, 0.18]).

Discussion

The manipulation at the centre of this experiment did not work. We presented participants with the same advice, labelled either as always coming from the same advisor or as coming from a different advisor on each trial. We had expected that presenting participants with a single advisor would allow them to learn about that advisor's ability to perform the task, and how well their confidence was calibrated (i.e. how strong the relationship between increased confidence and increased probability of being correct was). We did not observe differences in the influence of advice as a function of the presentation of the advisor. Furthermore, in a simulation that altered the participants' final decisions to implement a difference in the extent to which advice was discounted between advisors. The simulation also failed to detect the effects we were looking for.

The two reasons why we failed to detect the effects, even in simulation, are that the manipulation was not powerful enough or that there is no systematic relationship between calibration knowledge (and familiarity more generally) and advice influence. The manipulation may not have been strong enough because participants may have paid relatively little attention to the identity of the advisor and focused more on the content of the advice. Participants may also have learned about the calibration distributions just the same for individual and group advisors: statistically the experience is the same and some research has suggested that statistical mechanisms such as associative learning may be sufficient to explain social learning phenomena (Behrens et al. 2008; FeldmanHall and Dunsmoor 2019). It is also possible, though we suspect less likely given prior evidence of sensitivity to calibration (Pescetelli and Yeung 2021), that the participants were simply unable to distinguish the advisors even in principle, or that any distinction they did make failed to translate into differential influence.

9.3.2 Experiment 7: Identifiability of advice

The results of Experiment 6 were not promising. Consequently, we entirely re-designed the manipulation in an attempt to place the calibration of the advice in the forefront of participants' experience. To this end, we familiarised participants with two advisors whose accuracy, probability of agreement, and confidence calibration were identical, but whose use of the confidence scale differed. A Low confidence advisor used the bottom two-thirds of the confidence scale, while a High confidence advisor used the top two-thirds.

On the test trials, participants frequently saw advice where the confidence lay in the middle third of the scale. On some of these trials, the advice was labelled as coming from a specific advisor. If the advice came from the Low confidence advisor it would represent high confidence advice, and imply a high probability that it was correct. If the advice came from the High confidence advisor, it would represent low confidence advice, and imply a low probability that it was correct. On the remainder of these trials, the advice was labelled as coming from an unknown one of those two advisors. We expected that advice that was labelled in this way would be treated differently relative to advice that was identifiable because it was impossible for participants to appropriately interpret the confidence (and therefore probability of correctness) of the advice.

Open scholarship practices

This experiment was not preregistered because it was highly exploratory. The experiment data are available in the `esmData::` package for R from !TODO[Zenodo esmData], and also directly from !TODO[OSFify Dates task data]. A snapshot of the state of the code for running the experiment at the time the experiment was run can be obtained from `https://github.com/oxacclab/ExploringSocialMetacognition/blob/9ac26116e348f8c4ff5617a8e68710b889bc7e08/ACBin/ck.html`.

Method

Procedure 48 participants each completed 67 trials over 5 blocks of the binary version of the Dates task§3.1.3. The procedure for filling in responses and the way the advice was represented was the same as described for Experiment 6§9.3.1.

Participants started with 1 block of 10 trials that contained no advice to allow them to familiarise themselves with the task. All trials in this section included feedback for all participants indicating whether or not the participant's response was correct.

Participants then did 9 trials with a practice advisor to get used to receiving advice. They also received feedback on these trials. They were informed that

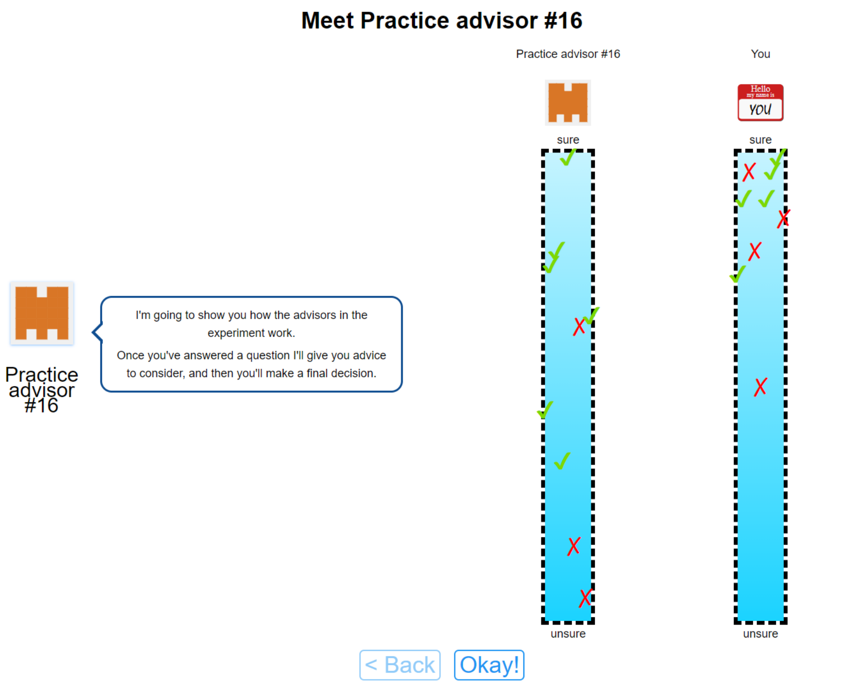

Figure 9.9: Advice scorecard for Experiment 7.
The scorecards indicated the relationship between confidence and accuracy, allowing participants to see how well calibrated they or an advisor was.

they would "get advice about the correct answer" and that the "advisors indicate their confidence in a similar way to you, by placing their marker higher on the scale if they are more sure".

Before performing trials with an advisor, participants saw the advisor's avatar alongside a speech bubble in which the advisor introduced themself. On the same screen, participants saw a scorecard for that advisor (Figure 9.9). Scorecards were introduced to the participant after the initial practice (without advice).

Participants then performed 3 blocks of trials that constituted the main experiment. In the first 2 blocks participants were familiarised with their advisors over 10 trials each (1 of which was an attention check). The order in which participants encountered the advisors was counterbalanced, and participants received feedback during these Familiarization trials.

Finally, participants performed 1 block of 28 Test trials that did not include feedback about the correct answer. Before this block, participants were reminded again about their advisors, and shown the introduction screen with the scorecards

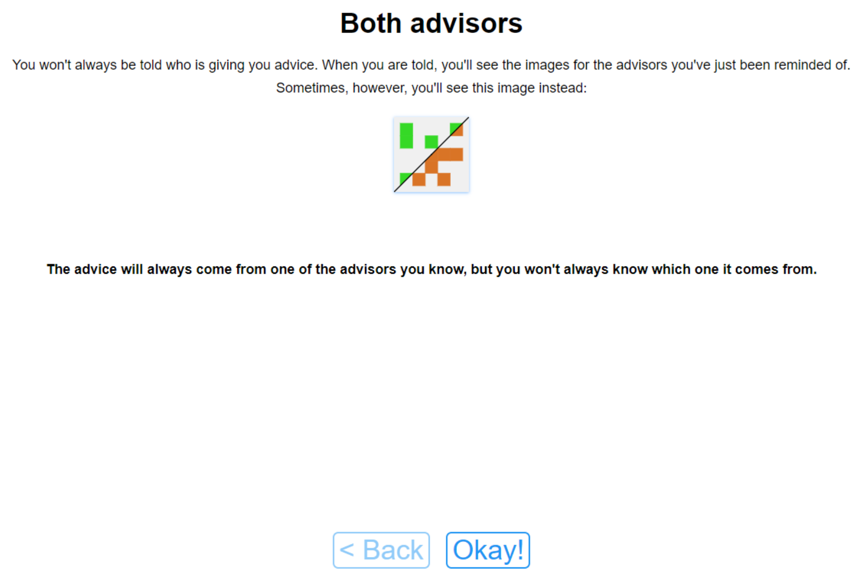

Figure 9.10: Hybrid advisor presentation.
The Hybrid advisor presentation was designed to indicate that one or other of the previously-encountered advisors would be giving advice, but that the participant did not know which. This was done by dividing the image diagonally and drawing half of one avatar in the top-left area and half of the other avatar in the bottom-right area.

for each advisor again. The scorecards were shown individually, and then displayed on the same screen next to one another, and the participants invited to "compare them with one another". During these Test trials, advice was sometimes labelled as coming from a specific one of the advisors encountered during the Familiarization trials, and sometimes the advice was labelled as coming from an unspecified one of those advisors (Figure 9.10).

Advice profiles There are two advisors in this experiment, a High confidence advisor and a Low confidence advisor. Before the Familiarization block with each advisor, the advisors introduced themselves with a statement about their advice-giving style. The High confidence advisor stated "I'm confident and outgoing. I tend to make up my mind and stick to it." The Low confidence advisor stated "I'm

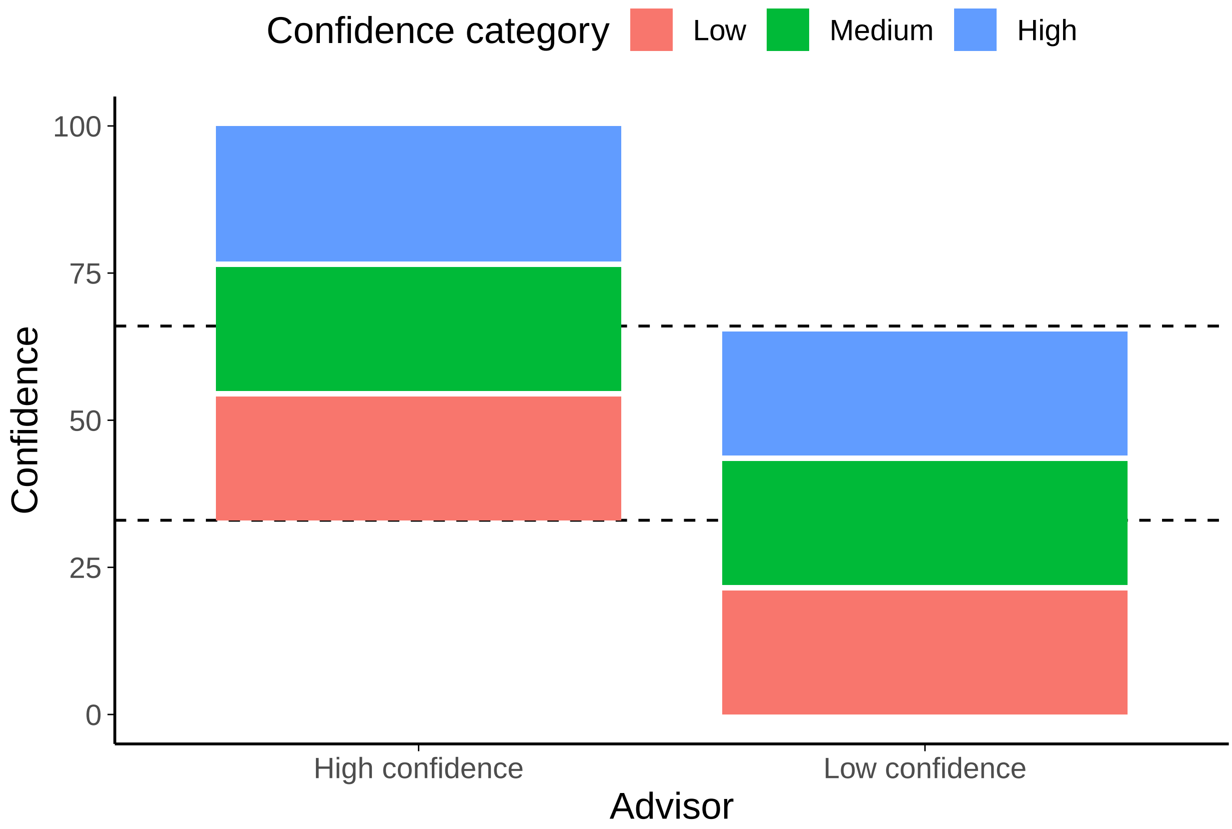

Figure 9.11: Advice distribution map for advisors in Experiment 7.
The High confidence equates to 100% correct, Medium to 66%, and Low to 33%. The dashed lines show the overlap area between the minimum confidence for the High confidence advisor and the maximum confidence for the Low confidence advisor. Note that the advisors are not equally correct on average in this zone: the Low confidence advisor is substantially more accurate because they are idiosyncratically more confidence.

cautious and careful. I try to keep an open mind even when I've made my choice."

The Advice profiles (Figure 9.11) for the two advisors were balanced in terms of the information provided by each advisor: both advisors had a subjectively high confidence at which they were 100% accurate, a medium confidence range within which they were 66% accurate, and a low accuracy range in which they were 33% accurate. Crucially, these ranges span different parts of the confidence scale; the Low confidence advisor never expressed confidence above 66%, and the High confidence advisor never expressed confidence below 33%.

The advisor's advice during the Familiarization phase was choreographed to provide all participants with an exact insight into the underlying calibration. Each advisor offered three estimates in each of their three confidence zones. All of the high-confidence advice was correct, two of the medium-confidence, and one of the low-confidence.

During the Test phase, advisors provided advice while either labelled unambiguously, as they had been during the Familiarization phase, with their avatar and advisor number, or ambiguously, with a hybrid avatar and question marks instead of the advisor number (Figure 9.10). Each advisor offered 7 pieces of advice while labelled unambiguously, and 7 while labelled ambiguously. Each of these sets of 7 advice had the following structure: three trials in the third of the confidence scale unique to the advisor that were correct, two trials in the ambiguous third of the confidence scale where the advisor agreed with the participant's initial estimate, and two trials in the ambiguous third where the advisor disagreed with the participant's initial estimate.

The decision to include advice outside the ambiguous third of the confidence scale (between the dashed lines in Figure 9.11)) was taken with the aim of ensuring participants' experience of the advice during the Test phase felt natural. Unfortunately, this meant that the advisors either had to degrade their calibration or they had to have different accuracy rates (either overall or within the ambiguous third of the confidence scale). Between these two lemmas we opted for the former, meaning that during the Test phase the Low confidence advisor's calibration changed such that all low-confidence advice was accurate.

The most important feature of this complex advice profile construction is that there are an equal number of trials that have advice with confidence in the ambiguous range for each advisor and for each presentation (ambiguous or identifiable). Within these trials, agreement is also tightly controlled meaning that each advisor-presentation combination agrees on half the trials and disagrees on half the trials. These 16 Key trials are central to testing our hypothesis that participants will be less influenced by advice when the source is ambiguous, so it is important that they are as balanced as possible for agreement and number.

Table 9.3: Participant exclusions for Experiment 5

Reason	Participants excluded
Multiple attempts	2
Not enough changes	13
Study incomplete	12
Too many outlying trials	0
Total excluded	**15**
Total remaining	**33**

Results

Exclusions Participants (total n = 48) were excluded for a number of reasons: having fewer than 16 Key trials which took less than 60s to complete, or providing final decisions which were the same as the initial estimate on more than 90% of trials. Table 9.3 shows the number of participants excluded and the reasons why.[1] Participants who failed attention checks do not reach this stage of the experiment.

The final participant list consists of 33 participants who completed an average of 45.91 trials each. Many participants were excluded using our typical exclusion criteria, but the results of the main statistical tests are the same when participants who seldom altered their responses are left in the sample, even where they did not complete the entire study.

Task performance

```
## Interaction calculation: Final decision - Initial estimate
```

```
## Interaction calculation: Final decision - Initial estimate
```

Participants increased their accuracy (Figure 9.12A) from initial estimate to final decision (F(1,32) = 29.53, p = <.001; $M_{FinalDecision}$ = 0.75 [0.71, 0.79], $M_{InitialEstimate}$ = 0.66 [0.60, 0.71]). There was no evidence of a difference between the advisors in either overall accuracy (F(1,32) = 0.09, p = .770; $M_{HighConfidence}$ = 0.70 [0.65, 0.75], $M_{LowConfidence}$ = 0.71 [0.65, 0.76]) or in the increase in accuracy following advice (F(1,32) = 0.28, p = .600; $M_{Improvement|HighConfidence}$ = 0.08 [0.03, 0.13], $M_{Improvement|LowConfidence}$ = 0.10 [0.06, 0.15]).

[1] Note that no participants were excluded only for failing to finish the study.

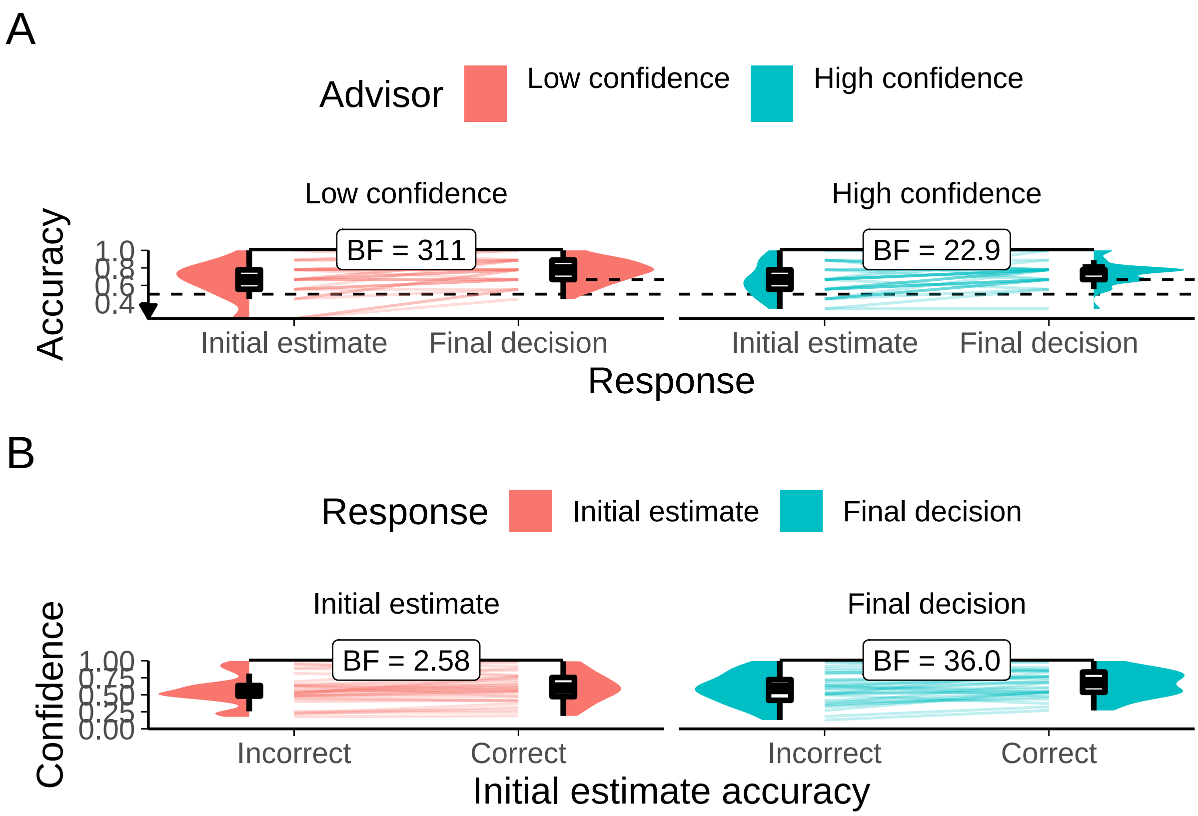

Figure 9.12: Task performance for Experiment 7.
A: Response accuracy. Faint lines show individual participant means, for which the violin and box plots show the distributions. The full width dashed line indicates chance performance, and the half width dashed lines indicate advisor accuracy. Because there were relatively few trials, the proportion of correct trials for a participant generally falls on one of a few specific values. This produces the lattice-like effect seen in the graph. Some participants had individual trials excluded for over-long response times, meaning that the denominator in the accuracy calculations is different, and thus producing accuracy values which are slightly offset from others'.
B: Confidence. Faint lines show individual participant means, for which the violin and box plots show the distributions.

Participants were more confident (Figure 9.12B) in their responses when their initial estimate was correct (F(1,32) = 12.22, p = .001; M_{Correct} = 0.63 [0.57, 0.70], $M_{\text{Incorrect}}$ = 0.58 [0.50, 0.65]). There was no evidence that they were systematically more confident on final decisions compared to initial estimates (F(1,32) = 3.12, p = .087; $M_{\text{FinalDecision}}$ = 0.63 [0.56, 0.70], $M_{\text{InitialEstimate}}$ = 0.58 [0.51, 0.66]), or that confidence change between decisions differed as a function of whether or not the initial estimate was correct (F(1,32) = 3.95, p = .055; $M_{\text{Increase|Correct}}$ = 0.06 [0.01,

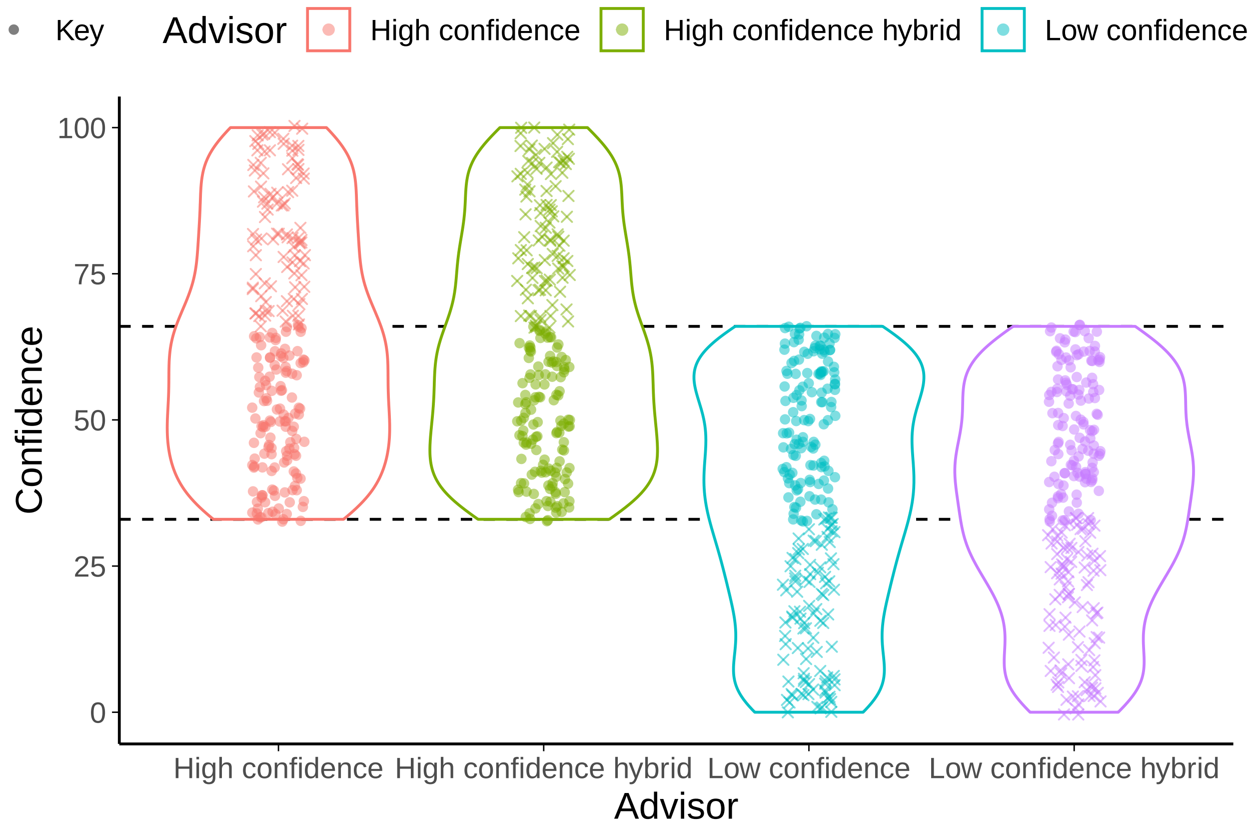

Figure 9.13: Key trials in Experiment 7.
Trials were matched for agreement and confidence across advisor presentation and advisor identity. The Flavour trials were included to avoid the feeling of the task changing dramatically from the Familiarization blocks to the Test block. The Key trials occur where advice could potentially come from either advisor. If the advice came from the High confidence advisor it represented a low confidence response; it if came from the Low confidence advisor it represented a high confidence response. If the advisor came from an ambigous (hybrid) source, the participant would not know whether it represented a high or low confidence response.

0.11], $M_{Increase|Incorrect}$ = 0.03 [-0.03, 0.08]). This pattern of participants' accuracy improving while their confidence does not may indicate a lack of metacognitive awareness of their own performance. This is a little strange given the number of trials in this experiment that included feedback, and the frequent inclusion of scorecards that indicated to participants how well they had been performing on recent trials.

Effect of advisor presentation

```
## Interaction calculation: Unambiguous - Hybrid
```

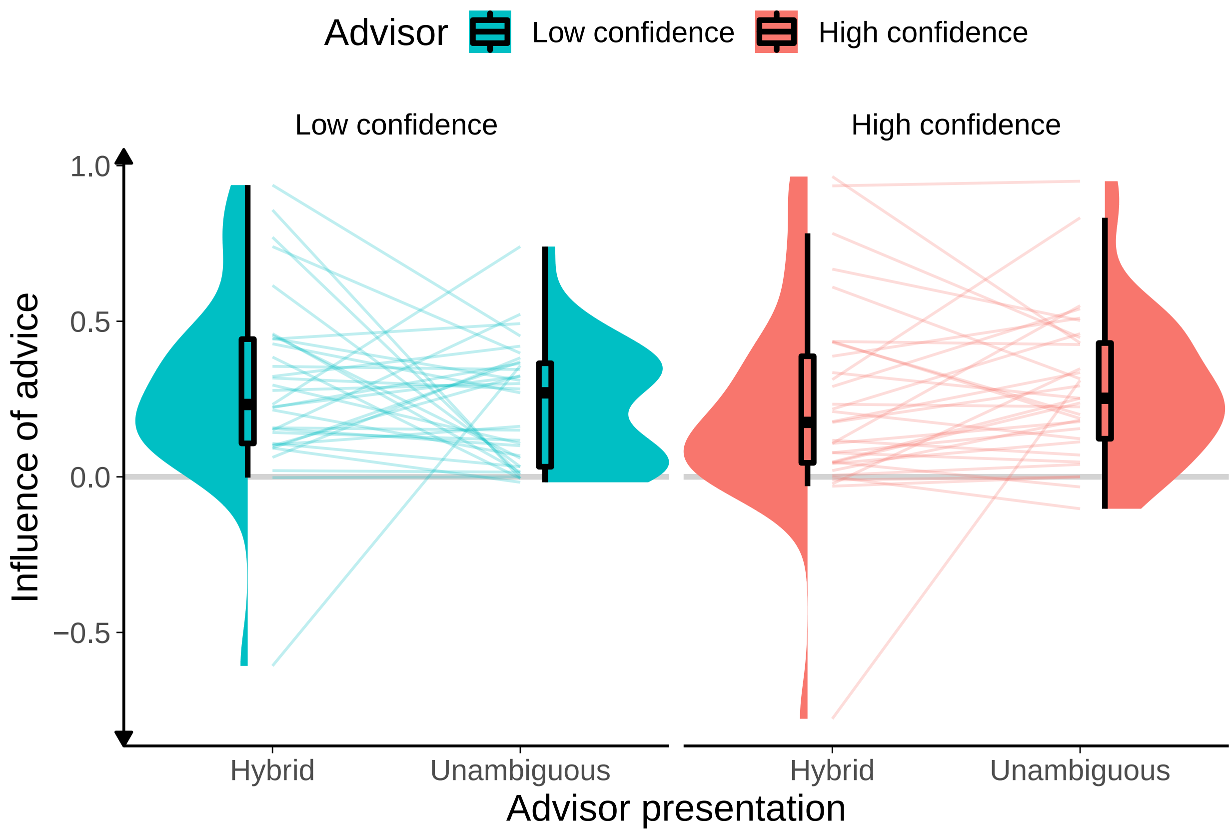

Figure 9.14: Advice influence in Experiment 7.
Participants' weight on the advice for advisors in the Test stage of the experiment. The shaded area and boxplots indicate the distribution of the individual participants' mean influence of advice. Individual means for each participant are shown with lines in the centre of the graph. The theoretical range for influence values is [-2, 2].

The main hypothesis in this experiment is that advice will be treated differently if it comes from an identifiable as opposed to an ambiguous source. Key trials occur in that band of the We thus expect that the High confidence advisor's advice will be more influential when the source is ambiguous, because participants who know that the advice comes from the High confidence advisor will know that the confidence score represents that advisor's low confidence advice. The opposite prediction is made for the Low confidence advisor.

These predictions were not borne out in the data. A 2x2 ANOVA of Advisor by presentation indicated no significant effects. Furthermore, the numerical direction of effects was against our prediction. As expected, there was no main effect to indicate that Hybrid presentation would be more or less influential than Unambiguous presentation ($F(1,32) = 0.00$, $p = .961$; $M_{\text{Unambiguous}} = 0.26$ [0.19, 0.32], $M_{\text{Hybrid}} = 0.26$ [0.16, 0.36]), and no main effect of advisor to indicate that, absent Unambiguous presentation, one advisor was more influential than the other ($F(1,32) = 0.02$, $p = .896$; $M_{\text{LowConfidence}} = 0.26$ [0.20, 0.32], $M_{\text{HighConfidence}} = 0.26$ [0.17, 0.34]). The interaction is not significant ($F(1,32) = 3.91$, $p = .057$; $M_{\text{Unambiguous-Hybrid|LowConfidence}} = -0.06$ [-0.19, 0.06], $M_{\text{Unambiguous-Hybrid|HighConfidence}} = 0.06$ [-0.04, 0.16]), and it goes in the opposite direction numerically to the one we would expect. The Unambiguous presentation of the Low confidence advisor is *less* influential, and the Unambiguous presentation of the High confidence advisor *more* influential than their Hybrid presentation counterparts.

Predictors of advice influence It is quite possible that participants failed to adequately grasp the calibration of the advisors, despite the lengths that we went to in order to highlight the importance of this feature. Indeed, a paired T-test of the influence of the advisors in Key trials where they were presented Unambiguously indicated that the High confidence advisor was numerically *more* influential than the Low confidence advisor, despite that advisor offering less useful advice. Statistically, there was not enough evidence to conclude whether or not the effect was present or absent.

This suggested that participants were forming a simple attachment to the more confident advisor, perhaps following a heuristic that the more confident advisor would give better advice. Indeed, linear mixed modelling indicated that, when predicting the influence on Key trials from the advisor, presentation, advisor-by-presentation interaction, and advice confidence (with random intercepts for participant), only advice confidence was a significant predictor (β=0.01 [0.01, 0.02], df=509.64, p=<.001, BF=2.5e4). The other main predictors were both demonstrably absent ($\beta_{\text{High confidence advisor}}$=-0.05 [-0.18, 0.07], df=490.18, p=.390, BF=1/5.11; $\beta_{\text{Unambiguous presentation}}$=-0.08 [-0.20, 0.05], df=490.11, p=.228, BF=1/5.11), and there was not sufficient evidence to adjudicate on the presence of an interaction ($\beta_{\text{High confidence advisor, Unambiguous presentation}}$=0.13 [-0.04, 0.31], df=490.14, p=.142, BF=1/1.86).

Discussion

We attempted to manipulate participants' ability to use their knowledge of advisors' idiosyncratic confidence expression to perform our Dates estimation task. We were unable to demonstrate that participants did so, and it seemed as though participants paid more attention to the advice than its source. While the effects were not statistically significant, they did trend in the opposite direction to that hypothesised, and it appeared as if participants had developed a mild positive association between the more confident advisor and more confident advice, treating that advisor as more influential even when they expressed a moderate level of confidence.

There are several features of this experiment that may have led to the underwhelming conclusion. First, the advisors' gravatar images may be harder for participants to differentiate than human faces. Second, without offering an incentive beyond the innate appeal of the task, participants may not have been motivated to absorb the complex confidence calibration information presented to them, especially where the time cost to do so is not offset by performance-based payment. Third, the somewhat artificial nature of the task may limit participants' interest in the character of the advisors. Adding additional information beyond the simple introductory speech bubble may have helped to alleviate this issue.

Whether or not additional appeal and character for the advisors would improve participants' sensitivity to the advisors' use of the confidence scale, these results cast doubt on a simple mechanistic implementation of our model of advice-taking.

9.4 General discussion

The results of the direct benevolence manipulation in Experiment 5 show that people are sensitive to the motivations of their advisors, and exercise appropriate caution where those motivations may mean the advice is misleading. They also demonstrate that, even where advice is considered trustworthy, it is discounted when coming from a less trusted advisor. Together with existing literature on the effects of advisor expertise, this supports the idea from the models that advice-taking can be flexibly adjusted according to the context.

The results of the confidence mapping experiments were less conclusive. We were unable to produce a manipulation which was sufficiently clear and strong to produce observable effects. It is plausible that people are sensitive to their knowledge of an advisor's confidence mapping, but it is also plausible that this degree of flexibility is beyond most people, and that simple heuristics such as relying on cultural norms about the meanings of metacognitive terms (**roseanoCommunicatingEpistemicStance2016**; Dhami and Wallsten 2005; but see Wallsten et al. 1986) are used instead of complex calculations. The lack of flexibility on the timescale of a behavioural experiment does not, of course, negate the possibility of flexibility on the timescale of a human interpersonal relationship (perhaps there are a certain number of people whose confidence mappings we can track, as suggested by Robin Dunbar's Social Brain Hypothesis (Dunbar 2010)[2]). Nor does it negate the possibility of a genetic or cultural evolutionary mechanism baking in advice discounting as a protection against unknown confidence mapping, although it is questionable whether advice has been occurring in human societies for long enough to allow the former to take place. We have a plausible explanation of egocentric which only relies on rational responses to environmental effects. These effects are ubiquitous, and thus a ubiquitous egocentric bias prior to engaging with the specifics of a situation makes sense. It is entirely plausible that experiments showing egocentric discounting behaviour fail to overcome the hyperpriors held by participants that taking advice is risky for various reasons.

We saw in the previous section that people are sensitive to the quality of advice and advisors, and will curate their information environments based on that

[2]Note that this hypothesis has been viewed increasingly sceptically in recent years, as neatly covered by Lindenfors et al. (2021)

sensitivity. We have seen in this section how the properties of the information environment can change people's sensitivity to advice.

There is one major feature of advice-taking that has received relatively little investigation, yet that would, if real, be entirely inexplicable in terms of evolutionarily optimal hyperpriors. This is the finding that people tend to consider all advisory estimates they receive as a single, multifaceted opinion, which is then integrated with their own individual opinion§7.2.2. Crucially, this seems to happen no matter how many advisory estimates are supplied (Hütter and Ache 2016; Yonah and Kessler 2021). A rational approach to egocentric discounting might attempt, as we have above, to explain how it is sensible to afford less weight to another's opinion compared to one's own, but it cannot explain how it is sensible to afford the same weight to one's opinion whether integrating it with the opinions of one advisor or of several advisors.

10
Conclusion

I haven't wasted 3 years of my life and Nick's time.

- I've demonstrated that the structure of information networks may be partially dependent upon psychological mechanisms of advisor evaluation.
- I've provided some evidence for a model of advisor evaluation in the absence of feedback and demonstrated how that model produces adverse network effects.
  - I've shown how some of these effects depend upon heterogeneity of agents' preferences/tendencies.
- I've shown how the information environment can affect advice taking, and I've provided and reviewed empirical evidence that humans can alter advice-taking flexibly depending on these contextual factors.
- I've modelled how the psychological mechanisms create an information network structure, and how that information network structure might use the flexibility to context to change advisor evaluation processes.
- I've demonstrated how those changes might attenuate or exacerbate the formation of pathological network structures.

10.1 Open questions

- People's social networks constrain whom they can receive advice from. Within these constraints, source selection processes operate to evaluate the relative (and absolute?) trustworthiness of each (or the most salient?) members. Is trustworthiness (or at least the ability component) modelled discretely for different domains, or is it unitary?
  - Perhaps a matter of association with a particular concept?
  - Quite possible associative processes drive source selection rather than direct comparisons.
- Relative magnitude of advice from selected individuals in a social network vs. adverts/web searches/consulting experts.
- Frequency of advice-seeking as characterised in the thesis.
- (Why) do people lump group advice together before discounting - this seems idiotic even with the adapativeness of discounting...

!TODO[If we have time, it'd be neat to see if we can get people's selection rates for advisors to recover after the advisors have been lying to them, showing that that how we curate information environment (source selection) also depends on the existing information environment properties.]

Appendices

A
The First Appendix

This first appendix includes an R chunk that was hidden in the document (using `echo = FALSE`) to help with readibility:

In 02-rmd-basics-code.Rmd

And here's another one from the same chapter, i.e. Chapter ??:

B
The Second Appendix, for Fun

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